Constructions With Sets

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June 18, 2025

This chapter develops some material relating to constructions with sets with an eye towards its categorical and higher-categorical counterparts to be introduced later in this work. Of particular interest are perhaps the following:

1. Explicit descriptions of the major types of co/limits in Sets, including in particular explicit descriptions of pushouts and coequalisers (see Definitions 2.4.1 and 2.5.1 and Remarks 2.4.4 and 2.5.4).

2. A discussion of powersets as decategorifications of categories of presheaves, including in particular results such as:

   (a) A discussion of the internal Hom of a powerset (Section 4.7).

   (b) A 0-categorical version of the Yoneda lemma (Presheaves and the Yoneda Lemma, **??**), which we term the *Yoneda lemma for sets* (Proposition 5.5.1).

   (c) A characterisation of powersets as free cocompletions (Section 4.5), mimicking the corresponding statement for categories of presheaves (**??**).

   (d) A characterisation of powersets as free completions (Section 4.6), mimicking the corresponding statement for categories of copresheaves (**??**).

   (e) A (-1)-categorical version of un/straightening (Item 2 of Proposition 5.1.4 and Remark 5.1.6).

   (f) A 0-categorical form of Isbell duality internal to powersets (Section 4.8).

3. A lengthy discussion of the adjoint triple

$$f_* \dashv f^{-1} \dashv f_! \colon \mathcal{P}(A) \overset{\rightleftarrows}{\to} \mathcal{P}(B)$$

of functors (i.e. morphisms of posets) between $\mathcal{P}(A)$ and $\mathcal{P}(B)$ induced by a map of sets $f\colon A \to B$, including in particular:

(a) How f^{-1} can be described as a precomposition while f_* and $f_!$ can be described as Kan extensions (Remarks 6.1.3, 6.2.2 and 6.3.3).

(b) An extensive list of the properties of f_*, f^{-1}, and $f_!$ (Propositions 6.1.4, 6.1.6, 6.2.3, 6.2.5, 6.3.6 and 6.3.8).

(c) How the functors f_*, f^{-1}, $f_!$, along with the functors

$$\begin{aligned} -_1 \cap -_2 &\colon \mathcal{P}(X) \times \mathcal{P}(X) \to \mathcal{P}(X), \\ [-_1, -_2]_X &\colon \mathcal{P}(X)^{\mathsf{op}} \times \mathcal{P}(X) \to \mathcal{P}(X) \end{aligned}$$

may be viewed as a six-functor formalism with the empty set $\emptyset$ as the dualising object (Section 6.4).

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1 Limits of Sets

1.1 The Terminal Set

Definition 1.1.1 ▶ The Terminal Set

The **terminal set** is the terminal object of Sets as in Limits and Colimits, **??**.

Construction 1.1.2 ▶ Construction of the Terminal Set

Concretely, the terminal set is the pair $\left(\mathrm{pt}, \{!_A\}_{A \in \mathrm{Obj}(\mathsf{Sets})}\right)$ consisting of:

1. *The Limit.* The punctual set $\mathrm{pt} \overset{\text{def}}{=} \{\star\}$.
2. *The Cone.* The collection of maps

$$\{!_A \colon A \longrightarrow \mathrm{pt}\}_{A \in \mathrm{Obj}(\mathsf{Sets})}$$

defined by

$$!_A(a) \overset{\text{def}}{=} \star$$

for each $a \in A$ and each $A \in \mathrm{Obj}(\mathsf{Sets})$.

Proof 1.1.3 ▶ Proof of Construction 1.1.2

We claim that pt is the terminal object of Sets. Indeed, suppose we have a diagram of the form

$$A \qquad \mathrm{pt}$$

in Sets. Then there exists a unique map $\phi \colon A \longrightarrow \mathrm{pt}$ making the diagram

$$A \xrightarrow[\exists!]{\phi} \mathrm{pt}$$

commute, namely $!_A$. ∎

1.2 Products of Families of Sets

Let $\{A_i\}_{i \in I}$ be a family of sets.

Definition 1.2.1 ▶ The Product of a Family of Sets

The **product**[1] **of** $\{A_i\}_{i\in I}$ is the product of $\{A_i\}_{i\in I}$ in Sets as in Limits and Colimits, **??**.

[1] *Further Terminology:* Also called the **Cartesian product of** $\{A_i\}_{i\in I}$.

Construction 1.2.2 ▶ Construction of the Product of a Family of Sets

Concretely, the product of $\{A_i\}_{i\in I}$ is the pair $\left(\prod_{i\in I} A_i, \{\mathrm{pr}_i\}_{i\in I}\right)$ consisting of:

1. *The Limit.* The set $\prod_{i\in I} A_i$ defined by

$$\prod_{i\in I} A_i \stackrel{\text{def}}{=} \left\{ f \in \mathrm{Sets}\left(I, \bigcup_{i\in I} A_i\right) \middle| \begin{array}{l}\text{for each } i \in I, \text{ we}\\ \text{have } f(i) \in A_i\end{array} \right\}.$$

2. *The Cone.* The collection

$$\left\{ \mathrm{pr}_i \colon \prod_{i\in I} A_i \to A_i \right\}_{i\in I}$$

of maps given by

$$\mathrm{pr}_i(f) \stackrel{\text{def}}{=} f(i)$$

for each $f \in \prod_{i\in I} A_i$ and each $i \in I$.

Proof 1.2.3 ▶ Proof of Construction 1.2.2

We claim that $\prod_{i\in I} A_i$ is the categorical product of $\{A_i\}_{i\in I}$ in Sets. Indeed, suppose we have, for each $i \in I$, a diagram of the form

$$\begin{tikzcd} P \arrow[dr, "p_i"] & \\ \prod_{i\in I} A_i \arrow[r, "\mathrm{pr}_i"'] & A_i \end{tikzcd}$$

in Sets. Then there exists a unique map $\phi\colon P \longrightarrow \prod_{i\in I} A_i$ making the diagram

$$\begin{tikzcd} P \arrow[d, dashed, "\phi"', "\exists!"] \arrow[dr, "p_i"] & \\ \prod_{i\in I} A_i \arrow[r, "\mathrm{pr}_i"'] & A_i \end{tikzcd}$$

commute, being uniquely determined by the condition $\mathrm{pr}_i \circ \phi = p_i$ for each $i \in I$ via

$$\phi(x) = (p_i(x))_{i\in I}$$

for each $x \in P$.

Remark 1.2.4 ▶ Unwinding Construction 1.2.2

Less formally, we may think of Cartesian products and projection maps as follows:

1. We think of $\prod_{i\in I} A_i$ as the set whose elements are I-indexed collections $(a_i)_{i\in I}$ with $a_i \in A_i$ for each $i \in I$.

2. We view the projection maps

$$\left\{\mathrm{pr}_i\colon \prod_{i\in I} A_i \longrightarrow A_i\right\}_{i\in I}$$

as being given by

$$\mathrm{pr}_i\Big(\big(a_j\big)_{j\in I}\Big) \overset{\text{def}}{=} a_i$$

for each $\big(a_j\big)_{j\in I} \in \prod_{i\in I} A_i$ and each $i \in I$.

Proposition 1.2.5 ▶ Properties of Products of Families of Sets

Let $\{A_i\}_{i\in I}$ be a family of sets.

1. *Functoriality.* The assignment $\{A_i\}_{i\in I} \mapsto \prod_{i\in I} A_i$ defines a functor

$$\prod_{i\in I} \colon \mathsf{Fun}(I_{\mathsf{disc}}, \mathsf{Sets}) \to \mathsf{Sets}$$

where

- *Action on Objects.* For each $(A_i)_{i\in I} \in \mathsf{Obj}(\mathsf{Fun}(I_{\mathsf{disc}}, \mathsf{Sets}))$, we have

$$\left[\prod_{i\in I}\right]((A_i)_{i\in I}) \overset{\mathsf{def}}{=} \prod_{i\in I} A_i$$

- *Action on Morphisms.* For each $(A_i)_{i\in I}, (B_i)_{i\in I} \in \mathsf{Obj}(\mathsf{Fun}(I_{\mathsf{disc}}, \mathsf{Sets}))$, the action on Hom-sets

$$\left(\prod_{i\in I}\right)_{(A_i)_{i\in I},(B_i)_{i\in I}} \colon \mathsf{Nat}((A_i)_{i\in I}, (B_i)_{i\in I}) \to \mathsf{Sets}\left(\prod_{i\in I} A_i, \prod_{i\in I} B_i\right)$$

of $\prod_{i\in I}$ at $((A_i)_{i\in I}, (B_i)_{i\in I})$ is defined by sending a map

$$\{f_i \colon A_i \to B_i\}_{i\in I}$$

in $\mathsf{Nat}((A_i)_{i\in I}, (B_i)_{i\in I})$ to the map of sets

$$\prod_{i\in I} f_i \colon \prod_{i\in I} A_i \to \prod_{i\in I} B_i$$

defined by

$$\left[\prod_{i\in I} f_i\right]((a_i)_{i\in I}) \overset{\mathsf{def}}{=} (f_i(a_i))_{i\in I}$$

for each $(a_i)_{i\in I} \in \prod_{i\in I} A_i$.

Proof 1.2.6 ▶ Proof of Proposition 1.2.5

Item 1: Functoriality

This follows from Limits and Colimits, **??** of **??**.

1.3 Binary Products of Sets

Let A and B be sets.

Definition 1.3.1 ▶ Binary Products of Sets

The **product of A and B**[1] is the product of A and B in Sets as in Limits and Colimits, **??**.

[1] *Further Terminology:* Also called the **Cartesian product of A and B**.

Construction 1.3.2 ▶ Construction of Binary Products of Sets

Concretely, the product of A and B is the pair $(A \times B, \{\mathrm{pr}_1, \mathrm{pr}_2\})$ consisting of:

1. *The Limit.* The set $A \times B$ defined by

$$\begin{aligned} A \times B &\overset{\text{def}}{=} \prod_{z \in \{A,B\}} z \\ &\overset{\text{def}}{=} \{f \in \mathrm{Sets}(\{0,1\}, A \cup B) \mid \text{we have } f(0) \in A \text{ and } f(1) \in B\} \\ &\cong \{\{\{a\},\{a,b\}\} \in \mathcal{P}(\mathcal{P}(A \cup B)) \mid \text{we have } a \in A \text{ and } b \in B\} \\ &\cong \left\{\begin{matrix}\text{ordered pairs } (a,b) \text{ with} \\ a \in A \text{ and } b \in B\end{matrix}\right\}. \end{aligned}$$

2. *The Cone.* The maps

$$\begin{aligned} \mathrm{pr}_1 &\colon A \times B \to A, \\ \mathrm{pr}_2 &\colon A \times B \to B \end{aligned}$$

defined by

$$\mathrm{pr}_1(a,b) \overset{\text{def}}{=} a,$$

$$\mathrm{pr}_2(a, b) \overset{\text{def}}{=} b$$

for each $(a, b) \in A \times B$.

Proof 1.3.3 ▶ Proof of Construction 1.3.2

We claim that $A \times B$ is the categorical product of A and B in the category of sets. Indeed, suppose we have a diagram of the form

$$\begin{tikzcd} & P \arrow[dl, bend right, "p_1"'] \arrow[dr, bend left, "p_2"] & \\ A & A \times B \arrow[l, two heads, "\mathrm{pr}_1"] \arrow[r, two heads, "\mathrm{pr}_2"'] & B \end{tikzcd}$$

in Sets. Then there exists a unique map $\phi\colon P \to A \times B$ making the diagram

$$\begin{tikzcd} & P \arrow[dl, bend right, "p_1"'] \arrow[d, dashed, "\phi"', "\exists!"] \arrow[dr, bend left, "p_2"] & \\ A & A \times B \arrow[l, two heads, "\mathrm{pr}_1"] \arrow[r, two heads, "\mathrm{pr}_2"'] & B \end{tikzcd}$$

commute, being uniquely determined by the conditions

$$\begin{aligned} \mathrm{pr}_1 \circ \phi &= p_1, \\ \mathrm{pr}_2 \circ \phi &= p_2 \end{aligned}$$

via

$$\phi(x) = (p_1(x), p_2(x))$$

for each $x \in P$. ∎

Proposition 1.3.4 ▶ Properties of Products of Sets

Let A, B, C, and X be sets.

1. *Functoriality.* The assignments $A, B, (A, B) \mapsto A \times B$ define functors

$$\begin{aligned} A \times -\colon &\quad \mathsf{Sets} \to \mathsf{Sets}, \\ - \times B\colon &\quad \mathsf{Sets} \to \mathsf{Sets}, \\ -_1 \times -_2\colon &\quad \mathsf{Sets} \times \mathsf{Sets} \to \mathsf{Sets}, \end{aligned}$$

where $-_1 \times -_2$ is the functor where

- *Action on Objects.* For each $(A, B) \in \mathrm{Obj}(\mathsf{Sets} \times \mathsf{Sets})$, we have

$$[-_1 \times -_2](A, B) \stackrel{\text{def}}{=} A \times B.$$

- *Action on Morphisms.* For each $(A, B), (X, Y) \in \mathrm{Obj}(\mathsf{Sets})$, the action on Hom-sets

$$\times_{(A,B),(X,Y)}\colon \mathsf{Sets}(A, X) \times \mathsf{Sets}(B, Y) \to \mathsf{Sets}(A \times B, X \times Y)$$

of $\times$ at $((A, B), (X, Y))$ is defined by sending (f, g) to the function

$$f \times g\colon A \times B \to X \times Y$$

defined by

$$[f \times g](a, b) \stackrel{\text{def}}{=} (f(a), g(b))$$

for each $(a, b) \in A \times B$.

and where $A \times -$ and $- \times B$ are the partial functors of $-_1 \times -_2$ at $A, B \in \mathrm{Obj}(\mathsf{Sets})$.

2. *Adjointness.* We have adjunctions

$$(A \times - \dashv \mathsf{Sets}(A, -))\colon \quad \begin{tikzcd} \mathsf{Sets} \arrow[r, bend left, "A\times-"] \arrow[r, phantom, "\perp"] & \mathsf{Sets}, \arrow[l, bend left, "\mathsf{Sets}(A,-)"] \end{tikzcd}$$

$$(- \times B \dashv \mathsf{Sets}(B, -))\colon \quad \begin{tikzcd} \mathsf{Sets} \arrow[r, bend left, "-\times B"] \arrow[r, phantom, "\perp"] & \mathsf{Sets}, \arrow[l, bend left, "\mathsf{Sets}(B,-)"] \end{tikzcd}$$

witnessed by bijections

$$\begin{aligned} \mathsf{Sets}(A \times B, C) &\cong \mathsf{Sets}(A, \mathsf{Sets}(B, C)), \\ \mathsf{Sets}(A \times B, C) &\cong \mathsf{Sets}(B, \mathsf{Sets}(A, C)), \end{aligned}$$

natural in $A, B, C \in \mathrm{Obj}(\mathsf{Sets})$.

3. *Associativity.* We have an isomorphism of sets

$$\alpha^{\mathsf{Sets}}_{A,B,C} \colon (A \times B) \times C \overset{\sim}{\dashrightarrow} A \times (B \times C),$$

natural in $A, B, C \in \mathrm{Obj}(\mathsf{Sets})$.

4. *Unitality.* We have isomorphisms of sets

$$\lambda^{\mathsf{Sets}}_{A} \colon \mathrm{pt} \times A \overset{\sim}{\dashrightarrow} A,$$
$$\rho^{\mathsf{Sets}}_{A} \colon A \times \mathrm{pt} \overset{\sim}{\dashrightarrow} A,$$

natural in $A \in \mathrm{Obj}(\mathsf{Sets})$.

5. *Commutativity.* We have an isomorphism of sets

$$\sigma^{\mathsf{Sets}}_{A,B} \colon A \times B \overset{\sim}{\dashrightarrow} B \times A,$$

natural in $A, B \in \mathrm{Obj}(\mathsf{Sets})$.

6. *Distributivity Over Coproducts.* We have isomorphisms of sets

$$\delta^{\mathsf{Sets}}_{\ell} \colon A \times (B \coprod C) \overset{\sim}{\dashrightarrow} (A \times B) \coprod (A \times C),$$
$$\delta^{\mathsf{Sets}}_{r} \colon (A \coprod B) \times C \overset{\sim}{\dashrightarrow} (A \times C) \coprod (B \times C),$$

natural in $A, B, C \in \mathrm{Obj}(\mathsf{Sets})$.

7. *Annihilation With the Empty Set.* We have isomorphisms of sets

$$\zeta^{\mathsf{Sets}}_{\ell} \colon \emptyset \times A \overset{\sim}{\dashrightarrow} \emptyset,$$
$$\zeta^{\mathsf{Sets}}_{r} \colon A \times \emptyset \overset{\sim}{\dashrightarrow} \emptyset,$$

natural in $A \in \mathrm{Obj}(\mathsf{Sets})$.

8. *Distributivity Over Unions.* Let X be a set. For each $U, V, W \in \mathcal{P}(X)$, we have equalities

$$U \times (V \cup W) = (U \times V) \cup (U \times W),$$
$$(U \cup V) \times W = (U \times W) \cup (V \times W)$$

of subsets of $\mathcal{P}(X \times X)$.

9. *Distributivity Over Intersections.* Let X be a set. For each $U, V, W \in \mathcal{P}(X)$, we have equalities

$$\begin{aligned} U \times (V \cap W) &= (U \times V) \cap (U \times W), \\ (U \cap V) \times W &= (U \times W) \cap (V \times W) \end{aligned}$$

of subsets of $\mathcal{P}(X \times X)$.

10. *Distributivity Over Differences.* Let X be a set. For each $U, V, W \in \mathcal{P}(X)$, we have equalities

$$\begin{aligned} U \times (V \setminus W) &= (U \times V) \setminus (U \times W), \\ (U \setminus V) \times W &= (U \times W) \setminus (V \times W) \end{aligned}$$

of subsets of $\mathcal{P}(X \times X)$.

11. *Distributivity Over Symmetric Differences.* Let X be a set. For each $U, V, W \in \mathcal{P}(X)$, we have equalities

$$\begin{aligned} U \times (V \triangle W) &= (U \times V) \triangle (U \times W), \\ (U \triangle V) \times W &= (U \times W) \triangle (V \times W) \end{aligned}$$

of subsets of $\mathcal{P}(X \times X)$.

12. *Middle-Four Exchange with Respect to Intersections.* The diagram

$$\begin{tikzcd} (\mathcal{P}(X) \times \mathcal{P}(X)) \times (\mathcal{P}(X) \times \mathcal{P}(X)) \arrow[r, "\cap\times\cap"] \arrow[d, "\mathcal{P}^{\times}_{X,X} \times \mathcal{P}^{\times}_{X,X}"'] & \mathcal{P}(X) \times \mathcal{P}(X) \arrow[d, "\mathcal{P}^{\times}_{X,X}"] \\ \mathcal{P}(X \times X) \times \mathcal{P}(X \times X) \arrow[r, "\cap"'] & \mathcal{P}(X \times X) \end{tikzcd}$$

commutes, i.e. we have

$$(U \times V) \cap (W \times T) = (U \cap V) \times (W \cap T).$$

for each $U, V, W, T \in \mathcal{P}(X)$.

13. *Symmetric Monoidality.* The 8-tuple $\big(\mathsf{Sets}, \times, \mathrm{pt}, \mathsf{Sets}(-_1, -_2), \alpha^{\mathsf{Sets}}, \lambda^{\mathsf{Sets}}, \rho^{\mathsf{Sets}}, \sigma^{\mathsf{Sets}}\big)$ is a closed symmetric monoidal category.

14. *Symmetric Bimonoidality.* The 18-tuple

$$\Big(\mathsf{Sets}, \coprod, \times, \emptyset, \mathrm{pt}, \mathsf{Sets}(-_1, -_2), \alpha^{\mathsf{Sets}}, \lambda^{\mathsf{Sets}}, \rho^{\mathsf{Sets}}, \sigma^{\mathsf{Sets}},$$
$$\alpha^{\mathsf{Sets},\coprod}, \lambda^{\mathsf{Sets},\coprod}, \rho^{\mathsf{Sets},\coprod}, \sigma^{\mathsf{Sets},\coprod}, \delta_\ell^{\mathsf{Sets}}, \delta_r^{\mathsf{Sets}}, \zeta_\ell^{\mathsf{Sets}}, \zeta_r^{\mathsf{Sets}}\Big),$$

is a symmetric closed bimonoidal category, where $\alpha^{\mathsf{Sets},\coprod}$, $\lambda^{\mathsf{Sets},\coprod}$, $\rho^{\mathsf{Sets},\coprod}$, and $\sigma^{\mathsf{Sets},\coprod}$ are the natural transformations from Items 2 to 4 of Proposition 2.3.4.

Proof 1.3.5 ▶ Proof of Proposition 1.3.4

Item 1: Functoriality

This follows from Limits and Colimits, **??** of **??**.

Item 2: Adjointness

We prove only that there,s an adjunction $- \times B \dashv \mathsf{Sets}(B, -)$, witnessed by a bijection

$$\mathsf{Sets}(A \times B, C) \cong \mathsf{Sets}(A, \mathsf{Sets}(B, C)),$$

natural in $B, C \in \mathrm{Obj}(\mathsf{Sets})$, as the proof of the existence of the adjunction $A \times - \dashv \mathsf{Sets}(A, -)$ follows almost exactly in the same way.

- *Map I.* We define a map

$$\Phi_{B,C} \colon \mathsf{Sets}(A \times B, C) \longrightarrow \mathsf{Sets}(A, \mathsf{Sets}(B, C)),$$

by sending a function

$$\xi \colon A \times B \longrightarrow C$$

to the function

$$\begin{aligned} \xi^\dagger \colon A &\longrightarrow \mathsf{Sets}(B, C), \\ a &\longmapsto \left(\xi_a^\dagger \colon B \longrightarrow C\right), \end{aligned}$$

where we define

$$\xi_a^\dagger(b) \stackrel{\text{def}}{=} \xi(a,b)$$

for each $b \in B$. In terms of the $[\![a \mapsto f(a)]\!]$ notation of Sets, Notation 1.1.2, we have

$$\xi^\dagger \stackrel{\text{def}}{=} [\![a \mapsto [\![b \mapsto \xi(a,b)]\!]]\!].$$

- *Map II.* We define a map

$$\Psi_{B,C}\colon \mathsf{Sets}(A, \mathsf{Sets}(B,C)), \to \mathsf{Sets}(A \times B, C)$$

given by sending a function

$$\begin{aligned} \xi\colon A &\longrightarrow \mathsf{Sets}(B,C), \\ a &\mapsto (\xi_a\colon B \to C), \end{aligned}$$

to the function

$$\xi^\dagger\colon A \times B \to C$$

defined by

$$\begin{aligned} \xi^\dagger(a,b) &\stackrel{\text{def}}{=} \mathsf{ev}_b(\mathsf{ev}_a(\xi)) \\ &\stackrel{\text{def}}{=} \mathsf{ev}_b(\xi_a) \\ &\stackrel{\text{def}}{=} \xi_a(b) \end{aligned}$$

for each $(a,b) \in A \times B$.

- *Invertibility I.* We claim that

$$\Psi_{A,B} \circ \Phi_{A,B} = \mathsf{id}_{\mathsf{Sets}(A \times B, C)}.$$

Indeed, given a function $\xi\colon A \times B \to C$, we have

$$\begin{aligned} \left[\Psi_{A,B} \circ \Phi_{A,B}\right](\xi) &= \Psi_{A,B}\big(\Phi_{A,B}(\xi)\big) \\ &= \Psi_{A,B}\big(\Phi_{A,B}([\![(a,b) \mapsto \xi(a,b)]\!])\big) \\ &= \Psi_{A,B}([\![a \mapsto [\![b \mapsto \xi(a,b)]\!]]\!]) \end{aligned}$$

$$
\begin{aligned}
&= \Psi_{A,B}([\![a' \mapsto [\![b' \mapsto \xi(a', b')]\!]]\!]) \\
&= [\![(a, b) \mapsto \mathrm{ev}_b(\mathrm{ev}_a([\![a' \mapsto [\![b' \mapsto \xi(a', b')]\!]]\!]))]\!] \\
&= [\![(a, b) \mapsto \mathrm{ev}_b([\![b' \mapsto \xi(a, b')]\!])]\!] \\
&= [\![(a, b) \mapsto \xi(a, b)]\!] \\
&= \xi.
\end{aligned}
$$

- *Invertibility II.* We claim that

$$\Phi_{A,B} \circ \Psi_{A,B} = \mathrm{id}_{\mathsf{Sets}(A,\mathsf{Sets}(B,C))}.$$

Indeed, given a function

$$
\begin{aligned}
\xi \colon A &\longrightarrow \mathsf{Sets}(B, C), \\
a &\mapsto (\xi_a \colon B \to C),
\end{aligned}
$$

we have

$$
\begin{aligned}
\left[\Phi_{A,B} \circ \Psi_{A,B}\right](\xi) &\stackrel{\text{def}}{=} \Phi_{A,B}\big(\Psi_{A,B}(\xi)\big) \\
&\stackrel{\text{def}}{=} \Phi_{A,B}([\![(a, b) \mapsto \xi_a(b)]\!]) \\
&\stackrel{\text{def}}{=} \Phi_{A,B}([\![(a', b') \mapsto \xi_{a'}(b')]\!]) \\
&\stackrel{\text{def}}{=} [\![a \mapsto [\![b \mapsto \mathrm{ev}_{(a,b)}([\![(a', b') \mapsto \xi_{a'}(b')]\!])]\!]]\!] \\
&\stackrel{\text{def}}{=} [\![a \mapsto [\![b \mapsto \xi_a(b)]\!]]\!] \\
&\stackrel{\text{def}}{=} [\![a \mapsto \xi_a]\!] \\
&\stackrel{\text{def}}{=} \xi.
\end{aligned}
$$

- *Naturality for Φ, Part I.* We need to show that, given a function $g \colon B \to B'$, the diagram

$$
\begin{tikzcd}
\mathsf{Sets}(A \times B', C) \arrow[r, "\Phi_{B',C}"] \arrow[d, "\mathrm{id}_A \times g^*"'] & \mathsf{Sets}(A, \mathsf{Sets}(B', C)), \arrow[d, "(g^*)_*"] \\
\mathsf{Sets}(A \times B, C) \arrow[r, "\Phi_{B,C}"'] & \mathsf{Sets}(A, \mathsf{Sets}(B, C))
\end{tikzcd}
$$

commutes. Indeed, given a function

$$\xi\colon A \times B' \to C,$$

we have

$$\begin{aligned}
\left[\Phi_{B,C} \circ (\mathrm{id}_A \times g^*)\right](\xi) &= \Phi_{B,C}([\mathrm{id}_A \times g^*](\xi)) \\
&= \Phi_{B,C}(\xi(-_1, g(-_2))) \\
&= [\xi(-_1, g(-_2))]^\dagger \\
&= \xi^\dagger_{-_1}(g(-_2)) \\
&= (g^*)_*\left(\xi^\dagger\right) \\
&= (g^*)_*\left(\Phi_{B',C}(\xi)\right) \\
&= \left[(g^*)_* \circ \Phi_{B',C}\right](\xi).
\end{aligned}$$

Alternatively, using the $[\![a \mapsto f(a)]\!]$ notation of Sets, Notation 1.1.2, we have

$$\begin{aligned}
\left[\Phi_{B,C} \circ (\mathrm{id}_A \times g^*)\right](\xi) &= \Phi_{B,C}([\mathrm{id}_A \times g^*](\xi)) \\
&= \Phi_{B,C}([\mathrm{id}_A \times g^*]([\![(a,b') \mapsto \xi(a,b')]\!])) \\
&= \Phi_{B,C}([\![(a,b) \mapsto \xi(a,g(b))]\!]) \\
&= [\![a \mapsto [\![b \mapsto \xi(a,g(b))]\!]]\!] \\
&= [\![a \mapsto g^*([\![b' \mapsto \xi(a,b')]\!])]\!] \\
&= (g^*)_*([\![a \mapsto [\![b' \mapsto \xi(a,b')]\!]]\!]) \\
&= (g^*)_*\left(\Phi_{B',C}([\![(a,b') \mapsto \xi(a,b')]\!])\right) \\
&= (g^*)_*\left(\Phi_{B',C}(\xi)\right) \\
&= \left[(g^*)_* \circ \Phi_{B',C}\right](\xi).
\end{aligned}$$

- *Naturality for Φ, Part II.* We need to show that, given a function $h\colon C \to$

C', the diagram

$$\begin{tikzcd}
\mathsf{Sets}(A \times B, C) \arrow[r, "\Phi_{B,C}"] \arrow[d, "h_*"'] & \mathsf{Sets}(A, \mathsf{Sets}(B, C)), \arrow[d, "(h_*)_*"] \\
\mathsf{Sets}(A \times B, C') \arrow[r, "\Phi_{B,C'}"'] & \mathsf{Sets}(A, \mathsf{Sets}(B, C'))
\end{tikzcd}$$

commutes. Indeed, given a function

$$\xi \colon A \times B \to C,$$

we have

$$\begin{aligned}
\left[\Phi_{B,C} \circ h_*\right](\xi) &= \Phi_{B,C}(h_*(\xi)) \\
&= \Phi_{B,C}(h_*([\![(a,b) \mapsto \xi(a,b)]\!])) \\
&= \Phi_{B,C}([\![(a,b) \mapsto h(\xi(a,b))]\!]) \\
&= [\![a \mapsto [\![b \mapsto h(\xi(a,b))]\!]]\!] \\
&= [\![a \mapsto h_*([\![b \mapsto \xi(a,b)]\!])]\!] \\
&= (h_*)_*([\![a \mapsto [\![b \mapsto \xi(a,b)]\!]]\!]) \\
&= (h_*)_*\big(\Phi_{B,C}([\![(a,b) \mapsto \xi(a,b)]\!])\big) \\
&= (h_*)_*\big(\Phi_{B,C}(\xi)\big) \\
&= \left[(h_*)_* \circ \Phi_{B,C}\right](\xi).
\end{aligned}$$

- *Naturality for* Ψ. Since Φ is natural in each argument and Φ is a componentwise inverse to Ψ in each argument, it follows from Categories, **??** of **??** that Ψ is also natural in each argument.

Item 3: Associativity

This is proved in the proof of Monoidal Structures on the Category of Sets, **??**.

Item 4: Unitality

This is proved in the proof of Monoidal Structures on the Category of Sets, **????**.

Item 5: Commutativity

This is proved in the proof of Monoidal Structures on the Category of Sets, **??**.

Item 6: Distributivity Over Coproducts

This is proved in the proof of Monoidal Structures on the Category of Sets, **????**.

Item 7: Annihilation With the Empty Set

This is proved in the proof of Monoidal Structures on the Category of Sets, **????**.

Item 8: Distributivity Over Unions

See [Pro24c].

Item 9: Distributivity Over Intersections

See [Pro24d, Corollary 1].

Item 10: Distributivity Over Differences

See [Pro24a].

Item 11: Distributivity Over Symmetric Differences

See [Pro24b].

Item 12: Middle-Four Exchange With Respect to Intersections

See [Pro24d, Corollary 1].

Item 13: Symmetric Monoidality

This is a repetition of Monoidal Structures on the Category of Sets, **??**, and is proved there.

Item 14: Symmetric Bimonoidality

This is a repetition of Monoidal Structures on the Category of Sets, **??**, and is proved there. ∎

Remark 1.3.6 ▶ The Cartesian Product of Sets as an $(\mathbb{E}_k, \mathbb{E}_\ell)$ Tensor Product

As shown in Item 1 of Proposition 1.3.4, the Cartesian product of sets defines a functor

$$-_1 \times -_2 \colon \mathsf{Sets} \times \mathsf{Sets} \to \mathsf{Sets}.$$

This functor is the $(k, \ell) = (-1, -1)$ case of a family of functors

$$\otimes_{k,\ell} \colon \mathsf{Mon}_{\mathbb{E}_k}(\mathsf{Sets}) \times \mathsf{Mon}_{\mathbb{E}_\ell}(\mathsf{Sets}) \to \mathsf{Mon}_{\mathbb{E}_{k+\ell}}(\mathsf{Sets})$$

of tensor products of $\mathbb{E}_k$-monoid objects on Sets with $\mathbb{E}_\ell$-monoid objects on Sets; see **??**.

We may state the equalities in Items 8 to 11 of Proposition 1.3.4 as the commutativity of the following diagrams:

$$\begin{tikzcd}
 & (\mathcal{P}(X)\times\mathcal{P}(X))\times(\mathcal{P}(X)\times\mathcal{P}(X)) \arrow[dr, "\mu_4^{\mathsf{Sets}}"] & \\
\mathcal{P}(X)\times(\mathcal{P}(X)\times\mathcal{P}(X)) \arrow[ur, "\Delta_{\mathcal{P}(X)}\times\mathrm{id}_{\mathcal{P}(X)}\times\mathrm{id}_{\mathcal{P}(X)}"] \arrow[d, "\mathrm{id}_{\mathcal{P}(X)}\times\cup"'] & & (\mathcal{P}(X)\times\mathcal{P}(X))\times(\mathcal{P}(X)\times\mathcal{P}(X)) \arrow[d, "\mathcal{P}^{\times}_{X,X}\times\mathcal{P}^{\times}_{X,X}"] \\
\mathcal{P}(X)\times\mathcal{P}(X) \arrow[dr, "\mathcal{P}^{\times}_{X,X}"'] & & \mathcal{P}(X\times X)\times\mathcal{P}(X\times X) \arrow[dl, "\cup"] \\
 & \mathcal{P}(X\times X) &
\end{tikzcd}$$

$$\begin{tikzcd}
 & (\mathcal{P}(X)\times\mathcal{P}(X))\times(\mathcal{P}(X)\times\mathcal{P}(X)) \arrow[dr, "\mu_4^{\mathsf{Sets}}"] & \\
(\mathcal{P}(X)\times\mathcal{P}(X))\times\mathcal{P}(X) \arrow[ur, "\mathrm{id}_{\mathcal{P}(X)}\times\mathrm{id}_{\mathcal{P}(X)}\times\Delta_{\mathcal{P}(X)}"] \arrow[d, "\cup\times\mathrm{id}_{\mathcal{P}(X)}"'] & & (\mathcal{P}(X)\times\mathcal{P}(X))\times(\mathcal{P}(X)\times\mathcal{P}(X)) \arrow[d, "\mathcal{P}^{\times}_{X,X}\times\mathcal{P}^{\times}_{X,X}"] \\
\mathcal{P}(X)\times\mathcal{P}(X) \arrow[dr, "\mathcal{P}^{\times}_{X,X}"'] & & \mathcal{P}(X\times X)\times\mathcal{P}(X\times X) \arrow[dl, "\cup"] \\
 & \mathcal{P}(X\times X) &
\end{tikzcd}$$

$$\begin{tikzcd}
 & (\mathcal{P}(X)\times\mathcal{P}(X))\times(\mathcal{P}(X)\times\mathcal{P}(X)) \arrow[dr, "\mu_4^{\mathsf{Sets}}"] & \\
\mathcal{P}(X)\times(\mathcal{P}(X)\times\mathcal{P}(X)) \arrow[ur, "\Delta_{\mathcal{P}(X)}\times\mathrm{id}_{\mathcal{P}(X)}\times\mathrm{id}_{\mathcal{P}(X)}"] \arrow[d, "\mathrm{id}_{\mathcal{P}(X)}\times\cap"'] & & (\mathcal{P}(X)\times\mathcal{P}(X))\times(\mathcal{P}(X)\times\mathcal{P}(X)) \arrow[d, "\mathcal{P}^{\times}_{X,X}\times\mathcal{P}^{\times}_{X,X}"] \\
\mathcal{P}(X)\times\mathcal{P}(X) \arrow[dr, "\mathcal{P}^{\times}_{X,X}"'] & & \mathcal{P}(X\times X)\times\mathcal{P}(X\times X) \arrow[dl, "\cap"] \\
 & \mathcal{P}(X\times X) &
\end{tikzcd}$$

$$\begin{tikzcd}
 & (\mathcal{P}(X)\times\mathcal{P}(X))\times(\mathcal{P}(X)\times\mathcal{P}(X)) \arrow[dr, "\mu_4^{\mathsf{Sets}}"] & \\
(\mathcal{P}(X)\times\mathcal{P}(X))\times\mathcal{P}(X) \arrow[ur, "\mathrm{id}_{\mathcal{P}(X)}\times\mathrm{id}_{\mathcal{P}(X)}\times\Delta_{\mathcal{P}(X)}"] \arrow[d, "\cap\times\mathrm{id}_{\mathcal{P}(X)}"'] & & (\mathcal{P}(X)\times\mathcal{P}(X))\times(\mathcal{P}(X)\times\mathcal{P}(X)) \arrow[d, "\mathcal{P}^{\times}_{X,X}\times\mathcal{P}^{\times}_{X,X}"] \\
\mathcal{P}(X)\times\mathcal{P}(X) \arrow[dr, "\mathcal{P}^{\times}_{X,X}"'] & & \mathcal{P}(X\times X)\times\mathcal{P}(X\times X) \arrow[dl, "\cap"] \\
 & \mathcal{P}(X\times X) &
\end{tikzcd}$$

$$\begin{tikzcd}
 & (\mathcal{P}(X)\times\mathcal{P}(X))\times(\mathcal{P}(X)\times\mathcal{P}(X)) \arrow[dr, "\mu_4^{\mathsf{Sets}}"] & \\
\mathcal{P}(X)\times(\mathcal{P}(X)\times\mathcal{P}(X)) \arrow[ur, "\Delta_{\mathcal{P}(X)}\times\mathrm{id}_{\mathcal{P}(X)}\times\mathrm{id}_{\mathcal{P}(X)}"] \arrow[d, "\mathrm{id}_{\mathcal{P}(X)}\times\setminus"'] & & (\mathcal{P}(X)\times\mathcal{P}(X))\times(\mathcal{P}(X)\times\mathcal{P}(X)) \arrow[d, "\mathcal{P}^{\times}_{X,X}\times\mathcal{P}^{\times}_{X,X}"] \\
\mathcal{P}(X)\times\mathcal{P}(X) \arrow[dr, "\mathcal{P}^{\times}_{X,X}"'] & & \mathcal{P}(X\times X)\times\mathcal{P}(X\times X) \arrow[dl, "\setminus"] \\
 & \mathcal{P}(X\times X) &
\end{tikzcd}$$

$$\begin{tikzcd}
 & (\mathcal{P}(X)\times\mathcal{P}(X))\times(\mathcal{P}(X)\times\mathcal{P}(X)) \arrow[dr, "\mu_4^{\mathsf{Sets}}"] & \\
(\mathcal{P}(X)\times\mathcal{P}(X))\times\mathcal{P}(X) \arrow[ur, "\mathrm{id}_{\mathcal{P}(X)}\times\mathrm{id}_{\mathcal{P}(X)}\times\Delta_{\mathcal{P}(X)}"] \arrow[d, "\setminus\times\mathrm{id}_{\mathcal{P}(X)}"'] & & (\mathcal{P}(X)\times\mathcal{P}(X))\times(\mathcal{P}(X)\times\mathcal{P}(X)) \arrow[d, "\mathcal{P}^{\times}_{X,X}\times\mathcal{P}^{\times}_{X,X}"] \\
\mathcal{P}(X)\times\mathcal{P}(X) \arrow[dr, "\mathcal{P}^{\times}_{X,X}"'] & & \mathcal{P}(X\times X)\times\mathcal{P}(X\times X) \arrow[dl, "\setminus"] \\
 & \mathcal{P}(X\times X) &
\end{tikzcd}$$

$$\begin{tikzcd}
 & (\mathcal{P}(X)\times\mathcal{P}(X))\times(\mathcal{P}(X)\times\mathcal{P}(X)) \arrow[dr, "\mu_4^{\mathsf{Sets}}"] & \\
\mathcal{P}(X)\times(\mathcal{P}(X)\times\mathcal{P}(X)) \arrow[ur, "\Delta_{\mathcal{P}(X)}\times\mathrm{id}_{\mathcal{P}(X)}\times\mathrm{id}_{\mathcal{P}(X)}"] \arrow[d, "\mathrm{id}_{\mathcal{P}(X)}\times\triangle"'] & & (\mathcal{P}(X)\times\mathcal{P}(X))\times(\mathcal{P}(X)\times\mathcal{P}(X)) \arrow[d, "\mathcal{P}^{\times}_{X,X}\times\mathcal{P}^{\times}_{X,X}"] \\
\mathcal{P}(X)\times\mathcal{P}(X) \arrow[dr, "\mathcal{P}^{\times}_{X,X}"'] & & \mathcal{P}(X\times X)\times\mathcal{P}(X\times X) \arrow[dl, "\triangle"] \\
 & \mathcal{P}(X\times X) &
\end{tikzcd}$$

$$\begin{tikzcd}
 & (\mathcal{P}(X)\times\mathcal{P}(X))\times(\mathcal{P}(X)\times\mathcal{P}(X)) \arrow[dr, "\mu_4^{\mathsf{Sets}}"] & \\
(\mathcal{P}(X)\times\mathcal{P}(X))\times\mathcal{P}(X) \arrow[ur, "\mathrm{id}_{\mathcal{P}(X)}\times\mathrm{id}_{\mathcal{P}(X)}\times\Delta_{\mathcal{P}(X)}"] \arrow[d, "\triangle\times\mathrm{id}_{\mathcal{P}(X)}"'] & & (\mathcal{P}(X)\times\mathcal{P}(X))\times(\mathcal{P}(X)\times\mathcal{P}(X)) \arrow[d, "\mathcal{P}^{\times}_{X,X}\times\mathcal{P}^{\times}_{X,X}"] \\
\mathcal{P}(X)\times\mathcal{P}(X) \arrow[dr, "\mathcal{P}^{\times}_{X,X}"'] & & \mathcal{P}(X\times X)\times\mathcal{P}(X\times X) \arrow[dl, "\triangle"] \\
 & \mathcal{P}(X\times X) &
\end{tikzcd}$$

1.4 Pullbacks

Let A, B, and C be sets and let $f\colon A \to C$ and $g\colon B \to C$ be functions.

Definition 1.4.1 ▶ Pullbacks of Sets

The **pullback of A and B over C along f and g**[1] is the pullback of A and B over C along f and g in Sets as in Limits and Colimits, **??**.

[1] *Further Terminology:* Also called the **fibre product of A and B over C along f and g**.

Construction 1.4.2 ▶ Construction of Pullbacks of Sets

Concretely, the pullback of A and B over C along f and g is the pair $(A \times_C B, \{\mathrm{pr}_1, \mathrm{pr}_2\})$ consisting of:

1. *The Limit.* The set $A \times_C B$ defined by

$$A \times_C B \overset{\text{def}}{=} \{(a, b) \in A \times B \mid f(a) = g(b)\}.$$

2. *The Cone.* The maps[1]

$$\begin{aligned} \mathrm{pr}_1 &\colon A \times_C B \to A, \\ \mathrm{pr}_2 &\colon A \times_C B \to B \end{aligned}$$

defined by

$$\begin{aligned} \mathrm{pr}_1(a, b) &\overset{\text{def}}{=} a, \\ \mathrm{pr}_2(a, b) &\overset{\text{def}}{=} b \end{aligned}$$

for each $(a, b) \in A \times_C B$.

[1] *Further Notation:* Also written $\mathrm{pr}_1^{A \times_C B}$ and $\mathrm{pr}_2^{A \times_C B}$.

Proof 1.4.3 ▶ Proof of Construction 1.4.2

We claim that $A \times_C B$ is the categorical pullback of A and B over C with respect to (f, g) in Sets. First we need to check that the relevant pullback diagram

commutes, i.e. that we have

$$f \circ \mathsf{pr}_1 = g \circ \mathsf{pr}_2, \qquad \begin{tikzcd} A \times_C B \arrow[r, two heads, "\mathsf{pr}_2"] \arrow[d, two heads, "\mathsf{pr}_1"'] & B \arrow[d, "g"] \\ A \arrow[r, "f"'] & C. \end{tikzcd}$$

Indeed, given $(a, b) \in A \times_C B$, we have

$$\begin{aligned} [f \circ \mathsf{pr}_1](a, b) &= f(\mathsf{pr}_1(a, b)) \\ &= f(a) \\ &= g(b) \\ &= g(\mathsf{pr}_2(a, b)) \\ &= [g \circ \mathsf{pr}_2](a, b), \end{aligned}$$

where $f(a) = g(b)$ since $(a, b) \in A \times_C B$. Next, we prove that $A \times_C B$ satisfies the universal property of the pullback. Suppose we have a diagram of the form

$$\begin{tikzcd} P \arrow[dr, bend left, "p_2"] \arrow[ddr, bend right, "p_1"'] & & \\ & A \times_C B \arrow[r, two heads, "\mathsf{pr}_2"] \arrow[d, two heads, "\mathsf{pr}_1"'] \arrow[dr, phantom, "\lrcorner", very near start] & B \arrow[d, "g"] \\ & A \arrow[r, "f"'] & C \end{tikzcd}$$

in Sets. Then there exists a unique map $\phi\colon P \longrightarrow A \times_C B$ making the diagram

$$\begin{tikzcd}
P \arrow[dr, dashed, "\phi", "\exists!"'] \arrow[drr, bend left, "p_2"] \arrow[ddr, bend right, "p_1"'] & & \\
& A \times_C B \arrow[r, two heads, "\mathrm{pr}_2"] \arrow[d, two heads, "\mathrm{pr}_1"'] \arrow[dr, phantom, "\lrcorner", very near start] & B \arrow[d, "g"] \\
& A \arrow[r, "f"'] & C
\end{tikzcd}$$

commute, being uniquely determined by the conditions

$$\begin{aligned}
\mathrm{pr}_1 \circ \phi &= p_1, \\
\mathrm{pr}_2 \circ \phi &= p_2
\end{aligned}$$

via

$$\phi(x) = (p_1(x), p_2(x))$$

for each $x \in P$, where we note that $(p_1(x), p_2(x)) \in A \times B$ indeed lies in $A \times_C B$ by the condition

$$f \circ p_1 = g \circ p_2,$$

which gives

$$f(p_1(x)) = g(p_2(x))$$

for each $x \in P$, so that $(p_1(x), p_2(x)) \in A \times_C B$. ∎

Remark 1.4.4 ▶ Pullbacks of Sets Depend on the Maps

It is common practice to write $A \times_C B$ for the pullback of A and B over C along f and g, omitting the maps f and g from the notation and instead leaving them implicit, to be understood from the context.
However, the set $A \times_C B$ depends very much on the maps f and g, and sometimes it is necessary or useful to note this dependence explicitly. In such situations, we will write $A \times_{f,C,g} B$ or $A \times_C^{f,g} B$ for $A \times_C B$.

Example 1.4.5 ▶ Examples of Pullbacks of Sets

Here are some examples of pullbacks of sets.

1. *Unions via Intersections.* Let X be a set. We have

$$A \cap B \cong A \times_{A \cup B} B, \qquad \begin{tikzcd} A \cap B \arrow[r, two heads] \arrow[d, two heads] \arrow[dr, phantom, "\lrcorner", very near start] & B \arrow[d, hook, "\iota_B"] \\ A \arrow[r, hook, "\iota_A"'] & A \cup B \end{tikzcd}$$

for each $A, B \in \mathcal{P}(X)$.

Proof 1.4.6 ▶ Proof of Example 1.4.5

Item 1: Unions via Intersections

Indeed, we have

$$\begin{aligned} A \times_{A \cup B} B &\cong \{(x, y) \in A \times B \mid x = y\} \\ &\cong A \cap B. \end{aligned}$$

This finishes the proof. ∎

Proposition 1.4.7 ▶ Properties of Pullbacks of Sets

Let A, B, C, and X be sets.

1. *Functoriality.* The assignment $(A, B, C, f, g) \mapsto A \times_{f,C,g} B$ defines a functor

$$-_1 \times_{-_3} -_1 \colon \mathrm{Fun}(\mathcal{P}, \mathrm{Sets}) \to \mathrm{Sets},$$

where $\mathcal{P}$ is the category that looks like this:

$$\begin{tikzcd} & \bullet \arrow[d] \\ \bullet \arrow[r] & \bullet. \end{tikzcd}$$

In particular, the action on morphisms of $-_1 \times_{-_3} -_1$ is given by sending

a morphism

$$\begin{tikzcd}
A \times_C B \arrow[rr] \arrow[dd] \arrow[dr, phantom, "\lrcorner", very near start] & & B \arrow[dr, "\psi"] \arrow[dd, "g"] & \\
& A' \times_{C'} B' \arrow[rr] \arrow[dd] \arrow[dr, phantom, "\lrcorner", very near start] & & B' \arrow[dd, "g'"] \\
A \arrow[rr, "f"] \arrow[dr, "\phi"'] & & C \arrow[dr, "\chi"] & \\
& A' \arrow[rr, "f'"'] & & C'
\end{tikzcd}$$

in $\mathsf{Fun}(\mathcal{P}, \mathsf{Sets})$ to the map $\xi\colon A \times_C B \xrightarrow{\exists!} A' \times_{C'} B'$ given by

$$\xi(a, b) \overset{\text{def}}{=} (\phi(a), \psi(b))$$

for each $(a, b) \in A \times_C B$, which is the unique map making the diagram

$$\begin{tikzcd}
A \times_C B \arrow[rr] \arrow[dd] \arrow[dr, dashed, phantom, "\lrcorner", very near start] \arrow[dr, dashed] & & B \arrow[dr, "\psi"] \arrow[dd, "g"] & \\
& A' \times_{C'} B' \arrow[rr] \arrow[dd] \arrow[dr, phantom, "\lrcorner", very near start] & & B' \arrow[dd, "g'"] \\
A \arrow[rr, "f"] \arrow[dr, "\phi"'] & & C \arrow[dr, "\chi"] & \\
& A' \arrow[rr, "f'"'] & & C'
\end{tikzcd}$$

commute.

2. *Adjointness.* We have adjunctions

$$\big(A \times_X - \dashv \mathbf{Sets}_{/X}(A, -)\big)\colon \quad \begin{tikzcd} \mathsf{Sets}_{-/X} \arrow[r, bend left, "A \times_X -"] \arrow[r, phantom, "\perp"] & \mathsf{Sets}_{-/X}, \arrow[l, bend left, "\mathbf{Sets}_{/X}(A,-)"] \end{tikzcd}$$

$$\big(- \times_X B \dashv \mathbf{Sets}_{/X}(B, -)\big)\colon \quad \begin{tikzcd} \mathsf{Sets}_{-/X} \arrow[r, bend left, "- \times_X B"] \arrow[r, phantom, "\perp"] & \mathsf{Sets}_{-/X}, \arrow[l, bend left, "\mathbf{Sets}_{/X}(B,-)"] \end{tikzcd}$$

witnessed by bijections

$$\begin{aligned}\mathsf{Sets}_{/X}(A \times_X B, C) &\cong \mathsf{Sets}_{/X}\big(A, \mathbf{Sets}_{/X}(B, C)\big),\\ \mathsf{Sets}_{/X}(A \times_X B, C) &\cong \mathsf{Sets}_{/X}\big(B, \mathbf{Sets}_{/X}(A, C)\big),\end{aligned}$$

natural in $(A, \phi_A), (B, \phi_B), (C, \phi_C) \in \mathrm{Obj}\big(\mathsf{Sets}_{/X}\big)$, where $\mathbf{Sets}_{/X}(A, B)$ is the object of $\mathsf{Sets}_{/X}$ consisting of (see Fibred Sets, **??**):

- *The Set.* The set $\mathbf{Sets}_{/X}(A, B)$ defined by

$$\mathbf{Sets}_{/X}(A, B) \stackrel{\text{def}}{=} \coprod_{x \in X} \mathsf{Sets}\big(\phi_A^{-1}(x), \phi_Y^{-1}(x)\big)$$

- *The Map to X.* The map

$$\phi_{\mathbf{Sets}_{/X}(A,B)} \colon \mathbf{Sets}_{/X}(A, B) \to X$$

  defined by

$$\phi_{\mathbf{Sets}_{/X}(A,B)}(x, f) \stackrel{\text{def}}{=} x$$

  for each $(x, f) \in \mathbf{Sets}_{/X}(A, B)$.

3. *Associativity.* Given a diagram

$$\begin{tikzcd}
A \arrow[dr, "f"'] & & B \arrow[dl, "g"] \arrow[dr, "h"'] & & C \arrow[dl, "k"] \\
& X & & Y &
\end{tikzcd}$$

in Sets, we have isomorphisms of sets

$$(A \times_X B) \times_Y C \cong (A \times_X B) \times_B (B \times_Y C) \cong A \times_X (B \times_Y C),$$

where these pullbacks are built as in the diagrams

$$\begin{tikzcd}
& (A \times_X B) \times_Y C \arrow[dl] \arrow[dr] \arrow[ddrrr] \arrow[d, phantom, "\lrcorner"] & & & \\
& A \times_X B \arrow[dl] \arrow[dr] \arrow[d, phantom, "\lrcorner"] & & & \\
A \arrow[dr, "f"'] & & B \arrow[dl, "g"] \arrow[dr, "h"'] & & C, \arrow[dl, "k"] \\
& X & & Y &
\end{tikzcd}
\qquad
\begin{tikzcd}
& (A \times_X B) \times_B (B \times_Y C) \arrow[dl] \arrow[dr] \arrow[d, phantom, "\lrcorner"] & & & \\
& A \times_X B \arrow[dl] \arrow[dr] \arrow[d, phantom, "\lrcorner"] & & B \times_Y C \arrow[dl] \arrow[dr] \arrow[d, phantom, "\lrcorner"] & \\
A \arrow[dr, "f"'] & & B \arrow[dl, "g"] \arrow[dr, "h"'] & & C, \arrow[dl, "k"] \\
& X & & Y &
\end{tikzcd}
\qquad
\begin{tikzcd}
& A \times_X (B \times_Y C) \arrow[dl] \arrow[dr] \arrow[ddlll] \arrow[d, phantom, "\lrcorner"] & & & \\
& & B \times_Y C \arrow[dl] \arrow[dr] \arrow[d, phantom, "\lrcorner"] & & \\
A \arrow[dr, "f"'] & & B \arrow[dl, "g"] \arrow[dr, "h"'] & & C. \arrow[dl, "k"] \\
& X & & Y &
\end{tikzcd}$$

4. *Interaction With Composition*. Given a diagram

$$\begin{tikzcd}
X \arrow[dr, "\phi"'] & & Y \arrow[dl, "\psi"] \\
& A \arrow[dr, "f"'] & B \arrow[dl, "g"] \\
& & K
\end{tikzcd}$$

in Sets, we have isomorphisms of sets

$$\begin{aligned}
X \times_K^{f\circ\phi, g\circ\psi} Y &\cong \left(X \times_A^{\phi,q_1} \left(A \times_K^{f,g} B\right)\right) \times_{A\times_K^{f,g} B}^{p_2,p_1} \left(\left(A \times_K^{f,g} B\right) \times_B^{q_2,\psi} Y\right) \\
&\cong X \times_A^{\phi,p} \left(\left(A \times_K^{f,g} B\right) \times_B^{q_2,\psi} Y\right) \\
&\cong \left(X \times_A^{\phi,q_1} \left(A \times_K^{f,g} B\right)\right) \times_B^{q,\psi} Y
\end{aligned}$$

where

$$\begin{aligned}
q_1 &= \mathrm{pr}_1^{A\times_K^{f,g} B}, & q_2 &= \mathrm{pr}_2^{A\times_K^{f,g} B}, \\
p_1 &= \mathrm{pr}_1^{\left(A\times_K^{f,g} B\right)\times_Y^{q_2,\psi}}, & p_2 &= \mathrm{pr}_2^{X\times_{A\times_K^{f,g} B}^{\phi,q_1}\left(A\times_K^{f,g} B\right)}, \\
p &= q_1 \circ \mathrm{pr}_1^{\left(A\times_K^{f,g} B\right)\times_B^{q_2,\psi} Y}, & q &= q_2 \circ \mathrm{pr}_2^{X\times_A^{\phi,q_1}\left(A\times_K^{f,g} B\right)},
\end{aligned}$$

and where these pullbacks are built as in the following diagrams:

$$\begin{tikzcd}
& (X \times_A (A \times_K B)) \times_{A\times_K B} ((A \times_K B) \times_B Y) \arrow[dl] \arrow[dr] \arrow[d, phantom, "\lrcorner"] & \\
& X \times_A (A \times_K B) \arrow[dl] \arrow[dr] \arrow[d, phantom, "\lrcorner"] & (A \times_K B) \times_B Y \arrow[dl] \arrow[dr] \arrow[d, phantom, "\lrcorner"] & \\
X \arrow[dr, "\phi"'] & & A \times_K B \arrow[dl] \arrow[dr] \arrow[d, phantom, "\lrcorner"] & & Y, \arrow[dl, "\psi"] \\
& A \arrow[dr, "f"'] & & B \arrow[dl, "g"] & \\
& & K & &
\end{tikzcd}
\qquad
\begin{tikzcd}
& X \times_A ((A \times_K B) \times_B Y) \arrow[dl] \arrow[dr] \arrow[d, phantom, "\lrcorner"] & \\
& (A \times_K B) \times_B Y \arrow[dl] \arrow[dr] \arrow[d, phantom, "\lrcorner"] & \\
X \arrow[dr, "\phi"'] & A \times_K B \arrow[dl] \arrow[dr] \arrow[d, phantom, "\lrcorner"] & Y, \arrow[dl, "\psi"] \\
& A \arrow[dr, "f"'] & B \arrow[dl, "g"] \\
& K &
\end{tikzcd}$$

$$
\begin{tikzcd}
 & (X \times_A (A \times_K B)) \times_B Y \arrow[dl] \arrow[dr] \arrow[d, phantom, "\lrcorner"] & \\
 & X \times_A (A \times_K B) \arrow[dl] \arrow[dr] \arrow[d, phantom, "\lrcorner"] & \\
X \arrow[dr, "\phi"'] & A \times_K B \arrow[dl] \arrow[dr] \arrow[d, phantom, "\lrcorner"] & Y, \arrow[dl, "\psi"] \\
 & A \arrow[dr, "f"'] & B \arrow[dl, "g"] \\
 & K &
\end{tikzcd}
\qquad
\begin{tikzcd}
 & X \times_K Y \arrow[dl] \arrow[dr] \arrow[d, phantom, "\lrcorner"] & \\
X \arrow[dr, "\phi"'] & & Y. \arrow[dl, "\psi"] \\
 & A \arrow[dr, "f"'] & B \arrow[dl, "g"] \\
 & K &
\end{tikzcd}
$$

5. *Unitality.* We have isomorphisms of sets

$$
\begin{tikzcd}
A \arrow[r, equal] \arrow[d, "f"'] \arrow[dr, phantom, "\lrcorner"] & A \arrow[d, "f"] \\
X \arrow[r, equal] & X
\end{tikzcd}
\qquad
\begin{aligned}
\lambda_A^{\mathsf{Sets}_{/X}} &\colon X \times_X A \overset{\sim}{\dashrightarrow} A, \\
\rho_A^{\mathsf{Sets}_{/X}} &\colon A \times_X X \overset{\sim}{\dashrightarrow} A,
\end{aligned}
\qquad
\begin{tikzcd}
A \arrow[r, "f"] \arrow[d, equal] \arrow[dr, phantom, "\lrcorner"] & X \arrow[d, equal] \\
X \arrow[r, "f"'] & X,
\end{tikzcd}
$$

natural in $(A, f) \in \mathrm{Obj}(\mathsf{Sets}_{/X})$.

6. *Commutativity.* We have an isomorphism of sets

$$
\begin{tikzcd}
A \times_X B \arrow[r] \arrow[d] \arrow[dr, phantom, "\lrcorner"] & B \arrow[d, "g"] \\
A \arrow[r, "f"'] & X,
\end{tikzcd}
\qquad
\sigma_{A,B}^{\mathsf{Sets}_{/X}} \colon A \times_X B \overset{\sim}{\dashrightarrow} B \times_X A
\qquad
\begin{tikzcd}
B \times_X A \arrow[r] \arrow[d] \arrow[dr, phantom, "\lrcorner"] & A \arrow[d, "f"] \\
B \arrow[r, "g"'] & X,
\end{tikzcd}
$$

natural in $(A, f), (B, g) \in \mathrm{Obj}(\mathsf{Sets}_{/X})$.

7. *Distributivity Over Coproducts.* Let A, B, and C be sets and let $\phi_A \colon A \to X$, $\phi_B \colon B \to X$, and $\phi_C \colon C \to X$ be morphisms of sets. We have isomorphisms of sets

$$
\begin{aligned}
\delta_\ell^{\mathsf{Sets}_{/X}} &\colon A \times_X (B \coprod C) \overset{\sim}{\dashrightarrow} (A \times_X B) \coprod (A \times_X C), \\
\delta_r^{\mathsf{Sets}_{/X}} &\colon (A \coprod B) \times_X C \overset{\sim}{\dashrightarrow} (A \times_X C) \coprod (B \times_X C),
\end{aligned}
$$

as in the diagrams

$$\begin{tikzcd} (A \times_X B) \coprod (A \times_X C) \arrow[r, two heads] \arrow[d, two heads] \arrow[dr, phantom, "\lrcorner", very near start] & B \coprod C \arrow[d, "\phi_B \coprod \phi_C"] \\ A \arrow[r, "\phi_A"'] & X \end{tikzcd} \qquad \begin{tikzcd} (A \times_X C) \coprod (B \times_X C) \arrow[r, two heads] \arrow[d, two heads] \arrow[dr, phantom, "\lrcorner", very near start] & C \arrow[d, "\phi_C"] \\ A \coprod B \arrow[r, "\phi_A \coprod \phi_B"'] & X \end{tikzcd}$$

natural in $A, B, C \in \mathrm{Obj}(\mathsf{Sets}_{/X})$.

8. *Annihilation With the Empty Set.* We have isomorphisms of sets

$$\begin{tikzcd} \emptyset \arrow[r] \arrow[d] \arrow[dr, phantom, "\lrcorner", very near start] & \emptyset \arrow[d] \\ A \arrow[r, "f"'] & X, \end{tikzcd} \qquad \begin{aligned} \zeta_\ell^{\mathsf{Sets}_{/X}} &: A \times_X \emptyset \overset{\sim}{\dashrightarrow} \emptyset, \\ \zeta_r^{\mathsf{Sets}_{/X}} &: \emptyset \times_X A \overset{\sim}{\dashrightarrow} \emptyset, \end{aligned} \qquad \begin{tikzcd} \emptyset \arrow[r] \arrow[d] \arrow[dr, phantom, "\lrcorner", very near start] & A \arrow[d, "f"] \\ \emptyset \arrow[r] & X, \end{tikzcd}$$

natural in $(A, f) \in \mathrm{Obj}(\mathsf{Sets}_{/X})$.

9. *Interaction With Products.* We have an isomorphism of sets

$$A \times_{\mathsf{pt}} B \cong A \times B, \qquad \begin{tikzcd} A \times B \arrow[r] \arrow[d] \arrow[dr, phantom, "\lrcorner", very near start] & B \arrow[d, "!_B"] \\ A \arrow[r, "!_A"'] & \mathsf{pt}. \end{tikzcd}$$

10. *Symmetric Monoidality.* The 8-tuple $(\mathsf{Sets}_{/X}, \times_X, X, \mathbf{Sets}_{/X}, \alpha^{\mathsf{Sets}_{/X}}, \lambda^{\mathsf{Sets}_{/X}}, \rho^{\mathsf{Sets}_{/X}}, \sigma^{\mathsf{Sets}_{/X}})$ is a symmetric closed monoidal category.

Proof 1.4.8 ▶ Proof of Proposition 1.4.7

Item 1: Functoriality

This is a special case of functoriality of co/limits, Limits and Colimits, **??** of **??**,

with the explicit expression for ξ following from the commutativity of the cube pullback diagram.

Item 2: Adjointness

This is a repetition of Fibred Sets, **??** of **??**, and is proved there.

Item 3: Associativity

We have

$$\begin{aligned}(A \times_X B) \times_Y C &\cong \{((a,b),c) \in (A \times_X B) \times C \mid h(b) = k(c)\} \\ &\cong \{((a,b),c) \in (A \times B) \times C \mid f(a) = g(b) \text{ and } h(b) = k(c)\} \\ &\cong \{(a,(b,c)) \in A \times (B \times C) \mid f(a) = g(b) \text{ and } h(b) = k(c)\} \\ &\cong \{(a,(b,c)) \in A \times (B \times_Y C) \mid f(a) = g(b)\} \\ &\cong A \times_X (B \times_Y C)\end{aligned}$$

and

$$\begin{aligned}(A \times_X B) \times_B (B \times_Y C) &\cong \{((a,b),(b',c)) \in (A \times_X B) \times (B \times_Y C) \mid b = b'\} \\ &\cong \left\{((a,b),(b',c)) \in (A \times B) \times (B \times C) \,\middle|\, \begin{array}{l} f(a) = g(b), b = b', \\ \text{and } h(b') = k(c) \end{array}\right\} \\ &\cong \left\{(a,(b,(b',c))) \in A \times (B \times (B \times C)) \,\middle|\, \begin{array}{l} f(a) = g(b), b = b', \\ \text{and } h(b') = k(c) \end{array}\right\} \\ &\cong \left\{(a,((b,b'),c)) \in A \times ((B \times B) \times C) \,\middle|\, \begin{array}{l} f(a) = g(b), b = b', \\ \text{and } h(b') = k(c) \end{array}\right\} \\ &\cong \left\{(a,((b,b'),c)) \in A \times ((B \times_B B) \times C) \,\middle|\, \begin{array}{l} f(a) = g(b) \text{ and} \\ h(b') = k(c) \end{array}\right\} \\ &\cong \{(a,(b,c)) \in A \times (B \times C) \mid f(a) = g(b) \text{ and } h(b) = k(c)\} \\ &\cong A \times_X (B \times_Y C),\end{aligned}$$

where we have used Item 5 for the isomorphism $B \times_B B \cong B$.

Item 4: Interaction With Composition

By Item 3, it suffices to construct only the isomorphism

$$X \times_K^{f \circ \phi, g \circ \psi} Y \cong \left(X \times_A^{\phi, q_1} \left(A \times_K^{f,g} B\right)\right) \times_{A \times_K^{f,g} B}^{p_2, p_1} \left(\left(A \times_K^{f,g} B\right) \times_B^{q_2, \psi} Y\right).$$

We have

$$\left(X \times_A^{\phi, q_1} \left(A \times_K^{f,g} B\right)\right) \overset{\text{def}}{=} \left\{(x,(a,b)) \in X \times \left(A \times_K^{f,g} B\right) \,\middle|\, \phi(x) = q_1(a,b)\right\}$$

$$
\begin{aligned}
&\overset{\text{def}}{=} \left\{(x,(a,b)) \in X \times \left(A \times_K^{f,g} B\right) \,\middle|\, \phi(x) = a\right\} \\
&\cong \{(x,(a,b)) \in X \times (A \times B) \mid \phi(x) = a \text{ and } f(a) = g(b)\}, \\
\left(\left(A \times_K^{f,g} B\right) \times_B^{q_2,\psi} Y\right) &\overset{\text{def}}{=} \left\{((a,b),y) \in \left(A \times_K^{f,g} B\right) \times Y \,\middle|\, q_2(a,b) = \psi(y)\right\} \\
&\overset{\text{def}}{=} \left\{((a,b),y) \in \left(A \times_K^{f,g} B\right) \times Y \,\middle|\, b = \psi(y)\right\} \\
&\cong \{((a,b),y) \in (A \times B) \times Y \mid b = \psi(y) \text{ and } f(a) = g(b)\},
\end{aligned}
$$

so writing

$$
\begin{aligned}
S &= \left(X \times_A^{\phi,q_1} \left(A \times_K^{f,g} B\right)\right) \\
S' &= \left(\left(A \times_K^{f,g} B\right) \times_B^{q_2,\psi} Y\right),
\end{aligned}
$$

we have

$$
\begin{aligned}
S \times_{A \times_K^{f,g} B}^{p_2,p_1} S' &\overset{\text{def}}{=} \{((x,(a,b)),((a',b'),y)) \in S \times S' \mid p_1(x,(a,b)) = p_2((a',b'),y)\} \\
&\overset{\text{def}}{=} \{((x,(a,b)),((a',b'),y)) \in S \times S' \mid (a,b) = (a',b')\} \\
&\cong \{((x,a,b,y)) \in X \times A \times B \times Y \mid \phi(x) = a, \psi(y) = b, \text{ and } f(a) = g(b)\} \\
&\cong \{((x,a,b,y)) \in X \times A \times B \times Y \mid f(\phi(x)) = g(\psi(y))\} \\
&\overset{\text{def}}{=} X \times_K Y.
\end{aligned}
$$

This finishes the proof.

Item 5: Unitality

We have

$$
\begin{aligned}
X \times_X A &\cong \{(x,a) \in X \times A \mid f(a) = x\}, \\
A \times_X X &\cong \{(a,x) \in X \times A \mid f(a) = x\},
\end{aligned}
$$

which are isomorphic to A via the maps $(x,a) \mapsto a$ and $(a,x) \mapsto a$. The proof of the naturality of $\lambda^{\mathsf{Sets}_{/X}}$ and $\rho^{\mathsf{Sets}_{/X}}$ is omitted.

Item 6: Commutativity

We have

$$
A \times_C B \overset{\text{def}}{=} \{(a,b) \in A \times B \mid f(a) = g(b)\}
$$

$$
\begin{aligned}
&= \{(a,b) \in A \times B \mid g(b) = f(a)\} \\
&\cong \{(b,a) \in B \times A \mid g(b) = f(a)\} \\
&\overset{\text{def}}{=} B \times_C A.
\end{aligned}
$$

The proof of the naturality of $\sigma^{\mathsf{Sets}_{/X}}$ is omitted.

Item 7: Distributivity Over Coproducts

We have

$$
\begin{aligned}
A \times_X (B \coprod C) &\overset{\text{def}}{=} \left\{(a,z) \in A \times (B \coprod C) \,\middle|\, \phi_A(a) = \phi_{B \coprod C}(z)\right\} \\
&= \left\{(a,z) \in A \times (B \coprod C) \,\middle|\, z = (0,b) \text{ and } \phi_A(a) = \phi_{B \coprod C}(z)\right\} \\
&\quad \cup \left\{(a,z) \in A \times (B \coprod C) \,\middle|\, z = (1,c) \text{ and } \phi_A(a) = \phi_{B \coprod C}(z)\right\} \\
&= \{(a,z) \in A \times (B \coprod C) \mid z = (0,b) \text{ and } \phi_A(a) = \phi_B(b)\} \\
&\quad \cup \{(a,z) \in A \times (B \coprod C) \mid z = (1,c) \text{ and } \phi_A(a) = \phi_C(c)\} \\
&\cong \{(a,b) \in A \times B \mid \phi_A(a) = \phi_B(b)\} \\
&\quad \cup \{(a,c) \in A \times C \mid \phi_A(a) = \phi_C(c)\} \\
&\overset{\text{def}}{=} (A \times_X B) \cup (A \times_X C) \\
&\cong (A \times_X B) \coprod (A \times_X C),
\end{aligned}
$$

with the construction of the isomorphism

$$
\delta_r^{\mathsf{Sets}_{/X}} \colon (A \coprod B) \times_X C \overset{\sim}{\dashrightarrow} (A \times_X C) \coprod (B \times_X C)
$$

being similar. The proof of the naturality of $\delta_\ell^{\mathsf{Sets}_{/X}}$ and $\delta_r^{\mathsf{Sets}_{/X}}$ is omitted.

Item 8: Annihilation With the Empty Set

We have

$$
\begin{aligned}
A \times_X \emptyset &\overset{\text{def}}{=} \{(a,b) \in A \times \emptyset \mid f(a) = g(b)\} \\
&= \{k \in \emptyset \mid f(a) = g(b)\} \\
&= \emptyset,
\end{aligned}
$$

and similarly for $\emptyset \times_X A$, where we have used Item 7 of Proposition 1.3.4. The proof of the naturality of $\zeta_\ell^{\mathsf{Sets}_{/X}}$ and $\zeta_r^{\mathsf{Sets}_{/X}}$ is omitted.

Item 9: Interaction With Products

We have

$$\begin{aligned} A \times_{\mathrm{pt}} B &\overset{\text{def}}{=} \{(a,b) \in A \times B \mid !_A(a) = !_B(b)\} \\ &\overset{\text{def}}{=} \{(a,b) \in A \times B \mid \star = \star\} \\ &= \{(a,b) \in A \times B\} \\ &= A \times B. \end{aligned}$$

Item 10: Symmetric Monoidality

Omitted.

1.5 Equalisers

Let A and B be sets and let $f, g \colon A \rightrightarrows B$ be functions.

Definition 1.5.1 ▶ Equalisers of Sets

The **equaliser of f and g** is the equaliser of f and g in Sets as in Limits and Colimits, **??**.

Construction 1.5.2 ▶ Construction of Equalisers of Sets

Concretely, the equaliser of f and g is the pair $(\mathrm{Eq}(f,g), \mathrm{eq}(f,g))$ consisting of:

1. *The Limit.* The set $\mathrm{Eq}(f,g)$ defined by

$$\mathrm{Eq}(f,g) \overset{\text{def}}{=} \{a \in A \mid f(a) = g(a)\}.$$

2. *The Cone.* The inclusion map

$$\mathrm{eq}(f,g) \colon \mathrm{Eq}(f,g) \hookrightarrow A.$$

Proof 1.5.3 ▶ Proof of Construction 1.5.2

We claim that $\mathsf{Eq}(f,g)$ is the categorical equaliser of f and g in Sets. First we need to check that the relevant equaliser diagram commutes, i.e. that we have

$$f \circ \mathsf{eq}(f,g) = g \circ \mathsf{eq}(f,g),$$

which indeed holds by the definition of the set $\mathsf{Eq}(f,g)$. Next, we prove that $\mathsf{Eq}(f,g)$ satisfies the universal property of the equaliser. Suppose we have a diagram of the form

$$\begin{tikzcd}
\mathsf{Eq}(f,g) \arrow[r, hook, "\mathsf{eq}(f,g)"] & A \arrow[r, shift left, "f"] \arrow[r, shift right, "g"'] & B \\
E \arrow[ur, "e"'] & &
\end{tikzcd}$$

in Sets. Then there exists a unique map $\phi\colon E \longrightarrow \mathsf{Eq}(f,g)$ making the diagram

$$\begin{tikzcd}
\mathsf{Eq}(f,g) \arrow[r, hook, "\mathsf{eq}(f,g)"] & A \arrow[r, shift left, "f"] \arrow[r, shift right, "g"'] & B \\
E \arrow[u, dashed, "\phi"] \arrow[ur, "e"'] \arrow[u, phantom, shift left=6ex, "\exists!"] & &
\end{tikzcd}$$

commute, being uniquely determined by the condition

$$\mathsf{eq}(f,g) \circ \phi = e$$

via

$$\phi(x) = e(x)$$

for each $x \in E$, where we note that $e(x) \in A$ indeed lies in $\mathsf{Eq}(f,g)$ by the condition

$$f \circ e = g \circ e,$$

which gives

$$f(e(x)) = g(e(x))$$

for each $x \in E$, so that $e(x) \in \mathsf{Eq}(f,g)$. ∎

Proposition 1.5.4 ► Properties of Equalisers of Sets

Let A, B, and C be sets.

1. *Associativity.* We have isomorphisms of sets[1]

$$\underbrace{\mathrm{Eq}(f \circ \mathrm{eq}(g,h), g \circ \mathrm{eq}(g,h))}_{=\mathrm{Eq}(f\circ\mathrm{eq}(g,h),h\circ\mathrm{eq}(g,h))} \cong \mathrm{Eq}(f,g,h) \cong \underbrace{\mathrm{Eq}(f \circ \mathrm{eq}(f,g), h \circ \mathrm{eq}(f,g))}_{=\mathrm{Eq}(g\circ\mathrm{eq}(f,g),h\circ\mathrm{eq}(f,g))},$$

   where $\mathrm{Eq}(f,g,h)$ is the limit of the diagram

$$\begin{tikzcd} A \arrow[r, shift left, "f"] \arrow[r, "g"] \arrow[r, shift right, "h"'] & B \end{tikzcd}$$

   in Sets, being explicitly given by

$$\mathrm{Eq}(f,g,h) \cong \{a \in A \mid f(a) = g(a) = h(a)\}.$$

2. *Unitality.* We have an isomorphism of sets

$$\mathrm{Eq}(f,f) \cong A.$$

3. *Commutativity.* We have an isomorphism of sets

$$\mathrm{Eq}(f,g) \cong \mathrm{Eq}(g,f).$$

4. *Interaction With Composition.* Let

$$\begin{tikzcd} A \arrow[r, shift left, "f"] \arrow[r, shift right, "g"'] & B \arrow[r, shift left, "h"] \arrow[r, shift right, "k"'] & C \end{tikzcd}$$

   be functions. We have an inclusion of sets

$$\mathrm{Eq}(h \circ f \circ \mathrm{eq}(f,g), k \circ g \circ \mathrm{eq}(f,g)) \subset \mathrm{Eq}(h \circ f, k \circ g),$$

   where $\mathrm{Eq}(h \circ f \circ \mathrm{eq}(f,g), k \circ g \circ \mathrm{eq}(f,g))$ is the equaliser of the composition

$$\mathrm{Eq}(f,g) \overset{\mathrm{eq}(f,g)}{\hookrightarrow} A \underset{g}{\overset{f}{\rightrightarrows}} B \underset{k}{\overset{h}{\rightrightarrows}} C.$$

[1]That is, the following three ways of forming "the' equaliser of (f, g, h) agree:

(a) Take the equaliser of (f, g, h), i.e. the limit of the diagram

$$A \underset{h}{\overset{f}{\underset{g}{\rightrightarrows}}} B$$

in Sets.

(b) First take the equaliser of f and g, forming a diagram

$$\mathrm{Eq}(f, g) \overset{\mathrm{eq}(f,g)}{\hookrightarrow} A \underset{g}{\overset{f}{\rightrightarrows}} B$$

and then take the equaliser of the composition

$$\mathrm{Eq}(f, g) \overset{\mathrm{eq}(f,g)}{\hookrightarrow} A \underset{h}{\overset{f}{\rightrightarrows}} B,$$

obtaining a subset

$$\mathrm{Eq}(f \circ \mathrm{eq}(f, g), h \circ \mathrm{eq}(f, g)) = \mathrm{Eq}(g \circ \mathrm{eq}(f, g), h \circ \mathrm{eq}(f, g))$$

of $\mathrm{Eq}(f, g)$.

(c) First take the equaliser of g and h, forming a diagram

$$\mathrm{Eq}(g, h) \overset{\mathrm{eq}(g,h)}{\hookrightarrow} A \underset{h}{\overset{g}{\rightrightarrows}} B$$

and then take the equaliser of the composition

$$\mathrm{Eq}(g, h) \overset{\mathrm{eq}(g,h)}{\hookrightarrow} A \underset{g}{\overset{f}{\rightrightarrows}} B,$$

obtaining a subset

$$\mathrm{Eq}(f \circ \mathrm{eq}(g, h), g \circ \mathrm{eq}(g, h)) = \mathrm{Eq}(f \circ \mathrm{eq}(g, h), h \circ \mathrm{eq}(g, h))$$

of $\mathrm{Eq}(g, h)$.

Proof 1.5.5 ► Proof of Proposition 1.5.4

Item 1: Associativity

We first prove that $\mathrm{Eq}(f, g, h)$ is indeed given by

$$\mathrm{Eq}(f, g, h) \cong \{a \in A \mid f(a) = g(a) = h(a)\}.$$

Indeed, suppose we have a diagram of the form

$$\begin{tikzcd}
\mathrm{Eq}(f,g,h) \arrow[r, hook, "\mathrm{eq}(f,g,h)"] & A \arrow[r, shift left, "f"] \arrow[r, "g"] \arrow[r, shift right, "h"'] & B \\
E \arrow[ur, "e"'] & &
\end{tikzcd}$$

in Sets. Then there exists a unique map $\phi\colon E \longrightarrow \mathrm{Eq}(f, g, h)$, uniquely determined by the condition

$$\mathrm{eq}(f, g) \circ \phi = e$$

being necessarily given by

$$\phi(x) = e(x)$$

for each $x \in E$, where we note that $e(x) \in A$ indeed lies in $\mathrm{Eq}(f, g, h)$ by the condition

$$f \circ e = g \circ e = h \circ e,$$

which gives

$$f(e(x)) = g(e(x)) = h(e(x))$$

for each $x \in E$, so that $e(x) \in \mathrm{Eq}(f, g, h)$.
We now check the equalities

$$\mathrm{Eq}(f \circ \mathrm{eq}(g, h), g \circ \mathrm{eq}(g, h)) \cong \mathrm{Eq}(f, g, h) \cong \mathrm{Eq}(f \circ \mathrm{eq}(f, g), h \circ \mathrm{eq}(f, g)).$$

Indeed, we have

$$\begin{aligned}
\mathrm{Eq}(f \circ \mathrm{eq}(g, h), g \circ \mathrm{eq}(g, h)) &\cong \{x \in \mathrm{Eq}(g, h) \mid [f \circ \mathrm{eq}(g, h)](a) = [g \circ \mathrm{eq}(g, h)](a)\} \\
&\cong \{x \in \mathrm{Eq}(g, h) \mid f(a) = g(a)\} \\
&\cong \{x \in A \mid f(a) = g(a) \text{ and } g(a) = h(a)\} \\
&\cong \{x \in A \mid f(a) = g(a) = h(a)\}
\end{aligned}$$

$$\cong \mathrm{Eq}(f, g, h).$$

Similarly, we have

$$\begin{aligned}
\mathrm{Eq}(f \circ \mathrm{eq}(f,g), h \circ \mathrm{eq}(f,g)) &\cong \{x \in \mathrm{Eq}(f,g) \mid [f \circ \mathrm{eq}(f,g)](a) = [h \circ \mathrm{eq}(f,g)](a)\} \\
&\cong \{x \in \mathrm{Eq}(f,g) \mid f(a) = h(a)\} \\
&\cong \{x \in A \mid f(a) = h(a) \text{ and } f(a) = g(a)\} \\
&\cong \{x \in A \mid f(a) = g(a) = h(a)\} \\
&\cong \mathrm{Eq}(f, g, h).
\end{aligned}$$

Item 2: Unitality

Indeed, we have

$$\begin{aligned}
\mathrm{Eq}(f, f) &\overset{\text{def}}{=} \{a \in A \mid f(a) = f(a)\} \\
&= A.
\end{aligned}$$

Item 3: Commutativity

Indeed, we have

$$\begin{aligned}
\mathrm{Eq}(f, g) &\overset{\text{def}}{=} \{a \in A \mid f(a) = g(a)\} \\
&= \{a \in A \mid g(a) = f(a)\} \\
&\overset{\text{def}}{=} \mathrm{Eq}(g, f).
\end{aligned}$$

Item 4: Interaction With Composition

Indeed, we have

$$\begin{aligned}
\mathrm{Eq}(h \circ f \circ \mathrm{eq}(f,g), k \circ g \circ \mathrm{eq}(f,g)) &\cong \{a \in \mathrm{Eq}(f,g) \mid h(f(a)) = k(g(a))\} \\
&\cong \{a \in A \mid f(a) = g(a) \text{ and } h(f(a)) = k(g(a))\}.
\end{aligned}$$

and

$$\mathrm{Eq}(h \circ f, k \circ g) \cong \{a \in A \mid h(f(a)) = k(g(a))\},$$

and thus there,s an inclusion from $\mathrm{Eq}(h \circ f \circ \mathrm{eq}(f,g), k \circ g \circ \mathrm{eq}(f,g))$ to $\mathrm{Eq}(h \circ f, k \circ g)$. ∎

1.6 Inverse Limits

Let $\left(X_\alpha, f_{\alpha\beta}\right)_{\alpha,\beta\in I}\colon (I, \preceq) \longrightarrow$ Sets be an inverse system of sets.

Definition 1.6.1 ► Inverse Limits of Sets

The **inverse limit of** $(X_\alpha, f_{\alpha\beta})_{\alpha,\beta\in I}$ is the inverse limit of $(X_\alpha, f_{\alpha\beta})_{\alpha,\beta\in I}$ in Sets as in Limits and Colimits, **??**.

Construction 1.6.2 ► Construction of Inverse Limits of Sets

Concretely, the inverse limit of $(X_\alpha, f_{\alpha\beta})_{\alpha,\beta\in I}$ is the pair $\left(\varprojlim_{\alpha\in I}(X_\alpha), \{\mathrm{pr}_\alpha\}_{\alpha\in I}\right)$ consisting of:

1. *The Limit.* The set $\varprojlim_{\alpha\in I}(X_\alpha)$ defined by

$$\varprojlim_{\alpha\in I}(X_\alpha) \stackrel{\text{def}}{=} \left\{ (x_\alpha)_{\alpha\in I} \in \prod_{\alpha\in I} X_\alpha \;\middle|\; \begin{array}{l}\text{for each } \alpha, \beta \in I, \text{ if } \alpha \preceq \beta, \\ \text{then we have } x_\alpha = f_{\alpha\beta}(x_\beta)\end{array} \right\}.$$

2. *The Cone.* The collection

$$\left\{ \mathrm{pr}_\gamma \colon \varprojlim_{\alpha\in I}(X_\alpha) \to X_\gamma \right\}_{\gamma\in I}$$

of maps of sets defined as the restriction of the maps

$$\left\{ \mathrm{pr}_\gamma \colon \prod_{\alpha\in I} X_\alpha \to X_\gamma \right\}_{\gamma\in I}$$

of Item 2 of Construction 1.2.2 to $\varprojlim_{\alpha\in I}(X_\alpha)$ and hence given by

$$\mathrm{pr}_\gamma((x_\alpha)_{\alpha\in I}) \stackrel{\text{def}}{=} x_\gamma$$

for each $\gamma \in I$ and each $(x_\alpha)_{\alpha\in I} \in \varprojlim_{\alpha\in I}(X_\alpha)$.

Proof 1.6.3 ▶ Proof of Construction 1.6.2

We claim that $\varprojlim_{\alpha\in I}(X_\alpha)$ is the limit of the inverse system of sets $\left(X_\alpha, f_{\alpha\beta}\right)_{\alpha,\beta\in I}$. First we need to check that the limit diagram defined by it commutes, i.e. that we have

$$f_{\alpha\beta}\circ \mathrm{pr}_\alpha = \mathrm{pr}_\beta, \qquad \begin{tikzcd} & \varprojlim_{\alpha\in I}(X_\alpha) \arrow[dl, "\mathrm{pr}_\alpha"'] \arrow[dr, "\mathrm{pr}_\beta"] & \\ X_\alpha \arrow[rr, "f_{\alpha\beta}"'] & & X_\beta \end{tikzcd}$$

for each $\alpha, \beta \in I$ with $\alpha \preceq \beta$. Indeed, given $\left(x_\gamma\right)_{\gamma\in I} \in \varprojlim_{\gamma\in I}\left(X_\gamma\right)$, we have

$$\begin{aligned} \left[f_{\alpha\beta}\circ \mathrm{pr}_\alpha\right]\left(\left(x_\gamma\right)_{\gamma\in I}\right) &\overset{\text{def}}{=} f_{\alpha\beta}\left(\mathrm{pr}_\alpha\left(\left(x_\gamma\right)_{\gamma\in I}\right)\right) \\ &\overset{\text{def}}{=} f_{\alpha\beta}(x_\alpha) \\ &= x_\beta \\ &\overset{\text{def}}{=} \mathrm{pr}_\beta\left(\left(x_\gamma\right)_{\gamma\in I}\right), \end{aligned}$$

where the third equality comes from the definition of $\varprojlim_{\alpha\in I}(X_\alpha)$. Next, we prove that $\varprojlim_{\alpha\in I}(X_\alpha)$ satisfies the universal property of an inverse limit. Suppose that we have, for each $\alpha, \beta \in I$ with $\alpha \preceq \beta$, a diagram of the form

$$\begin{tikzcd} & L \arrow[dl, bend right, "p_\alpha"'] \arrow[dr, bend left, "p_\beta"] & \\ & \varprojlim_{\alpha\in I}(X_\alpha) \arrow[dl, dashed, "\mathrm{pr}_\alpha"'] \arrow[dr, dashed, "\mathrm{pr}_\beta"] & \\ X_\alpha \arrow[rr, "f_{\alpha\beta}"'] & & X_\beta \end{tikzcd}$$

in Sets. Then there indeed exists a unique map $\phi : L \xrightarrow{\exists!} \varprojlim_{\alpha \in I}(X_\alpha)$ making the diagram

$$\begin{tikzcd}
 & L \arrow[d, dashed, "\phi"', "\exists!"] \arrow[ddl, bend right, "p_\alpha"'] \arrow[ddr, bend left, "p_\beta"] & \\
 & \varprojlim_{\alpha \in I}(X_\alpha) \arrow[dl, "\mathrm{pr}_\alpha"'] \arrow[dr, "\mathrm{pr}_\beta"] & \\
X_\alpha \arrow[rr, "f_{\alpha\beta}"'] & & X_\beta
\end{tikzcd}$$

commute, being uniquely determined by the family of conditions

$$\{p_\alpha = \mathrm{pr}_\alpha \circ \phi\}_{\alpha \in I}$$

via

$$\phi(\ell) = (p_\alpha(\ell))_{\alpha \in I}$$

for each $\ell \in L$, where we note that $(p_\alpha(\ell))_{\alpha \in I} \in \prod_{\alpha \in I} X_\alpha$ indeed lies in $\varprojlim_{\alpha \in I}(X_\alpha)$, as we have

$$\begin{aligned} f_{\alpha\beta}(p_\alpha(\ell)) &\overset{\text{def}}{=} [f_{\alpha\beta} \circ p_\alpha](\ell) \\ &\overset{\text{def}}{=} p_\beta(\ell) \end{aligned}$$

for each $\beta \in I$ with $\alpha \preceq \beta$ by the commutativity of the diagram for $(L, \{p_\alpha\}_{\alpha \in I})$. ∎

Example 1.6.4 ► Examples of Inverse Limits of Sets

Here are some examples of inverse limits of sets.

1. *The p-Adic Integers.* The ring of p-adic integers $\mathbb{Z}_p$ of **??** is the inverse limit

$$\mathbb{Z}_p \cong \varprojlim_{n \in \mathbb{N}}(\mathbb{Z}_{/p^n});$$

see **??**.

2. *Rings of Formal Power Series.* The ring $R[\![t]\!]$ of formal power series in a variable t is the inverse limit

$$R[\![t]\!] \cong \varprojlim_{n\in\mathbb{N}}(R[t]/t^nR[t]);$$

see **??**.

3. *Profinite Groups.* Profinite groups are inverse limits of finite groups; see **??**.

2 Colimits of Sets

2.1 The Initial Set

Definition 2.1.1 ▶ The Initial Set

The **initial set** is the initial object of Sets as in Limits and Colimits, **??**.

Construction 2.1.2 ▶ Construction of the Initial Set

Concretely, the initial set is the pair $\left(\emptyset, \{\iota_A\}_{A\in\mathrm{Obj}(\mathsf{Sets})}\right)$ consisting of:

1. *The Colimit.* The empty set $\emptyset$ of Definition 3.1.1.

2. *The Cocone.* The collection of maps

$$\{\iota_A\colon \emptyset \to A\}_{A\in\mathrm{Obj}(\mathsf{Sets})}$$

given by the inclusion maps from $\emptyset$ to A.

Proof 2.1.3 ▶ Proof of Construction 2.1.2

We claim that $\emptyset$ is the initial object of Sets. Indeed, suppose we have a diagram of the form

$$\emptyset \qquad A$$

in Sets. Then there exists a unique map $\phi\colon \emptyset \to A$ making the diagram

$$\begin{tikzcd} \emptyset \arrow[r, dashed, "\phi", "\exists!"'] & A \end{tikzcd}$$

commute, namely the inclusion map ι_A.

2.2 Coproducts of Families of Sets

Let $\{A_i\}_{i\in I}$ be a family of sets.

Definition 2.2.1 ▶ The Coproduct of a Family of Sets

The **coproduct of** $\{A_i\}_{i\in I}$[1] is the coproduct of $\{A_i\}_{i\in I}$ in Sets as in Limits and Colimits, **??**.

[1] *Further Terminology:* Also called the **disjoint union of the family** $\{A_i\}_{i\in I}$.

Construction 2.2.2 ▶ Construction of the Coproduct of a Family of Sets

Concretely, the disjoint union of $\{A_i\}_{i\in I}$ is the pair $\left(\coprod_{i\in I} A_i, \{\mathrm{inj}_i\}_{i\in I}\right)$ consisting of:

1. *The Colimit.* The set $\coprod_{i\in I} A_i$ defined by

$$\coprod_{i\in I} A_i \stackrel{\text{def}}{=} \left\{(i,x) \in I \times \left(\bigcup_{i\in I} A_i\right) \middle| x \in A_i\right\}.$$

2. *The Cocone.* The collection

$$\left\{\mathrm{inj}_i\colon A_i \to \coprod_{i\in I} A_i\right\}_{i\in I}$$

of maps given by

$$\mathrm{inj}_i(x) \stackrel{\text{def}}{=} (i,x)$$

for each $x \in A_i$ and each $i \in I$.

Proof 2.2.3 ▶ Proof of Construction 2.2.2

We claim that $\coprod_{i\in I} A_i$ is the categorical coproduct of $\{A_i\}_{i\in I}$ in Sets. Indeed, suppose we have, for each $i \in I$, a diagram of the form

$$\begin{tikzcd} & C \\ A_i \arrow[ur, "\iota_i"] \arrow[r, "\mathrm{inj}_i"'] & \coprod_{i\in I} A_i \end{tikzcd}$$

in Sets. Then there exists a unique map $\phi\colon \coprod_{i\in I} A_i \longrightarrow C$ making the diagram

$$\begin{tikzcd} & C \\ A_i \arrow[ur, "\iota_i"] \arrow[r, "\mathrm{inj}_i"'] & \coprod_{i\in I} A_i \arrow[u, dashed, "\phi"', "\exists!"] \end{tikzcd}$$

commute, being uniquely determined by the condition $\phi \circ \mathrm{inj}_i = \iota_i$ for each $i \in I$ via

$$\phi((i,x)) = \iota_i(x)$$

for each $(i,x) \in \coprod_{i\in I} A_i$. ∎

Proposition 2.2.4 ▶ Properties of Coproducts of Families of Sets

Let $\{A_i\}_{i\in I}$ be a family of sets.

1. *Functoriality.* The assignment $\{A_i\}_{i\in I} \mapsto \coprod_{i\in I} A_i$ defines a functor

$$\coprod_{i\in I}\colon \mathrm{Fun}(I_{\mathrm{disc}}, \mathrm{Sets}) \longrightarrow \mathrm{Sets}$$

   where

   - *Action on Objects.* For each $(A_i)_{i\in I} \in \mathrm{Obj}(\mathrm{Fun}(I_{\mathrm{disc}}, \mathrm{Sets}))$, we

have

$$\left[\coprod_{i\in I}\right]((A_i)_{i\in I}) \overset{\text{def}}{=} \coprod_{i\in I} A_i$$

- *Action on Morphisms.* For each $(A_i)_{i\in I}, (B_i)_{i\in I} \in \mathrm{Obj}(\mathrm{Fun}(I_{\mathrm{disc}}, \mathrm{Sets}))$, the action on Hom-sets

$$\left(\coprod_{i\in I}\right)_{(A_i)_{i\in I},(B_i)_{i\in I}} : \mathrm{Nat}((A_i)_{i\in I}, (B_i)_{i\in I}) \to \mathrm{Sets}\left(\coprod_{i\in I} A_i, \coprod_{i\in I} B_i\right)$$

of $\coprod_{i\in I}$ at $((A_i)_{i\in I}, (B_i)_{i\in I})$ is defined by sending a map

$$\{f_i \colon A_i \to B_i\}_{i\in I}$$

in $\mathrm{Nat}((A_i)_{i\in I}, (B_i)_{i\in I})$ to the map of sets

$$\coprod_{i\in I} f_i \colon \coprod_{i\in I} A_i \to \coprod_{i\in I} B_i$$

defined by

$$\left[\coprod_{i\in I} f_i\right](i, a) \overset{\text{def}}{=} f_i(a)$$

for each $(i, a) \in \coprod_{i\in I} A_i$.

Proof 2.2.5 ► Proof of Proposition 2.2.4

Item 1: Functoriality

This follows from Limits and Colimits, **??** of **??**.

2.3 Binary Coproducts

Let A and B be sets.

Definition 2.3.1 ▶ Coproducts of Sets

The **coproduct of** A **and** B[1] is the coproduct of A and B in Sets as in Limits and Colimits, **??**.

[1] *Further Terminology:* Also called the **disjoint union of** A **and** B.

Construction 2.3.2 ▶ Construction of Coproducts of Sets

Concretely, the coproduct of A and B is the pair $(A \coprod B, \{\mathrm{inj}_1, \mathrm{inj}_2\})$ consisting of:

1. *The Colimit.* The set $A \coprod B$ defined by

$$\begin{aligned} A \coprod B &\stackrel{\text{def}}{=} \coprod_{z \in \{A,B\}} z \\ &\stackrel{\text{def}}{=} \{(0,a) \in S \mid a \in A\} \cup \{(1,b) \in S \mid b \in B\}, \end{aligned}$$

   where $S = \{0,1\} \times (A \cup B)$.

2. *The Cocone.* The maps

$$\begin{aligned} \mathrm{inj}_1 &\colon A \to A \coprod B, \\ \mathrm{inj}_2 &\colon B \to A \coprod B, \end{aligned}$$

   given by

$$\begin{aligned} \mathrm{inj}_1(a) &\stackrel{\text{def}}{=} (0,a), \\ \mathrm{inj}_2(b) &\stackrel{\text{def}}{=} (1,b), \end{aligned}$$

   for each $a \in A$ and each $b \in B$.

Proof 2.3.3 ▶ Proof of Construction 2.3.2

We claim that $A \coprod B$ is the categorical coproduct of A and B in Sets. Indeed, suppose we have a diagram of the form

$$\begin{tikzcd} & C & \\ A \arrow[ur, bend left, "\iota_1"] \arrow[r, hook, "\mathrm{inj}_1"'] & A \coprod B \arrow[u, phantom] & B \arrow[ul, bend right, "\iota_2"'] \arrow[l, hook', "\mathrm{inj}_2"] \end{tikzcd}$$

in Sets. Then there exists a unique map $\phi\colon A \coprod B \longrightarrow C$ making the diagram

$$\begin{tikzcd} & C & \\ A \arrow[ur, bend left, "\iota_1"] \arrow[r, hook, "\mathrm{inj}_1"'] & A \coprod B \arrow[u, dashed, "\phi"', "\exists!"] & B \arrow[ul, bend right, "\iota_2"'] \arrow[l, hook', "\mathrm{inj}_2"] \end{tikzcd}$$

commute, being uniquely determined by the conditions

$$\begin{aligned} \phi \circ \mathrm{inj}_A &= \iota_A, \\ \phi \circ \mathrm{inj}_B &= \iota_B \end{aligned}$$

via

$$\phi(x) = \begin{cases} \iota_A(a) & \text{if } x = (0, a), \\ \iota_B(b) & \text{if } x = (1, b) \end{cases}$$

for each $x \in A \coprod B$. ∎

Proposition 2.3.4 ▶ Properties of Coproducts of Sets

Let A, B, C, and X be sets.

1. *Functoriality.* The assignment $A, B, (A, B) \mapsto A \coprod B$ defines functors

$$\begin{aligned} A \coprod -\colon &\quad \mathsf{Sets} \longrightarrow \mathsf{Sets}, \\ - \coprod B\colon &\quad \mathsf{Sets} \longrightarrow \mathsf{Sets}, \\ -_1 \coprod -_2\colon &\quad \mathsf{Sets} \times \mathsf{Sets} \longrightarrow \mathsf{Sets}, \end{aligned}$$

   where $-_1 \coprod -_2$ is the functor where

- *Action on Objects.* For each $(A, B) \in \mathrm{Obj}(\mathsf{Sets} \times \mathsf{Sets})$, we have
$$[-_1 \coprod -_2](A, B) \overset{\text{def}}{=} A \coprod B.$$
- *Action on Morphisms.* For each $(A, B), (X, Y) \in \mathrm{Obj}(\mathsf{Sets})$, the action on Hom-sets
$$\coprod_{(A,B),(X,Y)} \colon \mathsf{Sets}(A, X) \times \mathsf{Sets}(B, Y) \to \mathsf{Sets}(A \coprod B, X \coprod Y)$$
of $\coprod$ at $((A, B), (X, Y))$ is defined by sending (f, g) to the function
$$f \coprod g \colon A \coprod B \to X \coprod Y$$
defined by
$$[f \coprod g](x) \overset{\text{def}}{=} \begin{cases} (0, f(a)) & \text{if } x = (0, a), \\ (1, g(b)) & \text{if } x = (1, b), \end{cases}$$
for each $x \in A \coprod B$.

and where $A \coprod -$ and $- \coprod B$ are the partial functors of $-_1 \coprod -_2$ at $A, B \in \mathrm{Obj}(\mathsf{Sets})$.

2. *Associativity.* We have an isomorphism of sets
$$\alpha^{\mathsf{Sets},\coprod}_{X,Y,Z} \colon (X \coprod Y) \coprod Z \overset{\sim}{\dashrightarrow} X \coprod (Y \coprod Z),$$
natural in $X, Y, Z \in \mathrm{Obj}(\mathsf{Sets})$.

3. *Unitality.* We have isomorphisms of sets
$$\lambda^{\mathsf{Sets},\coprod}_{X} \colon \emptyset \coprod X \overset{\sim}{\dashrightarrow} X,$$
$$\rho^{\mathsf{Sets},\coprod}_{X} \colon X \coprod \emptyset \overset{\sim}{\dashrightarrow} X,$$
natural in $X \in \mathrm{Obj}(\mathsf{Sets})$.

4. *Commutativity.* We have an isomorphism of sets
$$\sigma^{\mathsf{Sets},\coprod}_{X,Y} \colon X \coprod Y \overset{\sim}{\dashrightarrow} Y \coprod X,$$
natural in $X, Y \in \mathrm{Obj}(\mathsf{Sets})$.

5. *Symmetric Monoidality.* The 7-tuple $\left(\mathsf{Sets}, \coprod, \emptyset, \alpha^{\mathsf{Sets}}_{\coprod}, \lambda^{\mathsf{Sets}}_{\coprod}, \rho^{\mathsf{Sets}}_{\coprod}, \sigma^{\mathsf{Sets}}\right)$ is a symmetric monoidal category.

Proof 2.3.5 ▶ Proof of Proposition 2.3.4

Item 1: Functoriality

This follows from Limits and Colimits, **??** of **??**.

Item 2: Associativity

This is proved in the proof of Monoidal Structures on the Category of Sets, **??**.

Item 3: Unitality

This is proved in the proof of Monoidal Structures on the Category of Sets, **????**.

Item 4: Commutativity

This is proved in the proof of Monoidal Structures on the Category of Sets, **??**.

Item 5: Symmetric Monoidality

This is a repetition of Monoidal Structures on the Category of Sets, **??**, and is proved there.

2.4 Pushouts

Let A, B, and C be sets and let $f\colon C \to A$ and $g\colon C \to B$ be functions.

Definition 2.4.1 ▶ Pushouts of Sets

The **pushout of A and B over C along f and g**[1] is the pushout of A and B over C along f and g in Sets as in Limits and Colimits, **??**.

[1] *Further Terminology:* Also called the **fibre coproduct of A and B over C along f and g**.

Construction 2.4.2 ▶ Construction of Pushouts of Sets

Concretely, the pushout of A and B over C along f and g is the pair $(A \coprod_C B, \{\mathrm{inj}_1, \mathrm{inj}_2\})$ consisting of:

1. *The Colimit.* The set $A \coprod_C B$ defined by

$$A \coprod_C B \overset{\text{def}}{=} A \coprod B/\sim_C,$$

where $\sim_C$ is the equivalence relation on $A \coprod B$ generated by $(0, f(c)) \sim_C (1, g(c))$.

2. *The Cocone.* The maps

$$\begin{aligned} \mathrm{inj}_1 &: A \longrightarrow A \coprod_C B, \\ \mathrm{inj}_2 &: B \longrightarrow A \coprod_C B \end{aligned}$$

given by

$$\begin{aligned} \mathrm{inj}_1(a) &\overset{\text{def}}{=} [(0, a)] \\ \mathrm{inj}_2(b) &\overset{\text{def}}{=} [(1, b)] \end{aligned}$$

for each $a \in A$ and each $b \in B$.

PROOF 2.4.3 ▶ PROOF OF CONSTRUCTION 2.4.2

We claim that $A \coprod_C B$ is the categorical pushout of A and B over C with respect to (f, g) in Sets. First we need to check that the relevant pushout diagram commutes, i.e. that we have

$$\mathrm{inj}_1 \circ f = \mathrm{inj}_2 \circ g, \qquad \begin{tikzcd} A \coprod_C B & B \arrow[l, "\mathrm{inj}_2"'] \\ A \arrow[u, "\mathrm{inj}_1"] & C. \arrow[l, "f"] \arrow[u, "g"'] \end{tikzcd}$$

Indeed, given $c \in C$, we have

$$\begin{aligned} [\mathrm{inj}_1 \circ f](c) &= \mathrm{inj}_1(f(c)) \\ &= [(0, f(c))] \\ &= [(1, g(c))] \\ &= \mathrm{inj}_2(g(c)) \\ &= [\mathrm{inj}_2 \circ g](c), \end{aligned}$$

where $[(0, f(c))] = [(1, g(c))]$ by the definition of the relation $\sim$ on $A \coprod B$. Next, we prove that $A \coprod_C B$ satisfies the universal property of the pushout.

Suppose we have a diagram of the form

$$\begin{tikzcd}
P & & \\
& A \coprod_C B \arrow[ul, bend right, "\iota_1"'] & B \arrow[l, "\mathsf{inj}_2"'] \arrow[ul, bend right, "\iota_2"'] \\
& A \arrow[u, "\mathsf{inj}_1"] \arrow[ur, phantom, "\ulcorner"] & C \arrow[l, "f"] \arrow[u, "g"']
\end{tikzcd}$$

in Sets. Then there exists a unique map $\phi\colon A \coprod_C B \longrightarrow P$ making the diagram

$$\begin{tikzcd}
P & & \\
& A \coprod_C B \arrow[ul, dashed, "\exists!"', "\phi"] & B \arrow[l, "\mathsf{inj}_2"'] \arrow[ul, bend right, "\iota_2"'] \\
& A \arrow[u, "\mathsf{inj}_1"] \arrow[ur, phantom, "\ulcorner"] \arrow[uul, bend left, "\iota_1"] & C \arrow[l, "f"] \arrow[u, "g"']
\end{tikzcd}$$

commute, being uniquely determined by the conditions

$$\begin{aligned}
\phi \circ \mathsf{inj}_1 &= \iota_1, \\
\phi \circ \mathsf{inj}_2 &= \iota_2
\end{aligned}$$

via

$$\phi(x) = \begin{cases} \iota_1(a) & \text{if } x = [(0, a)], \\ \iota_2(b) & \text{if } x = [(1, b)] \end{cases}$$

for each $x \in A \coprod_C B$, where the well-definedness of ϕ is guaranteed by the equality $\iota_1 \circ f = \iota_2 \circ g$ and the definition of the relation $\sim$ on $A \coprod B$ as follows:

1. *Case 1:* Suppose we have $x = [(0, a)] = [(0, a')]$ for some $a, a' \in A$. Then, by Remark 2.4.4, we have a sequence
$$(0, a) \sim' x_1 \sim' \cdots \sim' x_n \sim' (0, a').$$

2. *Case 2:* Suppose we have $x = [(1, b)] = [(1, b')]$ for some $b, b' \in B$. Then, by Remark 2.4.4, we have a sequence
$$(1, b) \sim' x_1 \sim' \cdots \sim' x_n \sim' (1, b').$$

3. *Case 3:* Suppose we have $x = [(0, a)] = [(1, b)]$ for some $a \in A$ and $b \in B$. Then, by Remark 2.4.4, we have a sequence
$$(0, a) \sim' x_1 \sim' \cdots \sim' x_n \sim' (1, b).$$

In all these cases, we declare $x \sim' y$ iff there exists some $c \in C$ such that $x = (0, f(c))$ and $y = (1, g(c))$ or $x = (1, g(c))$ and $y = (0, f(c))$. Then, the equality $\iota_1 \circ f = \iota_2 \circ g$ gives
$$\begin{aligned}\phi([x]) &= \phi([(0, f(c))]) \\ &\overset{\text{def}}{=} \iota_1(f(c)) \\ &= \iota_2(g(c)) \\ &\overset{\text{def}}{=} \phi([(1, g(c))]) \\ &= \phi([y]),\end{aligned}$$
with the case where $x = (1, g(c))$ and $y = (0, f(c))$ similarly giving $\phi([x]) = \phi([y])$. Thus, if $x \sim' y$, then $\phi([x]) = \phi([y])$. Applying this equality pairwise to the sequences
$$\begin{aligned}(0, a) &\sim' x_1 \sim' \cdots \sim' x_n \sim' (0, a'), \\ (1, b) &\sim' x_1 \sim' \cdots \sim' x_n \sim' (1, b'), \\ (0, a) &\sim' x_1 \sim' \cdots \sim' x_n \sim' (1, b)\end{aligned}$$
gives
$$\begin{aligned}\phi([(0, a)]) &= \phi([(0, a')]), \\ \phi([(1, b)]) &= \phi([(1, b')]), \\ \phi([(0, a)]) &= \phi([(1, b)]),\end{aligned}$$
showing ϕ to be well-defined. ∎

Remark 2.4.4 ► Unwinding Definition 2.4.1

In detail, by Conditions on Relations, **??**, the relation $\sim$ of Definition 2.4.1 is given by declaring $a \sim b$ iff one of the following conditions is satisfied:

1. We have $a, b \in A$ and $a = b$.
2. We have $a, b \in B$ and $a = b$.
3. There exist $x_1, \ldots, x_n \in A \coprod B$ such that $a \sim' x_1 \sim' \cdots \sim' x_n \sim' b$, where we declare $x \sim' y$ if one of the following conditions is satisfied:
   (a) There exists $c \in C$ such that $x = (0, f(c))$ and $y = (1, g(c))$.
   (b) There exists $c \in C$ such that $x = (1, g(c))$ and $y = (0, f(c))$.

   In other words, there exist $x_1, \ldots, x_n \in A \coprod B$ satisfying the following conditions:

   (c) There exists $c_0 \in C$ satisfying one of the following conditions:
      i. We have $a = f(c_0)$ and $x_1 = g(c_0)$.
      ii. We have $a = g(c_0)$ and $x_1 = f(c_0)$.
   (d) For each $1 \leq i \leq n-1$, there exists $c_i \in C$ satisfying one of the following conditions:
      i. We have $x_i = f(c_i)$ and $x_{i+1} = g(c_i)$.
      ii. We have $x_i = g(c_i)$ and $x_{i+1} = f(c_i)$.
   (e) There exists $c_n \in C$ satisfying one of the following conditions:
      i. We have $x_n = f(c_n)$ and $b = g(c_n)$.
      ii. We have $x_n = g(c_n)$ and $b = f(c_n)$.

Remark 2.4.5 ► Pushouts of Sets Depend on the Maps

It is common practice to write $A \coprod_C B$ for the pushout of A and B over C along f and g, omitting the maps f and g from the notation and instead leaving them implicit, to be understood from the context.
However, the set $A \coprod_C B$ depends very much on the maps f and g, and sometimes it is necessary or useful to note this dependence explicitly. In such situations, we will write $A \coprod_{f,C,g} B$ or $A \coprod_C^{f,g} B$ for $A \coprod_C B$.

Here are some examples of pushouts of sets.

1. *Wedge Sums of Pointed Sets.* The wedge sum of two pointed sets of Pointed Sets, **??** is an example of a pushout of sets.

2. *Intersections via Unions.* Let $A, B \subset X$. We have a bijection of sets

$$A \cup B \cong A \coprod_{A \cap B} B, \qquad \begin{tikzcd} A \cup B & B \arrow[l] \\ A \arrow[u] \arrow[ur, phantom, "\ulcorner"] & A \cap B. \arrow[l, hook] \arrow[u, hook] \end{tikzcd}$$

Proof 2.4.7 ▶ Proof of Example 2.4.6

Item 1: Wedge Sums of Pointed Sets

This follows by definition, as the wedge sum of two pointed sets is defined as a pushout.

Item 2: Intersections via Unions

Indeed, $A \coprod_{A \cap B} B$ is the quotient of $A \coprod B$ by the equivalence relation obtained by declaring $(0, a) \sim (1, b)$ iff $a = b \in A \cap B$, which is in bijection with $A \cup B$ via the map with $[(0, a)] \mapsto a$ and $[(1, b)] \mapsto b$. ∎

Proposition 2.4.8 ▶ Properties of Pushouts of Sets

Let A, B, C, and X be sets.

1. *Functoriality.* The assignment $(A, B, C, f, g) \mapsto A \coprod_{f,C,g} B$ defines a functor
$$-_1 \coprod_{-_3} -_1 \colon \mathrm{Fun}(\mathcal{P}, \mathrm{Sets}) \to \mathrm{Sets},$$
where $\mathcal{P}$ is the category that looks like this:
$$\begin{tikzcd} & \bullet \\ \bullet & \bullet. \arrow[l] \arrow[u] \end{tikzcd}$$
In particular, the action on morphisms of $-_1 \coprod_{-_3} -_1$ is given by sending

a morphism

$$\begin{tikzcd}
A \coprod_C B \arrow[rr, leftarrow] & & B \arrow[dr, "\psi"] & \\
& A' \coprod_{C'} B' \arrow[ur, phantom, "\ulcorner"] & & B' \arrow[ll, leftarrow] \\
A \arrow[uu] \arrow[rr, leftarrow] \arrow[dr, "\phi"'] & & C \arrow[uu, dashed, "g"] \arrow[dr, dashed, "\chi"] & \\
& A' \arrow[uu] \arrow[rr, leftarrow, "f'"'] & & C' \arrow[uu, "g'"']
\end{tikzcd}$$

in $\mathrm{Fun}(\mathcal{P}, \mathrm{Sets})$ to the map $\xi\colon A \coprod_C B \xrightarrow{\exists!} A' \coprod_{C'} B'$ given by

$$\xi(x) \overset{\text{def}}{=} \begin{cases} \phi(a) & \text{if } x = [(0,a)], \\ \psi(b) & \text{if } x = [(1,b)] \end{cases}$$

for each $x \in A \coprod_C B$, which is the unique map making the diagram

$$\begin{tikzcd}
A \coprod_C B \arrow[rr, leftarrow] \arrow[dr, dashed] & & B \arrow[dr, "\psi"] & \\
& A' \coprod_{C'} B' \arrow[ur, phantom, "\ulcorner"] & & B' \arrow[ll, leftarrow] \\
A \arrow[uu] \arrow[rr, leftarrow] \arrow[dr, "\phi"'] & & C \arrow[uu, dashed, "g"] \arrow[dr, dashed, "\chi"] & \\
& A' \arrow[uu] \arrow[rr, leftarrow, "f'"'] & & C' \arrow[uu, "g'"']
\end{tikzcd}$$

commute.

2. *Associativity.* Given a diagram

$$\begin{tikzcd}
A & & B & & C \\
& X \arrow[ul, "f"] \arrow[ur, "g"'] & & Y \arrow[ul, "h"] \arrow[ur, "k"'] &
\end{tikzcd}$$

in Sets, we have isomorphisms of sets

$$(A \coprod_X B) \coprod_Y C \cong (A \coprod_X B) \coprod_B (B \coprod_Y C) \cong A \coprod_X (B \coprod_Y C)$$

where these pullbacks are built as in the diagrams

$$\begin{tikzcd}
 & & (A \coprod_X B) \coprod_Y C \arrow[dr, phantom, "\wedge"] & & \\
 & A \coprod_X B \arrow[ur] \arrow[dr, phantom, "\wedge"] & & & \\
A \arrow[ur] & & B \arrow[ul] & & C, \arrow[uull] \\
 & X \arrow[ul, "f"] \arrow[ur, "g"] & & Y \arrow[ul, "h"] \arrow[ur, "k"] &
\end{tikzcd}
\quad
\begin{tikzcd}
 & & (A \coprod_X B) \coprod_B (B \coprod_Y C) \arrow[dr, phantom, "\wedge"] & & \\
 & A \coprod_X B \arrow[ur] \arrow[dr, phantom, "\wedge"] & & B \coprod_Y C \arrow[ul] \arrow[dr, phantom, "\wedge"] & \\
A \arrow[ur] & & B \arrow[ul] \arrow[ur] & & C, \arrow[ul] \\
 & X \arrow[ul, "f"] \arrow[ur, "g"] & & Y \arrow[ul, "h"] \arrow[ur, "k"] &
\end{tikzcd}
\quad
\begin{tikzcd}
 & & A \coprod_X (B \coprod_Y C) \arrow[dr, phantom, "\wedge"] & & \\
 & & & B \coprod_Y C \arrow[ul] \arrow[dr, phantom, "\wedge"] & \\
A \arrow[uurr] & & B \arrow[ur] & & C. \arrow[ul] \\
 & X \arrow[ul, "f"] \arrow[ur, "g"] & & Y \arrow[ul, "h"] \arrow[ur, "k"] &
\end{tikzcd}$$

3. *Interaction With Composition.* Given a diagram

$$\begin{tikzcd}
X & & & & Y \\
 & A \arrow[ul, "\phi"] & & B \arrow[ur, "\psi"] & \\
 & & K \arrow[ul, "f"] \arrow[ur, "g"] & &
\end{tikzcd}$$

in Sets, we have isomorphisms of sets

$$\begin{aligned}
X \coprod_K^{\phi\circ f, \psi\circ g} Y &\cong \left(X \coprod_A^{\phi, j_1} \left(A \coprod_K^{f,g} B\right)\right) \coprod_{A\coprod_K^{f,g} B}^{i_2, i_1} \left(\left(A \coprod_K^{f,g} B\right) \coprod_B^{j_2,\psi} Y\right) \\
&\cong X \coprod_A^{\phi, i} \left(\left(A \coprod_K^{f,g} B\right) \coprod_B^{j_2,\psi} Y\right) \\
&\cong \left(X \coprod_A^{\phi, i_1} \left(A \coprod_K^{f,g} B\right)\right) \coprod_B^{j,\psi} Y
\end{aligned}$$

where

$$\begin{aligned}
j_1 &= \mathrm{inj}_1^{A\times_K^{f,g} B}, & j_2 &= \mathrm{inj}_2^{A\times_K^{f,g} B}, \\
i_1 &= \mathrm{inj}_1^{\left(A\times_K^{f,g} B\right)\times_Y^{q_2,\psi}}, & i_2 &= \mathrm{inj}_2^{X\times_{A\times_K^{f,g} B}^{\phi,q_1} \left(A\times_K^{f,g} B\right)}, \\
i &= j_1 \circ \mathrm{inj}_1^{\left(A\times_K^{f,g} B\right)\times_B^{q_2,\psi} Y}, & j &= j_2 \circ \mathrm{inj}_2^{X\times_A^{\phi,q_1} \left(A\times_K^{f,g} B\right)},
\end{aligned}$$

and where these pullbacks are built as in the diagrams

$$\begin{tikzcd}
 & & (X \coprod_A (A \coprod_K B)) \coprod_{A \coprod_K B} ((A \coprod_K B) \coprod_B Y) & & \\
 & X \coprod_A (A \coprod_K B) \arrow[ur, hook] & \arrow[u, phantom, "\wedge"] & (A \coprod_K B) \coprod_B Y \arrow[ul, hook] & \\
X \arrow[ur, hook] & \arrow[u, phantom, "\wedge"] & A \coprod_K B \arrow[ul, hook] \arrow[ur, hook] & \arrow[u, phantom, "\wedge"] & Y, \arrow[ul, hook] \\
 & A \arrow[ul, "\phi"] \arrow[ur, hook] & \arrow[u, phantom, "\wedge"] & B \arrow[ul, hook] \arrow[ur, "\psi"] & \\
 & & K \arrow[ul, "f"] \arrow[ur, "g"] & &
\end{tikzcd}
\qquad
\begin{tikzcd}
 & & X \coprod_A ((A \coprod_K B) \coprod_B Y) & & \\
 & & \arrow[u, phantom, "\wedge"] & (A \coprod_K B) \coprod_B Y \arrow[ul, hook] & \\
X \arrow[uur, hook] & & A \coprod_K B \arrow[ur, hook] & \arrow[u, phantom, "\wedge"] & Y, \arrow[ul, hook] \\
 & A \arrow[ul, "\phi"] \arrow[ur, hook] & \arrow[u, phantom, "\wedge"] & B \arrow[ul, hook] \arrow[ur, "\psi"] & \\
 & & K \arrow[ul, "f"] \arrow[ur, "g"] & &
\end{tikzcd}$$

$$\begin{tikzcd}
 & & (X \coprod_A (A \coprod_K B)) \coprod_B Y & & \\
 & X \coprod_A (A \coprod_K B) \arrow[ur, hook] & \arrow[u, phantom, "\wedge"] & & \\
X \arrow[ur, hook] & \arrow[u, phantom, "\wedge"] & A \coprod_K B \arrow[ul, hook] \arrow[uur, hook] & \arrow[u, phantom, "\wedge"] & Y, \arrow[uul, hook] \\
 & A \arrow[ul, "\phi"] \arrow[ur, hook] & \arrow[u, phantom, "\wedge"] & B \arrow[ul, hook] \arrow[ur, "\psi"] & \\
 & & K \arrow[ul, "f"] \arrow[ur, "g"] & &
\end{tikzcd}
\qquad
\begin{tikzcd}
 & & X \coprod_K Y & & \\
 & & \arrow[u, phantom, "\wedge"] & & \\
X \arrow[uur, hook] & & & & Y. \arrow[uul, hook] \\
 & A \arrow[ul, "\phi"] \arrow[ur, hook] & & B \arrow[ul, hook] \arrow[ur, "\psi"] & \\
 & & K \arrow[ul, "f"] \arrow[ur, "g"] & &
\end{tikzcd}$$

4. *Unitality.* We have isomorphisms of sets

$$\begin{tikzcd}
A \arrow[r, equal] & A \\
X \arrow[u, "f"] \arrow[r, equal] \arrow[ur, phantom, "\ulcorner"] & X \arrow[u, "f"']
\end{tikzcd}
\qquad
\begin{aligned}
\lambda_A^{\mathsf{Sets}_{X/}} &: X \coprod_X A \xrightarrow{\sim} A, \\
\rho_A^{\mathsf{Sets}_{X/}} &: A \coprod_X X \xrightarrow{\sim} A,
\end{aligned}
\qquad
\begin{tikzcd}
A \arrow[d, equal] & X \arrow[l, "f"'] \arrow[d, equal] \arrow[dl, phantom, "\ulcorner"] \\
X & X, \arrow[l, "f"]
\end{tikzcd}$$

natural in $(A, f) \in \mathrm{Obj}(\mathsf{Sets}_{X/})$.

5. *Commutativity.* We have an isomorphism of sets

$$\begin{tikzcd} A \coprod_X B & B \arrow[l] \\ A \arrow[u] & X, \arrow[l, "f"] \arrow[u, "g"'] \arrow[ul, phantom, "\ulcorner"] \end{tikzcd} \qquad \sigma_A^{\mathsf{Sets}_{X/}} : A \coprod_X B \xrightarrow{\sim} B \coprod_X A \qquad \begin{tikzcd} B \coprod_X A & A \arrow[l] \\ B \arrow[u] & X. \arrow[l, "g"] \arrow[u, "f"'] \arrow[ul, phantom, "\ulcorner"] \end{tikzcd}$$

natural in $(A, f), (B, g) \in \mathrm{Obj}(\mathsf{Sets}_{X/})$.

6. *Interaction With Coproducts.* We have

$$A \coprod_\emptyset B \cong A \coprod B, \qquad \begin{tikzcd} A \coprod B & B \arrow[l] \\ A \arrow[u] & \emptyset. \arrow[l, hook, "\iota_A"] \arrow[u, hook, "\iota_B"'] \arrow[ul, phantom, "\ulcorner"] \end{tikzcd}$$

7. *Symmetric Monoidality.* The triple $(\mathsf{Sets}_{X/}, \coprod_X, X)$ is a symmetric monoidal category.

Proof 2.4.9 ► Proof of Proposition 2.4.8

Item 1: Functoriality

This is a special case of functoriality of co/limits, Limits and Colimits, **??** of **??**, with the explicit expression for ξ following from the commutativity of the cube pushout diagram.

Item 2: Associativity

Omitted.

Item 3: Interaction With Composition

Omitted.

Item 4: Unitality

Omitted.

Item 5: Commutativity

Omitted.

Item 6: Interaction With Coproducts

Omitted.

Item 7: Symmetric Monoidality

Omitted.

2.5 Coequalisers

Let A and B be sets and let $f, g\colon A \rightrightarrows B$ be functions.

Definition 2.5.1 ▶ Coequalisers of Sets

The **coequaliser of f and g** is the coequaliser of f and g in Sets as in Limits and Colimits, **??**.

Construction 2.5.2 ▶ Construction of Coequalisers of Sets

Concretely, the coequaliser of f and g is the pair $(\mathrm{CoEq}(f,g), \mathrm{coeq}(f,g))$ consisting of:

1. *The Colimit.* The set $\mathrm{CoEq}(f,g)$ defined by

$$\mathrm{CoEq}(f,g) \overset{\text{def}}{=} B/{\sim},$$

   where $\sim$ is the equivalence relation on B generated by $f(a) \sim g(a)$.

2. *The Cocone.* The map

$$\mathrm{coeq}(f,g)\colon B \twoheadrightarrow \mathrm{CoEq}(f,g)$$

   given by the quotient map $\pi\colon B \twoheadrightarrow B/{\sim}$ with respect to the equivalence relation generated by $f(a) \sim g(a)$.

Proof 2.5.3 ▶ Proof of Construction 2.5.2

We claim that $\mathrm{CoEq}(f,g)$ is the categorical coequaliser of f and g in Sets. First we need to check that the relevant coequaliser diagram commutes, i.e. that we have

$$\mathrm{coeq}(f,g) \circ f = \mathrm{coeq}(f,g) \circ g.$$

Indeed, we have

$$\begin{aligned}
[\mathrm{coeq}(f,g) \circ f](a) &\overset{\text{def}}{=} [\mathrm{coeq}(f,g)](f(a)) \\
&\overset{\text{def}}{=} [f(a)] \\
&= [g(a)] \\
&\overset{\text{def}}{=} [\mathrm{coeq}(f,g)](g(a)) \\
&\overset{\text{def}}{=} [\mathrm{coeq}(f,g) \circ g](a)
\end{aligned}$$

for each $a \in A$. Next, we prove that $\mathrm{CoEq}(f,g)$ satisfies the universal property of the coequaliser. Suppose we have a diagram of the form

$$\begin{tikzcd}
A \arrow[r, shift left, "f"] \arrow[r, shift right, "g"'] & B \arrow[r, hook, "\mathrm{coeq}(f,g)"] \arrow[dr, "c"'] & \mathrm{CoEq}(f,g) \\
& & C
\end{tikzcd}$$

in Sets. Then, since $c(f(a)) = c(g(a))$ for each $a \in A$, it follows from Conditions on Relations, **????** of **??** that there exists a unique map $\mathrm{CoEq}(f,g) \xrightarrow{\exists!} C$ making the diagram

$$\begin{tikzcd}
A \arrow[r, shift left, "f"] \arrow[r, shift right, "g"'] & B \arrow[r, hook, "\mathrm{coeq}(f,g)"] \arrow[dr, "c"'] & \mathrm{CoEq}(f,g) \arrow[d, dashed, "\exists!"] \\
& & C
\end{tikzcd}$$

commute. ∎

Remark 2.5.4 ► Unwinding Definition 2.5.1

In detail, by Conditions on Relations, **??**, the relation $\sim$ of Definition 2.5.1 is given by declaring $a \sim b$ iff one of the following conditions is satisfied:

1. We have $a = b$;
2. There exist $x_1, \ldots, x_n \in B$ such that $a \sim' x_1 \sim' \cdots \sim' x_n \sim' b$, where we declare $x \sim' y$ if one of the following conditions is satisfied:
   (a) There exists $z \in A$ such that $x = f(z)$ and $y = g(z)$.
   (b) There exists $z \in A$ such that $x = g(z)$ and $y = f(z)$.

   In other words, there exist $x_1, \ldots, x_n \in B$ satisfying the following conditions:

   (a) There exists $z_0 \in A$ satisfying one of the following conditions:
      i. We have $a = f(z_0)$ and $x_1 = g(z_0)$.
      ii. We have $a = g(z_0)$ and $x_1 = f(z_0)$.
   (b) For each $1 \leq i \leq n-1$, there exists $z_i \in A$ satisfying one of the following conditions:
      i. We have $x_i = f(z_i)$ and $x_{i+1} = g(z_i)$.
      ii. We have $x_i = g(z_i)$ and $x_{i+1} = f(z_i)$.
   (c) There exists $z_n \in A$ satisfying one of the following conditions:
      i. We have $x_n = f(z_n)$ and $b = g(z_n)$.
      ii. We have $x_n = g(z_n)$ and $b = f(z_n)$.

Example 2.5.5 ► Examples of Coequalisers of Sets

Here are some examples of coequalisers of sets.

1. *Quotients by Equivalence Relations.* Let R be an equivalence relation on a set X. We have a bijection of sets

$$X/{\sim_R} \cong \mathrm{CoEq}\left(R \hookrightarrow X \times X \underset{\mathrm{pr}_2}{\overset{\mathrm{pr}_1}{\rightrightarrows}} X\right).$$

Proof 2.5.6 ▶ Proof of Example 2.5.5

Item 1: Quotients by Equivalence Relations

See [Pro24w].

Proposition 2.5.7 ▶ Properties of Coequalisers of Sets

Let A, B, and C be sets.

1. *Associativity.* We have isomorphisms of sets[1]

$$\underbrace{\mathrm{CoEq}(\mathrm{coeq}(f,g)\circ f, \mathrm{coeq}(f,g)\circ h)}_{=\mathrm{CoEq}(\mathrm{coeq}(f,g)\circ g, \mathrm{coeq}(f,g)\circ h)} \cong \mathrm{CoEq}(f,g,h) \cong \underbrace{\mathrm{CoEq}(\mathrm{coeq}(g,h)\circ f, \mathrm{coeq}(g,h)\circ g)}_{=\mathrm{CoEq}(\mathrm{coeq}(g,h)\circ f, \mathrm{coeq}(g,h)\circ h)},$$

   where $\mathrm{CoEq}(f,g,h)$ is the colimit of the diagram

$$\begin{tikzcd} A \arrow[r, shift left, "f"] \arrow[r, "g"] \arrow[r, shift right, "h"'] & B \end{tikzcd}$$

   in Sets.

2. *Unitality.* We have an isomorphism of sets

$$\mathrm{CoEq}(f,f) \cong B.$$

3. *Commutativity.* We have an isomorphism of sets

$$\mathrm{CoEq}(f,g) \cong \mathrm{CoEq}(g,f).$$

4. *Interaction With Composition.* Let

$$\begin{tikzcd} A \arrow[r, shift left, "f"] \arrow[r, shift right, "g"'] & B \arrow[r, shift left, "h"] \arrow[r, shift right, "k"'] & C \end{tikzcd}$$

   be functions. We have a surjection

$$\mathrm{CoEq}(h\circ f, k\circ g) \twoheadrightarrow \mathrm{CoEq}(\mathrm{coeq}(h,k)\circ h\circ f, \mathrm{coeq}(h,k)\circ k\circ g)$$

   exhibiting $\mathrm{CoEq}(\mathrm{coeq}(h,k)\circ h\circ f, \mathrm{coeq}(h,k)\circ k\circ g)$ as a quotient of $\mathrm{CoEq}(h\circ f, k\circ g)$ by the relation generated by declaring $h(y) \sim k(y)$ for each $y \in B$.

[1]That is, the following three ways of forming "the' coequaliser of (f, g, h) agree:

(a) Take the coequaliser of (f, g, h), i.e. the colimit of the diagram

$$A \underset{h}{\overset{f}{\underset{g}{\rightrightarrows}}} B$$

in Sets.

(b) First take the coequaliser of f and g, forming a diagram

$$A \underset{g}{\overset{f}{\rightrightarrows}} B \overset{\mathrm{coeq}(f,g)}{\twoheadrightarrow} \mathrm{CoEq}(f, g)$$

and then take the coequaliser of the composition

$$A \underset{h}{\overset{f}{\rightrightarrows}} B \overset{\mathrm{coeq}(f,g)}{\twoheadrightarrow} \mathrm{CoEq}(f, g),$$

obtaining a quotient

$$\mathrm{CoEq}(\mathrm{coeq}(f, g) \circ f, \mathrm{coeq}(f, g) \circ h) = \mathrm{CoEq}(\mathrm{coeq}(f, g) \circ g, \mathrm{coeq}(f, g) \circ h)$$

of $\mathrm{CoEq}(f, g)$

(c) First take the coequaliser of g and h, forming a diagram

$$A \underset{h}{\overset{g}{\rightrightarrows}} B \overset{\mathrm{coeq}(g,h)}{\twoheadrightarrow} \mathrm{CoEq}(g, h)$$

and then take the coequaliser of the composition

$$A \underset{g}{\overset{f}{\rightrightarrows}} B \overset{\mathrm{coeq}(g,h)}{\twoheadrightarrow} \mathrm{CoEq}(g, h),$$

obtaining a quotient

$$\mathrm{CoEq}(\mathrm{coeq}(g, h) \circ f, \mathrm{coeq}(g, h) \circ g) = \mathrm{CoEq}(\mathrm{coeq}(g, h) \circ f, \mathrm{coeq}(g, h) \circ h)$$

of $\mathrm{CoEq}(g, h)$.

Proof 2.5.8 ▶ Proof of Proposition 2.5.7

Item 1: Associativity

Omitted.

Item 2: Unitality

Omitted.

Item 3: Commutativity

Omitted.

Item 4: Interaction With Composition

Omitted. ∎

2.6 Direct Colimits

Let $\left(X_\alpha, f_{\alpha\beta}\right)_{\alpha,\beta\in I} \colon (I, \preceq) \longrightarrow \mathbb{T}$ be a direct system of sets.

Definition 2.6.1 ▶ Direct Colimits of Sets

The **direct colimit of** $\left(X_\alpha, f_{\alpha\beta}\right)_{\alpha,\beta\in I}$ is the direct colimit of $\left(X_\alpha, f_{\alpha\beta}\right)_{\alpha,\beta\in I}$ in Sets as in Limits and Colimits, **??**.

Construction 2.6.2 ▶ Construction of Direct Colimits of Sets

Concretely, the direct colimit of $\left(X_\alpha, f_{\alpha\beta}\right)_{\alpha,\beta\in I}$ is the pair $\left(\varinjlim_{\alpha\in I}(X_\alpha), \{\mathrm{inj}_\alpha\}_{\alpha\in I}\right)$ consisting of:

1. *The Colimit.* The set $\varinjlim_{\alpha\in I}(X_\alpha)$ defined by

$$\varinjlim_{\alpha\in I}(X_\alpha) \overset{\text{def}}{=} \left(\coprod_{\alpha\in I} X_\alpha\right)\Big/\sim,$$

where $\sim$ is the equivalence relation on $\coprod_{\alpha\in I} X_\alpha$ generated by declaring $(\alpha, x) \sim (\beta, y)$ iff there exists some $\gamma \in I$ satisfying the following conditions:

(a) We have $\alpha \preceq \gamma$.

(b) We have $\beta \preceq \gamma$.

(c) We have $f_{\alpha\gamma}(x) = f_{\beta\gamma}(y)$.

2. *The Cocone.* The collection

$$\left\{ \mathrm{inj}_\gamma \colon X_\gamma \to \varinjlim_{\alpha \in I} (X_\alpha) \right\}_{\gamma \in I}$$

of maps of sets defined by

$$\mathrm{inj}_\gamma(x) \stackrel{\text{def}}{=} [(\gamma, x)]$$

for each $\gamma \in I$ and each $x \in X_\gamma$.

Proof 2.6.3 ▶ Proof of Construction 2.6.2

We will prove Construction 2.6.2 below in a bit, but first we need a lemma (which is interesting in its own right).

Lemma 2.6.4 ▶ Identification of x with $f_{\alpha\beta}(x)$ in Direct Colimits

For each $\alpha, \beta \in I$ and each $x \in X_\alpha$, if $\alpha \preceq \beta$, then we have

$$(\alpha, x) \sim \big(\beta, f_{\alpha\beta}(x)\big)$$

in $\varinjlim_{\alpha \in I} (X_\alpha)$.

Proof 2.6.5 ▶ Proof of Lemma 2.6.4

Taking $\gamma = \beta$, we have $f_{\alpha\gamma} = f_{\alpha\beta}$, we have $f_{\beta\gamma} = f_{\beta\beta} \stackrel{\text{def}}{=} \mathrm{id}_{X_\beta}$, and we have

$$\begin{aligned} f_{\alpha\beta}(x) &= f_{\beta\beta}\big(f_{\alpha\beta}(x)\big) \\ &\stackrel{\text{def}}{=} \mathrm{id}_{X_\beta}\big(f_{\alpha\beta}(x)\big), \end{aligned}$$

$$= f_{\alpha\beta}(x).$$

As a result, since $\alpha \preceq \beta$ and $\beta \preceq \beta$ as well, Items 1a to 1c of Construction 2.6.2 are met. Thus we have $(\alpha, x) \sim \big(\beta, f_{\alpha\beta}(x)\big)$.

We can now prove Construction 2.6.2:

PROOF 2.6.6 ▶ PROOF OF CONSTRUCTION 2.6.2

We claim that $\varinjlim_{\alpha \in I} (X_\alpha)$ is the colimit of the direct system of sets $\big(X_\alpha, f_{\alpha\beta}\big)_{\alpha,\beta \in I}$.

Commutativity of the Colimit Diagram

First, we need to check that the colimit diagram defined by $\varinjlim_{\alpha \in I} (X_\alpha)$ commutes, i.e. that we have

$$\mathrm{inj}_\alpha = \mathrm{inj}_\beta \circ f_{\alpha\beta}, \qquad \begin{tikzcd} & \varinjlim_{\alpha \in I}(X_\alpha) & \\ X_\alpha \arrow[ur, "\mathrm{inj}_\alpha"] \arrow[rr, "f_{\alpha\beta}"'] & & X_\beta \arrow[ul, "\mathrm{inj}_\beta"'] \end{tikzcd}$$

for each $\alpha, \beta \in I$ with $\alpha \preceq \beta$. Indeed, given $x \in X_\alpha$, we have

$$\begin{aligned} \big[\mathrm{inj}_\beta \circ f_{\alpha\beta}\big](x) &\overset{\text{def}}{=} \mathrm{inj}_\beta\big(f_{\alpha\beta}(x)\big) \\ &\overset{\text{def}}{=} \big[\big(\beta, f_{\alpha\beta}(x)\big)\big] \\ &= [(\alpha, x)] \\ &\overset{\text{def}}{=} \mathrm{inj}_\alpha(x), \end{aligned}$$

where we have used Lemma 2.6.4 for the third equality.

Proof of the Universal Property of the Colimit

Next, we prove that $\varinjlim_{\alpha \in I} (X_\alpha)$ as constructed in Construction 2.6.2 satisfies the universal property of a direct colimit. Suppose that we have, for each

$\alpha, \beta \in I$ with $\alpha \preceq \beta$, a diagram of the form

$$\begin{tikzcd}
 & & C & & \\
 & & \operatorname{colim}_{\alpha \in I}(X_\alpha) \arrow[u, dashed] & & \\
 & X_\alpha \arrow[ur, "\mathrm{inj}_\alpha"] \arrow[rr, "f_{\alpha\beta}"'] \arrow[uurr, bend left, "i_\alpha"] & & X_\beta \arrow[ul, "\mathrm{inj}_\beta"'] \arrow[uull, bend right, "i_\beta"'] &
\end{tikzcd}$$

in Sets. We claim that there exists a unique map $\phi\colon \operatorname{colim}_{\alpha \in I}(X_\alpha) \xrightarrow{\exists!} C$ making the diagram

$$\begin{tikzcd}
 & & C & & \\
 & & \operatorname{colim}_{\alpha \in I}(X_\alpha) \arrow[u, dashed, "\phi"', "\exists!"] & & \\
 & X_\alpha \arrow[ur, "\mathrm{inj}_\alpha"] \arrow[rr, "f_{\alpha\beta}"'] \arrow[uurr, bend left, "i_\alpha"] & & X_\beta \arrow[ul, "\mathrm{inj}_\beta"'] \arrow[uull, bend right, "i_\beta"'] &
\end{tikzcd}$$

commute. To this end, first consider the diagram

$$\begin{tikzcd}
\coprod_{\alpha \in I} X_\alpha \arrow[r, two heads, "\mathrm{pr}"] \arrow[dr, "\coprod_{\alpha \in I} i_\alpha"'] & \operatorname{colim}_{\alpha \in I}(X_\alpha) \\
 & C.
\end{tikzcd}$$

Lemma. If $(\alpha, x) \sim (\beta, y)$, then we have

$$\left[\coprod_{\alpha \in I} i_\alpha\right](x) = \left[\coprod_{\alpha \in I} i_\alpha\right](y).$$

Proof. Indeed, if $(\alpha, x) \sim (\beta, y)$, then there exists some $\gamma \in I$ satisfying the following conditions:

1. We have $\alpha \preceq \gamma$.
2. We have $\beta \preceq \gamma$.
3. We have $f_{\alpha\gamma}(x) = f_{\beta\gamma}(y)$.

We then have

$$\begin{aligned}
\left[\coprod_{\alpha \in I} i_\alpha\right](x) &\overset{\text{def}}{=} i_\alpha(x) \\
&\overset{\text{def}}{=} \left[i_\gamma \circ f_{\alpha\gamma}\right](x) \\
&\overset{\text{def}}{=} i_\gamma\big(f_{\alpha\gamma}(x)\big) \\
&= i_\gamma\big(f_{\beta\gamma}(x)\big) \\
&\overset{\text{def}}{=} \left[i_\gamma \circ f_{\beta\gamma}\right](x) \\
&= i_\beta(y) \\
&\overset{\text{def}}{=} \left[\coprod_{\alpha \in I} i_\alpha\right](y).
\end{aligned}$$

This finishes the proof of the lemma. Continuing, by Conditions on Relations, **??** of **??**, there then exists a map $\phi \colon \varinjlim_{\alpha \in I}(X_\alpha) \xrightarrow{\exists!} C$ making the diagram

$$\begin{tikzcd}
\coprod_{\alpha \in I} X_\alpha \arrow[r, two heads, "\mathrm{pr}"] \arrow[dr, "\coprod_{\alpha \in I} i_\alpha"'] & \varinjlim_{\alpha \in I}(X_\alpha) \arrow[d, "\phi"] \\
& C
\end{tikzcd}$$

commute. In particular, this implies that the diagram

$$\begin{tikzcd} X_\alpha \arrow[r, two heads, "\mathrm{inj}_\alpha"] \arrow[dr, "i_\alpha"'] & \varinjlim_{\alpha\in I}(X_\alpha) \arrow[d, "\phi"] \\ & C \end{tikzcd}$$

also commutes, and thus so does the diagram

$$\begin{tikzcd} & C & \\ & \varinjlim_{\alpha\in I}(X_\alpha) \arrow[u, dashed, "\phi"', "\exists!"] & \\ X_\alpha \arrow[uur, bend left, "i_\alpha"] \arrow[ur, "\mathrm{inj}_\alpha"] \arrow[rr, "f_{\alpha\beta}"'] & & X_\beta. \arrow[uul, bend right, "i_\beta"'] \arrow[ul, "\mathrm{inj}_\beta"'] \end{tikzcd}$$

This finishes the proof.[1] ∎

[1] Incidentally, the conditions

$$\{i_\alpha = \phi \circ \mathrm{inj}_\alpha\}_{\alpha\in I}$$

show that ϕ must be given by

$$\phi([(\alpha, x)]) = (i_\alpha(x))_{\alpha\in I}$$

for each $[(\alpha, x)] \in \varinjlim_{\alpha\in I}(X_\alpha)$, although we would need to show that this assignment is well-defined were we to prove Construction 2.6.2 in this way. Instead, invoking Conditions on Relations, **??** of **??** gave us a way to avoid having to prove this, leading to a cleaner alternative proof.

Example 2.6.7 ► Examples of Direct Colimits of Sets

Here are some examples of direct colimits of sets.

1. *The Prüfer Group.* The Prüfer group $\mathbb{Z}(p^\infty)$ is defined as the direct colimit

$$\mathbb{Z}(p^\infty) \overset{\text{def}}{=} \varinjlim_{n\in\mathbb{N}}\left(\mathbb{Z}_{/p^n}\right);$$

see **??**.

3 Operations With Sets

3.1 The Empty Set

Definition 3.1.1 ► The Empty Set

The **empty set** is the set $\emptyset$ defined by

$$\emptyset \overset{\text{def}}{=} \{x \in X \mid x \neq x\},$$

where X is the set in the set existence axiom, **??** of **??**.

3.2 Singleton Sets

Let X be a set.

Definition 3.2.1 ► Singleton Sets

The **singleton set containing** X is the set $\{X\}$ defined by

$$\{X\} \overset{\text{def}}{=} \{X, X\},$$

where $\{X, X\}$ is the pairing of X with itself of Definition 3.3.1.

3.3 Pairings of Sets

Let X and Y be sets.

Definition 3.3.1 ▶ Pairings of Sets

The **pairing of** X **and** Y is the set $\{X, Y\}$ defined by

$$\{X, Y\} \overset{\text{def}}{=} \{x \in A \mid x = X \text{ or } x = Y\},$$

where A is the set in the axiom of pairing, **??** of **??**.

3.4 Ordered Pairs

Let A and B be sets.

Definition 3.4.1 ▶ Ordered Pairs

The **ordered pair associated to** A **and** B is the set (A, B) defined by

$$(A, B) \overset{\text{def}}{=} \{\{A\}, \{A, B\}\}.$$

Proposition 3.4.2 ▶ Properties of Ordered Pairs

Let A and B be sets.

1. *Uniqueness.* Let A, B, C, and D be sets. The following conditions are equivalent:
   (a) We have $(A, B) = (C, D)$.
   (b) We have $A = C$ and $B = D$.

Proof 3.4.3 ▶ Proof of Proposition 3.4.2

Item 1: Uniqueness

See [Cie97, Theorem 1.2.3]. ∎

3.5 Sets of Maps

Let A and B be sets.

Definition 3.5.1 ▶ Sets of Maps

The **set of maps from** A **to** B[1] is the set $\mathsf{Sets}(A, B)$[2] whose elements are the functions from A to B.

[1] *Further Terminology:* Also called the **Hom set from** A **to** B.
[2] *Further Notation:* Also written $\mathrm{Hom}_{\mathsf{Sets}}(A, B)$.

Proposition 3.5.2 ▶ Properties of Sets of Maps

Let A and B be sets.

1. *Functoriality.* The assignments $X, Y, (X, Y) \mapsto \mathrm{Hom}_{\mathsf{Sets}}(X, Y)$ define functors

$$\begin{aligned} \mathsf{Sets}(X, -) &: \mathsf{Sets} \to \mathsf{Sets}, \\ \mathsf{Sets}(-, Y) &: \mathsf{Sets}^{\mathsf{op}} \to \mathsf{Sets}, \\ \mathsf{Sets}(-_1, -_2) &: \mathsf{Sets}^{\mathsf{op}} \times \mathsf{Sets} \to \mathsf{Sets}. \end{aligned}$$

2. *Adjointness.* We have adjunctions

$$(A \times - \dashv \mathsf{Sets}(A, -)): \quad \begin{tikzcd} \mathsf{Sets} \arrow[r, bend left, "A\times-"] \arrow[r, phantom, "\perp"] & \mathsf{Sets}, \arrow[l, bend left, "\mathsf{Sets}(A,-)"] \end{tikzcd}$$

$$(- \times B \dashv \mathsf{Sets}(B, -)): \quad \begin{tikzcd} \mathsf{Sets} \arrow[r, bend left, "-\times B"] \arrow[r, phantom, "\perp"] & \mathsf{Sets}, \arrow[l, bend left, "\mathsf{Sets}(B,-)"] \end{tikzcd}$$

   witnessed by bijections

$$\begin{aligned} \mathsf{Sets}(A \times B, C) &\cong \mathsf{Sets}(A, \mathsf{Sets}(B, C)), \\ \mathsf{Sets}(A \times B, C) &\cong \mathsf{Sets}(B, \mathsf{Sets}(A, C)), \end{aligned}$$

   natural in $A, B, C \in \mathrm{Obj}(\mathsf{Sets})$.

3. *Maps From the Punctual Set.* We have a bijection

$$\mathsf{Sets}(\mathrm{pt}, A) \cong A,$$

   natural in $A \in \mathrm{Obj}(\mathsf{Sets})$.

4. *Maps to the Punctual Set.* We have a bijection

$$\mathsf{Sets}(A, \mathrm{pt}) \cong \mathrm{pt},$$

   natural in $A \in \mathrm{Obj}(\mathsf{Sets})$.

Item 1: Functoriality

This follows from Categories, **????** of **??**.

Item 2: Adjointness

This is a repetition of Item 2 of Proposition 1.3.4 and is proved there.

Item 3: Maps From the Punctual Set

The bijection

$$\Phi_A \colon \mathsf{Sets}(\mathsf{pt}, A) \overset{\sim}{\dashrightarrow} A$$

is given by

$$\Phi_A(f) \overset{\text{def}}{=} f(\star)$$

for each $f \in \mathsf{Sets}(\mathsf{pt}, A)$, admitting an inverse

$$\Phi_A^{-1} \colon A \overset{\sim}{\dashrightarrow} \mathsf{Sets}(\mathsf{pt}, A)$$

given by

$$\Phi_A^{-1}(a) \overset{\text{def}}{=} [\![\star \mapsto a]\!]$$

for each $a \in A$. Indeed, we have

$$\begin{aligned}
\left[\Phi_A^{-1} \circ \Phi_A\right](f) &\overset{\text{def}}{=} \Phi_A^{-1}(\Phi_A(f)) \\
&\overset{\text{def}}{=} \Phi_A^{-1}(f(\star)) \\
&\overset{\text{def}}{=} [\![\star \mapsto f(\star)]\!] \\
&\overset{\text{def}}{=} f \\
&\overset{\text{def}}{=} \left[\mathrm{id}_{\mathsf{Sets}(\mathsf{pt},A)}\right](f)
\end{aligned}$$

for each $f \in \mathsf{Sets}(\mathsf{pt}, A)$ and

$$\begin{aligned}
\left[\Phi_A \circ \Phi_A^{-1}\right](a) &\overset{\text{def}}{=} \Phi_A\left(\Phi_A^{-1}(a)\right) \\
&\overset{\text{def}}{=} \Phi_A([\![\star \mapsto a]\!]) \\
&\overset{\text{def}}{=} \mathrm{ev}_\star([\![\star \mapsto a]\!]) \\
&\overset{\text{def}}{=} a \\
&\overset{\text{def}}{=} [\mathrm{id}_A](a)
\end{aligned}$$

for each $a \in A$, and thus we have

$$\begin{aligned}\Phi_A^{-1} \circ \Phi_A &= \mathrm{id}_{\mathsf{Sets}(\mathrm{pt},A)} \\ \Phi_A \circ \Phi_A^{-1} &= \mathrm{id}_A\,.\end{aligned}$$

To prove naturality, we need to show that the diagram

$$\begin{tikzcd} \mathsf{Sets}(\mathrm{pt}, A) \arrow[r, "f_*"] \arrow[d, dashed, "\Phi_A", "\wr"'] & \mathsf{Sets}(\mathrm{pt}, B) \arrow[d, dashed, "\Phi_B"', "\wr"] \\ A \arrow[r, "f"'] & B \end{tikzcd}$$

commutes. Indeed, we have

$$\begin{aligned}[f \circ \Phi_A](\phi) &\overset{\text{def}}{=} f(\Phi_A(\phi)) \\ &\overset{\text{def}}{=} f(\phi(\star)) \\ &\overset{\text{def}}{=} [f \circ \phi](\star) \\ &\overset{\text{def}}{=} \Phi_B(f \circ \phi) \\ &\overset{\text{def}}{=} \Phi_B(f_*(\phi)) \\ &\overset{\text{def}}{=} [\Phi_B \circ f_*](\phi)\end{aligned}$$

for each $\phi \in \mathsf{Sets}(\mathrm{pt}, A)$. This finishes the proof.

Item 4: Maps to the Punctual Set

This follows from the universal property of pt as the terminal set, Definition 1.1.1.

3.6 Unions of Families of Subsets

Let X be a set and let $\mathcal{U} \in \mathcal{P}(\mathcal{P}(X))$.

Definition 3.6.1 ▶ Unions of Families of Subsets

The **union of** $\mathcal{U}$ is the set $\bigcup_{U \in \mathcal{U}} U$ defined by

$$\bigcup_{U \in \mathcal{U}} U \overset{\text{def}}{=} \left\{ x \in X \;\middle|\; \begin{array}{l} \text{there exists some } U \in \mathcal{U} \\ \text{such that we have } x \in U \end{array} \right\}.$$

Proposition 3.6.2 ▶ Properties of Unions of Families of Subsets

Let X be a set.

1. *Functoriality.* The assignment $\mathcal{U} \mapsto \bigcup_{U \in \mathcal{U}} U$ defines a functor

$$\bigcup \colon (\mathcal{P}(\mathcal{P}(X)), \subset) \to (\mathcal{P}(X), \subset).$$

   In particular, for each $\mathcal{U}, \mathcal{V} \in \mathcal{P}(\mathcal{P}(X))$, the following condition is satisfied:

   ($\star$) If $\mathcal{U} \subset \mathcal{V}$, then $\displaystyle\bigcup_{U \in \mathcal{U}} U \subset \bigcup_{V \in \mathcal{V}} V$.

2. *Associativity.* The diagram

$$\begin{tikzcd} \mathcal{P}(\mathcal{P}(\mathcal{P}(X))) \arrow[r, "\mathrm{id}_{\mathcal{P}(X)} \star \bigcup"] \arrow[d, "\bigcup \star \mathrm{id}_{\mathcal{P}(X)}"'] & \mathcal{P}(\mathcal{P}(X)) \arrow[d, "\bigcup"] \\ \mathcal{P}(\mathcal{P}(X)) \arrow[r, "\bigcup"'] & \mathcal{P}(X) \end{tikzcd}$$

   commutes, i.e. we have

$$\bigcup_{U \in \bigcup_{A \in \mathcal{A}} A} U = \bigcup_{A \in \mathcal{A}} \left(\bigcup_{U \in A} U \right)$$

   for each $\mathcal{A} \in \mathcal{P}(\mathcal{P}(\mathcal{P}(X)))$.

3. *Left Unitality.* The diagram

$$\begin{tikzcd}
\mathcal{P}(X) \arrow[d, "\chi_{\mathcal{P}(X)}"'] \arrow[dr, "\mathrm{id}_{\mathcal{P}(X)}"] & \\
\mathcal{P}(\mathcal{P}(X)) \arrow[r, "\bigcup"'] & \mathcal{P}(X)
\end{tikzcd}$$

commutes, i.e. we have

$$\bigcup_{V \in \{U\}} V = U$$

for each $U \in \mathcal{P}(X)$.

4. *Right Unitality.* The diagram

$$\begin{tikzcd}
\mathcal{P}(X) \arrow[d, "\mathcal{P}(\chi_X)"'] \arrow[dr, "\mathrm{id}_{\mathcal{P}(X)}"] & \\
\mathcal{P}(\mathcal{P}(X)) \arrow[r, "\bigcup"'] & \mathcal{P}(X)
\end{tikzcd}$$

commutes, i.e. we have

$$\bigcup_{\{u\} \in \chi_X(U)} \{u\} = U$$

for each $U \in \mathcal{P}(X)$.

5. *Interaction With Unions I.* The diagram

$$\begin{tikzcd}
\mathcal{P}(\mathcal{P}(X)) \times \mathcal{P}(\mathcal{P}(X)) \arrow[r, "\cup"] \arrow[d, "\cup \times \cup"'] & \mathcal{P}(\mathcal{P}(X)) \arrow[d, "\cup"] \\
\mathcal{P}(X) \times \mathcal{P}(X) \arrow[r, "\cup"'] & \mathcal{P}(X)
\end{tikzcd}$$

commutes, i.e. we have

$$\bigcup_{W \in \mathcal{U} \cup \mathcal{V}} W = \left(\bigcup_{U \in \mathcal{U}} U \right) \cup \left(\bigcup_{V \in \mathcal{V}} V \right)$$

for each $\mathcal{U}, \mathcal{V} \in \mathcal{P}(\mathcal{P}(X))$.

6. *Interaction With Unions II.* The diagrams

$$\begin{tikzcd}
\mathcal{P}(\mathcal{P}(X)) \arrow[r, "\mathrm{id}_{\mathcal{P}(\mathcal{P}(X))} \times (U \cup -)"] \arrow[d, "\cup"'] & \mathcal{P}(\mathcal{P}(X)) \arrow[d, "\cup"] \\
\mathcal{P}(X) \arrow[r, "U \cup -"'] & \mathcal{P}(X)
\end{tikzcd}
\qquad
\begin{tikzcd}
\mathcal{P}(\mathcal{P}(X)) \arrow[r, "\mathrm{id}_{\mathcal{P}(\mathcal{P}(X))} \times (- \cup V)"] \arrow[d, "\cup"'] & \mathcal{P}(\mathcal{P}(X)) \arrow[d, "\cup"] \\
\mathcal{P}(X) \arrow[r, "- \cup V"'] & \mathcal{P}(X)
\end{tikzcd}$$

commute, i.e. we have

$$U \cup \left(\bigcup_{V \in \mathcal{V}} V \right) = \bigcup_{V \in \mathcal{V}} (U \cup V),$$
$$\left(\bigcup_{U \in \mathcal{U}} U \right) \cup V = \bigcup_{U \in \mathcal{U}} (U \cup V)$$

for each $\mathcal{U}, \mathcal{V} \in \mathcal{P}(\mathcal{P}(X))$ and each $U, V \in \mathcal{P}(X)$.

7. *Interaction With Intersections I.* We have a natural transformation

$$\begin{tikzcd}
\mathcal{P}(\mathcal{P}(X)) \times \mathcal{P}(\mathcal{P}(X)) \arrow[r, "\cap"] \arrow[d, "\cup \times \cup"'] \arrow[dr, phantom, "\supset"] & \mathcal{P}(\mathcal{P}(X)) \arrow[d, "\cup"] \\
\mathcal{P}(X) \times \mathcal{P}(X) \arrow[r, "\cap"'] & \mathcal{P}(X),
\end{tikzcd}$$

with components

$$\bigcup_{W \in \mathcal{U} \cap \mathcal{V}} W \subset \left(\bigcup_{U \in \mathcal{U}} U \right) \cap \left(\bigcup_{V \in \mathcal{V}} V \right)$$

for each $\mathcal{U}, \mathcal{V} \in \mathcal{P}(\mathcal{P}(X))$.

8. *Interaction With Intersections II.* The diagrams

$$\begin{tikzcd} \mathcal{P}(\mathcal{P}(X)) \arrow[r, "\mathrm{id}_{\mathcal{P}(\mathcal{P}(X))} \times (U \cap -)"] \arrow[d, "\cup"'] & \mathcal{P}(\mathcal{P}(X)) \arrow[d, "\cup"] \\ \mathcal{P}(X) \arrow[r, "U \cap -"'] & \mathcal{P}(X) \end{tikzcd} \qquad \begin{tikzcd} \mathcal{P}(\mathcal{P}(X)) \arrow[r, "\mathrm{id}_{\mathcal{P}(\mathcal{P}(X))} \times (- \cap V)"] \arrow[d, "\cup"'] & \mathcal{P}(\mathcal{P}(X)) \arrow[d, "\cup"] \\ \mathcal{P}(X) \arrow[r, "- \cap V"'] & \mathcal{P}(X) \end{tikzcd}$$

commute, i.e. we have

$$U \cup \left(\bigcup_{V \in \mathcal{V}} V \right) = \bigcup_{V \in \mathcal{V}} (U \cup V),$$
$$\left(\bigcup_{U \in \mathcal{U}} U \right) \cup V = \bigcup_{U \in \mathcal{U}} (U \cup V)$$

for each $\mathcal{U}, \mathcal{V} \in \mathcal{P}(\mathcal{P}(X))$ and each $U, V \in \mathcal{P}(X)$.

9. *Interaction With Differences.* The diagram

$$\begin{tikzcd} \mathcal{P}(\mathcal{P}(X)) \times \mathcal{P}(\mathcal{P}(X)) \arrow[r, "\setminus"] \arrow[d, "\cup \times \cup"'] \arrow[dr, phantom, "\times"] & \mathcal{P}(\mathcal{P}(X)) \arrow[d, "\cup"] \\ \mathcal{P}(X) \times \mathcal{P}(X) \arrow[r, "\setminus"'] & \mathcal{P}(X), \end{tikzcd}$$

does not commute in general, i.e. we may have

$$\bigcup_{W \in \mathcal{U} \setminus \mathcal{V}} W \neq \left(\bigcup_{U \in \mathcal{U}} U \right) \setminus \left(\bigcup_{V \in \mathcal{V}} V \right)$$

in general, where $\mathcal{U}, \mathcal{V} \in \mathcal{P}(\mathcal{P}(X))$.

10. *Interaction With Complements I.* The diagram

$$\begin{tikzcd} \mathcal{P}(\mathcal{P}(X))^{\mathrm{op}} \arrow[r, "(-)^{\mathrm{c}}"] \arrow[d, "\cup^{\mathrm{op}}"'] \arrow[dr, phantom, "\times"] & \mathcal{P}(\mathcal{P}(X)) \arrow[d, "\cup"] \\ \mathcal{P}(X)^{\mathrm{op}} \arrow[r, "(-)^{\mathrm{c}}"'] & \mathcal{P}(X), \end{tikzcd}$$

does not commute in general, i.e. we may have

$$\bigcup_{U \in \mathcal{U}^{\mathsf{c}}} U \neq \bigcup_{U \in \mathcal{U}} U^{\mathsf{c}}$$

in general, where $\mathcal{U} \in \mathcal{P}(\mathcal{P}(X))$.

11. *Interaction With Complements II.* The diagram

$$\begin{tikzcd}
 & \mathcal{P}(\mathcal{P}(X)^{\mathsf{op}}) \arrow[dr, dashed, "\sim"] & \\
\mathcal{P}(\mathcal{P}(X)) \arrow[ur, "\mathrm{id}_{\mathcal{P}(X)} \star (-)^{\mathsf{c}}"] \arrow[dd, "\cup"'] & & \mathcal{P}(\mathcal{P}(X))^{\mathsf{op}} \arrow[dd, "\cap^{\mathsf{op}}"] \\
 & & \\
\mathcal{P}(X) \arrow[rr, "(-)^{\mathsf{c}}"'] & & \mathcal{P}(X)^{\mathsf{op}},
\end{tikzcd}$$

commutes, i.e. we have

$$\left(\bigcup_{U \in \mathcal{U}} U \right)^{\mathsf{c}} = \bigcap_{U \in \mathcal{U}} U^{\mathsf{c}}$$

for each $\mathcal{U} \in \mathcal{P}(\mathcal{P}(X))$.

12. *Interaction With Complements III.* The diagram

$$\begin{tikzcd}
 & \mathcal{P}(\mathcal{P}(X)^{\mathsf{op}}) \arrow[dr, dashed, "\sim"] & \\
\mathcal{P}(\mathcal{P}(X)) \arrow[ur, "\mathrm{id}_{\mathcal{P}(X)} \star (-)^{\mathsf{c}}"] \arrow[dd, "\cap"'] & & \mathcal{P}(\mathcal{P}(X))^{\mathsf{op}} \arrow[dd, "\cup^{\mathsf{op}}"] \\
 & & \\
\mathcal{P}(X) \arrow[rr, "(-)^{\mathsf{c}}"'] & & \mathcal{P}(X)^{\mathsf{op}},
\end{tikzcd}$$

commutes, i.e. we have

$$\left(\bigcap_{U\in\mathcal{U}} U\right)^{\mathsf{c}} = \bigcup_{U\in\mathcal{U}} U^{\mathsf{c}}$$

for each $\mathcal{U} \in \mathcal{P}(\mathcal{P}(X))$.

13. *Interaction With Symmetric Differences.* The diagram

$$\begin{tikzcd}
\mathcal{P}(\mathcal{P}(X)) \times \mathcal{P}(\mathcal{P}(X)) \arrow[r, "\triangle"] \arrow[d, "\cup\times\cup"'] \arrow[dr, phantom, "\times"] & \mathcal{P}(\mathcal{P}(X)) \arrow[d, "\cup"] \\
\mathcal{P}(X) \times \mathcal{P}(X) \arrow[r, "\triangle"'] & \mathcal{P}(X),
\end{tikzcd}$$

does not commute in general, i.e. we may have

$$\bigcup_{W\in\mathcal{U}\triangle\mathcal{V}} W \neq \left(\bigcup_{U\in\mathcal{U}} U\right) \triangle \left(\bigcup_{V\in\mathcal{V}} V\right)$$

in general, where $\mathcal{U}, \mathcal{V} \in \mathcal{P}(\mathcal{P}(X))$.

14. *Interaction With Internal Homs I.* The diagram

$$\begin{tikzcd}
\mathcal{P}(\mathcal{P}(X))^{\mathsf{op}} \times \mathcal{P}(\mathcal{P}(X)) \arrow[r, "{[-_1,-_2]_{\mathcal{P}(X)}}"] \arrow[d, "\cup^{\mathsf{op}}\times\cup^{\mathsf{op}}"'] \arrow[dr, phantom, "\times"] & \mathcal{P}(\mathcal{P}(X)) \arrow[d, "\cup"] \\
\mathcal{P}(X)^{\mathsf{op}} \times \mathcal{P}(X) \arrow[r, "{[-_1,-_2]_X}"'] & \mathcal{P}(X),
\end{tikzcd}$$

does not commute in general, i.e. we may have

$$\bigcup_{W\in[\mathcal{U},\mathcal{V}]_{\mathcal{P}(X)}} W \neq \left[\bigcup_{U\in\mathcal{U}} U, \bigcup_{V\in\mathcal{V}} V\right]_X$$

in general, where $\mathcal{U} \in \mathcal{P}(\mathcal{P}(X))$.

15. *Interaction With Internal Homs II.* The diagram

$$\begin{tikzcd}
 & \mathcal{P}(\mathcal{P}(X))^{\mathsf{op}} \arrow[dr, "\bigcup^{\mathsf{op}}"] & \\
\mathcal{P}(\mathcal{P}(X)^{\mathsf{op}}) \arrow[ur, dashed, "\sim"] \arrow[dd, "\mathrm{id}_{\mathcal{P}(X)} \star [-,V]_X"'] & & \mathcal{P}(X)^{\mathsf{op}} \arrow[dd, "{[-,V]_X}"] \\
 & & \\
\mathcal{P}(\mathcal{P}(X)) \arrow[rr, "\bigcap"'] & & \mathcal{P}(X)
\end{tikzcd}$$

commutes, i.e. we have

$$\left[\bigcup_{U \in \mathcal{U}} U, V\right]_X = \bigcap_{U \in \mathcal{U}} [U, V]_X$$

for each $\mathcal{U} \in \mathcal{P}(\mathcal{P}(X))$ and each $V \in \mathcal{P}(X)$.

16. *Interaction With Internal Homs III.* The diagram

$$\begin{tikzcd}
\mathcal{P}(\mathcal{P}(X)) \arrow[r, "\bigcup"] \arrow[d, "\mathrm{id}_{\mathcal{P}(X)} \star [U,-]_X"'] & \mathcal{P}(X) \arrow[d, "{[U,-]_X}"] \\
\mathcal{P}(\mathcal{P}(X)) \arrow[r, "\bigcup"'] & \mathcal{P}(X)
\end{tikzcd}$$

commutes, i.e. we have

$$\left[U, \bigcup_{V \in \mathcal{V}} V\right]_X = \bigcup_{V \in \mathcal{V}} [U, V]_X$$

for each $U \in \mathcal{P}(X)$ and each $\mathcal{V} \in \mathcal{P}(\mathcal{P}(X))$.

17. *Interaction With Direct Images.* Let $f\colon X \to Y$ be a map of sets. The diagram

$$\begin{tikzcd}
\mathcal{P}(\mathcal{P}(X)) \arrow[r, "(f_*)_*"] \arrow[d, "\bigcup"'] & \mathcal{P}(\mathcal{P}(Y)) \arrow[d, "\bigcup"] \\
\mathcal{P}(X) \arrow[r, "f_*"'] & \mathcal{P}(Y)
\end{tikzcd}$$

commutes, i.e. we have

$$\bigcup_{U \in \mathcal{U}} f_*(U) = \bigcup_{V \in f_*(\mathcal{U})} V$$

for each $\mathcal{U} \in \mathcal{P}(X)$, where $f_*(\mathcal{U}) \overset{\text{def}}{=} (f_*)_*(\mathcal{U})$.

18. *Interaction With Inverse Images.* Let $f\colon X \longrightarrow Y$ be a map of sets. The diagram

$$\begin{tikzcd}
\mathcal{P}(\mathcal{P}(Y)) \arrow[r, "(f^{-1})^{-1}"] \arrow[d, "\cup"'] & \mathcal{P}(\mathcal{P}(X)) \arrow[d, "\cup"] \\
\mathcal{P}(Y) \arrow[r, "f^{-1}"'] & \mathcal{P}(X)
\end{tikzcd}$$

commutes, i.e. we have

$$\bigcup_{V \in \mathcal{V}} f^{-1}(V) = \bigcup_{U \in f^{-1}(\mathcal{U})} U$$

for each $\mathcal{V} \in \mathcal{P}(Y)$, where $f^{-1}(\mathcal{V}) \overset{\text{def}}{=} \left(f^{-1}\right)^{-1}(\mathcal{V})$.

19. *Interaction With Direct Images With Compact Support.* Let $f\colon X \longrightarrow Y$ be a map of sets. The diagram

$$\begin{tikzcd}
\mathcal{P}(\mathcal{P}(X)) \arrow[r, "(f_!)_!"] \arrow[d, "\cup"'] & \mathcal{P}(\mathcal{P}(Y)) \arrow[d, "\cup"] \\
\mathcal{P}(X) \arrow[r, "f_!"'] & \mathcal{P}(Y)
\end{tikzcd}$$

commutes, i.e. we have

$$\bigcup_{U \in \mathcal{U}} f_!(U) = \bigcup_{V \in f_!(\mathcal{U})} V$$

for each $\mathcal{U} \in \mathcal{P}(X)$, where $f_!(\mathcal{U}) \overset{\text{def}}{=} (f_!)_!(\mathcal{U})$.

20. *Interaction With Intersections of Families I.* The diagram

$$\begin{tikzcd}
\mathcal{P}(\mathcal{P}(X)) \arrow[r, "\mathrm{id}_{\mathcal{P}(X)} \star \bigcap"] \arrow[d, "\bigcup \star \mathrm{id}_{\mathcal{P}(X)}"'] & \mathcal{P}(\mathcal{P}(x)) \arrow[d, "\bigcap"] \\
\mathcal{P}(X) \arrow[r, "\bigcap"'] & X
\end{tikzcd}$$

commutes, i.e. we have

$$\bigcap_{U \in \bigcup_{A \in \mathcal{A}} A} U = \bigcap_{A \in \mathcal{A}} \left(\bigcap_{U \in A} U \right)$$

for each $\mathcal{A} \in \mathcal{P}(\mathcal{P}(X))$.

21. *Interaction With Intersections of Families II.* Let X be a set and consider the compositions

$$\begin{tikzcd}
 & & \mathcal{P}(\mathcal{P}(\mathcal{P}(X))) \arrow[dll, "\bigcap \star \mathrm{id}_{\mathcal{P}(X)}"'] \arrow[dl, "\bigcup \star \mathrm{id}_{\mathcal{P}(X)}"] \arrow[dr, "\mathrm{id}_{\mathcal{P}(X)} \star \bigcap"] \arrow[drr, "\mathrm{id}_{\mathcal{P}(X)} \star \bigcup"] & & \\
\mathcal{P}(\mathcal{P}(X)) \arrow[dr, "\bigcup"'] & \mathcal{P}(\mathcal{P}(X)) \arrow[d, "\bigcap"] & \mathcal{P}(\mathcal{P}(X)) \arrow[d, "\bigcup"] & \mathcal{P}(\mathcal{P}(X)) \arrow[dl, "\bigcap"] \\
 & \mathcal{P}(X) & &
\end{tikzcd}$$

given by

$$\mathcal{A} \mapsto \bigcup_{U \in \bigcap_{A \in \mathcal{A}} A} U, \qquad \mathcal{A} \mapsto \bigcap_{U \in \bigcup_{A \in \mathcal{A}} A} U,$$

$$\mathcal{A} \mapsto \bigcup_{A \in \mathcal{A}} \left(\bigcap_{U \in A} U \right), \quad \mathcal{A} \mapsto \bigcap_{A \in \mathcal{A}} \left(\bigcup_{U \in A} U \right)$$

for each $\mathcal{A} \in \mathcal{P}(\mathcal{P}(\mathcal{P}(X)))$. We have the following inclusions:

$$\begin{tikzcd}
 & \displaystyle\bigcup_{U \in \bigcap_{A \in \mathcal{A}} A} U \arrow[dr, hook] & \\
\displaystyle\bigcup_{A \in \mathcal{A}} \left(\bigcap_{U \in A} U \right) & & \displaystyle\bigcap_{A \in \mathcal{A}} \left(\bigcup_{U \in A} U \right) \\
 & \displaystyle\bigcap_{U \in \bigcup_{A \in \mathcal{A}} A} U \arrow[ul, hook] \arrow[u, hook] \arrow[ur, hook] &
\end{tikzcd}$$

All other possible inclusions fail to hold in general.

Proof 3.6.3 ► Proof of Proposition 3.6.2

Item 1: Functoriality

Since $\mathcal{P}(X)$ is posetal, it suffices to prove the condition $(\star)$. So let $\mathcal{U}, \mathcal{V} \in \mathcal{P}(\mathcal{P}(X))$ with $\mathcal{U} \subset \mathcal{V}$. We claim that

$$\bigcup_{U \in \mathcal{U}} U \subset \bigcup_{V \in \mathcal{V}} V.$$

Indeed, given $x \in \bigcup_{U \in \mathcal{U}} U$, there exists some $U \in \mathcal{U}$ such that $x \in U$, but since $\mathcal{U} \subset \mathcal{V}$, we have $U \in \mathcal{V}$ as well, and thus $x \in \bigcup_{V \in \mathcal{V}} V$, which gives our desired inclusion.

Item 2: Associativity

We have

$$\begin{aligned}
\bigcup_{U \in \bigcup_{A \in \mathcal{A}} A} U &\overset{\text{def}}{=} \left\{ x \in X \;\middle|\; \begin{array}{l} \text{there exists some } U \in \bigcup_{A \in \mathcal{A}} A \\ \text{such that we have } x \in U \end{array} \right\} \\
&= \left\{ x \in X \;\middle|\; \begin{array}{l} \text{there exists some } A \in \mathcal{A} \\ \text{and some } U \in A \text{ such that} \\ \text{we have } x \in U \end{array} \right\} \\
&= \left\{ x \in X \;\middle|\; \begin{array}{l} \text{there exists some } A \in \mathcal{A} \\ \text{such that we have } x \in \bigcup_{U \in A} U \end{array} \right\} \\
&\overset{\text{def}}{=} \bigcup_{A \in \mathcal{A}} \left(\bigcup_{U \in A} U \right).
\end{aligned}$$

This finishes the proof.

Item 3: Left Unitality

We have

$$\begin{aligned}
\bigcup_{V \in \{U\}} V &\overset{\text{def}}{=} \left\{ x \in X \;\middle|\; \begin{array}{l} \text{there exists some } V \in \{U\} \\ \text{such that we have } x \in U \end{array} \right\} \\
&= \{ x \in X \mid x \in U \} \\
&= U.
\end{aligned}$$

This finishes the proof.

Item 4: Right Unitality

We have

$$\begin{aligned}
\bigcup_{\{u\}\in\chi_X(U)} \{u\} &\overset{\text{def}}{=} \left\{x \in X \,\middle|\, \begin{array}{l}\text{there exists some } \{u\} \in \chi_X(U)\\ \text{such that we have } x \in \{u\}\end{array}\right\}\\
&= \left\{x \in X \,\middle|\, \begin{array}{l}\text{there exists some } \{u\} \in \chi_X(U)\\ \text{such that we have } x = u\end{array}\right\}\\
&= \left\{x \in X \,\middle|\, \begin{array}{l}\text{there exists some } u \in U\\ \text{such that we have } x = u\end{array}\right\}\\
&= \{x \in X \mid x \in U\}\\
&= U.
\end{aligned}$$

This finishes the proof.

Item 5: Interaction With Unions I

We have

$$\begin{aligned}
\bigcup_{W\in\mathcal{U}\cup\mathcal{V}} W &\overset{\text{def}}{=} \left\{x \in X \,\middle|\, \begin{array}{l}\text{there exists some } W \in \mathcal{U}\cup\mathcal{V}\\ \text{such that we have } x \in W\end{array}\right\}\\
&= \left\{x \in X \,\middle|\, \begin{array}{l}\text{there exists some } W \in \mathcal{U} \text{ or some}\\ W \in \mathcal{V} \text{ such that we have } x \in W\end{array}\right\}\\
&\overset{\text{def}}{=} \left\{x \in X \,\middle|\, \begin{array}{l}\text{there exists some } W \in \mathcal{U}\\ \text{such that we have } x \in W\end{array}\right\}\\
&\quad \cup \left\{x \in X \,\middle|\, \begin{array}{l}\text{there exists some } W \in \mathcal{V}\\ \text{such that we have } x \in W\end{array}\right\}\\
&\overset{\text{def}}{=} \left(\bigcup_{W\in\mathcal{U}} W\right) \cup \left(\bigcup_{W\in\mathcal{V}} W\right)\\
&= \left(\bigcup_{U\in\mathcal{U}} U\right) \cup \left(\bigcup_{V\in\mathcal{V}} V\right).
\end{aligned}$$

This finishes the proof.

Item 6: Interaction With Unions II

Omitted.

Item 7: Interaction With Intersections I

We have

$$\begin{aligned}
\bigcup_{W \in \mathcal{U} \cap \mathcal{V}} W &\overset{\text{def}}{=} \left\{ x \in X \;\middle|\; \begin{array}{l} \text{there exists some } W \in \mathcal{U} \cap \mathcal{V} \\ \text{such that we have } x \in W \end{array} \right\} \\
&\subset \left\{ x \in X \;\middle|\; \begin{array}{l} \text{there exists some } U \in \mathcal{U} \text{ and some } V \in \mathcal{V} \\ \text{such that we have } x \in U \text{ and } x \in V \end{array} \right\} \\
&= \left\{ x \in X \;\middle|\; \begin{array}{l} \text{there exists some } U \in \mathcal{U} \\ \text{such that we have } x \in U \end{array} \right\} \\
&\quad \cup \left\{ x \in X \;\middle|\; \begin{array}{l} \text{there exists some } V \in \mathcal{V} \\ \text{such that we have } x \in V \end{array} \right\} \\
&\overset{\text{def}}{=} \left(\bigcup_{U \in \mathcal{U}} U \right) \cap \left(\bigcup_{V \in \mathcal{V}} V \right).
\end{aligned}$$

This finishes the proof.

Item 8: Interaction With Intersections II

Omitted.

Item 9: Interaction With Differences

Let $X = \{0, 1\}$, let $\mathcal{U} = \{\{0, 1\}\}$, and let $\mathcal{V} = \{\{0\}\}$. We have

$$\begin{aligned}
\bigcup_{W \in \mathcal{U} \setminus \mathcal{V}} U &= \bigcup_{W \in \{\{0,1\}\}} W \\
&= \{0, 1\},
\end{aligned}$$

whereas

$$\left(\bigcup_{U \in \mathcal{U}} U \right) \setminus \left(\bigcup_{V \in \mathcal{V}} V \right) = \{0, 1\} \setminus \{0\}$$

$$= \{1\}.$$

Thus we have

$$\bigcup_{W \in \mathcal{U} \setminus \mathcal{V}} W = \{0, 1\} \neq \{1\} = \left(\bigcup_{U \in \mathcal{U}} U\right) \setminus \left(\bigcup_{V \in \mathcal{V}} V\right).$$

This finishes the proof.

Item 10: Interaction With Complements I

Let $X = \{0, 1\}$ and let $\mathcal{U} = \{0\}$. We have

$$\begin{aligned}\bigcup_{U \in \mathcal{U}^{\mathsf{c}}} U &= \bigcup_{U \in \{\emptyset, \{1\}, \{0,1\}\}} U \\ &= \{0, 1\},\end{aligned}$$

whereas

$$\begin{aligned}\bigcup_{U \in \mathcal{U}} U^{\mathsf{c}} &= \{0\}^{\mathsf{c}} \\ &= \{1\}.\end{aligned}$$

Thus we have

$$\bigcup_{U \in \mathcal{U}^{\mathsf{c}}} U = \{0, 1\} \neq \{1\} = \bigcup_{U \in \mathcal{U}} U^{\mathsf{c}}.$$

This finishes the proof.

Item 11: Interaction With Complements II

Omitted.

Item 12: Interaction With Complements III

Omitted.

Item 13: Interaction With Symmetric Differences

Let $X = \{0, 1\}$, let $\mathcal{U} = \{\{0, 1\}\}$, and let $\mathcal{V} = \{\{0\}, \{0, 1\}\}$. We have

$$\bigcup_{W \in \mathcal{U} \triangle \mathcal{V}} W = \bigcup_{W \in \{\{0\}\}} W$$

$$= \{0\},$$

whereas

$$\left(\bigcup_{U \in \mathcal{U}} U\right) \triangle \left(\bigcup_{V \in \mathcal{V}} V\right) = \{0, 1\} \triangle \{0, 1\}$$
$$= \emptyset,$$

Thus we have

$$\bigcup_{W \in \mathcal{U} \triangle \mathcal{V}} W = \{0\} \neq \emptyset = \left(\bigcup_{U \in \mathcal{U}} U\right) \triangle \left(\bigcup_{V \in \mathcal{V}} V\right).$$

This finishes the proof.

Item 14: Interaction With Internal Homs I

This is a repetition of Item 7 of Proposition 4.7.4 and is proved there.

Item 15: Interaction With Internal Homs II

This is a repetition of Item 8 of Proposition 4.7.4 and is proved there.

Item 16: Interaction With Internal Homs III

This is a repetition of Item 9 of Proposition 4.7.4 and is proved there.

Item 17: Interaction With Direct Images

This is a repetition of Item 3 of Proposition 6.1.4 and is proved there.

Item 18: Interaction With Inverse Images

This is a repetition of Item 3 of Proposition 6.2.3 and is proved there.

Item 19: Interaction With Direct Images With Compact Support

This is a repetition of Item 3 of Proposition 6.3.6 and is proved there.

Item 20: Interaction With Intersections of Families I

We have

$$\bigcap_{U \in \bigcup_{A \in \mathcal{A}} A} U \stackrel{\text{def}}{=} \left\{ x \in X \;\middle|\; \begin{array}{l} \text{for each } U \in \bigcup_{A \in \mathcal{A}} A, \\ \text{we have } x \in U \end{array} \right\}$$

$$= \left\{ x \in X \,\middle|\, \begin{array}{l} \text{for each } A \in \mathcal{A} \text{ and each} \\ U \in A, \text{ we have } x \in U \end{array} \right\}$$

$$\stackrel{\text{def}}{=} \bigcap_{A \in \mathcal{A}} \left(\bigcap_{U \in A} U \right).$$

This finishes the proof.

Item 21: Interaction With Intersections of Families II

Omitted.

3.7 Intersections of Families of Subsets

Let X be a set and let $\mathcal{U} \in \mathcal{P}(\mathcal{P}(X))$.

Definition 3.7.1 ▶ Intersections of Families of Subsets

The **intersection of** $\mathcal{U}$ is the set $\bigcap_{U \in \mathcal{U}} U$ defined by

$$\bigcap_{U \in \mathcal{U}} U \stackrel{\text{def}}{=} \left\{ x \in X \,\middle|\, \begin{array}{l} \text{for each } U \in \mathcal{U}, \\ \text{we have } x \in U \end{array} \right\}.$$

Proposition 3.7.2 ▶ Properties of Intersections of Families of Subsets

Let X be a set.

1. *Functoriality.* The assignment $\mathcal{U} \mapsto \bigcap_{U \in \mathcal{U}} U$ defines a functor

$$\bigcap \colon (\mathcal{P}(\mathcal{P}(X)), \supset) \to (\mathcal{P}(X), \subset).$$

   In particular, for each $\mathcal{U}, \mathcal{V} \in \mathcal{P}(\mathcal{P}(X))$, the following condition is satisfied:

   ($\star$) If $\mathcal{U} \subset \mathcal{V}$, then $\displaystyle\bigcap_{V \in \mathcal{V}} V \subset \bigcap_{U \in \mathcal{U}} U$.

2. *Oplax Associativity.* We have a natural transformation

$$\begin{tikzcd}
\mathcal{P}(\mathcal{P}(\mathcal{P}(X))) \arrow[r, "\mathrm{id}_{\mathcal{P}(X)} \star \bigcap"] \arrow[d, "\bigcap \star \mathrm{id}_{\mathcal{P}(X)}"'] \arrow[dr, phantom, "\supset"] & \mathcal{P}(\mathcal{P}(X)) \arrow[d, "\bigcap"] \\
\mathcal{P}(\mathcal{P}(X)) \arrow[r, "\bigcap"'] & \mathcal{P}(X)
\end{tikzcd}$$

with components

$$\bigcap_{A \in \mathcal{A}} \left(\bigcap_{U \in A} U \right) \subset \bigcap_{U \in \bigcap_{A \in \mathcal{A}} A} U$$

for each $\mathcal{A} \in \mathcal{P}(\mathcal{P}(\mathcal{P}(X)))$.

3. *Left Unitality.* The diagram

$$\begin{tikzcd}
\mathcal{P}(X) \arrow[d, "\chi_{\mathcal{P}(X)}"'] \arrow[dr, "\mathrm{id}_{\mathcal{P}(X)}"] & \\
\mathcal{P}(\mathcal{P}(X)) \arrow[r, "\bigcap"'] & \mathcal{P}(X)
\end{tikzcd}$$

commutes, i.e. we have

$$\bigcap_{V \in \{U\}} V = U.$$

for each $U \in \mathcal{P}(X)$.

4. *Oplax Right Unitality.* The diagram

$$\begin{tikzcd}
\mathcal{P}(X) \arrow[d, "\mathcal{P}(\chi_X)"'] \arrow[dr, "\mathrm{id}_{\mathcal{P}(X)}"] \arrow[dr, phantom, "\times"] & \\
\mathcal{P}(\mathcal{P}(X)) \arrow[r, "\bigcap"'] & \mathcal{P}(X)
\end{tikzcd}$$

does not commute in general, i.e. we may have

$$\bigcap_{\{x\}\in\chi_X(U)} \{x\} \neq U$$

in general, where $U \in \mathcal{P}(X)$. However, when U is nonempty, we have

$$\bigcap_{\{x\}\in\chi_X(U)} \{x\} \subset U.$$

5. *Interaction With Unions I.* The diagram

$$\begin{tikzcd}
\mathcal{P}(\mathcal{P}(X)) \times \mathcal{P}(\mathcal{P}(X)) \arrow[r, "\cup"] \arrow[d, "\cap\times\cap"'] & \mathcal{P}(\mathcal{P}(X)) \arrow[d, "\cap"] \\
\mathcal{P}(X) \times \mathcal{P}(X) \arrow[r, "\cap"'] & \mathcal{P}(X)
\end{tikzcd}$$

commutes, i.e. we have

$$\bigcap_{W\in\mathcal{U}\cup\mathcal{V}} W = \left(\bigcap_{U\in\mathcal{U}} U\right) \cap \left(\bigcap_{V\in\mathcal{V}} V\right)$$

for each $\mathcal{U}, \mathcal{V} \in \mathcal{P}(\mathcal{P}(X))$.

6. *Interaction With Unions II.* The diagram

$$\begin{tikzcd}
\mathcal{P}(\mathcal{P}(X)) \arrow[r, "\mathrm{id}_{\mathcal{P}(\mathcal{P}(X))}\times(U\cup-)"] \arrow[d, "\cap"'] & \mathcal{P}(\mathcal{P}(X)) \arrow[d, "\cap"] \\
\mathcal{P}(X) \arrow[r, "U\cup-"'] & \mathcal{P}(X)
\end{tikzcd}
\qquad
\begin{tikzcd}
\mathcal{P}(\mathcal{P}(X)) \arrow[r, "\mathrm{id}_{\mathcal{P}(\mathcal{P}(X))}\times(-\cup V)"] \arrow[d, "\cap"'] & \mathcal{P}(\mathcal{P}(X)) \arrow[d, "\cap"] \\
\mathcal{P}(X) \arrow[r, "-\cup V"'] & \mathcal{P}(X)
\end{tikzcd}$$

commute, i.e. we have

$$U \cup \left(\bigcap_{V\in\mathcal{V}} V\right) = \bigcap_{V\in\mathcal{V}} (U \cup V),$$

$$\left(\bigcap_{U\in\mathcal{U}} U\right) \cup V = \bigcap_{U\in\mathcal{U}} (U \cup V)$$

for each $\mathcal{U}, \mathcal{V} \in \mathcal{P}(\mathcal{P}(X))$ and each $U, V \in \mathcal{P}(X)$.

7. *Interaction With Intersections I.* We have a natural transformation

$$\begin{tikzcd}
\mathcal{P}(\mathcal{P}(X)) \times \mathcal{P}(\mathcal{P}(X)) \arrow[r, "\cap"] \arrow[d, "\cap\times\cap"'] \arrow[dr, phantom, "\supset"] & \mathcal{P}(\mathcal{P}(X)) \arrow[d, "\cap"] \\
\mathcal{P}(X) \times \mathcal{P}(X) \arrow[r, "\cap"'] & \mathcal{P}(X),
\end{tikzcd}$$

with components

$$\left(\bigcap_{U\in\mathcal{U}} U\right) \cap \left(\bigcap_{V\in\mathcal{V}} V\right) \subset \bigcap_{W\in\mathcal{U}\cap\mathcal{V}} W$$

for each $\mathcal{U}, \mathcal{V} \in \mathcal{P}(\mathcal{P}(X))$.

8. *Interaction With Intersections II.* The diagrams

$$\begin{tikzcd}
\mathcal{P}(\mathcal{P}(X)) \arrow[r, "\mathrm{id}_{\mathcal{P}(\mathcal{P}(X))}\times(U\cap-)"] \arrow[d, "\cap"'] & \mathcal{P}(\mathcal{P}(X)) \arrow[d, "\cap"] \\
\mathcal{P}(X) \arrow[r, "U\cap-"'] & \mathcal{P}(X)
\end{tikzcd}
\qquad
\begin{tikzcd}
\mathcal{P}(\mathcal{P}(X)) \arrow[r, "\mathrm{id}_{\mathcal{P}(\mathcal{P}(X))}\times(-\cap V)"] \arrow[d, "\cap"'] & \mathcal{P}(\mathcal{P}(X)) \arrow[d, "\cap"] \\
\mathcal{P}(X) \arrow[r, "-\cap V"'] & \mathcal{P}(X)
\end{tikzcd}$$

commute, i.e. we have

$$U \cup \left(\bigcap_{V\in\mathcal{V}} V\right) = \bigcap_{V\in\mathcal{V}} (U \cup V),$$
$$\left(\bigcap_{U\in\mathcal{U}} U\right) \cup V = \bigcap_{U\in\mathcal{U}} (U \cup V)$$

for each $\mathcal{U}, \mathcal{V} \in \mathcal{P}(\mathcal{P}(X))$ and each $U, V \in \mathcal{P}(X)$.

9. *Interaction With Differences.* The diagram

$$\begin{tikzcd}
\mathcal{P}(\mathcal{P}(X)) \times \mathcal{P}(\mathcal{P}(X)) \arrow[r, "\setminus"] \arrow[d, "\cap\times\cap"'] \arrow[dr, phantom, "\times"] & \mathcal{P}(\mathcal{P}(X)) \arrow[d, "\cap"] \\
\mathcal{P}(X) \times \mathcal{P}(X) \arrow[r, "\setminus"'] & \mathcal{P}(X),
\end{tikzcd}$$

does not commute in general, i.e. we may have

$$\bigcap_{W \in \mathcal{U} \setminus \mathcal{V}} W \neq \left(\bigcap_{U \in \mathcal{U}} U \right) \setminus \left(\bigcap_{V \in \mathcal{V}} V \right)$$

in general, where $\mathcal{U}, \mathcal{V} \in \mathcal{P}(\mathcal{P}(X))$.

10. *Interaction With Complements I.* The diagram

$$\begin{tikzcd}
\mathcal{P}(\mathcal{P}(X))^{\mathsf{op}} \arrow[r, "(-)^{\mathsf{c}}"] \arrow[d, "\bigcap^{\mathsf{op}}"'] \arrow[dr, phantom, "\times"] & \mathcal{P}(\mathcal{P}(X)) \arrow[d, "\bigcap"] \\
\mathcal{P}(X)^{\mathsf{op}} \arrow[r, "(-)^{\mathsf{c}}"'] & \mathcal{P}(X),
\end{tikzcd}$$

does not commute in general, i.e. we may have

$$\bigcap_{W \in \mathcal{U}^{\mathsf{c}}} W \neq \bigcap_{U \in \mathcal{U}} U^{\mathsf{c}}$$

in general, where $\mathcal{U} \in \mathcal{P}(\mathcal{P}(X))$.

11. *Interaction With Complements II.* The diagram

$$\begin{tikzcd}
 & \mathcal{P}(\mathcal{P}(X)^{\mathsf{op}}) \arrow[dr, dashed, "\sim"] & \\
\mathcal{P}(\mathcal{P}(X)) \arrow[ur, "\mathrm{id}_{\mathcal{P}(X)} \star (-)^{\mathsf{c}}"] \arrow[dr, "\bigcap"'] & & \mathcal{P}(\mathcal{P}(X))^{\mathsf{op}} \arrow[dl, "\bigcup^{\mathsf{op}}"] \\
 & \mathcal{P}(X) \arrow[r, "(-)^{\mathsf{c}}"'] & \mathcal{P}(X)^{\mathsf{op}},
\end{tikzcd}$$

commutes, i.e. we have

$$\left(\bigcap_{U \in \mathcal{U}} U \right)^{\mathsf{c}} = \bigcup_{U \in \mathcal{U}} U^{\mathsf{c}}$$

for each $\mathcal{U} \in \mathcal{P}(\mathcal{P}(X))$.

12. *Interaction With Complements III.* The diagram

$$\begin{tikzcd}
 & \mathcal{P}(\mathcal{P}(X)^{\mathsf{op}}) \arrow[dr, dashed, "\sim"] & \\
\mathcal{P}(\mathcal{P}(X)) \arrow[ur, "\mathrm{id}_{\mathcal{P}(X)} \star (-)^{\mathsf{c}}"] \arrow[dd, "\bigcup"'] & & \mathcal{P}(\mathcal{P}(X))^{\mathsf{op}} \arrow[dd, "\bigcap^{\mathsf{op}}"] \\
 & & \\
\mathcal{P}(X) \arrow[rr, "(-)^{\mathsf{c}}"'] & & \mathcal{P}(X)^{\mathsf{op}},
\end{tikzcd}$$

commutes, i.e. we have

$$\left(\bigcup_{U \in \mathcal{U}} U\right)^{\mathsf{c}} = \bigcap_{U \in \mathcal{U}} U^{\mathsf{c}}$$

for each $\mathcal{U} \in \mathcal{P}(\mathcal{P}(X))$.

13. *Interaction With Symmetric Differences.* The diagram

$$\begin{tikzcd}
\mathcal{P}(\mathcal{P}(X)) \times \mathcal{P}(\mathcal{P}(X)) \arrow[r, "\triangle"] \arrow[d, "\bigcap \times \bigcap"'] \arrow[dr, phantom, "\times"] & \mathcal{P}(\mathcal{P}(X)) \arrow[d, "\bigcap"] \\
\mathcal{P}(X) \times \mathcal{P}(X) \arrow[r, "\triangle"'] & \mathcal{P}(X),
\end{tikzcd}$$

does not commute in general, i.e. we may have

$$\bigcap_{W \in \mathcal{U} \triangle \mathcal{V}} W \neq \left(\bigcap_{U \in \mathcal{U}} U\right) \triangle \left(\bigcap_{V \in \mathcal{V}} V\right)$$

in general, where $\mathcal{U}, \mathcal{V} \in \mathcal{P}(\mathcal{P}(X))$.

14. *Interaction With Internal Homs I.* The diagram

$$\begin{tikzcd}
\mathcal{P}(\mathcal{P}(X))^{\mathsf{op}} \times \mathcal{P}(\mathcal{P}(X)) \arrow[r, "{[-_1,-_2]_{\mathcal{P}(X)}}"] \arrow[d, "\bigcap^{\mathsf{op}} \times \bigcap^{\mathsf{op}}"'] \arrow[dr, phantom, "\times"] & \mathcal{P}(\mathcal{P}(X)) \arrow[d, "\bigcap"] \\
\mathcal{P}(X)^{\mathsf{op}} \times \mathcal{P}(X) \arrow[r, "{[-_1,-_2]_X}"'] & \mathcal{P}(X),
\end{tikzcd}$$

does not commute in general, i.e. we may have

$$\bigcap_{W \in [\mathcal{U},\mathcal{V}]_{\mathcal{P}(X)}} W \neq \left[\bigcap_{U \in \mathcal{U}} U, \bigcap_{V \in \mathcal{V}} V\right]_X$$

in general, where $\mathcal{U} \in \mathcal{P}(\mathcal{P}(X))$.

15. *Interaction With Internal Homs II.* The diagram

$$\begin{tikzcd}
 & \mathcal{P}(\mathcal{P}(X))^{\mathsf{op}} \arrow[dr, "\bigcap^{\mathsf{op}}"] & \\
\mathcal{P}(\mathcal{P}(X)^{\mathsf{op}}) \arrow[ur, dashed, "\sim"] \arrow[dd, "\mathrm{id}_{\mathcal{P}(X)} \star [-,V]_X"'] & & \mathcal{P}(X)^{\mathsf{op}} \arrow[dd, "{[-,V]_X}"] \\
 & & \\
\mathcal{P}(\mathcal{P}(X)) \arrow[rr, "\bigcup"'] & & \mathcal{P}(X)
\end{tikzcd}$$

commutes, i.e. we have

$$\left[\bigcap_{U \in \mathcal{U}} U, V\right]_X = \bigcup_{U \in \mathcal{U}} [U, V]_X$$

for each $\mathcal{U} \in \mathcal{P}(\mathcal{P}(X))$ and each $V \in \mathcal{P}(X)$.

16. *Interaction With Internal Homs III.* The diagram

$$\begin{tikzcd}
\mathcal{P}(\mathcal{P}(X)) \arrow[r, "\bigcap"] \arrow[d, "\mathrm{id}_{\mathcal{P}(X)} \star [U,-]_X"'] & \mathcal{P}(X) \arrow[d, "{[U,-]_X}"] \\
\mathcal{P}(\mathcal{P}(X)) \arrow[r, "\bigcap"'] & \mathcal{P}(X)
\end{tikzcd}$$

commutes, i.e. we have

$$\left[U, \bigcap_{V \in \mathcal{V}} V\right]_X = \bigcap_{V \in \mathcal{V}} [U, V]_X$$

for each $U \in \mathcal{P}(X)$ and each $\mathcal{V} \in \mathcal{P}(\mathcal{P}(X))$.

17. *Interaction With Direct Images.* Let $f\colon X \longrightarrow Y$ be a map of sets. The diagram

$$\begin{tikzcd}
\mathcal{P}(\mathcal{P}(X)) \arrow[r, "(f_*)_*"] \arrow[d, "\cap"'] & \mathcal{P}(\mathcal{P}(Y)) \arrow[d, "\cap"] \\
\mathcal{P}(X) \arrow[r, "f_*"'] & \mathcal{P}(Y)
\end{tikzcd}$$

commutes, i.e. we have

$$\bigcap_{U \in \mathcal{U}} f_*(U) = \bigcap_{V \in f_*(\mathcal{U})} V$$

for each $\mathcal{U} \in \mathcal{P}(X)$, where $f_*(\mathcal{U}) \overset{\text{def}}{=} (f_*)_*(\mathcal{U})$.

18. *Interaction With Inverse Images.* Let $f\colon X \longrightarrow Y$ be a map of sets. The diagram

$$\begin{tikzcd}
\mathcal{P}(\mathcal{P}(Y)) \arrow[r, "(f^{-1})^{-1}"] \arrow[d, "\cap"'] & \mathcal{P}(\mathcal{P}(X)) \arrow[d, "\cap"] \\
\mathcal{P}(Y) \arrow[r, "f^{-1}"'] & \mathcal{P}(X)
\end{tikzcd}$$

commutes, i.e. we have

$$\bigcap_{V \in \mathcal{V}} f^{-1}(V) = \bigcap_{U \in f^{-1}(\mathcal{U})} U$$

for each $\mathcal{V} \in \mathcal{P}(Y)$, where $f^{-1}(\mathcal{V}) \overset{\text{def}}{=} \left(f^{-1}\right)^{-1}(\mathcal{V})$.

19. *Interaction With Direct Images With Compact Support.* Let $f\colon X \longrightarrow Y$ be a map of sets. The diagram

$$\begin{tikzcd}
\mathcal{P}(\mathcal{P}(X)) \arrow[r, "(f_!)_!"] \arrow[d, "\cap"'] & \mathcal{P}(\mathcal{P}(Y)) \arrow[d, "\cap"] \\
\mathcal{P}(X) \arrow[r, "f_!"'] & \mathcal{P}(Y)
\end{tikzcd}$$

commutes, i.e. we have

$$\bigcap_{U \in \mathcal{U}} f_!(U) = \bigcap_{V \in f_!(\mathcal{U})} V$$

for each $\mathcal{U} \in \mathcal{P}(X)$, where $f_!(\mathcal{U}) \overset{\text{def}}{=} (f_!)_!(\mathcal{U})$.

20. *Interaction With Unions of Families I.* The diagram

$$\begin{tikzcd}
\mathcal{P}(\mathcal{P}(X)) \arrow[r, "\mathrm{id}_{\mathcal{P}(X)} \star \bigcap"] \arrow[d, "\bigcup \star \mathrm{id}_{\mathcal{P}(X)}"'] & \mathcal{P}(\mathcal{P}(x)) \arrow[d, "\bigcap"] \\
\mathcal{P}(X) \arrow[r, "\bigcap"'] & X
\end{tikzcd}$$

commutes, i.e. we have

$$\bigcap_{U \in \bigcup_{A \in \mathcal{A}} A} U = \bigcap_{A \in \mathcal{A}} \left(\bigcap_{U \in A} U \right)$$

for each $\mathcal{A} \in \mathcal{P}(\mathcal{P}(X))$.

21. *Interaction With Unions of Families II.* Let X be a set and consider the compositions

$$\begin{tikzcd}
& \mathcal{P}(\mathcal{P}(\mathcal{P}(X))) \arrow[dl, "\bigcap \star \mathrm{id}_{\mathcal{P}(X)}"'] \arrow[dl, shift left=3, "\bigcup \star \mathrm{id}_{\mathcal{P}(X)}"] \arrow[dl, shift right=3, "\mathrm{id}_{\mathcal{P}(X)} \star \bigcap"'] \arrow[dr, "\mathrm{id}_{\mathcal{P}(X)} \star \bigcup"] \\
\mathcal{P}(\mathcal{P}(X)) \arrow[dr, "\bigcup"'] & \mathcal{P}(\mathcal{P}(X)) \arrow[d, "\bigcap"] & \mathcal{P}(\mathcal{P}(X)) \arrow[d, "\bigcup"'] & \mathcal{P}(\mathcal{P}(X)) \arrow[dl, "\bigcap"] \\
& \mathcal{P}(X) & &
\end{tikzcd}$$

given by

$$\mathcal{A} \mapsto \bigcup_{U \in \bigcap_{A \in \mathcal{A}} A} U, \qquad \mathcal{A} \mapsto \bigcap_{U \in \bigcup_{A \in \mathcal{A}} A} U,$$

$$\mathcal{A} \mapsto \bigcup_{A \in \mathcal{A}} \left(\bigcap_{U \in A} U \right), \quad \mathcal{A} \mapsto \bigcap_{A \in \mathcal{A}} \left(\bigcup_{U \in A} U \right)$$

for each $\mathcal{A} \in \mathcal{P}(\mathcal{P}(\mathcal{P}(X)))$. We have the following inclusions:

$$\begin{tikzcd}
 & \displaystyle\bigcup_{U \in \bigcap_{A \in \mathcal{A}} A} U \arrow[dr, hook] & \\
\displaystyle\bigcup_{A \in \mathcal{A}} \left(\bigcap_{U \in A} U \right) & & \displaystyle\bigcap_{A \in \mathcal{A}} \left(\bigcup_{U \in A} U \right) \\
 & \displaystyle\bigcap_{U \in \bigcup_{A \in \mathcal{A}} A} U \arrow[ul, hook] \arrow[uu, hook] \arrow[ur, hook] &
\end{tikzcd}$$

All other possible inclusions fail to hold in general.

Proof 3.7.3 ▶ Proof of Proposition 3.6.2

Item 1: Functoriality

Since $\mathcal{P}(X)$ is posetal, it suffices to prove the condition $(\star)$. So let $\mathcal{U}, \mathcal{V} \in \mathcal{P}(\mathcal{P}(X))$ with $\mathcal{U} \subset \mathcal{V}$. We claim that

$$\bigcap_{V \in \mathcal{V}} V \subset \bigcap_{U \in \mathcal{U}} U.$$

Indeed, if $x \in \bigcap_{V \in \mathcal{V}} V$, then $x \in V$ for all $V \in \mathcal{V}$. But since $\mathcal{U} \subset \mathcal{V}$, it follows that $x \in U$ for all $U \in \mathcal{U}$ as well. Thus $x \in \bigcap_{U \in \mathcal{U}} U$, which gives our desired inclusion.

Item 2: Oplax Associativity

We have

$$\begin{aligned}
\bigcap_{A \in \mathcal{A}} \left(\bigcap_{U \in A} U \right) &\overset{\text{def}}{=} \left\{ x \in X \;\middle|\; \begin{array}{l} \text{for each } A \in \mathcal{A}, \\ \text{we have } x \in \displaystyle\bigcap_{U \in A} U \end{array} \right\} \\
&\overset{\text{def}}{=} \left\{ x \in X \;\middle|\; \begin{array}{l} \text{for each } A \in \mathcal{A} \text{ and each} \\ U \in A, \text{ we have } x \in U \end{array} \right\} \\
&= \left\{ x \in X \;\middle|\; \begin{array}{l} \text{for each } U \in \displaystyle\bigcup_{A \in \mathcal{A}} A, \\ \text{we have } x \in U \end{array} \right\} \\
&\subset \left\{ x \in X \;\middle|\; \begin{array}{l} \text{for each } U \in \displaystyle\bigcap_{A \in \mathcal{A}} A, \\ \text{we have } x \in U \end{array} \right\} \\
&\overset{\text{def}}{=} \bigcap_{U \in \bigcap_{A \in \mathcal{A}} A} U.
\end{aligned}$$

Since $\mathcal{P}(X)$ is posetal, naturality is automatic (**??**). This finishes the proof.

Item 3: Left Unitality

We have

$$\begin{aligned}
\bigcap_{V \in \{U\}} V &\overset{\text{def}}{=} \left\{ x \in X \;\middle|\; \begin{array}{l} \text{for each } V \in \{U\}, \\ \text{we have } x \in U \end{array} \right\} \\
&= \{ x \in X \mid x \in U \} \\
&= U.
\end{aligned}$$

This finishes the proof.

Item 4: Oplax Right Unitality

If $U = \emptyset$, then we have

$$\begin{aligned}\bigcap_{\{u\}\in\chi_X(U)} \{u\} &= \bigcap_{\{u\}\in\emptyset} \{u\} \\ &= X,\end{aligned}$$

so $\bigcap_{\{u\}\in\chi_X(U)} \{u\} = X \neq \emptyset = U$. When U is nonempty, we have two cases:

1. If U is a singleton, say $U = \{u\}$, we have

$$\begin{aligned}\bigcap_{\{u\}\in\chi_X(U)} \{u\} &= \{u\} \\ &\overset{\text{def}}{=} U.\end{aligned}$$

2. If U contains at least two elements, we have

$$\begin{aligned}\bigcap_{\{u\}\in\chi_X(U)} \{u\} &= \emptyset \\ &\subset U.\end{aligned}$$

This finishes the proof.

Item 5: Interaction With Unions I

We have

$$\begin{aligned}\bigcap_{W\in\mathcal{U}\cup\mathcal{V}} W &\overset{\text{def}}{=} \left\{x \in X \,\middle|\, \begin{array}{l}\text{for each } W \in \mathcal{U}\cup\mathcal{V}, \\ \text{we have } x \in W\end{array}\right\} \\ &= \left\{x \in X \,\middle|\, \begin{array}{l}\text{for each } W \in \mathcal{U} \text{ and each} \\ W \in \mathcal{V}, \text{ we have } x \in W\end{array}\right\} \\ &\overset{\text{def}}{=} \left\{x \in X \,\middle|\, \begin{array}{l}\text{for each } W \in \mathcal{U}, \\ \text{we have } x \in W\end{array}\right\} \\ &\quad \cap \left\{x \in X \,\middle|\, \begin{array}{l}\text{for each } W \in \mathcal{V}, \\ \text{we have } x \in W\end{array}\right\}\end{aligned}$$

$$\begin{aligned}
&\overset{\text{def}}{=} \left(\bigcap_{W\in\mathcal{U}} W\right) \cap \left(\bigcap_{W\in\mathcal{V}} W\right) \\
&= \left(\bigcap_{U\in\mathcal{U}} U\right) \cap \left(\bigcap_{V\in\mathcal{V}} V\right).
\end{aligned}$$

This finishes the proof.

Item 6: Interaction With Unions II

Omitted.

Item 7: Interaction With Intersections I

We have

$$\begin{aligned}
\left(\bigcap_{U\in\mathcal{U}} U\right) \cap \left(\bigcap_{V\in\mathcal{V}} V\right) &\overset{\text{def}}{=} \left\{x \in X \,\middle|\, \begin{array}{l}\text{for each } U \in \mathcal{U}, \\ \text{we have } x \in U\end{array}\right\} \\
&\quad \cup \left\{x \in X \,\middle|\, \begin{array}{l}\text{for each } V \in \mathcal{V}, \\ \text{we have } x \in V\end{array}\right\} \\
&= \left\{x \in X \,\middle|\, \begin{array}{l}\text{for each } W \in \mathcal{U} \cap \mathcal{V}, \\ \text{we have } x \in W\end{array}\right\} \\
&\subset \left\{x \in X \,\middle|\, \begin{array}{l}\text{for each } W \in \mathcal{U} \cup \mathcal{V}, \\ \text{we have } x \in W\end{array}\right\} \\
&\overset{\text{def}}{=} \bigcap_{W\in\mathcal{U}\cap\mathcal{V}} W.
\end{aligned}$$

Since $\mathcal{P}(X)$ is posetal, naturality is automatic (**??**). This finishes the proof.

Item 8: Interaction With Intersections II

Omitted.

Item 9: Interaction With Differences

Let $X = \{0, 1\}$, let $\mathcal{U} = \{\{0\}, \{0, 1\}\}$, and let $\mathcal{V} = \{\{0\}\}$. We have

$$\bigcap_{W\in\mathcal{U}\setminus\mathcal{V}} U = \bigcap_{W\in\{\{0,1\}\}} W$$

$$= \{0, 1\},$$

whereas

$$\left(\bigcap_{U \in \mathcal{U}} U\right) \setminus \left(\bigcap_{V \in \mathcal{V}} V\right) = \{0\} \setminus \{0\}$$
$$= \emptyset.$$

Thus we have

$$\bigcap_{W \in \mathcal{U} \setminus \mathcal{V}} W = \{0, 1\} \neq \emptyset = \left(\bigcap_{U \in \mathcal{U}} U\right) \setminus \left(\bigcap_{V \in \mathcal{V}} V\right).$$

This finishes the proof.

Item 10: Interaction With Complements I

Let $X = \{0, 1\}$ and let $\mathcal{U} = \{\{0\}\}$. We have

$$\bigcap_{W \in \mathcal{U}^c} U = \bigcap_{W \in \{\emptyset, \{1\}, \{0,1\}\}} W$$
$$= \emptyset,$$

whereas

$$\bigcap_{U \in \mathcal{U}} U^c = \{0\}^c$$
$$= \{1\}.$$

Thus we have

$$\bigcap_{W \in \mathcal{U}^c} U = \emptyset \neq \{1\} = \bigcap_{U \in \mathcal{U}} U^c.$$

This finishes the proof.

Item 11: Interaction With Complements II

This is a repetition of Item 12 of Proposition 3.6.2 and is proved there.

Item 12: Interaction With Complements III

This is a repetition of Item 11 of Proposition 3.6.2 and is proved there.

Item 13: Interaction With Symmetric Differences

Let $X = \{0, 1\}$, let $\mathcal{U} = \{\{0, 1\}\}$, and let $\mathcal{V} = \{\{0\}, \{0, 1\}\}$. We have

$$\begin{aligned}\bigcap_{W \in \mathcal{U} \triangle \mathcal{V}} W &= \bigcap_{W \in \{\{0\}\}} W \\ &= \{0\},\end{aligned}$$

whereas

$$\begin{aligned}\left(\bigcap_{U \in \mathcal{U}} U\right) \triangle \left(\bigcap_{V \in \mathcal{V}} V\right) &= \{0, 1\} \triangle \{0\} \\ &= \emptyset,\end{aligned}$$

Thus we have

$$\bigcap_{W \in \mathcal{U} \triangle \mathcal{V}} W = \{0\} \neq \emptyset = \left(\bigcap_{U \in \mathcal{U}} U\right) \triangle \left(\bigcap_{V \in \mathcal{V}} V\right).$$

This finishes the proof.

Item 14: Interaction With Internal Homs I

This is a repetition of Item 10 of Proposition 4.7.4 and is proved there.

Item 15: Interaction With Internal Homs II

This is a repetition of Item 11 of Proposition 4.7.4 and is proved there.

Item 16: Interaction With Internal Homs III

This is a repetition of Item 12 of Proposition 4.7.4 and is proved there.

Item 17: Interaction With Direct Images

This is a repetition of Item 4 of Proposition 6.1.4 and is proved there.

Item 18: Interaction With Inverse Images

This is a repetition of Item 4 of Proposition 6.2.3 and is proved there.

Item 19: Interaction With Direct Images With Compact Support

This is a repetition of Item 4 of Proposition 6.3.6 and is proved there.

Item 20: Interaction With Unions of Families I

This is a repetition of Item 20 of Proposition 3.6.2 and is proved there.

Item 21: Interaction With Unions of Families II

This is a repetition of Item 21 of Proposition 3.6.2 and is proved there.

3.8 Binary Unions

Let X be a set and let $U, V \in \mathcal{P}(X)$.

Definition 3.8.1 ▶ Binary Unions

The **union of** U **and** V is the set $U \cup V$ defined by

$$\begin{aligned} U \cup V &\overset{\text{def}}{=} \bigcup_{z \in \{U,V\}} z \\ &\overset{\text{def}}{=} \{x \in X \mid x \in U \text{ or } x \in V\}. \end{aligned}$$

Proposition 3.8.2 ▶ Properties of Binary Unions

Let X be a set.

1. *Functoriality.* The assignments $U, V, (U, V) \mapsto U \cup V$ define functors

$$\begin{aligned} U \cup -\colon &\quad (\mathcal{P}(X), \subset) &&\to (\mathcal{P}(X), \subset), \\ - \cup V\colon &\quad (\mathcal{P}(X), \subset) &&\to (\mathcal{P}(X), \subset), \\ -_1 \cup -_2\colon &\quad (\mathcal{P}(X) \times \mathcal{P}(X), \subset \times \subset) &&\to (\mathcal{P}(X), \subset). \end{aligned}$$

   In particular, the following statements hold for each $U, V, A, B \in \mathcal{P}(X)$:

   (a) If $U \subset A$, then $U \cup V \subset A \cup V$.

   (b) If $V \subset B$, then $U \cup V \subset U \cup B$.

   (c) If $U \subset A$ and $V \subset B$, then $U \cup V \subset A \cup B$.

2. *Associativity.* The diagram

$$\begin{tikzcd}
 & \mathcal{P}(X) \times (\mathcal{P}(X) \times \mathcal{P}(X)) \arrow[dr, "\mathrm{id}_{\mathcal{P}(X)} \times \cup"] & \\
(\mathcal{P}(X) \times \mathcal{P}(X)) \times \mathcal{P}(X) \arrow[ur, dashed, "\alpha^{\mathsf{Sets}}_{\mathcal{P}(X),\mathcal{P}(X),\mathcal{P}(X)}", "\sim"'] \arrow[dr, "\cup \times \mathrm{id}_{\mathcal{P}(X)}"'] & & \mathcal{P}(X) \times \mathcal{P}(X) \arrow[dl, "\cup"] \\
 & \mathcal{P}(X) \times \mathcal{P}(X) \underset{\cup}{\longrightarrow} \mathcal{P}(X), &
\end{tikzcd}$$

commutes, i.e. we have an equality of sets

$$(U \cup V) \cup W = U \cup (V \cup W)$$

for each $U, V, W \in \mathcal{P}(X)$.

3. *Unitality.* The diagrams

$$\begin{tikzcd}
\mathsf{pt} \times \mathcal{P}(X) \arrow[r, "{[\emptyset] \times \mathrm{id}_{\mathcal{P}(X)}}"] \arrow[dr, dashed, "\lambda^{\mathsf{Sets}}_{\mathcal{P}(X)}", "\sim"'] & \mathcal{P}(X) \times \mathcal{P}(X) \arrow[d, "\cup"] \\
 & \mathcal{P}(X)
\end{tikzcd}
\qquad
\begin{tikzcd}
\mathcal{P}(X) \times \mathsf{pt} \arrow[r, "{\mathrm{id}_{\mathcal{P}(X)} \times [\emptyset]}"] \arrow[dr, dashed, "\rho^{\mathsf{Sets}}_{\mathcal{P}(X)}", "\sim"'] & \mathcal{P}(X) \times \mathcal{P}(X) \arrow[d, "\cup"] \\
 & \mathcal{P}(X)
\end{tikzcd}$$

commute, i.e. we have equalities of sets

$$\begin{aligned} \emptyset \cup U &= U, \\ U \cup \emptyset &= U \end{aligned}$$

for each $U \in \mathcal{P}(X)$.

4. *Commutativity.* The diagram

$$\begin{tikzcd}
\mathcal{P}(X) \times \mathcal{P}(X) \arrow[r, "\sigma^{\mathsf{Sets}}_{\mathcal{P}(X),\mathcal{P}(X)}"] \arrow[dr, "\cup"'] & \mathcal{P}(X) \times \mathcal{P}(X) \arrow[d, "\cup"] \\
 & \mathcal{P}(X)
\end{tikzcd}$$

commutes, i.e. we have an equality of sets

$$U \cup V = V \cup U$$

for each $U, V \in \mathcal{P}(X)$.

5. *Annihilation With X.* The diagrams

$$\begin{tikzcd}
 & \mathrm{pt} \times \mathrm{pt} \arrow[dr, "\mu^{\mathsf{Sets}}_{4|\mathcal{P}(X),\mathcal{P}(X),\mathcal{P}(X),\mathcal{P}(X)}"] & \\
\mathrm{pt} \times \mathcal{P}(X) \arrow[ur, "\mathrm{id}_{\mathrm{pt}} \times \epsilon^{\mathsf{Sets}}_{\mathcal{P}(X)}"] \arrow[d, "{[X] \times \mathrm{id}_{\mathcal{P}(X)}}"'] & & \mathrm{pt} \arrow[d, "{[X]}"] \\
\mathcal{P}(X) \times \mathcal{P}(X) \arrow[rr, "\cup"'] & & \mathcal{P}(X)
\end{tikzcd}
\qquad
\begin{tikzcd}
 & \mathrm{pt} \times \mathrm{pt} \arrow[dr, "\mu^{\mathsf{Sets}}_{4|\mathcal{P}(X),\mathcal{P}(X),\mathcal{P}(X),\mathcal{P}(X)}"] & \\
\mathcal{P}(X) \times \mathrm{pt} \arrow[ur, "\epsilon^{\mathsf{Sets}}_{\mathcal{P}(X)} \times \mathrm{id}_{\mathrm{pt}}"] \arrow[d, "{\mathrm{id}_{\mathcal{P}(X)} \times [X]}"'] & & \mathrm{pt} \arrow[d, "{[X]}"] \\
\mathcal{P}(X) \times \mathcal{P}(X) \arrow[rr, "\cup"'] & & \mathcal{P}(X)
\end{tikzcd}$$

commute, i.e. we have equalities of sets

$$U \cup X = X,$$
$$X \cup V = X$$

for each $U, V \in \mathcal{P}(X)$.

6. *Distributivity of Unions Over Intersections.* The diagrams

$$\begin{tikzcd}
 & (\mathcal{P}(X) \times \mathcal{P}(X)) \times (\mathcal{P}(X) \times \mathcal{P}(X)) \arrow[dr, "\mu^{\mathsf{Sets}}_{4|\mathcal{P}(X),\mathcal{P}(X),\mathcal{P}(X),\mathcal{P}(X)}"] & \\
\mathcal{P}(X) \times (\mathcal{P}(X) \times \mathcal{P}(X)) \arrow[ur, "{\Delta_{\mathcal{P}(X)} \times (\mathrm{id}_{\mathcal{P}(X)} \times \mathrm{id}_{\mathcal{P}(X)})}"] \arrow[d, "{\mathrm{id}_{\mathcal{P}(X)} \times \cap}"'] & & (\mathcal{P}(X) \times \mathcal{P}(X)) \times (\mathcal{P}(X) \times \mathcal{P}(X)) \arrow[d, "{\cup \times \cup}"] \\
\mathcal{P}(X) \times \mathcal{P}(X) \arrow[dr, "\cup"'] & & \mathcal{P}(X) \times \mathcal{P}(X), \arrow[dl, "\cap"] \\
 & \mathcal{P}(X) &
\end{tikzcd}
\qquad
\begin{tikzcd}
 & (\mathcal{P}(X) \times \mathcal{P}(X)) \times (\mathcal{P}(X) \times \mathcal{P}(X)) \arrow[dr, "\mu^{\mathsf{Sets}}_{4|\mathcal{P}(X),\mathcal{P}(X),\mathcal{P}(X),\mathcal{P}(X)}"] & \\
(\mathcal{P}(X) \times \mathcal{P}(X)) \times \mathcal{P}(X) \arrow[ur, "{(\mathrm{id}_{\mathcal{P}(X)} \times \mathrm{id}_{\mathcal{P}(X)}) \times \Delta_{\mathcal{P}(X)}}"] \arrow[d, "{\cap \times \mathrm{id}_{\mathcal{P}(X)}}"'] & & (\mathcal{P}(X) \times \mathcal{P}(X)) \times (\mathcal{P}(X) \times \mathcal{P}(X)) \arrow[d, "{\cup \times \cup}"] \\
\mathcal{P}(X) \times \mathcal{P}(X) \arrow[dr, "\cup"'] & & \mathcal{P}(X) \times \mathcal{P}(X), \arrow[dl, "\cap"] \\
 & \mathcal{P}(X) &
\end{tikzcd}$$

commute, i.e. we have equalities of sets

$$U \cup (V \cap W) = (U \cup V) \cap (U \cup W),$$
$$(U \cap V) \cup W = (U \cup W) \cap (V \cup W)$$

for each $U, V, W \in \mathcal{P}(X)$.

7. *Distributivity of Intersections Over Unions.* The diagrams

$$\begin{tikzcd}
 & (\mathcal{P}(X)\times\mathcal{P}(X))\times(\mathcal{P}(X)\times\mathcal{P}(X)) \arrow[dr, "\mu^{\mathsf{Sets}}_{4(\mathcal{P}(X),\mathcal{P}(X),\mathcal{P}(X),\mathcal{P}(X)}"] & \\
\mathcal{P}(X)\times(\mathcal{P}(X)\times\mathcal{P}(X)) \arrow[ur, "\Delta_{\mathcal{P}(X)}\times(\mathrm{id}_{\mathcal{P}(X)}\times\mathrm{id}_{\mathcal{P}(X)})"] \arrow[d, "\mathrm{id}_{\mathcal{P}(X)}\times\cup"'] & & (\mathcal{P}(X)\times\mathcal{P}(X))\times(\mathcal{P}(X)\times\mathcal{P}(X)) \arrow[d, "\cap\times\cap"] \\
\mathcal{P}(X)\times\mathcal{P}(X) \arrow[dr, "\cap"'] & & \mathcal{P}(X)\times\mathcal{P}(X), \arrow[dl, "\cup"] \\
 & \mathcal{P}(X) &
\end{tikzcd}$$

$$\begin{tikzcd}
 & (\mathcal{P}(X)\times\mathcal{P}(X))\times(\mathcal{P}(X)\times\mathcal{P}(X)) \arrow[dr, "\mu^{\mathsf{Sets}}_{4(\mathcal{P}(X),\mathcal{P}(X),\mathcal{P}(X),\mathcal{P}(X)}"] & \\
(\mathcal{P}(X)\times\mathcal{P}(X))\times\mathcal{P}(X) \arrow[ur, "(\mathrm{id}_{\mathcal{P}(X)}\times\mathrm{id}_{\mathcal{P}(X)})\times\Delta_{\mathcal{P}(X)}"] \arrow[d, "\cup\times\mathrm{id}_{\mathcal{P}(X)}"'] & & (\mathcal{P}(X)\times\mathcal{P}(X))\times(\mathcal{P}(X)\times\mathcal{P}(X)) \arrow[d, "\cap\times\cap"] \\
\mathcal{P}(X)\times\mathcal{P}(X) \arrow[dr, "\cap"'] & & \mathcal{P}(X)\times\mathcal{P}(X), \arrow[dl, "\cup"] \\
 & \mathcal{P}(X) &
\end{tikzcd}$$

commute, i.e. we have equalities of sets

$$\begin{aligned}
U\cap(V\cup W) &= (U\cap V)\cup(U\cap W),\\
(U\cup V)\cap W &= (U\cap W)\cup(V\cap W)
\end{aligned}$$

for each $U, V, W \in \mathcal{P}(X)$.

8. *Idempotency.* The diagram

$$\begin{tikzcd}
\mathcal{P}(X) \arrow[r, "\Delta_{\mathcal{P}(X)}"] \arrow[dr, "\mathrm{id}_{\mathcal{P}(X)}"'] & \mathcal{P}(X)\times\mathcal{P}(X) \arrow[d, "\cup"] \\
 & \mathcal{P}(X)
\end{tikzcd}$$

commutes, i.e. we have an equality of sets

$$U\cup U = U$$

for each $U \in \mathcal{P}(X)$.

9. *Via Intersections and Symmetric Differences.* The diagram

$$\begin{tikzcd}
(\mathcal{P}(X)\times\mathcal{P}(X))\times(\mathcal{P}(X)\times\mathcal{P}(X)) \arrow[r, "\triangle\times\cap"] & \mathcal{P}(X)\times\mathcal{P}(X) \arrow[dr, "\triangle"] & \\
\mathcal{P}(X)\times\mathcal{P}(X) \arrow[u, "\Delta_{\mathcal{P}(X)\times\mathcal{P}(X)}"] \arrow[rr, "\cup"'] & & \mathcal{P}(X)
\end{tikzcd}$$

commutes, i.e. we have an equality of sets

$$U\cup V = (U\triangle V)\triangle(U\cap V)$$

for each $U, V \in \mathcal{P}(X)$.

10. *Interaction With Characteristic Functions I.* We have

$$\chi_{U\cup V} = \max(\chi_U, \chi_V)$$

for each $U, V \in \mathcal{P}(X)$.

11. *Interaction With Characteristic Functions II.* We have

$$\chi_{U\cup V} = \chi_U + \chi_V - \chi_{U\cap V}$$

for each $U, V \in \mathcal{P}(X)$.

12. *Interaction With Direct Images.* Let $f\colon X \longrightarrow Y$ be a function. The diagram

$$\begin{tikzcd}
\mathcal{P}(X) \times \mathcal{P}(X) \arrow[r, "f_* \times f_*"] \arrow[d, "\cup"'] & \mathcal{P}(Y) \times \mathcal{P}(Y) \arrow[d, "\cup"] \\
\mathcal{P}(X) \arrow[r, "f_*"'] & \mathcal{P}(Y)
\end{tikzcd}$$

commutes, i.e. we have

$$f_*(U \cup V) = f_*(U) \cup f_*(V)$$

for each $U, V \in \mathcal{P}(X)$.

13. *Interaction With Inverse Images.* Let $f\colon X \longrightarrow Y$ be a function. The diagram

$$\begin{tikzcd}
\mathcal{P}(Y) \times \mathcal{P}(Y) \arrow[r, "f^{-1} \times f^{-1}"] \arrow[d, "\cup"'] & \mathcal{P}(X) \times \mathcal{P}(X) \arrow[d, "\cup"] \\
\mathcal{P}(Y) \arrow[r, "f^{-1}"'] & \mathcal{P}(X)
\end{tikzcd}$$

commutes, i.e. we have

$$f^{-1}(U \cup V) = f^{-1}(U) \cup f^{-1}(V)$$

for each $U, V \in \mathcal{P}(Y)$.

14. *Interaction With Direct Images With Compact Support.* Let $f\colon X \to Y$ be a function. We have a natural transformation

$$\begin{tikzcd}
\mathcal{P}(X) \times \mathcal{P}(X) \arrow[r, "f_! \times f_!"] \arrow[d, "\cup"'] \arrow[dr, phantom, "\circlearrowright"] & \mathcal{P}(Y) \times \mathcal{P}(Y) \arrow[d, "\cup"] \\
\mathcal{P}(X) \arrow[r, "f_!"'] & \mathcal{P}(Y)
\end{tikzcd}$$

with components

$$f_!(U) \cup f_!(V) \subset f_!(U \cup V)$$

indexed by $U, V \in \mathcal{P}(X)$.

15. *Interaction With Powersets and Semirings.* The quintuple $(\mathcal{P}(X), \cup, \cap, \emptyset, X)$ is an idempotent commutative semiring.

Proof 3.8.3 ▶ Proof of Proposition 3.8.2

Item 1: Functoriality

See [Pro24ak].

Item 2: Associativity

See [Pro24ax].

Item 3: Unitality

This follows from [Pro24ba] and Item 4.

Item 4: Commutativity

See [Pro24ay].

Item 5: Annihilation With X

We have

$$\begin{aligned} U \cup X &\overset{\text{def}}{=} \{x \in X \mid x \in U \text{ or } x \in X\} \\ &= \{x \in X \mid x \in X\}, \end{aligned}$$

$$= X$$

and

$$\begin{aligned} X \cup V &\overset{\text{def}}{=} \{x \in X \mid x \in X \text{ or } x \in V\} \\ &= \{x \in X \mid x \in X\} \\ &= X. \end{aligned}$$

This finishes the proof.

Item 6: Distributivity of Unions Over Intersections

See [Pro24aw].

Item 7: Distributivity of Intersections Over Unions

See [Pro24ag].

Item 8: Idempotency

See [Pro24aj].

Item 9: Via Intersections and Symmetric Differences

See [Pro24av].

Item 10: Interaction With Characteristic Functions I

See [Pro24h].

Item 11: Interaction With Characteristic Functions II

See [Pro24h].

Item 12: Interaction With Direct Images

See [Pro24n].

Item 13: Interaction With Inverse Images

See [Pro24v].

Item 14: Interaction With Direct Images With Compact Support

This is a repetition of Item 5 of Proposition 6.3.6 and is proved there.

Item 15: Interaction With Powersets and Semirings

This follows from Items 2 to 4 and 8 of this propostition and Items 3 to 6 and 8 of Proposition 3.9.2.

3.9 Binary Intersections

Let X be a set and let $U, V \in \mathcal{P}(X)$.

Definition 3.9.1 ▶ Binary Intersections

The **intersection of** U **and** V is the set $U \cap V$ defined by

$$\begin{aligned} U \cap V &\overset{\text{def}}{=} \bigcap_{z \in \{U,V\}} z \\ &\overset{\text{def}}{=} \{x \in X \mid x \in U \text{ or } x \in V\}. \end{aligned}$$

Proposition 3.9.2 ▶ Properties of Binary Intersections

Let X be a set.

1. *Functoriality.* The assignments $U, V, (U, V) \mapsto U \cap V$ define functors

$$\begin{aligned} U \cap -\colon &\quad (\mathcal{P}(X), \subset) &&\to (\mathcal{P}(X), \subset), \\ - \cap V\colon &\quad (\mathcal{P}(X), \subset) &&\to (\mathcal{P}(X), \subset), \\ -_1 \cap -_2\colon &\quad (\mathcal{P}(X) \times \mathcal{P}(X), \subset \times \subset) &&\to (\mathcal{P}(X), \subset). \end{aligned}$$

   In particular, the following statements hold for each $U, V, A, B \in \mathcal{P}(X)$:

   (a) If $U \subset A$, then $U \cap V \subset A \cap V$.

   (b) If $V \subset B$, then $U \cap V \subset U \cap B$.

   (c) If $U \subset A$ and $V \subset B$, then $U \cap V \subset A \cap B$.

2. *Adjointness.* We have adjunctions

$$(U \cap - \dashv [U, -]_X)\colon \quad \mathcal{P}(X) \underset{[U,-]_X}{\overset{U\cap -}{\underset{\longleftarrow}{\overset{\longrightarrow}{\perp}}}} \mathcal{P}(X),$$

$$(- \cap V \dashv [V, -]_X)\colon \quad \mathcal{P}(X) \underset{[V,-]_X}{\overset{-\cap V}{\underset{\longleftarrow}{\overset{\longrightarrow}{\perp}}}} \mathcal{P}(X),$$

   witnessed by bijections

$$\mathrm{Hom}_{\mathcal{P}(X)}(U \cap V, W) \cong \mathrm{Hom}_{\mathcal{P}(X)}(U, [V, W]_X),$$

$$\mathrm{Hom}_{\mathcal{P}(X)}(U \cap V, W) \cong \mathrm{Hom}_{\mathcal{P}(X)}(V, [U, W]_X),$$

natural in $U, V, W \in \mathcal{P}(X)$, where

$$[-_1, -_2]_X \colon \mathcal{P}(X)^{\mathsf{op}} \times \mathcal{P}(X) \to \mathcal{P}(X)$$

is the bifunctor of Section 4.7. In particular, the following statements hold for each $U, V, W \in \mathcal{P}(X)$:

(a) The following conditions are equivalent:

i. We have $U \cap V \subset W$.

ii. We have $U \subset [V, W]_X$.

(b) The following conditions are equivalent:

i. We have $U \cap V \subset W$.

ii. We have $V \subset [U, W]_X$.

3. *Associativity.* The diagram

$$\begin{tikzcd}
 & \mathcal{P}(X) \times (\mathcal{P}(X) \times \mathcal{P}(X)) \arrow[dr, "\mathrm{id}_{\mathcal{P}(X)} \times \cap"] & \\
(\mathcal{P}(X) \times \mathcal{P}(X)) \times \mathcal{P}(X) \arrow[ur, dashed, "\alpha^{\mathsf{Sets}}_{\mathcal{P}(X),\mathcal{P}(X),\mathcal{P}(X)}", "\sim"'] \arrow[dd, "\cap \times \mathrm{id}_{\mathcal{P}(X)}"'] & & \mathcal{P}(X) \times \mathcal{P}(X) \arrow[dd, "\cap"] \\
 & & \\
\mathcal{P}(X) \times \mathcal{P}(X) \arrow[rr, "\cap"'] & & \mathcal{P}(X),
\end{tikzcd}$$

commutes, i.e. we have an equality of sets

$$(U \cap V) \cap W = U \cap (V \cap W)$$

for each $U, V, W \in \mathcal{P}(X)$.

4. *Unitality.* The diagrams

$$\begin{tikzcd}
\mathsf{pt} \times \mathcal{P}(X) \arrow[r, "{[X] \times \mathrm{id}_{\mathcal{P}(X)}}"] \arrow[dr, dashed, "\lambda^{\mathsf{Sets}}_{\mathcal{P}(X)}", "\sim"'] & \mathcal{P}(X) \times \mathcal{P}(X) \arrow[d, "\cap"] \\
 & \mathcal{P}(X)
\end{tikzcd}
\qquad
\begin{tikzcd}
\mathcal{P}(X) \times \mathsf{pt} \arrow[r, "{\mathrm{id}_{\mathcal{P}(X)} \times [X]}"] \arrow[dr, dashed, "\rho^{\mathsf{Sets}}_{\mathcal{P}(X)}", "\sim"'] & \mathcal{P}(X) \times \mathcal{P}(X) \arrow[d, "\cap"] \\
 & \mathcal{P}(X)
\end{tikzcd}$$

commute, i.e. we have equalities of sets

$$X \cap U = U,$$
$$U \cap X = U$$

for each $U \in \mathcal{P}(X)$.

5. *Commutativity.* The diagram

$$\begin{tikzcd}
\mathcal{P}(X) \times \mathcal{P}(X) \arrow[r, "\sigma^{\mathsf{Sets}}_{\mathcal{P}(X),\mathcal{P}(X)}"] \arrow[dr, "\cap"'] & \mathcal{P}(X) \times \mathcal{P}(X) \arrow[d, "\cap"] \\
& \mathcal{P}(X)
\end{tikzcd}$$

commutes, i.e. we have an equality of sets

$$U \cap V = V \cap U$$

for each $U, V \in \mathcal{P}(X)$.

6. *Annihilation With the Empty Set.* The diagrams

$$\begin{tikzcd}
& \mathsf{pt} \times \mathsf{pt} \arrow[dr, "\mu^{\mathsf{Sets}}_{4|\mathcal{P}(X),\mathcal{P}(X),\mathcal{P}(X),\mathcal{P}(X)}"] & \\
\mathsf{pt} \times \mathcal{P}(X) \arrow[ur, "\mathsf{id}_{\mathsf{pt}} \times \epsilon^{\mathsf{Sets}}_{\mathcal{P}(X)}"] \arrow[d, "[\emptyset] \times \mathsf{id}_{\mathcal{P}(X)}"'] & & \mathsf{pt} \arrow[d, "{[\emptyset]}"] \\
\mathcal{P}(X) \times \mathcal{P}(X) \arrow[r, "\cap"'] & & \mathcal{P}(X)
\end{tikzcd}
\qquad
\begin{tikzcd}
& \mathsf{pt} \times \mathsf{pt} \arrow[dr, "\mu^{\mathsf{Sets}}_{4|\mathcal{P}(X),\mathcal{P}(X),\mathcal{P}(X),\mathcal{P}(X)}"] & \\
\mathcal{P}(X) \times \mathsf{pt} \arrow[ur, "\epsilon^{\mathsf{Sets}}_{\mathcal{P}(X)} \times \mathsf{id}_{\mathsf{pt}}"] \arrow[d, "\mathsf{id}_{\mathcal{P}(X)} \times [\emptyset]"'] & & \mathsf{pt} \arrow[d, "{[\emptyset]}"] \\
\mathcal{P}(X) \times \mathcal{P}(X) \arrow[r, "\cap"'] & & \mathcal{P}(X)
\end{tikzcd}$$

commute, i.e. we have equalities of sets

$$\emptyset \cap X = \emptyset,$$
$$X \cap \emptyset = \emptyset$$

for each $U \in \mathcal{P}(X)$.

7. *Distributivity of Unions Over Intersections.* The diagrams

$$\begin{tikzcd}
 & (\mathcal{P}(X)\times\mathcal{P}(X))\times(\mathcal{P}(X)\times\mathcal{P}(X)) \arrow[dr, "\mu^{\mathsf{Sets}}_{4(\mathcal{P}(X),\mathcal{P}(X),\mathcal{P}(X),\mathcal{P}(X)}"] & \\
\mathcal{P}(X)\times(\mathcal{P}(X)\times\mathcal{P}(X)) \arrow[ur, "\Delta_{\mathcal{P}(X)}\times(\mathrm{id}_{\mathcal{P}(X)}\times\mathrm{id}_{\mathcal{P}(X)})"] \arrow[d, "\mathrm{id}_{\mathcal{P}(X)}\times\cap"'] & & (\mathcal{P}(X)\times\mathcal{P}(X))\times(\mathcal{P}(X)\times\mathcal{P}(X)) \arrow[d, "\cup\times\cup"] \\
\mathcal{P}(X)\times\mathcal{P}(X) \arrow[dr, "\cup"'] & & \mathcal{P}(X)\times\mathcal{P}(X), \arrow[dl, "\cap"] \\
 & \mathcal{P}(X) &
\end{tikzcd}
\qquad
\begin{tikzcd}
 & (\mathcal{P}(X)\times\mathcal{P}(X))\times(\mathcal{P}(X)\times\mathcal{P}(X)) \arrow[dr, "\mu^{\mathsf{Sets}}_{4(\mathcal{P}(X),\mathcal{P}(X),\mathcal{P}(X),\mathcal{P}(X)}"] & \\
(\mathcal{P}(X)\times\mathcal{P}(X))\times\mathcal{P}(X) \arrow[ur, "(\mathrm{id}_{\mathcal{P}(X)}\times\mathrm{id}_{\mathcal{P}(X)})\times\Delta_{\mathcal{P}(X)}"] \arrow[d, "\cap\times\mathrm{id}_{\mathcal{P}(X)}"'] & & (\mathcal{P}(X)\times\mathcal{P}(X))\times(\mathcal{P}(X)\times\mathcal{P}(X)) \arrow[d, "\cup\times\cup"] \\
\mathcal{P}(X)\times\mathcal{P}(X) \arrow[dr, "\cup"'] & & \mathcal{P}(X)\times\mathcal{P}(X), \arrow[dl, "\cap"] \\
 & \mathcal{P}(X) &
\end{tikzcd}$$

commute, i.e. we have equalities of sets

$$\begin{aligned}
U\cup(V\cap W) &= (U\cup V)\cap(U\cup W),\\
(U\cap V)\cup W &= (U\cup W)\cap(V\cup W)
\end{aligned}$$

for each $U, V, W \in \mathcal{P}(X)$.

8. *Distributivity of Intersections Over Unions.* The diagrams

$$\begin{tikzcd}
 & (\mathcal{P}(X)\times\mathcal{P}(X))\times(\mathcal{P}(X)\times\mathcal{P}(X)) \arrow[dr, "\mu^{\mathsf{Sets}}_{4(\mathcal{P}(X),\mathcal{P}(X),\mathcal{P}(X),\mathcal{P}(X)}"] & \\
\mathcal{P}(X)\times(\mathcal{P}(X)\times\mathcal{P}(X)) \arrow[ur, "\Delta_{\mathcal{P}(X)}\times(\mathrm{id}_{\mathcal{P}(X)}\times\mathrm{id}_{\mathcal{P}(X)})"] \arrow[d, "\mathrm{id}_{\mathcal{P}(X)}\times\cup"'] & & (\mathcal{P}(X)\times\mathcal{P}(X))\times(\mathcal{P}(X)\times\mathcal{P}(X)) \arrow[d, "\cap\times\cap"] \\
\mathcal{P}(X)\times\mathcal{P}(X) \arrow[dr, "\cap"'] & & \mathcal{P}(X)\times\mathcal{P}(X), \arrow[dl, "\cup"] \\
 & \mathcal{P}(X) &
\end{tikzcd}
\qquad
\begin{tikzcd}
 & (\mathcal{P}(X)\times\mathcal{P}(X))\times(\mathcal{P}(X)\times\mathcal{P}(X)) \arrow[dr, "\mu^{\mathsf{Sets}}_{4(\mathcal{P}(X),\mathcal{P}(X),\mathcal{P}(X),\mathcal{P}(X)}"] & \\
(\mathcal{P}(X)\times\mathcal{P}(X))\times\mathcal{P}(X) \arrow[ur, "(\mathrm{id}_{\mathcal{P}(X)}\times\mathrm{id}_{\mathcal{P}(X)})\times\Delta_{\mathcal{P}(X)}"] \arrow[d, "\cup\times\mathrm{id}_{\mathcal{P}(X)}"'] & & (\mathcal{P}(X)\times\mathcal{P}(X))\times(\mathcal{P}(X)\times\mathcal{P}(X)) \arrow[d, "\cap\times\cap"] \\
\mathcal{P}(X)\times\mathcal{P}(X) \arrow[dr, "\cap"'] & & \mathcal{P}(X)\times\mathcal{P}(X), \arrow[dl, "\cup"] \\
 & \mathcal{P}(X) &
\end{tikzcd}$$

commute, i.e. we have equalities of sets

$$\begin{aligned}
U\cap(V\cup W) &= (U\cap V)\cup(U\cap W),\\
(U\cup V)\cap W &= (U\cap W)\cup(V\cap W)
\end{aligned}$$

for each $U, V, W \in \mathcal{P}(X)$.

9. *Idempotency.* The diagram

$$\begin{tikzcd}
\mathcal{P}(X) \arrow[r, "\Delta_{\mathcal{P}(X)}"] \arrow[dr, "\mathrm{id}_{\mathcal{P}(X)}"'] & \mathcal{P}(X)\times\mathcal{P}(X) \arrow[d, "\cap"] \\
 & \mathcal{P}(X)
\end{tikzcd}$$

commutes, i.e. we have an equality of sets

$$U \cap U = U$$

for each $U \in \mathcal{P}(X)$.

10. *Interaction With Characteristic Functions I.* We have

$$\chi_{U \cap V} = \chi_U \chi_V$$

for each $U, V \in \mathcal{P}(X)$.

11. *Interaction With Characteristic Functions II.* We have

$$\chi_{U \cap V} = \min(\chi_U, \chi_V)$$

for each $U, V \in \mathcal{P}(X)$.

12. *Interaction With Direct Images.* Let $f \colon X \to Y$ be a function. We have a natural transformation

$$\begin{tikzcd}
\mathcal{P}(X) \times \mathcal{P}(X) \arrow[r, "f_* \times f_*"] \arrow[d, "\cap"'] \arrow[dr, phantom, "\circlearrowright"] & \mathcal{P}(Y) \times \mathcal{P}(Y) \arrow[d, "\cap"] \\
\mathcal{P}(X) \arrow[r, "f_*"'] & \mathcal{P}(Y)
\end{tikzcd}$$

with components

$$f_*(U \cap V) \subset f_*(U) \cap f_*(V)$$

indexed by $U, V \in \mathcal{P}(X)$.

13. *Interaction With Inverse Images.* Let $f \colon X \to Y$ be a function. The diagram

$$\begin{tikzcd}
\mathcal{P}(Y) \times \mathcal{P}(Y) \arrow[r, "f^{-1} \times f^{-1}"] \arrow[d, "\cap"'] & \mathcal{P}(X) \times \mathcal{P}(X) \arrow[d, "\cap"] \\
\mathcal{P}(Y) \arrow[r, "f^{-1}"'] & \mathcal{P}(X)
\end{tikzcd}$$

commutes, i.e. we have

$$f^{-1}(U \cap V) = f^{-1}(U) \cap f^{-1}(V)$$

for each $U, V \in \mathcal{P}(Y)$.

14. *Interaction With Direct Images With Compact Support.* Let $f\colon X \longrightarrow Y$ be a function. The diagram

$$\begin{tikzcd}
\mathcal{P}(X) \times \mathcal{P}(X) \arrow[r, "f_! \times f_!"] \arrow[d, "\cap"'] & \mathcal{P}(Y) \times \mathcal{P}(Y) \arrow[d, "\cap"] \\
\mathcal{P}(X) \arrow[r, "f_!"'] & \mathcal{P}(Y)
\end{tikzcd}$$

commutes, i.e. we have

$$f_!(U) \cap f_!(V) = f_!(U \cap V)$$

for each $U, V \in \mathcal{P}(X)$.

15. *Interaction With Powersets and Monoids With Zero.* The quadruple $((\mathcal{P}(X), \emptyset), \cap, X)$ is a commutative monoid with zero.

16. *Interaction With Powersets and Semirings.* The quintuple $(\mathcal{P}(X), \cup, \cap, \emptyset, X)$ is an idempotent commutative semiring.

Proof 3.9.3 ► Proof of Proposition 3.9.2

Item 1: Functoriality

See [Pro24ai].

Item 2: Adjointness

See [MSE 267469].

Item 3: Associativity

See [Pro24p].

Item 4: Unitality

This follows from [Pro24t] and Item 5.

Item 5: Commutativity

See [Pro24q].

Item 6: Annihilation With the Empty Set

This follows from [Pro24r] and Item 5.

Item 7: Distributivity of Unions Over Intersections

See [Pro24aw].

Item 8: Distributivity of Intersections Over Unions

See [Pro24ag].

Item 9: Idempotency

See [Pro24ah].

Item 10: Interaction With Characteristic Functions I

See [Pro24e].

Item 11: Interaction With Characteristic Functions II

See [Pro24e].

Item 12: Interaction With Direct Images

See [Pro24m].

Item 13: Interaction With Inverse Images

See [Pro24u].

Item 14: Interaction With Direct Images With Compact Support

This is a repetition of Item 6 of Proposition 6.3.6 and is proved there.

Item 15: Interaction With Powersets and Monoids With Zero

This follows from Items 3 to 6.

Item 16: Interaction With Powersets and Semirings

This follows from Items 2 to 4 and 8 and Items 3 to 6 and 8 of Proposition 3.9.2.

3.10 Differences

Let X and Y be sets.

The **difference of** X **and** Y is the set $X \setminus Y$ defined by

$$X \setminus Y \overset{\text{def}}{=} \{a \in X \mid a \notin Y\}.$$

Proposition 3.10.2 ► Properties of Differences

Let X be a set.

1. *Functoriality.* The assignments $U, V, (U, V) \mapsto U \cap V$ define functors

$$\begin{aligned} U \setminus -\colon & \quad (\mathcal{P}(X), \supset) & \to (\mathcal{P}(X), \subset), \\ - \setminus V\colon & \quad (\mathcal{P}(X), \subset) & \to (\mathcal{P}(X), \subset), \\ -_1 \setminus -_2\colon & \quad (\mathcal{P}(X) \times \mathcal{P}(X), \subset \times \supset) & \to (\mathcal{P}(X), \subset). \end{aligned}$$

   In particular, the following statements hold for each $U, V, A, B \in \mathcal{P}(X)$:

   (a) If $U \subset A$, then $U \setminus V \subset A \setminus V$.

   (b) If $V \subset B$, then $U \setminus B \subset U \setminus V$.

   (c) If $U \subset A$ and $V \subset B$, then $U \setminus B \subset A \setminus V$.

2. *De Morgan's Laws.* We have equalities of sets

$$\begin{aligned} X \setminus (U \cup V) &= (X \setminus U) \cap (X \setminus V), \\ X \setminus (U \cap V) &= (X \setminus U) \cup (X \setminus V) \end{aligned}$$

   for each $U, V \in \mathcal{P}(X)$.

3. *Interaction With Unions I.* We have equalities of sets

$$U \setminus (V \cup W) = (U \setminus V) \cap (U \setminus W)$$

   for each $U, V, W \in \mathcal{P}(X)$.

4. *Interaction With Unions II.* We have equalities of sets

$$(U \setminus V) \cup W = (U \cup W) \setminus (V \setminus W)$$

   for each $U, V, W \in \mathcal{P}(X)$.

5. *Interaction With Unions III.* We have equalities of sets

$$\begin{aligned} U \setminus (V \cup W) &= (U \cup W) \setminus (V \cup W) \\ &= (U \setminus V) \setminus W \\ &= (U \setminus W) \setminus V \end{aligned}$$

for each $U, V, W \in \mathcal{P}(X)$.

6. *Interaction With Unions IV.* We have equalities of sets

$$(U \cup V) \setminus W = (U \setminus W) \cup (V \setminus W)$$

for each $U, V, W \in \mathcal{P}(X)$.

7. *Interaction With Intersections.* We have equalities of sets

$$\begin{aligned} (U \setminus V) \cap W &= (U \cap W) \setminus V \\ &= U \cap (W \setminus V) \end{aligned}$$

for each $U, V, W \in \mathcal{P}(X)$.

8. *Interaction With Complements.* We have an equality of sets

$$U \setminus V = U \cap V^{\mathsf{c}}$$

for each $U, V \in \mathcal{P}(X)$.

9. *Interaction With Symmetric Differences.* We have an equality of sets

$$U \setminus V = U \triangle (U \cap V)$$

for each $U, V \in \mathcal{P}(X)$.

10. *Triple Differences.* We have

$$U \setminus (V \setminus W) = (U \cap W) \cup (U \setminus V)$$

for each $U, V, W \in \mathcal{P}(X)$.

11. *Left Annihilation.* We have

$$\emptyset \setminus U = \emptyset$$

for each $U \in \mathcal{P}(X)$.

12. *Right Unitality.* We have

$$U \setminus \emptyset = U$$

for each $U \in \mathcal{P}(X)$.

13. *Right Annihilation.* We have

$$U \setminus X = U$$

for each $U \in \mathcal{P}(X)$.

14. *Invertibility.* We have

$$U \setminus U = \emptyset$$

for each $U \in \mathcal{P}(X)$.

15. *Interaction With Containment.* The following conditions are equivalent:

    (a) We have $V \setminus U \subset W$.

    (b) We have $V \setminus W \subset U$.

16. *Interaction With Characteristic Functions.* We have

$$\chi_{U \setminus V} = \chi_U - \chi_{U \cap V}$$

for each $U, V \in \mathcal{P}(X)$.

17. *Interaction With Direct Images.* We have a natural transformation

$$\begin{tikzcd}
\mathcal{P}(X)^{\mathsf{op}} \times \mathcal{P}(X) \arrow[r, "f_*^{\mathsf{op}} \times f_*"] \arrow[d, "\setminus"'] \arrow[dr, phantom, "\supset"] & \mathcal{P}(Y)^{\mathsf{op}} \times \mathcal{P}(Y) \arrow[d, "\setminus"] \\
\mathcal{P}(X) \arrow[r, "f_*"'] & \mathcal{P}(Y)
\end{tikzcd}$$

with components

$$f_*(U) \setminus f_*(V) \subset f_*(U \setminus V)$$

indexed by $U, V \in \mathcal{P}(X)$.

18. *Interaction With Inverse Images.* The diagram

$$\begin{tikzcd}
\mathcal{P}(Y)^{\mathrm{op}} \times \mathcal{P}(Y) \arrow[r, "f^{\mathrm{op},-1} \times f^{-1}"] \arrow[d, "\setminus"'] & \mathcal{P}(X)^{\mathrm{op}} \times \mathcal{P}(X) \arrow[d, "\setminus"] \\
\mathcal{P}(Y) \arrow[r, "f^{-1}"'] & \mathcal{P}(X)
\end{tikzcd}$$

commutes, i.e. we have

$$f^{-1}(U \setminus V) = f^{-1}(U) \setminus f^{-1}(V)$$

for each $U, V \in \mathcal{P}(X)$.

19. *Interaction With Direct Images With Compact Support.* We have a natural transformation

$$\begin{tikzcd}
\mathcal{P}(X)^{\mathrm{op}} \times \mathcal{P}(X) \arrow[r, "f_*^{\mathrm{op}} \times f_*"] \arrow[d, "\setminus"'] \arrow[dr, phantom, "\supset"] & \mathcal{P}(Y)^{\mathrm{op}} \times \mathcal{P}(Y) \arrow[d, "\setminus"] \\
\mathcal{P}(X) \arrow[r, "f_*"'] & \mathcal{P}(Y)
\end{tikzcd}$$

with components

$$f_*(U) \setminus f_*(V) \subset f_*(U \setminus V)$$

indexed by $U, V \in \mathcal{P}(X)$.

Proof 3.10.3 ► Proof of Proposition 3.10.2

Item 1: Functoriality

See [Pro24aa] and [Pro24ae].

Item 2: De Morgan,s Laws

See [Pro24j].

Item 3: Interaction With Unions I

See [Pro24k].

Item 4: Interaction With Unions II

Omitted.

Item 5: Interaction With Unions III

See [Pro24af].

Item 6: Interaction With Unions IV

See [Pro24z].

Item 7: Interaction With Intersections

See [Pro24s].

Item 8: Interaction With Complements

See [Pro24x].

Item 9: Interaction With Symmetric Differences

See [Pro24y].

Item 10: Triple Differences

See [Pro24ad].

Item 11: Left Annihilation

Omitted.

Item 12: Right Unitality

See [Pro24ab].

Item 13: Right Annihilation

Omitted.

Item 14: Invertibility

See [Pro24ac].

Item 15: Interaction With Containment

Omitted.

Item 16: Interaction With Characteristic Functions

See [Pro24f].

Item 17: Interaction With Direct Images

See [**proof-wiki:image-of-set-difference-under-mapping**].

Item 18: Interaction With Inverse Images

See [**proof-wiki:preimage-of-set-difference-under-mapping**].

3.11 Complements

Let X be a set and let $U \in \mathcal{P}(X)$.

Definition 3.11.1 ▶ Complements

The **complement of** U is the set U^{c} defined by

$$\begin{aligned} U^{\mathsf{c}} &\overset{\text{def}}{=} X \setminus U \\ &\overset{\text{def}}{=} \{a \in X \mid a \notin U\}. \end{aligned}$$

Proposition 3.11.2 ▶ Properties of Complements

Let X be a set.

1. *Functoriality.* The assignment $U \mapsto U^{\mathsf{c}}$ defines a functor

$$(-)^{\mathsf{c}} \colon \mathcal{P}(X)^{\mathsf{op}} \to \mathcal{P}(X).$$

   In particular, the following statements hold for each $U, V \in \mathcal{P}(X)$:

   ($\star$) If $U \subset V$, then $V^{\mathsf{c}} \subset U^{\mathsf{c}}$.

2. *De Morgan's Laws.* The diagrams

$$\begin{tikzcd} \mathcal{P}(X)^{\mathsf{op}} \times \mathcal{P}(X)^{\mathsf{op}} \arrow[r, "\cup^{\mathsf{op}}"] \arrow[d, "(-)^{\mathsf{c}} \times (-)^{\mathsf{c}}"'] & \mathcal{P}(X)^{\mathsf{op}} \arrow[d, "(-)^{\mathsf{c}}"] \\ \mathcal{P}(X) \times \mathcal{P}(X) \arrow[r, "\cap"'] & \mathcal{P}(X) \end{tikzcd} \qquad \begin{tikzcd} \mathcal{P}(X)^{\mathsf{op}} \times \mathcal{P}(X)^{\mathsf{op}} \arrow[r, "\cap^{\mathsf{op}}"] \arrow[d, "(-)^{\mathsf{c}} \times (-)^{\mathsf{c}}"'] & \mathcal{P}(X)^{\mathsf{op}} \arrow[d, "(-)^{\mathsf{c}}"] \\ \mathcal{P}(X) \times \mathcal{P}(X) \arrow[r, "\cup"'] & \mathcal{P}(X) \end{tikzcd}$$

   commute, i.e. we have equalities of sets

$$(U \cup V)^{\mathsf{c}} = U^{\mathsf{c}} \cap V^{\mathsf{c}},$$

$$(U \cap V)^{\mathsf{c}} = U^{\mathsf{c}} \cup V^{\mathsf{c}}$$

for each $U, V \in \mathcal{P}(X)$.

3. *Involutority.* The diagram

$$\begin{tikzcd}
\mathcal{P}(X)^{\mathsf{op}} \arrow[r, "(-)^{\mathsf{c}}"] \arrow[dr, "\mathrm{id}_{\mathcal{P}(X)^{\mathsf{op}}}"'] & \mathcal{P}(X) \arrow[d, "(-)^{\mathsf{c},\mathsf{op}}"] \\
& \mathcal{P}(X)^{\mathsf{op}}
\end{tikzcd}$$

commutes, i.e. we have

$$(U^{\mathsf{c}})^{\mathsf{c}} = U$$

for each $U \in \mathcal{P}(X)$.

4. *Interaction With Characteristic Functions.* We have

$$\chi_{U^{\mathsf{c}}} \equiv 1 - \chi_U \pmod 2$$

for each $U \in \mathcal{P}(X)$.

5. *Interaction With Direct Images.* Let $f \colon X \longrightarrow Y$ be a function. The diagram

$$\begin{tikzcd}
\mathcal{P}(X)^{\mathsf{op}} \arrow[r, "f_!^{\mathsf{op}}"] \arrow[d, "(-)^{\mathsf{c}}"'] & \mathcal{P}(Y)^{\mathsf{op}} \arrow[d, "(-)^{\mathsf{c}}"] \\
\mathcal{P}(X) \arrow[r, "f_*"'] & \mathcal{P}(Y)
\end{tikzcd}$$

commutes, i.e. we have

$$f_*(U^{\mathsf{c}}) = f_!(U)^{\mathsf{c}}$$

for each $U \in \mathcal{P}(X)$.

6. *Interaction With Inverse Images.* Let $f\colon X \longrightarrow Y$ be a function. The diagram

$$\begin{tikzcd} \mathcal{P}(Y)^{\mathsf{op}} \arrow[r, "f^{-1,\mathsf{op}}"] \arrow[d, "(-)^{\mathsf{c}}"'] & \mathcal{P}(X)^{\mathsf{op}} \arrow[d, "(-)^{\mathsf{c}}"] \\ \mathcal{P}(Y) \arrow[r, "f^{-1}"'] & \mathcal{P}(X) \end{tikzcd}$$

commutes, i.e. we have

$$f^{-1}(U^{\mathsf{c}}) = f^{-1}(U)^{\mathsf{c}}$$

for each $U \in \mathcal{P}(X)$.

7. *Interaction With Direct Images With Compact Support.* Let $f\colon X \longrightarrow Y$ be a function. The diagram

$$\begin{tikzcd} \mathcal{P}(X)^{\mathsf{op}} \arrow[r, "f_*^{\mathsf{op}}"] \arrow[d, "(-)^{\mathsf{c}}"'] & \mathcal{P}(Y)^{\mathsf{op}} \arrow[d, "(-)^{\mathsf{c}}"] \\ \mathcal{P}(X) \arrow[r, "f_!"'] & \mathcal{P}(Y) \end{tikzcd}$$

commutes, i.e. we have

$$f_!(U^{\mathsf{c}}) = f_*(U)^{\mathsf{c}}$$

for each $U \in \mathcal{P}(X)$.

Proof 3.11.3 ► Proof of Proposition 3.11.2

Item 1: Functoriality

This follows from Item 1 of Proposition 3.10.2.

Item 2: De Morgan,s Laws

See [Pro24j].

Item 3: Involutority

See [Pro24i].

Item 4: Interaction With Characteristic Functions

Omitted.

Item 5: Interaction With Direct Images

This is a repetition of Item 8 of Proposition 6.1.4 and is proved there.

Item 6: Interaction With Inverse Images

This is a repetition of Item 8 of Proposition 6.2.3 and is proved there.

Item 7: Interaction With Direct Images With Compact Support

This is a repetition of Item 7 of Proposition 6.3.6 and is proved there. ∎

3.12 Symmetric Differences

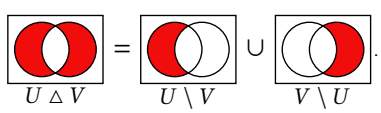
Let X be a set and let $U, V \in \mathcal{P}(X)$.

Definition 3.12.1 ▶ Symmetric Differences

The **symmetric difference of** U **and** V is the set $U \triangle V$ defined by[1]

$$U \triangle V \overset{\text{def}}{=} (U \setminus V) \cup (V \setminus U).$$

[1] *Illustration:*



Proposition 3.12.2 ▶ Properties of Symmetric Differences

Let X be a set.

1. *Lack of Functoriality.* The assignment $(U, V) \mapsto U \triangle V$ ***does not*** in general define functors

$$\begin{aligned} U \triangle -\colon & (\mathcal{P}(X), \subset) \to (\mathcal{P}(X), \subset), \\ - \triangle V\colon & (\mathcal{P}(X), \subset) \to (\mathcal{P}(X), \subset), \\ -_1 \triangle -_2\colon & (\mathcal{P}(X) \times \mathcal{P}(X), \subset \times \subset) \to (\mathcal{P}(X), \subset). \end{aligned}$$

2. *Via Unions and Intersections.* We have

$$U \triangle V = (U \cup V) \setminus (U \cap V)$$

for each $U, V \in \mathcal{P}(X)$, as in the Venn diagram

3. *Symmetric Differences of Disjoint Sets.* If U and V are disjoint, then we have

$$U \triangle V = U \cup V.$$

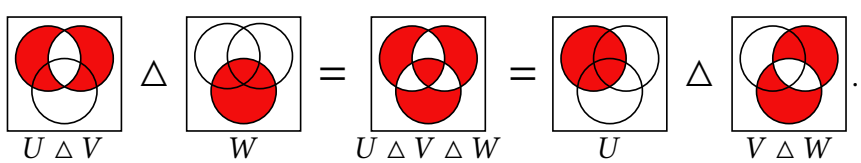
4. *Associativity.* The diagram

$$\begin{tikzcd}
 & \mathcal{P}(X) \times (\mathcal{P}(X) \times \mathcal{P}(X)) \arrow[dr, "\mathrm{id}_{\mathcal{P}(X)} \times \triangle"] & \\
(\mathcal{P}(X) \times \mathcal{P}(X)) \times \mathcal{P}(X) \arrow[ur, dashed, "\alpha^{\mathsf{Sets}}_{\mathcal{P}(X),\mathcal{P}(X),\mathcal{P}(X)}", "\sim"'] \arrow[d, "\triangle \times \mathrm{id}_{\mathcal{P}(X)}"'] & & \mathcal{P}(X) \times \mathcal{P}(X) \arrow[d, "\triangle"] \\
\mathcal{P}(X) \times \mathcal{P}(X) \arrow[rr, "\triangle"'] & & \mathcal{P}(X),
\end{tikzcd}$$

commutes, i.e. we have

$$(U \triangle V) \triangle W = U \triangle (V \triangle W)$$

for each $U, V, W \in \mathcal{P}(X)$, as in the Venn diagram



5. *Unitality.* The diagrams

$$\begin{tikzcd}
\mathsf{pt} \times \mathcal{P}(X) \arrow[r, "[\emptyset] \times \mathrm{id}_{\mathcal{P}(X)}"] \arrow[dr, dashed, "\lambda^{\mathsf{Sets}}_{\mathcal{P}(X)}", "\sim"'] & \mathcal{P}(X) \times \mathcal{P}(X) \arrow[d, "\triangle"] \\
 & \mathcal{P}(X)
\end{tikzcd}
\qquad
\begin{tikzcd}
\mathcal{P}(X) \times \mathsf{pt} \arrow[r, "\mathrm{id}_{\mathcal{P}(X)} \times [\emptyset]"] \arrow[dr, dashed, "\rho^{\mathsf{Sets}}_{\mathcal{P}(X)}", "\sim"'] & \mathcal{P}(X) \times \mathcal{P}(X) \arrow[d, "\triangle"] \\
 & \mathcal{P}(X)
\end{tikzcd}$$

commute, i.e. we have

$$U \triangle \emptyset = U,$$
$$\emptyset \triangle U = U$$

for each $U \in \mathcal{P}(X)$.

6. *Commutativity.* The diagram

$$\begin{tikzcd} \mathcal{P}(X) \times \mathcal{P}(X) \arrow[r, "\sigma^{\mathsf{Sets}}_{\mathcal{P}(X),\mathcal{P}(X)}"] \arrow[dr, "\triangle"'] & \mathcal{P}(X) \times \mathcal{P}(X) \arrow[d, "\triangle"] \\ & \mathcal{P}(X) \end{tikzcd}$$

commutes, i.e. we have

$$U \triangle V = V \triangle U$$

for each $U, V \in \mathcal{P}(X)$.

7. *Invertibility.* We have

$$U \triangle U = \emptyset$$

for each $U \in \mathcal{P}(X)$.

8. *Interaction With Unions.* We have

$$(U \triangle V) \cup (V \triangle T) = (U \cup V \cup W) \setminus (U \cap V \cap W)$$

for each $U, V, W \in \mathcal{P}(X)$.

9. *Interaction With Complements I.* We have

$$U \triangle U^{\mathsf{c}} = X$$

for each $U \in \mathcal{P}(X)$.

10. *Interaction With Complements II.* We have

$$U \triangle X = U^{\mathsf{c}},$$
$$X \triangle U = U^{\mathsf{c}}$$

for each $U \in \mathcal{P}(X)$.

11. *Interaction With Complements III.* The diagram

$$\begin{tikzcd}
\mathcal{P}(X) \times \mathcal{P}(X) \arrow[r, "\triangle"] \arrow[d, "(-)^{\mathsf{c}} \times (-)^{\mathsf{c}}"'] & \mathcal{P}(X) \arrow[d, "(-)^{\mathsf{c}}"] \\
\mathcal{P}(X) \times \mathcal{P}(X) \arrow[r, "\triangle"'] & \mathcal{P}(X)
\end{tikzcd}$$

commutes, i.e. we have

$$U^{\mathsf{c}} \triangle V^{\mathsf{c}} = U \triangle V$$

for each $U, V \in \mathcal{P}(X)$.

12. *"Transitivity".* We have

$$(U \triangle V) \triangle (V \triangle W) = U \triangle W$$

for each $U, V, W \in \mathcal{P}(X)$.

13. *The Triangle Inequality for Symmetric Differences.* We have

$$U \triangle W \subset U \triangle V \cup V \triangle W$$

for each $U, V, W \in \mathcal{P}(X)$.

14. *Distributivity Over Intersections.* We have

$$\begin{aligned} U \cap (V \triangle W) &= (U \cap V) \triangle (U \cap W), \\ (U \triangle V) \cap W &= (U \cap W) \triangle (V \cap W) \end{aligned}$$

for each $U, V, W \in \mathcal{P}(X)$.

15. *Interaction With Characteristic Functions.* We have

$$\chi_{U \triangle V} = \chi_U + \chi_V - 2\chi_{U \cap V}$$

and thus, in particular, we have

$$\chi_{U \triangle V} \equiv \chi_U + \chi_V \pmod 2$$

for each $U, V \in \mathcal{P}(X)$.

16. *Bijectivity.* Given $U, V \in \mathcal{P}(X)$, the maps

$$\begin{aligned} U \triangle - &\colon \mathcal{P}(X) \to \mathcal{P}(X), \\ - \triangle V &\colon \mathcal{P}(X) \to \mathcal{P}(X) \end{aligned}$$

are bijections with inverses given by

$$\begin{aligned} (U \triangle -)^{-1} &= - \cup (U \cap -), \\ (- \triangle V)^{-1} &= - \cup (V \cap -). \end{aligned}$$

Moreover, the map

$$\begin{aligned} \mathcal{P}(X) &\longrightarrow \mathcal{P}(X) \\ C &\longmapsto C \triangle (U \triangle V) \end{aligned}$$

is a bijection of $\mathcal{P}(X)$ onto itself sending U to V and V to U.

17. *Interaction With Powersets and Groups.* Let X be a set.

    (a) The quadruple $\left(\mathcal{P}(X), \triangle, \emptyset, \mathrm{id}_{\mathcal{P}(X)}\right)$ is an abelian group.[1]

    (b) Every element of $\mathcal{P}(X)$ has order 2 with respect to $\triangle$, and thus $\mathcal{P}(X)$ is a *Boolean group* (i.e. an abelian 2-group).

18. *Interaction With Powersets and Vector Spaces I.* The pair $\left(\mathcal{P}(X), \alpha_{\mathcal{P}(X)}\right)$ consisting of

    - The group $\mathcal{P}(X)$ of Item 17;
    - The map $\alpha_{\mathcal{P}(X)} \colon \mathbb{F}_2 \times \mathcal{P}(X) \to \mathcal{P}(X)$ defined by

$$\begin{aligned} 0 \cdot U &\stackrel{\text{def}}{=} \emptyset, \\ 1 \cdot U &\stackrel{\text{def}}{=} U; \end{aligned}$$

    is an $\mathbb{F}_2$-vector space.

19. *Interaction With Powersets and Vector Spaces II.* If X is finite, then:

    (a) The set of singletons sets on the elements of X forms a basis for the $\mathbb{F}_2$-vector space $\left(\mathcal{P}(X), \alpha_{\mathcal{P}(X)}\right)$ of Item 18.

(b) We have

$$\dim(\mathcal{P}(X)) = \#X.$$

20. *Interaction With Powersets and Rings.* The quintuple $(\mathcal{P}(X), \triangle, \cap, \emptyset, X)$ is a commutative ring.[2]

21. *Interaction With Direct Images.* We have a natural transformation

$$\begin{tikzcd}
\mathcal{P}(X)^{\mathsf{op}} \times \mathcal{P}(X) \arrow[r, "f_*^{\mathsf{op}} \times f_*"] \arrow[d, "\triangle"'] \arrow[dr, phantom, "\supset"] & \mathcal{P}(Y)^{\mathsf{op}} \times \mathcal{P}(Y) \arrow[d, "\triangle"] \\
\mathcal{P}(X) \arrow[r, "f_*"'] & \mathcal{P}(Y)
\end{tikzcd}$$

with components

$$f_*(U) \triangle f_*(V) \subset f_*(U \triangle V)$$

indexed by $U, V \in \mathcal{P}(X)$.

22. *Interaction With Inverse Images.* The diagram

$$\begin{tikzcd}
\mathcal{P}(Y)^{\mathsf{op}} \times \mathcal{P}(Y) \arrow[r, "f^{\mathsf{op},-1} \times f^{-1}"] \arrow[d, "\triangle"'] & \mathcal{P}(X)^{\mathsf{op}} \times \mathcal{P}(X) \arrow[d, "\triangle"] \\
\mathcal{P}(Y) \arrow[r, "f^{-1}"'] & \mathcal{P}(X)
\end{tikzcd}$$

i.e. we have

$$f^{-1}(U) \triangle f^{-1}(V) = f^{-1}(U \triangle V)$$

for each $U, V \in \mathcal{P}(Y)$.

23. *Interaction With Direct Images With Compact Support.* We have a natural transformation

$$\begin{tikzcd}
\mathcal{P}(X)^{\mathsf{op}} \times \mathcal{P}(X) \arrow[r, "f_!^{\mathsf{op}} \times f_!"] \arrow[d, "\triangle"'] \arrow[dr, phantom, "\supset"] & \mathcal{P}(Y)^{\mathsf{op}} \times \mathcal{P}(Y) \arrow[d, "\triangle"] \\
\mathcal{P}(X) \arrow[r, "f_!"'] & \mathcal{P}(Y)
\end{tikzcd}$$

with components

$$f_!(U \triangle V) \subset f_1(U) \triangle f_!(V)$$

indexed by $U, V \in \mathcal{P}(X)$.

[1]Here are some examples:

i. When $X = \emptyset$, we have an isomorphism of groups between $\mathcal{P}(\emptyset)$ and the trivial group:

$$\left(\mathcal{P}(\emptyset), \triangle, \emptyset, \mathrm{id}_{\mathcal{P}(\emptyset)}\right) \cong \mathrm{pt}.$$

ii. When $X = \mathrm{pt}$, we have an isomorphism of groups between $\mathcal{P}(\mathrm{pt})$ and $\mathbb{Z}_{/2}$:

$$\left(\mathcal{P}(\mathrm{pt}), \triangle, \emptyset, \mathrm{id}_{\mathcal{P}(\mathrm{pt})}\right) \cong \mathbb{Z}_{/2}.$$

iii. When $X = \{0, 1\}$, we have an isomorphism of groups between $\mathcal{P}(\{0, 1\})$ and $\mathbb{Z}_{/2} \times \mathbb{Z}_{/2}$:

$$\left(\mathcal{P}(\{0, 1\}), \triangle, \emptyset, \mathrm{id}_{\mathcal{P}(\{0,1\})}\right) \cong \mathbb{Z}_{/2} \times \mathbb{Z}_{/2}.$$

[2] *Warning:* The analogous statement replacing intersections by unions (i.e. that the quintuple $(\mathcal{P}(X), \triangle, \cup, \emptyset, X)$ is a ring) is false, however. See [Pro24at] for a proof.

Proof 3.12.3 ▶ Proof of Proposition 3.12.2

Item 1: Lack of Functoriality

Omitted.

Item 2: Via Unions and Intersections

See [Pro24l].

Item 3: Symmetric Differences of Disjoint Sets

Since U and V are disjoint, we have $U \cap V = \emptyset$, and therefore we have

$$\begin{aligned} U \triangle V &= (U \cup V) \setminus (U \cap V) \\ &= (U \cup V) \setminus \emptyset \\ &= U \cup V, \end{aligned}$$

where we,ve used Item 2 and Item 12 of Proposition 3.10.2.

Item 4: Associativity

See [Pro24al].

Item 5: Unitality

This follows from Item 6 and [Pro24aq].

Item 6: Commutativity

See [Pro24am].

Item 7: Invertibility

See [Pro24as].

Item 8: Interaction With Unions

See [Pro24az].

Item 9: Interaction With Complements I

See [Pro24ap].

Item 10: Interaction With Complements II

This follows from Item 6 and [Pro24au].

Item 11: Interaction With Complements III

See [Pro24an].

Item 12: "Transitivity'

We have

$$\begin{aligned}
(U \triangle V) \triangle (V \triangle W) &= U \triangle (V \triangle (V \triangle W)) && \text{(by Item 4)} \\
&= U \triangle ((V \triangle V) \triangle W) && \text{(by Item 4)} \\
&= U \triangle (\emptyset \triangle W) && \text{(by Item 7)} \\
&= U \triangle W. && \text{(by Item 5)}
\end{aligned}$$

This finishes the proof.

Item 13: The Triangle Inequality for Symmetric Differences

This follows from Items 2 and 12.

Item 14: Distributivity Over Intersections

See [Pro24o].

Item 15: Interaction With Characteristic Functions

See [Pro24g].

Item 16: Bijectivity

Omitted.

Item 17: Interaction With Powersets and Groups

Item 17a follows from Items 4 to 7, while Item 17b follows from Item 7.[1]

Item 18: Interaction With Powersets and Vector Spaces I

See [MSE 2719059].

Item 19: Interaction With Powersets and Vector Spaces II

See [MSE 2719059].

Item 20: Interaction With Powersets and Rings

This follows from Items 6 and 15 of Proposition 3.9.2 and Items 14 and 17.[2]

Item 21: Interaction With Direct Images

This is a repetition of Item 9 of Proposition 6.1.4 and is proved there.

Item 22: Interaction With Inverse Images

This is a repetition of Item 9 of Proposition 6.2.3 and is proved there.

Item 23: Interaction With Direct Images With Compact Support

This is a repetition of Item 8 of Proposition 6.3.6 and is proved there.

[1] *Reference:* [Pro24ao].
[2] *Reference:* [Pro24ar].

4 Powersets

4.1 Foundations

Let X be a set.

Definition 4.1.1 ▶ Powersets

The **powerset of** X is the set $\mathcal{P}(X)$ defined by

$$\mathcal{P}(X) \overset{\text{def}}{=} \{U \in P \mid U \subset X\},$$

where P is the set in the axiom of powerset, **??** of **??**.

Remark 4.1.2 ► Powersets as Decategorifications of Co/Presheaf Categories

Under the analogy that $\{\mathrm{t}, \mathrm{f}\}$ should be the (-1)-categorical analogue of Sets, we may view the powerset of a set as a decategorification of the category of presheaves of a category (or of the category of copresheaves):

- The powerset of a set X is equivalently (Item 2 of Proposition 5.1.4) the set
$$\mathrm{Sets}(X, \{\mathrm{t}, \mathrm{f}\})$$
of functions from X to the set $\{\mathrm{t}, \mathrm{f}\}$ of classical truth values.
- The category of presheaves on a category $\mathcal{C}$ is the category
$$\mathrm{Fun}(\mathcal{C}^{\mathrm{op}}, \mathrm{Sets})$$
of functors from $\mathcal{C}^{\mathrm{op}}$ to the category Sets of sets.

Notation 4.1.3 ► Further Notation for Powersets

Let X be a set.

1. We write $\mathcal{P}_0(X)$ for the set of nonempty subsets of X.
2. We write $\mathcal{P}_{\mathrm{fin}}(X)$ for the set of finite subsets of X.

Proposition 4.1.4 ► Elementary Properties of Powersets

Let X be a set.

1. *Co/Completeness.* The (posetal) category (associated to) $(\mathcal{P}(X), \subset)$ is complete and cocomplete:
   (a) *Products.* The products in $\mathcal{P}(X)$ are given by intersection of subsets.
   (b) *Coproducts.* The coproducts in $\mathcal{P}(X)$ are given by union of subsets.
   (c) *Co/Equalisers.* Being a posetal category, $\mathcal{P}(X)$ only has at most one morphisms between any two objects, so co/equalisers are trivial.
2. *Cartesian Closedness.* The category $\mathcal{P}(X)$ is Cartesian closed.

3. *Powersets as Sets of Relations.* We have bijections

$$\begin{aligned}\mathcal{P}(X) &\cong \mathsf{Rel}(\mathrm{pt}, X),\\ \mathcal{P}(X) &\cong \mathsf{Rel}(X, \mathrm{pt}),\end{aligned}$$

natural in $X \in \mathrm{Obj}(\mathsf{Sets})$.

4. *Interaction With Products I.* The map

$$\begin{aligned}\mathcal{P}(X) \times \mathcal{P}(Y) &\longrightarrow \mathcal{P}(X \coprod Y)\\ (U, V) &\longmapsto U \cup V\end{aligned}$$

is an isomorphism of sets, natural in $X, Y \in \mathrm{Obj}(\mathsf{Sets})$ with respect to each of the functor structures $\mathcal{P}_*$, $\mathcal{P}^{-1}$, and $\mathcal{P}_!$ on $\mathcal{P}$ of Proposition 4.2.1. Moreover, this makes each $\mathcal{P}_*$, $\mathcal{P}^{-1}$, and $\mathcal{P}_!$ into a symmetric monoidal functor.

5. *Interaction With Products II.* The map

$$\begin{aligned}\mathcal{P}(X) \times \mathcal{P}(Y) &\longrightarrow \mathcal{P}(X \coprod Y)\\ (U, V) &\longmapsto U \boxtimes_{X \times Y} V,\end{aligned}$$

where

$$U \boxtimes_{X \times Y} V \stackrel{\text{def}}{=} \{(u, v) \in X \times Y \mid u \in U \text{ and } v \in V\}$$

is an inclusion of sets, natural in $X, Y \in \mathrm{Obj}(\mathsf{Sets})$ with respect to each of the functor structures $\mathcal{P}_*$, $\mathcal{P}^{-1}$, and $\mathcal{P}_!$ on $\mathcal{P}$ of Proposition 4.2.1. Moreover, this makes each of $\mathcal{P}_*$, $\mathcal{P}^{-1}$, and $\mathcal{P}_!$ into a symmetric monoidal functor.

6. *Interaction With Products III.* We have an isomorphism

$$\mathcal{P}(X) \otimes \mathcal{P}(Y) \cong \mathcal{P}(X \times Y),$$

natural in $X, Y \in \mathrm{Obj}(\mathsf{Sets})$ with respect to each of the functor structures $\mathcal{P}_*$, $\mathcal{P}^{-1}$, and $\mathcal{P}_!$ on $\mathcal{P}$ of Proposition 4.2.1, where $\otimes$ denotes the tensor product of suplattices of **??**. Moreover, this makes each of $\mathcal{P}_*$, $\mathcal{P}^{-1}$, and $\mathcal{P}_!$ into a symmetric monoidal functor.

Proof 4.1.5 ▶ Proof of Proposition 4.1.4

Item 1: Co/Completeness

Omitted.

Item 2: Cartesian Closedness

See Section 4.7.

Item 3: Powersets as Sets of Relations

Indeed, we have

$$\begin{aligned}\mathrm{Rel}(\mathrm{pt}, X) &\overset{\text{def}}{=} \mathcal{P}(\mathrm{pt} \times X)\\ &\cong \mathcal{P}(X)\end{aligned}$$

and

$$\begin{aligned}\mathrm{Rel}(X, \mathrm{pt}) &\overset{\text{def}}{=} \mathcal{P}(X \times \mathrm{pt})\\ &\cong \mathcal{P}(X),\end{aligned}$$

where we have used Item 4 of Proposition 1.3.4.

Item 4: Interaction With Products I

The inverse of the map in the statement is the map

$$\Phi\colon \mathcal{P}(X \coprod Y) \to \mathcal{P}(X) \times \mathcal{P}(Y)$$

defined by

$$\Phi(S) \overset{\text{def}}{=} (S_X, S_Y)$$

for each $S \in \mathcal{P}(X \coprod Y)$, where

$$\begin{aligned}S_X &\overset{\text{def}}{=} \{x \in X \mid (0, x) \in S\}\\ S_Y &\overset{\text{def}}{=} \{y \in Y \mid (1, y) \in S\}.\end{aligned}$$

The rest of the proof is omitted.

Item 5: Interaction With Products II

Omitted.

Item 6: Interaction With Products III

Omitted. ∎

4.2 Functoriality of Powersets

Proposition 4.2.1 ▶ Functoriality of Powersets

Let X be a set.

1. *Functoriality I.* The assignment $X \mapsto \mathcal{P}(X)$ defines a functor

$$\mathcal{P}_*\colon \mathsf{Sets} \to \mathsf{Sets},$$

where

- *Action on Objects.* For each $A \in \mathrm{Obj}(\mathsf{Sets})$, we have

$$\mathcal{P}_*(A) \overset{\text{def}}{=} \mathcal{P}(A).$$

- *Action on Morphisms.* For each $A, B \in \mathrm{Obj}(\mathsf{Sets})$, the action on morphisms

$$\mathcal{P}_{*|A,B}\colon \mathsf{Sets}(A, B) \to \mathsf{Sets}(\mathcal{P}(A), \mathcal{P}(B))$$

of $\mathcal{P}_*$ at (A, B) is the map defined by by sending a map of sets $f\colon A \to B$ to the map

$$\mathcal{P}_*(f)\colon \mathcal{P}(A) \to \mathcal{P}(B)$$

defined by

$$\mathcal{P}_*(f) \overset{\text{def}}{=} f_*,$$

as in Definition 6.1.1.

2. *Functoriality II.* The assignment $X \mapsto \mathcal{P}(X)$ defines a functor

$$\mathcal{P}^{-1}\colon \mathsf{Sets}^{\mathsf{op}} \to \mathsf{Sets},$$

where

- *Action on Objects.* For each $A \in \mathrm{Obj}(\mathsf{Sets})$, we have

$$\mathcal{P}^{-1}(A) \overset{\text{def}}{=} \mathcal{P}(A).$$

- *Action on Morphisms.* For each $A, B \in \mathrm{Obj}(\mathsf{Sets})$, the action on morphisms

$$\mathcal{P}^{-1}_{A,B} \colon \mathsf{Sets}(A, B) \to \mathsf{Sets}(\mathcal{P}(B), \mathcal{P}(A))$$

of $\mathcal{P}^{-1}$ at (A, B) is the map defined by by sending a map of sets $f \colon A \to B$ to the map

$$\mathcal{P}^{-1}(f) \colon \mathcal{P}(B) \to \mathcal{P}(A)$$

defined by

$$\mathcal{P}^{-1}(f) \stackrel{\text{def}}{=} f^{-1},$$

as in Definition 6.2.1.

3. *Functoriality III.* The assignment $X \mapsto \mathcal{P}(X)$ defines a functor

$$\mathcal{P}_! \colon \mathsf{Sets} \to \mathsf{Sets},$$

where

- *Action on Objects.* For each $A \in \mathrm{Obj}(\mathsf{Sets})$, we have

$$\mathcal{P}_!(A) \stackrel{\text{def}}{=} \mathcal{P}(A).$$

- *Action on Morphisms.* For each $A, B \in \mathrm{Obj}(\mathsf{Sets})$, the action on morphisms

$$\mathcal{P}_{!|A,B} \colon \mathsf{Sets}(A, B) \to \mathsf{Sets}(\mathcal{P}(A), \mathcal{P}(B))$$

of $\mathcal{P}_!$ at (A, B) is the map defined by by sending a map of sets $f \colon A \to B$ to the map

$$\mathcal{P}_!(f) \colon \mathcal{P}(A) \to \mathcal{P}(B)$$

defined by

$$\mathcal{P}_!(f) \stackrel{\text{def}}{=} f_!,$$

as in Definition 6.3.1.

Proof 4.2.2 ▶ Proof of Proposition 4.2.1

Item 1: Functoriality I

This follows from Items 3 and 4 of Proposition 6.1.6.

Item 2: Functoriality II

This follows from Items 3 and 4 of Proposition 6.2.5.

Item 3: Functoriality III

This follows from Items 3 and 4 of Proposition 6.3.8. ∎

4.3 Adjointness of Powersets I

Proposition 4.3.1 ▶ Adjointness of Powersets I

We have an adjunction

$$\left(\mathcal{P}^{-1} \dashv \mathcal{P}^{-1,\mathsf{op}}\right): \quad \begin{tikzcd} \mathsf{Sets}^{\mathsf{op}} \arrow[r, bend left, "\mathcal{P}^{-1}"] \arrow[r, phantom, "\perp"] & \mathsf{Sets}, \arrow[l, bend left, "\mathcal{P}^{-1,\mathsf{op}}"] \end{tikzcd}$$

witnessed by a bijection

$$\underbrace{\mathsf{Sets}^{\mathsf{op}}(\mathcal{P}(X), Y)}_{\overset{\mathsf{def}}{=}\mathsf{Sets}(Y,\mathcal{P}(X))} \cong \mathsf{Sets}(X, \mathcal{P}(Y)),$$

natural in $X \in \mathsf{Obj}(\mathsf{Sets})$ and $Y \in \mathsf{Obj}(\mathsf{Sets}^{\mathsf{op}})$.

Proof 4.3.2 ▶ Proof of Proposition 4.3.1

We have

$$
\begin{aligned}
\mathsf{Sets}^{\mathsf{op}}(\mathcal{P}(A), B) &\overset{\text{def}}{=} \mathsf{Sets}(B, \mathcal{P}(A)) \\
&\cong \mathsf{Sets}(B, \mathsf{Sets}(A, \{\mathsf{t}, \mathsf{f}\})) && \text{(by Item 2 of Proposition 5.1.4)} \\
&\cong \mathsf{Sets}(A \times B, \{\mathsf{t}, \mathsf{f}\}) && \text{(by Item 2 of Proposition 1.3.4)} \\
&\cong \mathsf{Sets}(A, \mathsf{Sets}(B, \{\mathsf{t}, \mathsf{f}\})) && \text{(by Item 2 of Proposition 1.3.4)} \\
&\cong \mathsf{Sets}(A, \mathcal{P}(B)), && \text{(by Item 2 of Proposition 5.1.4)}
\end{aligned}
$$

where all bijections are natural in A and B.[1]

[1] Here we are using Item 3 of Proposition 5.1.4.

4.4 Adjointness of Powersets II

Proposition 4.4.1 ▶ Adjointness of Powersets II

We have an adjunction

$$
(\mathsf{Gr} \dashv \mathcal{P}_*) \colon \quad \mathsf{Sets} \underset{\mathcal{P}_*}{\overset{\mathsf{Gr}}{\underset{\longleftarrow}{\overset{\longrightarrow}{\perp}}}} \mathsf{Rel},
$$

witnessed by a bijection of sets

$$
\mathsf{Rel}(\mathsf{Gr}(X), Y) \cong \mathsf{Sets}(X, \mathcal{P}(Y))
$$

natural in $X \in \mathsf{Obj}(\mathsf{Sets})$ and $Y \in \mathsf{Obj}(\mathsf{Rel})$, where Gr is the graph functor of Constructions With Relations, **??** of **??** and $\mathcal{P}_*$ is the functor of Constructions With Relations, **??**.

Proof 4.4.2 ▶ Proof of Proposition 4.4.1

We have

$$
\begin{aligned}
\mathsf{Rel}(\mathsf{Gr}(A), B) &\cong \mathcal{P}(A \times B) \\
&\cong \mathsf{Sets}(A \times B, \{\mathsf{t}, \mathsf{f}\}) && \text{(by Item 2 of Proposition 5.1.4)} \\
&\cong \mathsf{Sets}(A, \mathsf{Sets}(B, \{\mathsf{t}, \mathsf{f}\})) && \text{(by Item 2 of Proposition 1.3.4)} \\
&\cong \mathsf{Sets}(A, \mathcal{P}(B)), && \text{(by Item 2 of Proposition 5.1.4)}
\end{aligned}
$$

where all bijections are natural in A, (where we are using Item 3 of Proposition 5.1.4). Explicitly, this isomorphism is given by sending a relation $R\colon \mathrm{Gr}(A) \nrightarrow B$ to the map $R^{\dagger}\colon A \to \mathcal{P}(B)$ sending a to the subset $R(a)$ of B, as in Relations, **??**.
Naturality in B is then the statement that given a relation $R\colon B \nrightarrow B'$, the diagram

$$\begin{tikzcd}
\mathrm{Rel}(\mathrm{Gr}(A), B) \arrow[r, "R\diamond -"] \arrow[d, dashed, "\wr"'] & \mathrm{Rel}(\mathrm{Gr}(A), B') \arrow[d, dashed, "\wr"] \\
\mathrm{Sets}(A, \mathcal{P}(B)) \arrow[r, "R_*"'] & \mathrm{Sets}(A, \mathcal{P}(B'))
\end{tikzcd}$$

commutes, which follows from Constructions With Relations, **??**.

4.5 Powersets as Free Cocompletions

Let X be a set.

Proposition 4.5.1 ▶ Powersets as Free Cocompletions: Universal Property

The pair $(\mathcal{P}(X), \chi_{(-)})$ consisting of

- The powerset $(\mathcal{P}(X), \subset)$ of X of Definition 4.1.1;
- The characteristic embedding $\chi_{(-)}\colon X \hookrightarrow \mathcal{P}(X)$ of X into $\mathcal{P}(X)$ of Definition 5.4.1;

satisfies the following universal property:

($\star$) Given another pair (Y, f) consisting of

  - A suplattice $(Y, \preceq)$;
  - A function $f\colon X \to Y$;

  there exists a unique morphism of suplattices

$$(\mathcal{P}(X), \subset) \xrightarrow{\exists!} (Y, \preceq)$$

making the diagram

$$\begin{tikzcd} & \mathcal{P}(X) \arrow[d, dashed, "\exists!"] \\ X \arrow[ur, "\chi_X"] \arrow[r, "f"'] & Y \end{tikzcd}$$

commute.

Proof 4.5.2 ▶ Proof of Proposition 4.5.1

This is a rephrasing of Proposition 4.5.3, which we prove below.[1]

[1] Here we only remark that the unique morphism of suplattices in the statement is given by the left Kan extension $\mathrm{Lan}_{\chi_X}(f)$ of f along χ_X.

Proposition 4.5.3 ▶ Powersets as Free Cocompletions: Adjointness

We have an adjunction

$$(\mathcal{P} \dashv 忘) : \quad \begin{tikzcd} \mathsf{Sets} \arrow[r, bend left, "\mathcal{P}"] \arrow[r, phantom, "\perp"] & \mathsf{SupLat}, \arrow[l, bend left, "忘"] \end{tikzcd}$$

witnessed by a bijection

$$\mathsf{SupLat}((\mathcal{P}(X), \subset), (Y, \preceq)) \cong \mathsf{Sets}(X, Y),$$

natural in $X \in \mathrm{Obj}(\mathsf{Sets})$ and $(Y, \preceq) \in \mathrm{Obj}(\mathsf{SupLat})$, where:

- The category SupLat is the category of suplattices of **??**.
- The map

  $$\chi_X^* : \mathsf{SupLat}((\mathcal{P}(X), \subset), (Y, \preceq)) \to \mathsf{Sets}(X, Y)$$

  witnessing the above bijection is defined by

  $$\chi_X^*(f) \overset{\text{def}}{=} f \circ \chi_X,$$

  i.e. by sending a morphism of suplattices $f : \mathcal{P}(X) \to Y$ to the composition

  $$X \overset{\chi_X}{\hookrightarrow} \mathcal{P}(X) \xrightarrow{f} Y.$$

- The map

$$\mathsf{Lan}_{\chi_X} \colon \mathsf{Sets}(X, Y) \to \mathsf{SupLat}((\mathcal{P}(X), \subset), (Y, \preceq))$$

witnessing the above bijection is given by sending a function $f \colon X \to Y$ to its left Kan extension along χ_X,

$$\mathsf{Lan}_{\chi_X}(f) \colon \mathcal{P}(X) \to Y, \qquad \begin{tikzcd} & \mathcal{P}(X) \arrow[d, dashed, "\mathsf{Lan}_{\chi_X}(f)"] \\ X \arrow[ur, "\chi_X"] \arrow[r, "f"'] \arrow[ur, phantom, shift right=3ex, "\Uparrow"] & Y. \end{tikzcd}$$

Moreover, invoking the bijection $\mathcal{P}(X) \cong \mathsf{Sets}(X, \{\mathsf{t}, \mathsf{f}\})$ of Item 2 of Proposition 5.1.4, $\mathsf{Lan}_{\chi_X}(f)$ can be explicitly computed by

$$\begin{aligned}
\left[\mathsf{Lan}_{\chi_X}(f)\right](U) &= \int^{x \in X} \chi_{\mathcal{P}(X)}(\chi_x, U) \odot f(x) \\
&= \int^{x \in X} \chi_U(x) \odot f(x) \\
&= \bigvee_{x \in X} (\chi_U(x) \odot f(x)) \\
&= \left(\bigvee_{x \in U} (\chi_U(x) \odot f(x))\right) \vee \left(\bigvee_{x \in U^c} (\chi_U(x) \odot f(x))\right) \\
&= \left(\bigvee_{x \in U} f(x)\right) \vee \left(\bigvee_{x \in U^c} \varnothing_Y\right) \\
&= \bigvee_{x \in U} f(x)
\end{aligned}$$

for each $U \in \mathcal{P}(X)$, where:

- We have used **??** for the first equality.
- We have used Proposition 5.5.1 for the second equality.
- We have used **??** for the third equality.
- The symbol $\bigvee$ denotes the join in $(Y, \preceq)$.

– The symbol $\odot$ denotes the tensor of an element of Y by a truth value as in **??**. In particular, we have

$$\begin{aligned} \text{true} \odot f(x) &\overset{\text{def}}{=} f(x), \\ \text{false} \odot f(x) &\overset{\text{def}}{=} \varnothing_Y, \end{aligned}$$

where $\varnothing_Y$ is the bottom element of $(Y, \preceq)$.

In particular, when $(Y, \preceq_Y) = (\mathcal{P}(B), \subset)$ for some set B, the Kan extension $\mathsf{Lan}_{\chi_X}(f)$ is given by

$$\begin{aligned} \left[\mathsf{Lan}_{\chi_X}(f)\right](U) &= \bigvee_{x \in U} f(x) \\ &= \bigcup_{x \in U} f(x) \end{aligned}$$

for each $U \in \mathcal{P}(X)$.

Proof 4.5.4 ► Proof of Proposition 4.5.3

Map I

We define a map

$$\Phi_{X,Y} \colon \mathsf{SupLat}((\mathcal{P}(X), \subset), (Y, \preceq)) \to \mathsf{Sets}(X, Y)$$

as in the statement, i.e. by

$$\Phi_{X,Y}(f) \overset{\text{def}}{=} f \circ \chi_X$$

for each $f \in \mathsf{SupLat}((\mathcal{P}(X), \subset), (Y, \preceq))$.

Map II

We define a map

$$\Psi_{X,Y} \colon \mathsf{Sets}(X, Y) \to \mathsf{SupLat}((\mathcal{P}(X), \subset), (Y, \preceq))$$

as in the statement, i.e. by

$$\Psi_{X,Y}(f) \overset{\text{def}}{=} \mathsf{Lan}_{\chi_X}(f), \qquad \begin{tikzcd} & \mathcal{P}(X) \arrow[d, dashed, "\mathsf{Lan}_{\chi_X}(f)"] \\ X \arrow[ur, "\chi_X"] \arrow[r, "f"'] \arrow[ur, phantom, shift right=3ex, "\Uparrow"] & Y, \end{tikzcd}$$

for each $f \in \mathsf{Sets}(X, Y)$.

Invertibility I

We claim that

$$\Psi_{X,Y} \circ \Phi_{X,Y} = \mathsf{id}_{\mathsf{SupLat}((\mathcal{P}(X),\subset),(Y,\preceq))}.$$

We have

$$\begin{aligned} \left[\Psi_{X,Y} \circ \Phi_{X,Y}\right](f) &\overset{\text{def}}{=} \Psi_{X,Y}\left(\Phi_{X,Y}(f)\right) \\ &\overset{\text{def}}{=} \Psi_{X,Y}(f \circ \chi_X) \\ &\overset{\text{def}}{=} \mathsf{Lan}_{\chi_X}(f \circ \chi_X) \end{aligned}$$

for each $f \in \mathsf{SupLat}((\mathcal{P}(X), \subset), (Y, \preceq))$. We now claim that

$$\mathsf{Lan}_{\chi_X}(f \circ \chi_X) = f$$

for each $f \in \mathsf{SupLat}((\mathcal{P}(X), \subset), (Y, \preceq))$. Indeed, we have

$$\begin{aligned} \left[\mathsf{Lan}_{\chi_X}(f \circ \chi_X)\right](U) &= \bigvee_{x \in U} f(\chi_X(x)) \\ &= f\left(\bigvee_{x \in U} \chi_X(x)\right) \\ &= f\left(\bigcup_{x \in U} \{x\}\right) \\ &= f(U) \end{aligned}$$

for each $U \in \mathcal{P}(X)$, where we have used that f is a morphism of suplattices and hence preserves joins for the second equality. This proves our claim. Since we have shown that

$$\left[\Psi_{X,Y} \circ \Phi_{X,Y}\right](f) = f$$

for each $f \in \mathsf{SupLat}((\mathcal{P}(X), \subset), (Y, \preceq))$, it follows that $\Psi_{X,Y} \circ \Phi_{X,Y}$ must be equal to the identity map $\mathrm{id}_{\mathsf{SupLat}((\mathcal{P}(X),\subset),(Y,\preceq))}$ of $\mathsf{SupLat}((\mathcal{P}(X), \subset), (Y, \preceq))$.

Invertibility II

We claim that

$$\Phi_{X,Y} \circ \Psi_{X,Y} = \mathrm{id}_{\mathsf{Sets}(X,Y)}\,.$$

We have

$$\begin{aligned}\left[\Phi_{X,Y} \circ \Psi_{X,Y}\right](f) &\overset{\text{def}}{=} \Phi_{X,Y}\left(\Psi_{X,Y}(f)\right)\\ &\overset{\text{def}}{=} \Phi_{X,Y}\left(\mathsf{Lan}_{\chi_X}(f)\right)\\ &\overset{\text{def}}{=} \mathsf{Lan}_{\chi_X}(f) \circ \chi_X\end{aligned}$$

for each $f \in \mathsf{Sets}(X, Y)$. We now claim that

$$\mathsf{Lan}_{\chi_X}(f) \circ \chi_X = f$$

for each $f \in \mathsf{Sets}(X, Y)$. Indeed, we have

$$\begin{aligned}\left[\mathsf{Lan}_{\chi_X}(f) \circ \chi_X\right](x) &= \bigvee_{y \in \{x\}} f(y)\\ &= f(x)\end{aligned}$$

for each $x \in X$. This proves our claim. Since we have shown that

$$\left[\Phi_{X,Y} \circ \Psi_{X,Y}\right](f) = f$$

for each $f \in \mathsf{Sets}(X, Y)$, it follows that $\Phi_{X,Y} \circ \Psi_{X,Y}$ must be equal to the identity map $\mathrm{id}_{\mathsf{Sets}(X,Y)}$ of $\mathsf{Sets}(X, Y)$.

Naturality for Φ, Part I

We need to show that, given a function $f\colon X \to X'$, the diagram

$$\begin{tikzcd}
\mathsf{SupLat}((\mathcal{P}(X'), \subset), (Y, \preceq)) \arrow[r, "\Phi_{X',Y}"] \arrow[d, "\mathcal{P}_*(f)^*"'] & \mathsf{Sets}(X', Y) \arrow[d, "f^*"] \\
\mathsf{SupLat}((\mathcal{P}(X), \subset), (Y, \preceq)) \arrow[r, "\Phi_{X,Y}"'] & \mathsf{Sets}(X, Y)
\end{tikzcd}$$

commutes. Indeed, we have

$$\begin{aligned}
\left[\Phi_{X,Y} \circ \mathcal{P}_*(f)^*\right](\xi) &\overset{\text{def}}{=} \Phi_{X,Y}\left(\mathcal{P}_*(f)^*(\xi)\right) \\
&\overset{\text{def}}{=} \Phi_{X,Y}(\xi \circ f_*) \\
&\overset{\text{def}}{=} (\xi \circ f_*) \circ \chi_X \\
&= \xi \circ (f_* \circ \chi_X) \\
&\overset{(\dagger)}{=} \xi \circ (\chi_{X'} \circ f) \\
&= (\xi \circ \chi_{X'}) \circ f \\
&\overset{\text{def}}{=} \Phi_{X',Y}(\xi) \circ f \\
&\overset{\text{def}}{=} f^*\left(\Phi_{X',Y}(\xi)\right) \\
&\overset{\text{def}}{=} \left[f^* \circ \Phi_{X',Y}\right](\xi),
\end{aligned}$$

for each $\xi \in \mathsf{SupLat}((\mathcal{P}(X'), \subset), (Y, \preceq))$, where we have used Item 1 of Proposition 5.4.3 for the fifth equality above.

Naturality for Φ, Part II

We need to show that, given a morphism of suplattices

$$g\colon (Y, \preceq_Y) \to (Y', \preceq_{Y'}),$$

the diagram

$$\begin{tikzcd}
\mathsf{SupLat}((\mathcal{P}(X), \subset), (Y, \preceq)) \arrow[r, "\Phi_{X,Y}"] \arrow[d, "g_*"'] & \mathsf{Sets}(X, Y) \arrow[d, "g_*"] \\
\mathsf{SupLat}((\mathcal{P}(X), \subset), (Y', \preceq)) \arrow[r, "\Phi_{X,Y'}"'] & \mathsf{Sets}(X, Y')
\end{tikzcd}$$

commutes. Indeed, we have

$$\begin{aligned}
\left[\Phi_{X,Y'} \circ g_*\right](\xi) &\overset{\text{def}}{=} \Phi_{X,Y'}(g_*(\xi)) \\
&\overset{\text{def}}{=} \Phi_{X,Y'}(g \circ \xi) \\
&\overset{\text{def}}{=} (g \circ \xi) \circ \chi_X \\
&= g \circ (\xi \circ \chi_X) \\
&\overset{\text{def}}{=} g \circ \left(\Phi_{X,Y}(\xi)\right) \\
&\overset{\text{def}}{=} g_*\left(\Phi_{X,Y}(\xi)\right) \\
&\overset{\text{def}}{=} \left[g_* \circ \Phi_{X,Y}\right](\xi).
\end{aligned}$$

for each $\xi \in \mathsf{SupLat}((\mathcal{P}(X), \subset), (Y, \preceq))$.

Naturality for Ψ

Since Φ is natural in each argument and Φ is a componentwise inverse to Ψ in each argument, it follows from Categories, **??** of **??** that Ψ is also natural in each argument.

WARNING 4.5.5 ▶ FREE COCOMPLETION IS NOT AN IDEMPOTENT OPERATION

A lthough the assignment $X \mapsto \mathcal{P}(X)$ is called the *free cocompletion of* X, it is not an idempotent operation, i.e. we have $\mathcal{P}(\mathcal{P}(X)) \neq \mathcal{P}(X)$.

4.6 Powersets as Free Completions

Let X be a set.

PROPOSITION 4.6.1 ▶ POWERSETS AS FREE COMPLETIONS: UNIVERSAL PROPERTY

The pair $\left(\mathcal{P}(X), \chi_{(-)}\right)$ consisting of

- The powerset of X together with reverse inclusion $\mathcal{P}(X)^{\mathrm{op}} = (\mathcal{P}(X), \supset)$ of Definition 4.1.1;
- The characteristic embedding $\chi_{(-)} \colon X \hookrightarrow \mathcal{P}(X)$ of X into $\mathcal{P}(X)$ of Definition 5.4.1;

satisfies the following universal property:

($\star$) Given another pair (Y, f) consisting of

- An inflattice $(Y, \preceq)$;
- A function $f\colon X \to Y$;

there exists a unique morphism of inflattices

$$(\mathcal{P}(X), \supset) \xrightarrow{\exists!} (Y, \preceq)$$

making the diagram

$$\begin{tikzcd} & \mathcal{P}(X)^{\mathrm{op}} \arrow[d, dashed, "\exists!"] \\ X \arrow[ur, "\chi_X"] \arrow[r, "f"'] & Y \end{tikzcd}$$

commute.

Proof 4.6.2 ► Proof of Proposition 4.6.1

This is a rephrasing of Proposition 4.6.3, which we prove below.[1]

[1] Here we only remark that the unique morphism of inflattices in the statement is given by the right Kan extension $\mathrm{Ran}_{\chi_X}(f)$ of f along χ_X.

Proposition 4.6.3 ► Powersets as Free Completions: Adjointness

We have an adjunction

$$(\mathcal{P} \dashv 忘)\colon \quad \begin{tikzcd} \mathsf{Sets} \arrow[r, bend left, "\mathcal{P}"] \arrow[r, phantom, "\perp"] & \mathsf{InfLat}, \arrow[l, bend left, "忘"] \end{tikzcd}$$

witnessed by a bijection

$$\mathsf{InfLat}((\mathcal{P}(X), \supset), (Y, \preceq)) \cong \mathsf{Sets}(X, Y),$$

natural in $X \in \mathrm{Obj}(\mathsf{Sets})$ and $(Y, \preceq) \in \mathrm{Obj}(\mathsf{InfLat})$, where:

- The category InfLat is the category of inflattices of **??**.
- The map

$$\chi_X^* \colon \mathsf{InfLat}((\mathcal{P}(X), \supset), (Y, \preceq)) \to \mathsf{Sets}(X, Y)$$

  witnessing the above bijection is defined by

$$\chi_X^*(f) \overset{\text{def}}{=} f \circ \chi_X,$$

  i.e. by sending a morphism of inflattices $f \colon \mathcal{P}(X)^{\mathsf{op}} \to Y$ to the composition

$$X \overset{\chi_X}{\hookrightarrow} \mathcal{P}(X)^{\mathsf{op}} \xrightarrow{f} Y.$$

- The map

$$\mathsf{Ran}_{\chi_X} \colon \mathsf{Sets}(X, Y) \to \mathsf{InfLat}((\mathcal{P}(X), \supset), (Y, \preceq))$$

  witnessing the above bijection is given by sending a function $f \colon X \to Y$ to its right Kan extension along χ_X,

$$\mathsf{Ran}_{\chi_X}(f) \colon \mathcal{P}(X)^{\mathsf{op}} \to Y, \qquad \begin{tikzcd} & \mathcal{P}(X)^{\mathsf{op}} \arrow[d, dashed, "\mathsf{Ran}_{\chi_X}(f)"] \\ X \arrow[ur, "\chi_X"] \arrow[r, "f"'] \arrow[ur, phantom, shift right=3ex, "\Downarrow"] & Y. \end{tikzcd}$$

  Moreover, invoking the bijection $\mathcal{P}(X) \cong \mathsf{Sets}(X, \{\mathsf{t}, \mathsf{f}\})$ of Item 2 of Proposition 5.1.4, $\mathsf{Ran}_{\chi_X}(f)$ can be explicitly computed by

$$\begin{aligned} \left[\mathsf{Ran}_{\chi_X}(f)\right](U) &= \int_{x \in X} \chi_{\mathcal{P}(X)^{\mathsf{op}}}(\chi_x, U) \pitchfork f(x) \\ &= \int_{x \in X} \chi_{\mathcal{P}(X)}(U, \chi_x) \pitchfork f(x) \\ &= \int_{x \in X} \chi_U(x) \pitchfork f(x) \\ &= \bigwedge_{x \in X} \chi_U(x) \pitchfork f(x) \end{aligned}$$

$$
\begin{aligned}
&= \left(\bigwedge_{x\in U} \chi_U(x) \pitchfork f(x)\right) \wedge \left(\bigwedge_{x\in U^c} \chi_U(x) \pitchfork f(x)\right) \\
&= \left(\bigwedge_{x\in U} f(x)\right) \wedge \left(\bigwedge_{x\in U^c} \infty_Y\right) \\
&= \left(\bigwedge_{x\in U} f(x)\right) \wedge \infty_Y \\
&= \bigwedge_{x\in U} f(x)
\end{aligned}
$$

for each $U \in \mathcal{P}(X)$, where:

- We have used **??** for the first equality.
- We have used Proposition 5.5.1 for the second equality.
- We have used **??** for the third equality.
- The symbol $\bigwedge$ denotes the meet in $(Y, \preceq)$.
- The symbol $\pitchfork$ denotes the cotensor of an element of Y by a truth value as in **??**. In particular, we have

  $$
  \begin{aligned}
  \text{true} \pitchfork f(x) &\overset{\text{def}}{=} f(x), \\
  \text{false} \pitchfork f(x) &\overset{\text{def}}{=} \infty_Y,
  \end{aligned}
  $$

  where ∞_Y is the top element of $(Y, \preceq)$.

In particular, when $(Y, \preceq_Y) = (\mathcal{P}(B), \subset)$ for some set B, the Kan extension $\mathrm{Ran}_{\chi_X}(f)$ is given by

$$
\begin{aligned}
\left[\mathrm{Ran}_{\chi_X}(f)\right](U) &= \bigwedge_{x\in U} f(x) \\
&= \bigcap_{x\in U} f(x)
\end{aligned}
$$

for each $U \in \mathcal{P}(X)$.

Proof 4.6.4 ► Proof of Proposition 4.5.3

Map I

We define a map

$$\Phi_{X,Y}\colon \mathsf{InfLat}((\mathcal{P}(X),\supset),(Y,\preceq)) \to \mathsf{Sets}(X,Y)$$

as in the statement, i.e. by

$$\Phi_{X,Y}(f) \overset{\text{def}}{=} f \circ \chi_X$$

for each $f \in \mathsf{InfLat}((\mathcal{P}(X),\supset),(Y,\preceq))$.

Map II

We define a map

$$\Psi_{X,Y}\colon \mathsf{Sets}(X,Y) \to \mathsf{InfLat}((\mathcal{P}(X),\supset),(Y,\preceq))$$

as in the statement, i.e. by

$$\Psi_{X,Y}(f) \overset{\text{def}}{=} \mathsf{Ran}_{\chi_X}(f), \qquad \begin{tikzcd} & \mathcal{P}(X) \arrow[d, dashed, "\mathsf{Ran}_{\chi_X}(f)"] \arrow[dl, phantom, "\Downarrow"] \\ X \arrow[ur, "\chi_X"] \arrow[r, "f"'] & Y, \end{tikzcd}$$

for each $f \in \mathsf{Sets}(X,Y)$.

Invertibility I

We claim that

$$\Psi_{X,Y} \circ \Phi_{X,Y} = \mathsf{id}_{\mathsf{InfLat}((\mathcal{P}(X),\supset),(Y,\preceq))}.$$

We have

$$\begin{aligned} \left[\Psi_{X,Y} \circ \Phi_{X,Y}\right](f) &\overset{\text{def}}{=} \Psi_{X,Y}\big(\Phi_{X,Y}(f)\big) \\ &\overset{\text{def}}{=} \Psi_{X,Y}(f \circ \chi_X) \\ &\overset{\text{def}}{=} \mathsf{Ran}_{\chi_X}(f \circ \chi_X) \end{aligned}$$

for each $f \in \mathsf{InfLat}((\mathcal{P}(X), \supset), (Y, \preceq))$. We now claim that

$$\mathsf{Ran}_{\chi_X}(f \circ \chi_X) = f$$

for each $f \in \mathsf{InfLat}((\mathcal{P}(X), \supset), (Y, \preceq))$. Indeed, we have

$$\begin{aligned}\left[\mathsf{Ran}_{\chi_X}(f \circ \chi_X)\right](U) &= \bigwedge_{x \in U} f(\chi_X(x)) \\ &= f\left(\bigwedge_{x \in U} \chi_X(x)\right) \\ &= f\left(\bigcup_{x \in U} \{x\}\right) \\ &= f(U)\end{aligned}$$

for each $U \in \mathcal{P}(X)$, where we have used that f is a morphism of inflattices and hence preserves meets in $(\mathcal{P}(X), \supset)$ (i.e. joins in $(\mathcal{P}(X), \subset)$) for the second equality. This proves our claim. Since we have shown that

$$\left[\Psi_{X,Y} \circ \Phi_{X,Y}\right](f) = f$$

for each $f \in \mathsf{InfLat}((\mathcal{P}(X), \supset), (Y, \preceq))$, it follows that $\Psi_{X,Y} \circ \Phi_{X,Y}$ must be equal to the identity map $\mathsf{id}_{\mathsf{InfLat}((\mathcal{P}(X),\supset),(Y,\preceq))}$ of $\mathsf{InfLat}((\mathcal{P}(X), \supset), (Y, \preceq))$.

Invertibility II

We claim that

$$\Phi_{X,Y} \circ \Psi_{X,Y} = \mathsf{id}_{\mathsf{Sets}(X,Y)}\,.$$

We have

$$\begin{aligned}\left[\Phi_{X,Y} \circ \Psi_{X,Y}\right](f) &\overset{\text{def}}{=} \Phi_{X,Y}\big(\Psi_{X,Y}(f)\big) \\ &\overset{\text{def}}{=} \Phi_{X,Y}\big(\mathsf{Ran}_{\chi_X}(f)\big) \\ &\overset{\text{def}}{=} \mathsf{Ran}_{\chi_X}(f) \circ \chi_X\end{aligned}$$

for each $f \in \mathsf{Sets}(X, Y)$. We now claim that

$$\mathsf{Ran}_{\chi_X}(f) \circ \chi_X = f$$

for each $f \in \mathsf{Sets}(X, Y)$. Indeed, we have

$$\begin{aligned}\left[\mathsf{Ran}_{\chi_X}(f) \circ \chi_X\right](x) &= \bigwedge_{y \in \{x\}} f(y) \\ &= f(x)\end{aligned}$$

for each $x \in X$. This proves our claim. Since we have shown that

$$\left[\Phi_{X,Y} \circ \Psi_{X,Y}\right](f) = f$$

for each $f \in \mathsf{Sets}(X, Y)$, it follows that $\Phi_{X,Y} \circ \Psi_{X,Y}$ must be equal to the identity map $\mathrm{id}_{\mathsf{Sets}(X,Y)}$ of $\mathsf{Sets}(X, Y)$.

Naturality for Φ, Part I

We need to show that, given a function $f\colon X \to X'$, the diagram

$$\begin{tikzcd}
\mathsf{InfLat}((\mathcal{P}(X'), \supset), (Y, \preceq)) \arrow[r, "\Phi_{X',Y}"] \arrow[d, "\mathcal{P}_*(f)^*"'] & \mathsf{Sets}(X', Y) \arrow[d, "f^*"] \\
\mathsf{InfLat}((\mathcal{P}(X), \supset), (Y, \preceq)) \arrow[r, "\Phi_{X,Y}"'] & \mathsf{Sets}(X, Y)
\end{tikzcd}$$

commutes. Indeed, we have

$$\begin{aligned}\left[\Phi_{X,Y} \circ \mathcal{P}_*(f)^*\right](\xi) &\overset{\text{def}}{=} \Phi_{X,Y}\left(\mathcal{P}_*(f)^*(\xi)\right) \\ &\overset{\text{def}}{=} \Phi_{X,Y}(\xi \circ f_*) \\ &\overset{\text{def}}{=} (\xi \circ f_*) \circ \chi_X \\ &= \xi \circ (f_* \circ \chi_X) \\ &\overset{(\dagger)}{=} \xi \circ (\chi_{X'} \circ f) \\ &= (\xi \circ \chi_{X'}) \circ f \\ &\overset{\text{def}}{=} \Phi_{X',Y}(\xi) \circ f \\ &\overset{\text{def}}{=} f^*\left(\Phi_{X',Y}(\xi)\right) \\ &\overset{\text{def}}{=} \left[f^* \circ \Phi_{X',Y}\right](\xi),\end{aligned}$$

for each $\xi \in \mathsf{InfLat}((\mathcal{P}(X'), \supset), (Y, \preceq))$, where we have used Item 1 of Proposition 5.4.3 for the fifth equality above.

Naturality for Φ, Part II

We need to show that, given a cocontinuous morphism of posets

$$g\colon (Y, \preceq_Y) \to (Y', \preceq_{Y'}),$$

the diagram

$$\begin{tikzcd}
\mathsf{InfLat}((\mathcal{P}(X), \supset), (Y, \preceq)) \arrow[r, "\Phi_{X,Y}"] \arrow[d, "g_*"'] & \mathsf{Sets}(X, Y) \arrow[d, "g_*"] \\
\mathsf{InfLat}((\mathcal{P}(X), \supset), (Y', \preceq)) \arrow[r, "\Phi_{X,Y'}"'] & \mathsf{Sets}(X, Y')
\end{tikzcd}$$

commutes. Indeed, we have

$$\begin{aligned}
\left[\Phi_{X,Y'} \circ g_*\right](\xi) &\overset{\text{def}}{=} \Phi_{X,Y'}(g_*(\xi)) \\
&\overset{\text{def}}{=} \Phi_{X,Y'}(g \circ \xi) \\
&\overset{\text{def}}{=} (g \circ \xi) \circ \chi_X \\
&= g \circ (\xi \circ \chi_X) \\
&\overset{\text{def}}{=} g \circ \left(\Phi_{X,Y}(\xi)\right) \\
&\overset{\text{def}}{=} g_*\left(\Phi_{X,Y}(\xi)\right) \\
&\overset{\text{def}}{=} \left[g_* \circ \Phi_{X,Y}\right](\xi).
\end{aligned}$$

for each $\xi \in \mathsf{InfLat}((\mathcal{P}(X), \supset), (Y, \preceq))$.

Naturality for Ψ

Since Φ is natural in each argument and Φ is a componentwise inverse to Ψ in each argument, it follows from Categories, **??** of **??** that Ψ is also natural in each argument.

WARNING 4.6.5 ▶ FREE COMPLETION IS NOT AN IDEMPOTENT OPERATION

A lthough the assignment $X \mapsto \mathcal{P}(X)^{\mathsf{op}}$ is called the *free completion of* X, it is not an idempotent operation, i.e. we have $\mathcal{P}(\mathcal{P}(X)^{\mathsf{op}})^{\mathsf{op}} \neq \mathcal{P}(X)^{\mathsf{op}}$.

4.7 The Internal Hom of a Powerset

Let X be a set and let $U, V \in \mathcal{P}(X)$.

DEFINITION 4.7.1 ▶ THE INTERNAL HOM OF A POWERSET

The **internal Hom of** $\mathcal{P}(X)$ **from** U **to** V is the subset $[U, V]_X$[1] of X defined by

$$\begin{aligned}[U, V]_X &\overset{\text{def}}{=} U^{\mathsf{c}} \cup V \\ &= (U \setminus V)^{\mathsf{c}}\end{aligned}$$

where U^{c} is the complement of U of Definition 3.11.1.

[1] *Further Notation:* Also written $\mathbf{Hom}_{\mathcal{P}(X)}(U, V)$.

PROOF 4.7.2 ▶ PROOF OF DEFINITION 4.7.1

We have

$$\begin{aligned}(U \setminus V)^{\mathsf{c}} &\overset{\text{def}}{=} X \setminus (U \setminus V) \\ &= (X \cap V) \cup (X \setminus U) \\ &= V \cup (X \setminus U) \\ &\overset{\text{def}}{=} V \cup U^{\mathsf{c}} \\ &= U^{\mathsf{c}} \cup V,\end{aligned}$$

where we have used:

1. Item 10 of Proposition 3.10.2 for the second equality.

2. Item 4 of Proposition 3.9.2 for the third equality.

3. Item 4 of Proposition 3.8.2 for the last equality.

This finishes the proof.

Remark 4.7.3 ▶ Intuition for the Internal Hom of $\mathcal{P}(X)$

Henning Makholm suggests the following heuristic intuition for the internal Hom of $\mathcal{P}(X)$ from U to V ([**MSE267365**]):

1. Since products in $\mathcal{P}(X)$ are given by binary intersections (Item 1 of Proposition 4.1.4), the right adjoint $\mathbf{Hom}_{\mathcal{P}(X)}(U, -)$ of $U \cap -$ may be thought of as a function type $[U, V]$.

2. Under the Curry–Howard correspondence (**??**), the function type $[U, V]$ corresponds to implication $U \implies V$.

3. Implication $U \Rightarrow V$ is logically equivalent to $\neg U \vee V$.

4. The expression $\neg U \vee V$ then corresponds to the set $U^c \cup V$ in $\mathcal{P}(X)$.

5. The set $U^c \vee V$ turns out to indeed be the internal Hom of $\mathcal{P}(X)$.

Proposition 4.7.4 ▶ Properties of Internal Homs of Powersets

Let X be a set.

1. *Functoriality.* The assignments $U, V, (U, V) \mapsto \mathbf{Hom}_{\mathcal{P}(X)}$ define functors

$$\begin{aligned}
[U, -]_X &: (\mathcal{P}(X), \supset) \to (\mathcal{P}(X), \subset), \\
[-, V]_X &: (\mathcal{P}(X), \subset) \to (\mathcal{P}(X), \subset), \\
[-_1, -_2]_X &: (\mathcal{P}(X) \times \mathcal{P}(X), \subset \times \supset) \to (\mathcal{P}(X), \subset).
\end{aligned}$$

   In particular, the following statements hold for each $U, V, A, B \in \mathcal{P}(X)$:

   (a) If $U \subset A$, then $[A, V]_X \subset [U, V]_X$.

   (b) If $V \subset B$, then $[U, V]_X \subset [U, B]_X$.

   (c) If $U \subset A$ and $V \subset B$, then $[A, V]_X \subset [U, B]_X$.

2. *Adjointness.* We have adjunctions

$$(U \cap - \dashv [U,-]_X)\colon \quad \begin{tikzcd} \mathcal{P}(X) \arrow[r, bend left, "U\cap -"] \arrow[r, phantom, "\perp"] & \mathcal{P}(X), \arrow[l, bend left, "{[U,-]_X}"] \end{tikzcd}$$

$$(- \cap V \dashv [V,-]_X)\colon \quad \begin{tikzcd} \mathcal{P}(X) \arrow[r, bend left, "-\cap V"] \arrow[r, phantom, "\perp"] & \mathcal{P}(X), \arrow[l, bend left, "{[V,-]_X}"] \end{tikzcd}$$

witnessed by bijections

$$\begin{aligned} \mathrm{Hom}_{\mathcal{P}(X)}(U \cap V, W) &\cong \mathrm{Hom}_{\mathcal{P}(X)}(U, [V,W]_X), \\ \mathrm{Hom}_{\mathcal{P}(X)}(U \cap V, W) &\cong \mathrm{Hom}_{\mathcal{P}(X)}(V, [U,W]_X). \end{aligned}$$

In particular, the following statements hold for each $U, V, W \in \mathcal{P}(X)$:

(a) The following conditions are equivalent:

i. We have $U \cap V \subset W$.

ii. We have $U \subset [V,W]_X$.

(b) The following conditions are equivalent:

i. We have $U \cap V \subset W$.

ii. We have $V \subset [U,W]_X$.

3. *Interaction With the Empty Set I.* We have

$$\begin{aligned} [U,\emptyset]_X &= U^{\mathsf{c}}, \\ [\emptyset,V]_X &= X, \end{aligned}$$

natural in $U, V \in \mathcal{P}(X)$.

4. *Interaction With X.* We have

$$\begin{aligned} [U,X]_X &= X, \\ [X,V]_X &= V, \end{aligned}$$

natural in $U, V \in \mathcal{P}(X)$.

5. *Interaction With the Empty Set II.* The functor

$$D_X \colon \mathcal{P}(X)^{\mathsf{op}} \to \mathcal{P}(X)$$

defined by

$$\begin{aligned} D_X &\overset{\text{def}}{=} [-, \emptyset]_X \\ &= (-)^{\mathsf{c}} \end{aligned}$$

is an involutory isomorphism of categories, making $\emptyset$ into a dualising object for $(\mathcal{P}(X), \cap, X, [-,-]_X)$ in the sense of **??**. In particular:

(a) The diagram

$$\begin{tikzcd} \mathcal{P}(X)^{\mathsf{op}} \arrow[r, "D_X"] \arrow[dr, "\mathrm{id}_{\mathcal{P}(X)^{\mathsf{op}}}"'] & \mathcal{P}(X) \arrow[d, "D_X"] \\ & \mathcal{P}(X)^{\mathsf{op}} \end{tikzcd}$$

commutes, i.e. we have

$$\underbrace{D_X(D_X(U))}_{\overset{\text{def}}{=}[[U,\emptyset]_X,\emptyset]_X} = U$$

for each $U \in \mathcal{P}(X)$.

(b) The diagram

$$\begin{tikzcd} & \mathcal{P}(X)^{\mathsf{op}} \times \mathcal{P}(X)^{\mathsf{op}} \arrow[r, "\cap^{\mathsf{op}}"] & \mathcal{P}(X)^{\mathsf{op}} \arrow[dr, "D_X"] & \\ \mathcal{P}(X)^{\mathsf{op}} \times \mathcal{P}(X) \arrow[ur, "\mathrm{id}_{\mathcal{P}(X)^{\mathsf{op}}} \times D_X"] \arrow[rrr, "{[-_1,-_2]_X}"'] & & & \mathcal{P}(X) \end{tikzcd}$$

commutes, i.e. we have

$$\underbrace{D_X(U \cap D_X(V))}_{\overset{\text{def}}{=}[U\cap[V,\emptyset]_X,\emptyset]_X} = [U, V]_X$$

for each $U, V \in \mathcal{P}(X)$.

6. *Interaction With the Empty Set III.* Let $f\colon X \to Y$ be a function.

   (a) *Interaction With Direct Images.* The diagram

$$\begin{tikzcd} \mathcal{P}(X)^{\mathsf{op}} \arrow[r, "f_!^{\mathsf{op}}"] \arrow[d, "D_X"'] & \mathcal{P}(Y)^{\mathsf{op}} \arrow[d, "D_Y"] \\ \mathcal{P}(X) \arrow[r, "f_*"'] & \mathcal{P}(Y) \end{tikzcd}$$

   commutes, i.e. we have

$$f_*(D_X(U)) = D_Y(f_!(U))$$

   for each $U \in \mathcal{P}(X)$.

   (b) *Interaction With Inverse Images.* The diagram

$$\begin{tikzcd} \mathcal{P}(Y)^{\mathsf{op}} \arrow[r, "f^{-1,\mathsf{op}}"] \arrow[d, "D_Y"'] & \mathcal{P}(X)^{\mathsf{op}} \arrow[d, "D_X"] \\ \mathcal{P}(Y) \arrow[r, "f^{-1}"'] & \mathcal{P}(X) \end{tikzcd}$$

   commutes, i.e. we have

$$f^{-1}(D_Y(U)) = D_X\big(f^{-1}(U)\big)$$

   for each $U \in \mathcal{P}(X)$.

   (c) *Interaction With Direct Images With Compact Support.* The diagram

$$\begin{tikzcd} \mathcal{P}(X)^{\mathsf{op}} \arrow[r, "f_*^{\mathsf{op}}"] \arrow[d, "D_X"'] & \mathcal{P}(Y)^{\mathsf{op}} \arrow[d, "D_Y"] \\ \mathcal{P}(X) \arrow[r, "f_!"'] & \mathcal{P}(Y) \end{tikzcd}$$

   commutes, i.e. we have

$$f_!(D_X(U)) = D_Y(f_*(U))$$

   for each $U \in \mathcal{P}(X)$.

7. *Interaction With Unions of Families of Subsets I.* The diagram

$$\begin{tikzcd}
\mathcal{P}(\mathcal{P}(X))^{\mathsf{op}} \times \mathcal{P}(\mathcal{P}(X)) \arrow[r, "{[-_1,-_2]_{\mathcal{P}(X)}}"] \arrow[d, "\bigcup^{\mathsf{op}} \times \bigcup^{\mathsf{op}}"'] \arrow[dr, phantom, "\times"] & \mathcal{P}(\mathcal{P}(X)) \arrow[d, "\bigcup"] \\
\mathcal{P}(X)^{\mathsf{op}} \times \mathcal{P}(X) \arrow[r, "{[-_1,-_2]_X}"'] & \mathcal{P}(X),
\end{tikzcd}$$

does not commute in general, i.e. we may have

$$\bigcup_{W \in [\mathcal{U},\mathcal{V}]_{\mathcal{P}(X)}} W \neq \left[\bigcup_{U \in \mathcal{U}} U, \bigcup_{V \in \mathcal{V}} V\right]_X$$

in general, where $\mathcal{U} \in \mathcal{P}(\mathcal{P}(X))$.

8. *Interaction With Unions of Families of Subsets II.* The diagram

$$\begin{tikzcd}
 & \mathcal{P}(\mathcal{P}(X))^{\mathsf{op}} \arrow[dr, "\bigcup^{\mathsf{op}}"] & \\
\mathcal{P}(\mathcal{P}(X)^{\mathsf{op}}) \arrow[ur, dashed, "\sim"] \arrow[dr, "\mathrm{id}_{\mathcal{P}(X)} \star [-,V]_X"'] & & \mathcal{P}(X)^{\mathsf{op}} \arrow[dl, "{[-,V]_X}"] \\
 & \mathcal{P}(\mathcal{P}(X)) \arrow[r, "\cap"'] & \mathcal{P}(X)
\end{tikzcd}$$

commutes, i.e. we have

$$\left[\bigcup_{U \in \mathcal{U}} U, V\right]_X = \bigcap_{U \in \mathcal{U}} [U, V]_X$$

for each $\mathcal{U} \in \mathcal{P}(\mathcal{P}(X))$ and each $V \in \mathcal{P}(X)$.

9. *Interaction With Unions of Families of Subsets III.* The diagram

$$\begin{tikzcd}
\mathcal{P}(\mathcal{P}(X)) \arrow[r, "\bigcup"] \arrow[d, "\mathrm{id}_{\mathcal{P}(X)} \star [U,-]_X"'] & \mathcal{P}(X) \arrow[d, "{[U,-]_X}"] \\
\mathcal{P}(\mathcal{P}(X)) \arrow[r, "\bigcup"'] & \mathcal{P}(X)
\end{tikzcd}$$

commutes, i.e. we have

$$\left[U, \bigcup_{V \in \mathcal{V}} V\right]_X = \bigcup_{V \in \mathcal{V}} [U, V]_X$$

for each $U \in \mathcal{P}(X)$ and each $\mathcal{V} \in \mathcal{P}(\mathcal{P}(X))$.

10. *Interaction With Intersections of Families of Subsets I.* The diagram

$$\begin{tikzcd}
\mathcal{P}(\mathcal{P}(X))^{\mathsf{op}} \times \mathcal{P}(\mathcal{P}(X)) \arrow[r, "{[-_1,-_2]_{\mathcal{P}(X)}}"] \arrow[d, "\bigcap^{\mathsf{op}} \times \bigcap^{\mathsf{op}}"'] \arrow[dr, phantom, "\times"] & \mathcal{P}(\mathcal{P}(X)) \arrow[d, "\bigcap"] \\
\mathcal{P}(X)^{\mathsf{op}} \times \mathcal{P}(X) \arrow[r, "{[-_1,-_2]_X}"'] & \mathcal{P}(X),
\end{tikzcd}$$

does not commute in general, i.e. we may have

$$\bigcap_{W \in [\mathcal{U}, \mathcal{V}]_{\mathcal{P}(X)}} W \neq \left[\bigcap_{U \in \mathcal{U}} U, \bigcap_{V \in \mathcal{V}} V\right]_X$$

in general, where $\mathcal{U} \in \mathcal{P}(\mathcal{P}(X))$.

11. *Interaction With Intersections of Families of Subsets II.* The diagram

$$\begin{tikzcd}
 & \mathcal{P}(\mathcal{P}(X))^{\mathsf{op}} \arrow[dr, "\bigcap^{\mathsf{op}}"] & \\
\mathcal{P}(\mathcal{P}(X)^{\mathsf{op}}) \arrow[ur, dashed, "\sim"] \arrow[dd, "{\mathsf{id}_{\mathcal{P}(X)} \star [-,V]_X}"'] & & \mathcal{P}(X)^{\mathsf{op}} \arrow[dd, "{[-,V]_X}"] \\
 & & \\
\mathcal{P}(\mathcal{P}(X)) \arrow[rr, "\bigcup"'] & & \mathcal{P}(X)
\end{tikzcd}$$

commutes, i.e. we have

$$\left[\bigcap_{U \in \mathcal{U}} U, V\right]_X = \bigcup_{U \in \mathcal{U}} [U, V]_X$$

for each $\mathcal{U} \in \mathcal{P}(\mathcal{P}(X))$ and each $V \in \mathcal{P}(X)$.

12. *Interaction With Intersections of Families of Subsets III.* The diagram

$$\begin{tikzcd}
\mathcal{P}(\mathcal{P}(X)) \arrow[r, "\bigcap"] \arrow[d, "\mathrm{id}_{\mathcal{P}(X)} \star [U,-]_X"'] & \mathcal{P}(X) \arrow[d, "{[U,-]_X}"] \\
\mathcal{P}(\mathcal{P}(X)) \arrow[r, "\bigcap"'] & \mathcal{P}(X)
\end{tikzcd}$$

commutes, i.e. we have

$$\left[U, \bigcap_{V \in \mathcal{V}} V\right]_X = \bigcap_{V \in \mathcal{V}} [U, V]_X$$

for each $U \in \mathcal{P}(X)$ and each $\mathcal{V} \in \mathcal{P}(\mathcal{P}(X))$.

13. *Interaction With Binary Unions.* We have equalities of sets

$$\begin{aligned}
[U \cap V, W]_X &= [U, W]_X \cup [V, W]_X, \\
[U, V \cap W]_X &= [U, V]_X \cap [U, W]_X
\end{aligned}$$

for each $U, V, W \in \mathcal{P}(X)$.

14. *Interaction With Binary Intersections.* We have equalities of sets

$$\begin{aligned}
[U \cup V, W]_X &= [U, W]_X \cap [V, W]_X, \\
[U, V \cup W]_X &= [U, V]_X \cup [U, W]_X
\end{aligned}$$

for each $U, V, W \in \mathcal{P}(X)$.

15. *Interaction With Differences.* We have equalities of sets

$$\begin{aligned}
[U \setminus V, W]_X &= [U, W]_X \cup [V^{\mathsf{c}}, W]_X \\
&= [U, W]_X \cup [U, V]_X, \\
[U, V \setminus W]_X &= [U, V]_X \setminus (U \cap W)
\end{aligned}$$

for each $U, V, W \in \mathcal{P}(X)$.

16. *Interaction With Complements.* We have equalities of sets

$$\begin{aligned}[U^{\mathsf{c}}, V]_X &= U \cup V,\\ [U, V^{\mathsf{c}}]_X &= U \cap V,\\ [U, V]_X^{\mathsf{c}} &= U \setminus V\end{aligned}$$

for each $U, V \in \mathcal{P}(X)$.

17. *Interaction With Characteristic Functions.* We have

$$\chi_{[U,V]_{\mathcal{P}(X)}}(x) = \max(1 - \chi_U \pmod 2, \chi_V)$$

for each $U, V \in \mathcal{P}(X)$.

18. *Interaction With Direct Images.* Let $f\colon X \to Y$ be a function. The diagram

$$\begin{tikzcd}
\mathcal{P}(X)^{\mathsf{op}} \times \mathcal{P}(X) \arrow[r, "f_!^{\mathsf{op}} \times f_*"] \arrow[d, "{[-_1,-_2]_X}"'] & \mathcal{P}(Y)^{\mathsf{op}} \times \mathcal{P}(Y) \arrow[d, "{[-_1,-_2]_Y}"] \\
\mathcal{P}(X) \arrow[r, "f_*"'] & \mathcal{P}(Y)
\end{tikzcd}$$

commutes, i.e. we have an equality of sets

$$f_*([U, V]_X) = [f_!(U), f_*(V)]_Y,$$

natural in $U, V \in \mathcal{P}(X)$.

19. *Interaction With Inverse Images.* Let $f\colon X \to Y$ be a function. The diagram

$$\begin{tikzcd}
\mathcal{P}(Y)^{\mathsf{op}} \times \mathcal{P}(Y) \arrow[r, "f^{-1,\mathsf{op}} \times f^{-1}"] \arrow[d, "{[-_1,-_2]_Y}"'] & \mathcal{P}(X)^{\mathsf{op}} \times \mathcal{P}(X) \arrow[d, "{[-_1,-_2]_X}"] \\
\mathcal{P}(Y) \arrow[r, "f^{-1}"'] & \mathcal{P}(X)
\end{tikzcd}$$

commutes, i.e. we have an equality of sets

$$f^{-1}([U, V]_Y) = \left[f^{-1}(U), f^{-1}(V)\right]_X,$$

natural in $U, V \in \mathcal{P}(X)$.

20. *Interaction With Direct Images With Compact Support.* Let $f\colon X \to Y$ be a function. We have a natural transformation

$$\begin{tikzcd}
\mathcal{P}(X)^{\mathsf{op}} \times \mathcal{P}(X) \arrow[r, "f_*^{\mathsf{op}} \times f_!"] \arrow[d, "{[-_1,-_2]_X}"'] \arrow[dr, phantom, "\supset"] & \mathcal{P}(Y)^{\mathsf{op}} \times \mathcal{P}(Y) \arrow[d, "{[-_1,-_2]_Y}"] \\
\mathcal{P}(X) \arrow[r, "f_!"'] & \mathcal{P}(Y)
\end{tikzcd}$$

with components

$$[f_*(U), f_!(V)]_Y \subset f_!([U,V]_X)$$

indexed by $U, V \in \mathcal{P}(X)$.

Proof 4.7.5 ▶ Proof of Proposition 4.7.4

Item 1: Functoriality

Since $\mathcal{P}(X)$ is posetal, it suffices to prove Items 1a to 1c.

1. *Proof of Item 1a:* We have

$$\begin{aligned}
[A,V]_X &\overset{\text{def}}{=} A^{\mathsf{c}} \cup V \\
&\subset U^{\mathsf{c}} \cup V \\
&\overset{\text{def}}{=} [U,V]_X,
\end{aligned}$$

where we have used:

(a) Item 1 of Proposition 3.11.2, which states that if $U \subset A$, then $A^{\mathsf{c}} \subset U^{\mathsf{c}}$.

(b) Item 1a of Item 1 of Proposition 3.11.2, which states that if $A^{\mathsf{c}} \subset U^{\mathsf{c}}$, then $A^{\mathsf{c}} \cup K \subset U^{\mathsf{c}} \cup K$ for any $K \in \mathcal{P}(X)$.

2. *Proof of Item 1b:* We have

$$[U,V]_X \overset{\text{def}}{=} U^{\mathsf{c}} \cup V$$

$$\begin{aligned} &\subset U^c \cup B \\ &\overset{\text{def}}{=} [U, B]_X, \end{aligned}$$

where we have used Item 1b of Item 1 of Proposition 3.11.2, which states that if $V \subset B$, then $K \cup V \subset K \cup B$ for any $K \in \mathcal{P}(X)$.

3. *Proof of Item 1c*: We have

$$\begin{aligned} [A, V]_X &\subset [U, V]_X \\ &\subset [U, B]_X, \end{aligned}$$

where we have used Items 1a and 1b.

This finishes the proof.

Item 2: Adjointness

This is a repetition of Item 2 of Proposition 3.9.2 and is proved there.

Item 3: Interaction With the Empty Set I

We have

$$\begin{aligned} [U, \emptyset]_X &\overset{\text{def}}{=} U^c \cup \emptyset \\ &= U^c, \end{aligned}$$

where we have used Item 3 of Proposition 3.8.2, and we have

$$\begin{aligned} [\emptyset, V]_X &\overset{\text{def}}{=} \emptyset^c \cup V \\ &\overset{\text{def}}{=} (X \setminus \emptyset) \cup V \\ &= X \cup V \\ &= X, \end{aligned}$$

where we have used:

1. Item 12 of Proposition 3.10.2 for the first equality.
2. Item 5 of Proposition 3.8.2 for the last equality.

Since $\mathcal{P}(X)$ is posetal, naturality is automatic (**??**).

Item 4: Interaction With X

We have

$$\begin{aligned}[U, X]_X &\overset{\text{def}}{=} U^{\mathsf{c}} \cup X \\ &= X,\end{aligned}$$

where we have used Item 5 of Proposition 3.8.2, and we have

$$\begin{aligned}[X, V]_X &\overset{\text{def}}{=} X^{\mathsf{c}} \cup V \\ &\overset{\text{def}}{=} (X \setminus X) \cup V \\ &= \emptyset \cup V \\ &= V,\end{aligned}$$

where we have used Item 3 of Proposition 3.8.2 for the last equality. Since $\mathcal{P}(X)$ is posetal, naturality is automatic (**??**).

Item 5: Interaction With the Empty Set II

We have

$$\begin{aligned}D_X(D_X(U)) &\overset{\text{def}}{=} [[U, \emptyset]_X, \emptyset]_X \\ &= [U^{\mathsf{c}}, \emptyset]_X \\ &= (U^{\mathsf{c}})^{\mathsf{c}} \\ &= U,\end{aligned}$$

where we have used:

1. Item 3 for the second and third equalities.
2. Item 3 of Proposition 3.11.2 for the fourth equality.

Since $\mathcal{P}(X)$ is posetal, naturality is automatic (**??**), and thus we have

$$[[-, \emptyset]_X, \emptyset]_X \cong \mathrm{id}_{\mathcal{P}(X)}$$

This finishes the proof.

Item 6: Interaction With the Empty Set III

Since $D_X = (-)^{\mathsf{c}}$, this is essentially a repetition of the corresponding results for $(-)^{\mathsf{c}}$, namely Items 5 to 7 of Proposition 3.11.2.

Item 7: Interaction With Unions of Families of Subsets I

By Item 3 of Proposition 4.7.4, we have

$$\begin{aligned}[\mathcal{U}, \emptyset]_{\mathcal{P}(X)} &= \mathcal{U}^{\mathsf{c}}, \\ [U, \emptyset]_X &= U^{\mathsf{c}}.\end{aligned}$$

With this, the counterexample given in the proof of Item 10 of Proposition 3.6.2 then applies.

Item 8: Interaction With Unions of Families of Subsets II

We have

$$\begin{aligned}\left[\bigcup_{U \in \mathcal{U}} U, V\right]_X &\overset{\text{def}}{=} \left(\bigcup_{U \in \mathcal{U}} U\right)^{\mathsf{c}} \cup V \\ &= \left(\bigcap_{U \in \mathcal{U}} U^{\mathsf{c}}\right) \cup V \\ &= \bigcap_{U \in \mathcal{U}} (U^{\mathsf{c}} \cup V) \\ &\overset{\text{def}}{=} \bigcap_{U \in \mathcal{U}} [U, V]_X,\end{aligned}$$

where we have used:

1. Item 11 of Proposition 3.6.2 for the second equality.
2. Item 6 of Proposition 3.7.2 for the third equality.

This finishes the proof.

Item 9: Interaction With Unions of Families of Subsets III

We have

$$\bigcup_{V \in \mathcal{V}} [U, V]_X \overset{\text{def}}{=} \bigcup_{V \in \mathcal{V}} (U^{\mathsf{c}} \cup V)$$

$$
\begin{aligned}
&= U^{\mathsf{c}} \cup \left(\bigcup_{V \in \mathcal{V}} V \right) \\
&\overset{\text{def}}{=} \left[U, \bigcup_{V \in \mathcal{V}} V \right]_X .
\end{aligned}
$$

where we have used Item 6. This finishes the proof.

Item 10: Interaction With Intersections of Families of Subsets I

Let $X = \{0, 1\}$, let $\mathcal{U} = \{\{0, 1\}\}$, and let $\mathcal{V} = \{\{0\}, \{0, 1\}\}$. We have

$$
\begin{aligned}
\bigcap_{W \in [\mathcal{U}, \mathcal{V}]_{\mathcal{P}(X)}} W &= \bigcap_{W \in \mathcal{P}(X)} W \\
&= \{0, 1\},
\end{aligned}
$$

whereas

$$
\begin{aligned}
\left[\bigcap_{U \in \mathcal{U}} U, \bigcap_{V \in \mathcal{V}} V \right]_X &= [\{0, 1\}, \{0\}] \\
&= \{0\},
\end{aligned}
$$

Thus we have

$$
\bigcap_{W \in [\mathcal{U}, \mathcal{V}]_{\mathcal{P}(X)}} W = \{0, 1\} \neq \{0\} = \left[\bigcap_{U \in \mathcal{U}} U, \bigcap_{V \in \mathcal{V}} V \right]_X .
$$

This finishes the proof.

Item 11: Interaction With Intersections of Families of Subsets II

We have

$$
\begin{aligned}
\left[\bigcap_{U \in \mathcal{U}} U, V \right]_X &\overset{\text{def}}{=} \left(\bigcap_{U \in \mathcal{U}} U \right)^{\mathsf{c}} \cup V \\
&= \left(\bigcup_{U \in \mathcal{U}} U^{\mathsf{c}} \right) \cup V
\end{aligned}
$$

$$
\begin{aligned}
&= \bigcup_{U \in \mathcal{U}} (U^{\mathsf{c}} \cup V) \\
&\overset{\text{def}}{=} \bigcup_{U \in \mathcal{U}} [U, V]_X,
\end{aligned}
$$

where we have used:

1. Item 12 of Proposition 3.6.2 for the second equality.
2. Item 6 of Proposition 3.7.2 for the third equality.

This finishes the proof.

Item 12: Interaction With Intersections of Families of Subsets III

We have

$$
\begin{aligned}
\bigcap_{V \in \mathcal{V}} [U, V]_X &\overset{\text{def}}{=} \bigcap_{V \in \mathcal{V}} (U^{\mathsf{c}} \cup V) \\
&= U^{\mathsf{c}} \cup \left(\bigcap_{V \in \mathcal{V}} V \right) \\
&\overset{\text{def}}{=} \left[U, \bigcap_{V \in \mathcal{V}} V \right]_X.
\end{aligned}
$$

where we have used Item 6. This finishes the proof.

Item 13: Interaction With Binary Unions

We have

$$
\begin{aligned}
[U \cap V, W]_X &\overset{\text{def}}{=} (U \cap V)^{\mathsf{c}} \cup W \\
&= (U^{\mathsf{c}} \cup V^{\mathsf{c}}) \cup W \\
&= (U^{\mathsf{c}} \cup V^{\mathsf{c}}) \cup (W \cup W) \\
&= (U^{\mathsf{c}} \cup W) \cup (V^{\mathsf{c}} \cup W) \\
&\overset{\text{def}}{=} [U, W]_X \cup [V, W]_X,
\end{aligned}
$$

where we have used:

1. Item 2 of Proposition 3.11.2 for the second equality.

2. Item 8 of Proposition 3.8.2 for the third equality.

3. Several applications of Items 2 and 4 of Proposition 3.8.2 and for the fourth equality.

For the second equality in the statement, we have

$$\begin{aligned}[U, V \cap W]_X &\overset{\text{def}}{=} U^c \cup (V \cap W) \\ &= (U^c \cup V) \cap (U^c \cap W) \\ &\overset{\text{def}}{=} [U, V]_X \cap [U, W]_X,\end{aligned}$$

where we have used Item 6 of Proposition 3.8.2 for the second equality.

Item 14: Interaction With Binary Intersections

We have

$$\begin{aligned}[U \cup V, W]_X &\overset{\text{def}}{=} (U \cup V)^c \cup W \\ &= (U^c \cap V^c) \cup W \\ &= (U^c \cup W) \cap (V^c \cup W) \\ &\overset{\text{def}}{=} [U, W]_X \cap [V, W]_X,\end{aligned}$$

where we have used:

1. Item 2 of Proposition 3.11.2 for the second equality.

2. Item 6 of Proposition 3.8.2 for the third equality.

Now, for the second equality in the statement, we have

$$\begin{aligned}[U, V \cup W]_X &\overset{\text{def}}{=} U^c \cup (V \cup W) \\ &= (U^c \cup U^c) \cup (V \cup W) \\ &= (U^c \cup V) \cup (U^c \cup W) \\ &\overset{\text{def}}{=} [U, V]_X \cup [U, W]_X,\end{aligned}$$

where we have used:

1. Item 8 of Proposition 3.8.2 for the second equality.

2. Several applications of Items 2 and 4 of Proposition 3.8.2 and for the third equality.

This finishes the proof.

Item 15: Interaction With Differences

We have

$$\begin{aligned}
[U \setminus V, W]_X &\overset{\text{def}}{=} (U \setminus V)^{\mathsf{c}} \cup W \\
&\overset{\text{def}}{=} (X \setminus (U \setminus V)) \cup W \\
&= ((X \cap V) \cup (X \setminus U)) \cup W \\
&= (V \cup (X \setminus U)) \cup W \\
&\overset{\text{def}}{=} (V \cup U^{\mathsf{c}}) \cup W \\
&= (V \cup (U^{\mathsf{c}} \cup U^{\mathsf{c}})) \cup W \\
&= (U^{\mathsf{c}} \cup W) \cup (U^{\mathsf{c}} \cup V) \\
&\overset{\text{def}}{=} [U, W]_X \cup [U, V]_X,
\end{aligned}$$

where we have used:

1. Item 10 of Proposition 3.10.2 for the third equality.
2. Item 4 of Proposition 3.9.2 for the fourth equality.
3. Item 8 of Proposition 3.8.2 for the sixth equality.
4. Several applications of Items 2 and 4 of Proposition 3.8.2 and for the seventh equality.

We also have

$$\begin{aligned}
[U \setminus V, W]_X &\overset{\text{def}}{=} (U \setminus V)^{\mathsf{c}} \cup W \\
&\overset{\text{def}}{=} (X \setminus (U \setminus V)) \cup W \\
&= ((X \cap V) \cup (X \setminus U)) \cup W \\
&= (V \cup (X \setminus U)) \cup W \\
&\overset{\text{def}}{=} (V \cup U^{\mathsf{c}}) \cup W \\
&= (V \cup U^{\mathsf{c}}) \cup (W \cup W)
\end{aligned}$$

$$
\begin{aligned}
&= (U^c \cup W) \cup (V \cup W) \\
&= (U^c \cup W) \cup \left((V^c)^c \cup W\right) \\
&\overset{\text{def}}{=} [U, W]_X \cup [V^c, W]_X,
\end{aligned}
$$

where we have used:

1. Item 10 of Proposition 3.10.2 for the third equality.
2. Item 4 of Proposition 3.9.2 for the fourth equality.
3. Item 8 of Proposition 3.8.2 for the sixth equality.
4. Several applications of Items 2 and 4 of Proposition 3.8.2 and for the seventh equality.
5. Item 3 of Proposition 3.11.2 for the eighth equality.

Now, for the second equality in the statement, we have

$$
\begin{aligned}
[U, V \setminus W]_X &\overset{\text{def}}{=} U^c \cup (V \setminus W) \\
&= (V \setminus W) \cup U^c \\
&= (V \cup U^c) \setminus (W \setminus U^c) \\
&\overset{\text{def}}{=} (V \cup U^c) \setminus (W \setminus (X \setminus U)) \\
&= (V \cup U^c) \setminus ((W \cap U) \cup (W \setminus X)) \\
&= (V \cup U^c) \setminus ((W \cap U) \cup \emptyset) \\
&= (V \cup U^c) \setminus (W \cap U) \\
&= (V \cup U^c) \setminus (U \cap W) \\
&\overset{\text{def}}{=} [U, V]_X \setminus (U \cap W)
\end{aligned}
$$

where we have used:

1. Item 4 of Proposition 3.8.2 for the second equality.
2. Item 4 of Proposition 3.10.2 for the third equality.
3. Item 10 of Proposition 3.10.2 for the fifth equality.

4. Item 13 of Proposition 3.10.2 for the sixth equality.
5. Item 3 of Proposition 3.8.2 for the seventh equality.
6. Item 5 of Proposition 3.9.2 for the eighth equality.

This finishes the proof.

Item 16: Interaction With Complements

We have

$$\begin{aligned}[U^c, V]_X &\stackrel{\text{def}}{=} (U^c)^c \cup V, \\ &= U \cup V,\end{aligned}$$

where we have used Item 3 of Proposition 3.11.2. We also have

$$\begin{aligned}[U, V^c]_X &\stackrel{\text{def}}{=} U^c \cup V^c \\ &= U \cap V\end{aligned}$$

where we have used Item 2 of Proposition 3.11.2. Finally, we have

$$\begin{aligned}[U, V]_X^c &= \left((U \setminus V)^c\right)^c \\ &= U \setminus V,\end{aligned}$$

where we have used Item 2 of Proposition 3.11.2.

Item 17: Interaction With Characteristic Functions

We have

$$\begin{aligned}\chi_{[U,V]_{\mathcal{P}(X)}}(x) &\stackrel{\text{def}}{=} \chi_{U^c \cup V}(x) \\ &= \max(\chi_{U^c}, \chi_V) \\ &= \max(1 - \chi_U \pmod 2, \chi_V),\end{aligned}$$

where we have used:

1. Item 10 of Proposition 3.8.2 for the second equality.

2. Item 4 of Proposition 3.11.2 for the third equality.

This finishes the proof.

Item 18: Interaction With Direct Images

This is a repetition of Item 10 of Proposition 6.1.4 and is proved there.

Item 19: Interaction With Inverse Images

This is a repetition of Item 10 of Proposition 6.2.3 and is proved there.

Item 20: Interaction With Direct Images With Compact Support

This is a repetition of Item 9 of Proposition 6.3.6 and is proved there.

4.8 Isbell Duality for Sets

Let X be a set.

Definition 4.8.1 ▶ The Isbell Function

The **Isbell function** of X is the map

$$\mathsf{I}\colon \mathcal{P}(X) \longrightarrow \mathsf{Sets}(X, \mathcal{P}(X))$$

defined by

$$\mathsf{I}(U) \overset{\text{def}}{=} [\![x \mapsto [U, \{x\}]_X]\!]$$

for each $U \in \mathcal{P}(X)$.

Remark 4.8.2 ▶ Motivation for the Isbell Function

Recall from Remark 4.1.2 that we may view the powerset $\mathcal{P}(X)$ of a set X as the decategorification of the category of presheaves $\mathsf{PSh}(\mathcal{C})$ of a category $\mathcal{C}$. Building upon this analogy, we want to mimic the definition of the Isbell Spec functor, which is given on objects by

$$\mathsf{Spec}(\mathcal{F}) \overset{\text{def}}{=} \mathsf{Nat}\big(\mathcal{F}, h_{(-)}\big)$$

for each $\mathcal{F} \in \mathsf{Obj}(\mathsf{PSh}(\mathcal{C}))$. To this end, we could define

$$\mathsf{I}(U) \overset{\text{def}}{=} \big[U, \chi_{(-)}\big]_X,$$

replacing:

- The Yoneda embedding $X \mapsto h_X$ of $\mathcal{C}$ into $\mathsf{PSh}(\mathcal{C})$ with the characteristic embedding $x \mapsto \chi_x$ of X into $\mathcal{P}(X)$ of Definition 5.4.1.
- The internal Hom Nat of $\mathsf{PSh}(\mathcal{C})$ with the internal Hom $[-,-]_X$ of $\mathcal{P}(X)$ of Definition 4.7.1.

However, since $[U, \chi_x]_X$ is a subset of U instead of a truth value, we get a function

$$\mathsf{I}\colon \mathcal{P}(X) \to \mathsf{Sets}(X, \mathcal{P}(X))$$

instead of a function

$$\mathsf{I}\colon \mathcal{P}(X) \to \mathcal{P}(X).$$

This makes some of the properties involving I a bit more cumbersome to state, although we still have an analogue of Isbell duality in that $\mathsf{I}_* \circ \mathsf{I}$ evaluates to $\mathrm{id}_{\mathcal{P}(X)}$ in the sense of Proposition 4.8.3.

Proposition 4.8.3 ► Isbell Duality for Sets

The diagram

$$\begin{tikzcd}
\mathcal{P}(X) \arrow[r, "\mathsf{I}"] \arrow[dr, "\Delta_{\Delta_{\mathrm{id}_{\mathcal{P}(X)}}}"'] & \mathsf{Sets}(X, \mathcal{P}(X)) \arrow[d, "\mathsf{I}_*"] \\
& \mathsf{Sets}(X, \mathsf{Sets}(X, \mathcal{P}(X)))
\end{tikzcd}$$

commutes, i.e. we have

$$\mathsf{I}_*(\mathsf{I}(U)) = [\![x \mapsto [\![y \mapsto U]\!]]\!]$$

for each $U \in \mathcal{P}(X)$.

Proof 4.8.4 ► Proof of Proposition 4.8.3

We have

$$\begin{aligned}
\mathsf{I}_*(\mathsf{I}(U)) &\overset{\text{def}}{=} \mathsf{I}_*([\![x \mapsto U^{\mathsf{c}} \cup \{x\}]\!]) \\
&\overset{\text{def}}{=} [\![x \mapsto \mathsf{I}(U^{\mathsf{c}} \cup \{x\})]\!]
\end{aligned}$$

$$
\begin{aligned}
&\stackrel{\text{def}}{=} [\![x \mapsto [\![y \mapsto (U^{\mathsf{c}} \cup \{x\})^{\mathsf{c}} \cup \{x\}]\!]]\!] \\
&= [\![x \mapsto [\![y \mapsto (U \cap (X \setminus \{x\})) \cup \{x\}]\!]]\!] \\
&= [\![x \mapsto [\![y \mapsto (U \setminus \{x\}) \cup \{x\}]\!]]\!] \\
&= [\![x \mapsto [\![y \mapsto U]\!]]\!],
\end{aligned}
$$

where we have used Item 2 of Proposition 3.11.2 for the fourth equality above.

5 Characteristic Functions

5.1 The Characteristic Function of a Subset

Let X be a set and let $U \in \mathcal{P}(X)$.

Definition 5.1.1 ▶ The Characteristic Function of a Subset

The **characteristic function of** U[1] is the function $\chi_U \colon X \to \{\mathsf{t}, \mathsf{f}\}$[2] defined by

$$
\chi_U(x) \stackrel{\text{def}}{=} \begin{cases} \text{true} & \text{if } x \in U, \\ \text{false} & \text{if } x \notin U \end{cases}
$$

for each $x \in X$.

[1] *Further Terminology:* Also called the **indicator function of** U.
[2] *Further Notation:* Also written $\chi_X(U, -)$ or $\chi_X(-, U)$.

Remark 5.1.2 ▶ Characteristic Functions of Subsets as Decategorifications of Presheaves

Under the analogy that $\{\mathsf{t}, \mathsf{f}\}$ should be the (-1)-categorical analogue of Sets, we may view a function

$$
f \colon X \to \{\mathsf{t}, \mathsf{f}\}
$$

as a decategorification of presheaves and copresheaves

$$
\begin{aligned}
\mathcal{F} &\colon \mathcal{C}^{\mathsf{op}} \to \mathsf{Sets}, \\
F &\colon \mathcal{C} \to \mathsf{Sets}.
\end{aligned}
$$

The characteristic functions χ_U of the subsets of X are then the primordial examples of such functions (and, in fact, all of them).

Notation 5.1.3 ► Further Notation for Characteristic Functions

We will often employ the bijection $\{\mathrm{t}, \mathrm{f}\} \cong \{0, 1\}$ to make use of the arithmetical operations defined on $\{0, 1\}$ when disucssing characteristic functions. Examples of this include Items 4 to 11 of Proposition 5.1.4 below.

Proposition 5.1.4 ► Properties of Characteristic Functions of Subsets

Let X be a set.

1. *Functionality.* The assignment $U \mapsto \chi_U$ defines a function
$$\chi_{(-)} \colon \mathcal{P}(X) \to \mathsf{Sets}(X, \{\mathrm{t}, \mathrm{f}\}).$$
2. *Bijectivity.* The function $\chi_{(-)}$ from Item 1 is bijective.
3. *Naturality.* The collection
$$\left\{\chi_{(-)} \colon \mathcal{P}(X) \to \mathsf{Sets}(X, \{\mathrm{t}, \mathrm{f}\})\right\}_{X \in \mathrm{Obj}(\mathsf{Sets})}$$
defines a natural isomorphism between $\mathcal{P}^{-1}$ and $\mathsf{Sets}(-, \{\mathrm{t}, \mathrm{f}\})$. In particular, given a function $f \colon X \to Y$, the diagram
$$\begin{tikzcd} \mathcal{P}(Y) \arrow[r, "f^{-1}"] \arrow[d, dashed, "\chi_{(-)}"'] \arrow[dr, phantom, "\wr"] & \mathcal{P}(X) \arrow[d, dashed, "\chi_{(-)}"] \arrow[dl, phantom, "\wr"] \\ \mathsf{Sets}(Y, \{\mathrm{t}, \mathrm{f}\}) \arrow[r, "f^*"'] & \mathsf{Sets}(X, \{\mathrm{t}, \mathrm{f}\}) \end{tikzcd}$$
commutes, i.e. we have
$$\chi_V \circ f = \chi_{f^{-1}(V)}$$
for each $V \in \mathcal{P}(Y)$.

4. *Interaction With Unions I.* We have
$$\chi_{U\cup V} = \max(\chi_U, \chi_V)$$
for each $U, V \in \mathcal{P}(X)$.

5. *Interaction With Unions II.* We have
$$\chi_{U\cup V} = \chi_U + \chi_V - \chi_{U\cap V}$$
for each $U, V \in \mathcal{P}(X)$.

6. *Interaction With Intersections I.* We have
$$\chi_{U\cap V} = \chi_U \chi_V$$
for each $U, V \in \mathcal{P}(X)$.

7. *Interaction With Intersections II.* We have
$$\chi_{U\cap V} = \min(\chi_U, \chi_V)$$
for each $U, V \in \mathcal{P}(X)$.

8. *Interaction With Differences.* We have
$$\chi_{U\setminus V} = \chi_U - \chi_{U\cap V}$$
for each $U, V \in \mathcal{P}(X)$.

9. *Interaction With Complements.* We have
$$\chi_{U^c} \equiv 1 - \chi_U \pmod 2$$
for each $U \in \mathcal{P}(X)$.

10. *Interaction With Symmetric Differences.* We have
$$\chi_{U\triangle V} = \chi_U + \chi_V - 2\chi_{U\cap V}$$
and thus, in particular, we have
$$\chi_{U\triangle V} \equiv \chi_U + \chi_V \pmod 2$$
for each $U, V \in \mathcal{P}(X)$.

11. *Interaction With Internal Homs.* We have
$$\chi_{[U,V]_{\mathcal{P}(X)}} = \max(1 - \chi_U \pmod 2, \chi_V)$$
for each $U, V \in \mathcal{P}(X)$.

Proof 5.1.5 ► Proof of Proposition 5.1.4

Item 1: Functionality

There is nothing to prove.

Item 2: Bijectivity

We proceed in three steps:

1. *The Inverse of $\chi_{(-)}$.* The inverse of $\chi_{(-)}$ is the map

$$\Phi\colon \mathrm{Sets}(X, \{\mathrm{t}, \mathrm{f}\}) \overset{\sim}{\dashrightarrow} \mathcal{P}(X),$$

defined by

$$\begin{aligned}\Phi(f) &\overset{\text{def}}{=} U_f \\ &\overset{\text{def}}{=} f^{-1}(\mathrm{true}) \\ &\overset{\text{def}}{=} \{x \in X \mid f(x) = \mathrm{true}\}\end{aligned}$$

for each $f \in \mathrm{Sets}(X, \{\mathrm{t}, \mathrm{f}\})$.

2. *Invertibility I.* We have

$$\begin{aligned}\left[\Phi \circ \chi_{(-)}\right](U) &\overset{\text{def}}{=} \Phi(\chi_U) \\ &\overset{\text{def}}{=} \chi_U^{-1}(\mathrm{true}) \\ &\overset{\text{def}}{=} \{x \in X \mid \chi_U(x) = \mathrm{true}\} \\ &\overset{\text{def}}{=} \{x \in X \mid x \in U\} \\ &= U \\ &\overset{\text{def}}{=} \left[\mathrm{id}_{\mathcal{P}(X)}\right](U)\end{aligned}$$

for each $U \in \mathcal{P}(X)$. Thus, we have

$$\Phi \circ \chi_{(-)} = \mathrm{id}_{\mathcal{P}(X)}.$$

3. *Invertibility II.* We have

$$\left[\chi_{(-)} \circ \Phi\right](U) \overset{\text{def}}{=} \chi_{\Phi(f)}$$

$$
\begin{aligned}
&\overset{\text{def}}{=} \chi_{f^{-1}(\text{true})} \\
&\overset{\text{def}}{=} \left[\!\left[x \mapsto \begin{cases} \text{true} & \text{if } x \in f^{-1}(\text{true}) \\ \text{false} & \text{otherwise} \end{cases} \right]\!\right] \\
&= \left[\!\left[x \mapsto f(x) \right]\!\right] \\
&= f \\
&\overset{\text{def}}{=} \left[\mathrm{id}_{\mathsf{Sets}(X,\{\mathsf{t},\mathsf{f}\})}\right](f)
\end{aligned}
$$

for each $f \in \mathsf{Sets}(X, \{\mathsf{t}, \mathsf{f}\})$. Thus, we have

$$\chi_{(-)} \circ \Phi = \mathrm{id}_{\mathsf{Sets}(X,\{\mathsf{t},\mathsf{f}\})} .$$

This finishes the proof.

Item 3: Naturality

We proceed in two steps:

1. *Naturality of* $\chi_{(-)}$. We have

$$
\begin{aligned}
[\chi_V \circ f](v) &\overset{\text{def}}{=} \chi_V(f(v)) \\
&= \begin{cases} \text{true} & \text{if } f(v) \in V, \\ \text{false} & \text{otherwise} \end{cases} \\
&= \begin{cases} \text{true} & \text{if } v \in f^{-1}(V), \\ \text{false} & \text{otherwise} \end{cases} \\
&\overset{\text{def}}{=} \chi_{f^{-1}(V)}(v)
\end{aligned}
$$

   for each $v \in V$.

2. *Naturality of* Φ. Since $\chi_{(-)}$ is natural and a componentwise inverse to Φ, it follows from Categories, **??** of **??** that Φ is also natural in each argument.

This finishes the proof.

Item 4: Interaction With Unions I

This is a repetition of Item 10 of Proposition 3.8.2 and is proved there.

Item 5: Interaction With Unions II

This is a repetition of Item 11 of Proposition 3.8.2 and is proved there.

Item 6: Interaction With Intersections I

This is a repetition of Item 10 of Proposition 3.9.2 and is proved there.

Item 7: Interaction With Intersections II

This is a repetition of Item 11 of Proposition 3.9.2 and is proved there.

Item 8: Interaction With Differences

This is a repetition of Item 16 of Proposition 3.10.2 and is proved there.

Item 9: Interaction With Complements

This is a repetition of Item 4 of Proposition 3.11.2 and is proved there.

Item 10: Interaction With Symmetric Differences

This is a repetition of Item 15 of Proposition 3.12.2 and is proved there.

Item 11: Interaction With Internal Homs

This is a repetition of Item 17 of Proposition 4.7.4 and is proved there.

Remark 5.1.6 ► Powersets as Sets of Functions and Un/Straightening

The bijection

$$\mathcal{P}(X) \cong \mathsf{Sets}(X, \{\mathsf{t}, \mathsf{f}\})$$

of Item 2 of Proposition 5.1.4, which

- Takes a subset $U \hookrightarrow X$ of X and *straightens* it to a function $\chi_U \colon X \to \{\text{true}, \text{false}\}$;
- Takes a function $f \colon X \to \{\text{true}, \text{false}\}$ and *unstraightens* it to a subset $f^{-1}(\text{true}) \hookrightarrow X$ of X;

may be viewed as the (-1)-categorical version of the 0-categorical un/straightening isomorphism between indexed and fibred sets

$$\underbrace{\mathsf{FibSets}_X}_{\overset{\text{def}}{=}\mathsf{Sets}_{/X}} \cong \underbrace{\mathsf{ISets}_X}_{\overset{\text{def}}{=}\mathsf{Fun}(X_{\mathsf{disc}},\mathsf{Sets})}$$

of Un/Straightening for Indexed and Fibred Sets, **??**. Here we view:

- Subsets $U \hookrightarrow X$ as being analogous to X-fibred sets $\phi_X \colon A \to X$.
- Functions $f \colon X \to \{\mathrm{t}, \mathrm{f}\}$ as being analogous to X-indexed sets $A \colon X_{\mathrm{disc}} \to \mathrm{Sets}$.

5.2 The Characteristic Function of a Point

Let X be a set and let $x \in X$.

Definition 5.2.1 ▶ The Characteristic Function of a Point

The **characteristic function of** x is the function[1]

$$\chi_x \colon X \to \{\mathrm{t}, \mathrm{f}\}$$

defined by

$$\chi_x \overset{\text{def}}{=} \chi_{\{x\}},$$

i.e. by

$$\chi_x(y) \overset{\text{def}}{=} \begin{cases} \text{true} & \text{if } x = y, \\ \text{false} & \text{if } x \neq y \end{cases}$$

for each $y \in X$.

[1] *Further Notation:* Also written χ^x, $\chi_X(x, -)$, or $\chi_X(-, x)$.

Remark 5.2.2 ▶ Characteristic Functions of Points as Decategorifications of Representable Presheaves

Expanding upon Remark 5.1.2, we may think of the characteristic function

$$\chi_x \colon X \to \{\mathrm{t}, \mathrm{f}\}$$

of an *element* x of X as a decategorification of the representable presheaf and of the representable copresheaf

$$h_X \colon C^{\mathrm{op}} \to \mathrm{Sets},$$
$$h^X \colon C \to \mathrm{Sets}$$

associated of an *object* X of a category C.

5.3 The Characteristic Relation of a Set

Let X be a set.

Definition 5.3.1 ► The Characteristic Relation of a Set

The **characteristic relation on** X[1] is the relation[2]

$$\chi_X(-_1, -_2) \colon X \times X \to \{\mathsf{t}, \mathsf{f}\}$$

on X defined by[3]

$$\chi_X(x, y) \overset{\text{def}}{=} \begin{cases} \text{true} & \text{if } x = y, \\ \text{false} & \text{if } x \neq y \end{cases}$$

for each $x, y \in X$.

[1] *Further Terminology:* Also called the **identity relation on** X.

[2] *Further Notation:* Also written $\chi_{-_2}^{-_1}$, or $\sim_{\mathrm{id}}$ in the context of relations.

[3] Under the bijection $\mathrm{Sets}(X \times X, \{\mathsf{t}, \mathsf{f}\}) \cong \mathcal{P}(X \times X)$ of Item 2 of Proposition 5.1.4, the relation χ_X corresponds to the diagonal $\Delta_X \subset X \times X$ of X.

Remark 5.3.2 ► The Characteristic Relation of a Set as a Decategorification of the Hom Profunctor

Expanding upon Remarks 5.1.2 and 5.2.2, we may view the characteristic relation

$$\chi_X(-_1, -_2) \colon X \times X \to \{\mathsf{t}, \mathsf{f}\}$$

of X as a decategorification of the Hom profunctor

$$\mathrm{Hom}_{\mathcal{C}}(-_1, -_2) \colon \mathcal{C}^{\mathrm{op}} \times \mathcal{C} \to \mathrm{Sets}$$

of a category $\mathcal{C}$.

Proposition 5.3.3 ► Properties of Characteristic Relations

Let $f \colon X \to Y$ be a function.

1. *The Inclusion of Characteristic Relations Associated to a Function.* Let $f \colon A \to$

B be a function. We have an inclusion[1]

$$\chi_B \circ (f \times f) \subset \chi_A, \qquad \begin{tikzcd} A \times A \arrow[rr, "f \times f"] \arrow[dr, "\chi_A"'] & \arrow[d, phantom, "\supset"] & B \times B \arrow[dl, "\chi_B"] \\ & \{\mathrm{t}, \mathrm{f}\}. & \end{tikzcd}$$

[1] *Note:* This is the 0-categorical version of Categories, **??**.

Proof 5.3.4 ▶ Proof of Proposition 5.3.3

Item 1: The Inclusion of Characteristic Relations Associated to a Function

The inclusion $\chi_B(f(a), f(b)) \subset \chi_A(a, b)$ is equivalent to the statement "if $a = b$, then $f(a) = f(b)$", which is true. ∎

5.4 The Characteristic Embedding of a Set

Let X be a set.

Definition 5.4.1 ▶ The Characteristic Embedding of a Set

The **characteristic embedding**[1] **of** X **into** $\mathcal{P}(X)$ is the function

$$\chi_{(-)} \colon X \hookrightarrow \mathcal{P}(X)$$

defined by[2]

$$\begin{aligned} \chi_{(-)}(x) &\overset{\text{def}}{=} \chi_x \\ &= \{x\} \end{aligned}$$

for each $x \in X$.

[1] The name "characteristic *embedding*" is justified by Corollary 5.5.3, which gives an analogue of fully faithfulness for $\chi_{(-)}$.

[2] Here we are identifying $\mathcal{P}(X)$ with $\mathrm{Sets}(X, \{\mathrm{t}, \mathrm{f}\})$ as per Item 2 of Proposition 5.1.4.

Remark 5.4.2 ▶ The Characteristic Embedding of a Set as a Decategorification of the Yoneda Embedding

Expanding upon Remarks 5.1.2, 5.2.2 and 5.3.2, we may view the characteristic embedding

$$\chi_{(-)} \colon X \hookrightarrow \mathcal{P}(X)$$

of X into $\mathcal{P}(X)$ as a decategorification of the Yoneda embedding

$$よ \colon \mathcal{C}^{\mathrm{op}} \hookrightarrow \mathsf{PSh}(\mathcal{C})$$

of a category $\mathcal{C}$ into $\mathsf{PSh}(\mathcal{C})$.

Proposition 5.4.3 ▶ Properties of Characteristic Embeddings

Let $f \colon X \longrightarrow Y$ be a map of sets.

1. *Interaction With Functions.* We have

$$f_* \circ \chi_X = \chi_Y \circ f, \qquad \begin{tikzcd} X \arrow[r, "f"] \arrow[d, "\chi_X"'] & Y \arrow[d, "\chi_Y"] \\ \mathcal{P}(X) \arrow[r, "f_*"'] & \mathcal{P}(Y). \end{tikzcd}$$

Proof 5.4.4 ▶ Proof of Proposition 5.4.3

Item 1: Interaction With Functions

Indeed, we have

$$\begin{aligned} [f_* \circ \chi_X](x) &\overset{\text{def}}{=} f_*(\chi_X(x)) \\ &\overset{\text{def}}{=} f_*(\{x\}) \\ &= \{f(x)\} \\ &\overset{\text{def}}{=} \chi_{X'}(f(x)) \\ &\overset{\text{def}}{=} [\chi_{X'} \circ f](x), \end{aligned}$$

for each $x \in X$, showing the desired equality. ∎

5.5 The Yoneda Lemma for Sets

Let X be a set and let $U \subset X$ be a subset of X.

Proposition 5.5.1 ▶ The Yoneda Lemma for Sets

We have

$$\chi_{\mathcal{P}(X)}(\chi_x, \chi_U) = \chi_U(x)$$

for each $x \in X$, giving an equality of functions

$$\chi_{\mathcal{P}(X)}\big(\chi_{(-)}, \chi_U\big) = \chi_U,$$

where

$$\chi_{\mathcal{P}(X)}(U, V) \overset{\text{def}}{=} \begin{cases} \text{true} & \text{if } U \subset V, \\ \text{false} & \text{otherwise.} \end{cases}$$

Proof 5.5.2 ▶ Proof of Proposition 5.5.1

We have

$$\begin{aligned} \chi_{\mathcal{P}(X)}(\chi_x, \chi_U) &\overset{\text{def}}{=} \begin{cases} \text{true} & \text{if } \{x\} \subset U, \\ \text{false} & \text{otherwise} \end{cases} \\ &= \begin{cases} \text{true} & \text{if } x \in U \\ \text{false} & \text{otherwise} \end{cases} \\ &\overset{\text{def}}{=} \chi_U(x). \end{aligned}$$

This finishes the proof. ∎

Corollary 5.5.3 ▶ The Characteristic Embedding Is Fully Faithful

The characteristic embedding is fully faithful, i.e., we have

$$\chi_{\mathcal{P}(X)}\big(\chi_x, \chi_y\big) \cong \chi_X(x, y)$$

for each $x, y \in X$.

Proof 5.5.4 ▶ Proof of Corollary 5.5.3

We have

$$\begin{aligned}\chi_{\mathcal{P}(X)}(\chi_x, \chi_y) &= \chi_y(x) \\ &\overset{\text{def}}{=} \begin{cases} \text{true} & \text{if } x \in \{y\} \\ \text{false} & \text{otherwise} \end{cases} \\ &= \begin{cases} \text{true} & \text{if } x = y \\ \text{false} & \text{otherwise} \end{cases} \\ &\overset{\text{def}}{=} \chi_X(x, y).\end{aligned}$$

where we have used Proposition 5.5.1 for the first equality. ∎

6 The Adjoint Triple $f_* \dashv f^{-1} \dashv f_!$

6.1 Direct Images

Let $f\colon X \to Y$ be a function.

Definition 6.1.1 ▶ Direct Images

The **direct image function associated to** f is the function

$$f_*\colon \mathcal{P}(X) \to \mathcal{P}(Y)$$

defined by[1]

$$\begin{aligned}f_*(U) &\overset{\text{def}}{=} f(U) \\ &\overset{\text{def}}{=} \left\{ y \in Y \;\middle|\; \begin{array}{l} \text{there exists some } x \in U \\ \text{such that } y = f(x) \end{array} \right\} \\ &= \{f(x) \in Y \mid x \in U\}\end{aligned}$$

for each $U \in \mathcal{P}(X)$.

[1] *Further Terminology*: The set $f(U)$ is called the **direct image of** U **by** f.

Notation 6.1.2 ► Further Notation for Direct Images

Sometimes one finds the notation

$$\exists_f \colon \mathcal{P}(X) \to \mathcal{P}(Y)$$

for f_*. This notation comes from the fact that the following statements are equivalent, where $y \in Y$ and $U \in \mathcal{P}(X)$:

- We have $y \in \exists_f(U)$.
- There exists some $x \in U$ such that $f(x) = y$.

We will not make use of this notation in the present work.

Remark 6.1.3 ► Unwinding Definition 6.1.1

Identifying $\mathcal{P}(X)$ with $\mathsf{Sets}(X, \{\mathsf{t}, \mathsf{f}\})$ via Item 2 of Proposition 5.1.4, we see that the direct image function associated to f is equivalently the function

$$f_* \colon \mathcal{P}(X) \to \mathcal{P}(Y)$$

defined by

$$\begin{aligned} f_*(\chi_U) &\overset{\text{def}}{=} \mathsf{Ran}_f(\chi_U) \\ &= \mathsf{colim}\left(\left(f \overset{\rightarrow}{\times} \underline{(-_1)}\right) \overset{\text{pr}}{\twoheadrightarrow} A \xrightarrow{\chi_U} \{\mathsf{t}, \mathsf{f}\}\right) \\ &= \operatorname*{colim}_{\substack{x \in X \\ f(x) = -_1}} (\chi_U(x)) \end{aligned}$$

where we have used Kan Extensions, **??** for the second equality. In other words, we have

$$\begin{aligned} [f_*(\chi_U)](y) &= \bigvee_{\substack{x \in X \\ f(x) = y}} (\chi_U(x)) \\ &= \begin{cases} \mathsf{true} & \text{if there exists some } x \in X \text{ such} \\ & \text{that } f(x) = y \text{ and } x \in U, \\ \mathsf{false} & \text{otherwise} \end{cases} \end{aligned}$$

$$= \begin{cases} \text{true} & \text{if there exists some } x \in U \\ & \text{such that } f(x) = y, \\ \text{false} & \text{otherwise} \end{cases}$$

for each $y \in Y$.

Proposition 6.1.4 ▶ Properties of Direct Images I

Let $f\colon X \to Y$ be a function.

1. *Functoriality.* The assignment $U \mapsto f_*(U)$ defines a functor

$$f_*\colon (\mathcal{P}(X), \subset) \to (\mathcal{P}(Y), \subset).$$

   In particular, for each $U, V \in \mathcal{P}(X)$, the following condition is satisfied:

   ($\star$) If $U \subset V$, then $f_*(U) \subset f_*(V)$.

2. *Triple Adjointness.* We have a triple adjunction

$$(f_* \dashv f^{-1} \dashv f_!)\colon \quad \begin{tikzcd} \mathcal{P}(X) \arrow[r, bend left=60, "f_*"] \arrow[r, bend right=60, "f_!"'] \arrow[r, phantom, "\perp"] \arrow[r, phantom, "\perp"] & \mathcal{P}(Y) \arrow[l, "f^{-1}"'] \end{tikzcd},$$

   witnessed by:

   (a) Units and counits of the form

$$\begin{aligned} \mathrm{id}_{\mathcal{P}(X)} &\hookrightarrow f^{-1} \circ f_*, & \mathrm{id}_{\mathcal{P}(Y)} &\hookrightarrow f_! \circ f^{-1}, \\ f_* \circ f^{-1} &\hookrightarrow \mathrm{id}_{\mathcal{P}(Y)}, & f^{-1} \circ f_! &\hookrightarrow \mathrm{id}_{\mathcal{P}(X)}, \end{aligned}$$

   having components of the form

$$\begin{aligned} U &\subset f^{-1}(f_*(U)), & V &\subset f_!\big(f^{-1}(V)\big), \\ f_*\big(f^{-1}(V)\big) &\subset V, & f^{-1}(f_!(U)) &\subset U \end{aligned}$$

   indexed by $U \in \mathcal{P}(X)$ and $V \in \mathcal{P}(Y)$.

(b) Bijections of sets

$$\begin{aligned}\mathrm{Hom}_{\mathcal{P}(Y)}(f_*(U), V) &\cong \mathrm{Hom}_{\mathcal{P}(X)}\big(U, f^{-1}(V)\big),\\ \mathrm{Hom}_{\mathcal{P}(X)}\big(f^{-1}(U), V\big) &\cong \mathrm{Hom}_{\mathcal{P}(X)}(U, f_!(V)),\end{aligned}$$

natural in $U \in \mathcal{P}(X)$ and $V \in \mathcal{P}(Y)$ and (respectively) $V \in \mathcal{P}(X)$ and $U \in \mathcal{P}(Y)$. In particular:

i. The following conditions are equivalent:
   A. We have $f_*(U) \subset V$.
   B. We have $U \subset f^{-1}(V)$.

ii. The following conditions are equivalent:
   A. We have $f^{-1}(U) \subset V$.
   B. We have $U \subset f_!(V)$.

3. *Interaction With Unions of Families of Subsets.* The diagram

$$\begin{tikzcd}\mathcal{P}(\mathcal{P}(X)) \arrow[r, "(f_*)_*"] \arrow[d, "\cup"'] & \mathcal{P}(\mathcal{P}(Y)) \arrow[d, "\cup"] \\ \mathcal{P}(X) \arrow[r, "f_*"'] & \mathcal{P}(Y)\end{tikzcd}$$

commutes, i.e. we have

$$\bigcup_{U \in \mathcal{U}} f_*(U) = \bigcup_{V \in f_*(\mathcal{U})} V$$

for each $\mathcal{U} \in \mathcal{P}(X)$, where $f_*(\mathcal{U}) \overset{\text{def}}{=} (f_*)_*(\mathcal{U})$.

4. *Interaction With Intersections of Families of Subsets.* The diagram

$$\begin{tikzcd}\mathcal{P}(\mathcal{P}(X)) \arrow[r, "(f_*)_*"] \arrow[d, "\cap"'] & \mathcal{P}(\mathcal{P}(Y)) \arrow[d, "\cap"] \\ \mathcal{P}(X) \arrow[r, "f_*"'] & \mathcal{P}(Y)\end{tikzcd}$$

commutes, i.e. we have

$$\bigcap_{U \in \mathcal{U}} f_*(U) = \bigcap_{V \in f_*(\mathcal{U})} V$$

for each $\mathcal{U} \in \mathcal{P}(X)$, where $f_*(\mathcal{U}) \overset{\text{def}}{=} (f_*)_*(\mathcal{U})$.

5. *Interaction With Binary Unions.* The diagram

$$\begin{tikzcd}
\mathcal{P}(X) \times \mathcal{P}(X) \arrow[r, "f_* \times f_*"] \arrow[d, "\cup"'] & \mathcal{P}(Y) \times \mathcal{P}(Y) \arrow[d, "\cup"] \\
\mathcal{P}(X) \arrow[r, "f_*"'] & \mathcal{P}(Y)
\end{tikzcd}$$

commutes, i.e. we have

$$f_*(U \cup V) = f_*(U) \cup f_*(V)$$

for each $U, V \in \mathcal{P}(X)$.

6. *Interaction With Binary Intersections.* We have a natural transformation

$$\begin{tikzcd}
\mathcal{P}(X) \times \mathcal{P}(X) \arrow[r, "f_* \times f_*"] \arrow[d, "\cap"'] \arrow[dr, phantom, "\circlearrowright"] & \mathcal{P}(Y) \times \mathcal{P}(Y) \arrow[d, "\cap"] \\
\mathcal{P}(X) \arrow[r, "f_*"'] & \mathcal{P}(Y)
\end{tikzcd}$$

with components

$$f_*(U \cap V) \subset f_*(U) \cap f_*(V)$$

indexed by $U, V \in \mathcal{P}(X)$.

7. *Interaction With Differences.* We have a natural transformation

$$\begin{tikzcd}
\mathcal{P}(X)^{\mathsf{op}} \times \mathcal{P}(X) \arrow[r, "f_*^{\mathsf{op}} \times f_*"] \arrow[d, "\setminus"'] \arrow[dr, phantom, "\supset"] & \mathcal{P}(Y)^{\mathsf{op}} \times \mathcal{P}(Y) \arrow[d, "\setminus"] \\
\mathcal{P}(X) \arrow[r, "f_*"'] & \mathcal{P}(Y)
\end{tikzcd}$$

with components

$$f_*(U) \setminus f_*(V) \subset f_*(U \setminus V)$$

indexed by $U, V \in \mathcal{P}(X)$.

8. *Interaction With Complements.* The diagram

$$\begin{tikzcd}
\mathcal{P}(X)^{\mathsf{op}} \arrow[r, "f_!^{\mathsf{op}}"] \arrow[d, "(-)^{\mathsf{c}}"'] & \mathcal{P}(Y)^{\mathsf{op}} \arrow[d, "(-)^{\mathsf{c}}"] \\
\mathcal{P}(X) \arrow[r, "f_*"'] & \mathcal{P}(Y)
\end{tikzcd}$$

commutes, i.e. we have

$$f_*(U^{\mathsf{c}}) = f_!(U)^{\mathsf{c}}$$

for each $U \in \mathcal{P}(X)$.

9. *Interaction With Symmetric Differences.* We have a natural transformation

$$\begin{tikzcd}
\mathcal{P}(X)^{\mathsf{op}} \times \mathcal{P}(X) \arrow[r, "f_*^{\mathsf{op}} \times f_*"] \arrow[d, "\triangle"'] \arrow[dr, phantom, "\supset"] & \mathcal{P}(Y)^{\mathsf{op}} \times \mathcal{P}(Y) \arrow[d, "\triangle"] \\
\mathcal{P}(X) \arrow[r, "f_*"'] & \mathcal{P}(Y)
\end{tikzcd}$$

with components

$$f_*(U) \triangle f_*(V) \subset f_*(U \triangle V)$$

indexed by $U, V \in \mathcal{P}(X)$.

10. *Interaction With Internal Homs of Powersets.* The diagram

$$\begin{tikzcd}
\mathcal{P}(X)^{\mathsf{op}} \times \mathcal{P}(X) \arrow[r, "f_!^{\mathsf{op}} \times f_*"] \arrow[d, "{[-_1,-_2]_X}"'] & \mathcal{P}(Y)^{\mathsf{op}} \times \mathcal{P}(Y) \arrow[d, "{[-_1,-_2]_Y}"] \\
\mathcal{P}(X) \arrow[r, "f_*"'] & \mathcal{P}(Y)
\end{tikzcd}$$

commutes, i.e. we have an equality of sets

$$f_*([U,V]_X) = [f_!(U), f_*(V)]_Y,$$

natural in $U, V \in \mathcal{P}(X)$.

11. *Preservation of Colimits.* We have an equality of sets

$$f_*\left(\bigcup_{i\in I} U_i\right) = \bigcup_{i\in I} f_*(U_i),$$

natural in $\{U_i\}_{i\in I} \in \mathcal{P}(X)^{\times I}$. In particular, we have equalities

$$f_*(U) \cup f_*(V) = f_*(U \cup V),$$
$$f_*(\emptyset) = \emptyset,$$

natural in $U, V \in \mathcal{P}(X)$.

12. *Oplax Preservation of Limits.* We have an inclusion of sets

$$f_*\left(\bigcap_{i\in I} U_i\right) \subset \bigcap_{i\in I} f_*(U_i),$$

natural in $\{U_i\}_{i\in I} \in \mathcal{P}(X)^{\times I}$. In particular, we have inclusions

$$f_*(U \cap V) \subset f_*(U) \cap f_*(V),$$
$$f_*(X) \subset Y,$$

natural in $U, V \in \mathcal{P}(X)$.

13. *Symmetric Strict Monoidality With Respect to Unions.* The direct image function of Item 1 has a symmetric strict monoidal structure

$$\left(f_*, f_*^{\otimes}, f_{*|\mathbb{1}}^{\otimes}\right) \colon (\mathcal{P}(X), \cup, \emptyset) \to (\mathcal{P}(Y), \cup, \emptyset),$$

being equipped with equalities

$$f_{*|U,V}^{\otimes} \colon f_*(U) \cup f_*(V) \xrightarrow{=} f_*(U \cup V),$$
$$f_{*|\mathbb{1}}^{\otimes} \colon \emptyset \xrightarrow{=} \emptyset,$$

natural in $U, V \in \mathcal{P}(X)$.

14. *Symmetric Oplax Monoidality With Respect to Intersections.* The direct image function of Item 1 has a symmetric oplax monoidal structure

$$\left(f_*, f_*^{\otimes}, f_{*|\mathbb{1}}^{\otimes}\right) \colon (\mathcal{P}(X), \cap, X) \to (\mathcal{P}(Y), \cap, Y),$$

being equipped with inclusions

$$\begin{aligned} f_{*|U,V}^{\otimes} &\colon f_*(U \cap V) \hookrightarrow f_*(U) \cap f_*(V), \\ f_{*|\mathbb{1}}^{\otimes} &\colon f_*(X) \hookrightarrow Y, \end{aligned}$$

natural in $U, V \in \mathcal{P}(X)$.

15. *Interaction With Coproducts.* Let $f \colon X \to X'$ and $g \colon Y \to Y'$ be maps of sets. We have

$$(f \coprod g)_*(U \coprod V) = f_*(U) \coprod g_*(V)$$

for each $U \in \mathcal{P}(X)$ and each $V \in \mathcal{P}(Y)$.

16. *Interaction With Products.* Let $f \colon X \to X'$ and $g \colon Y \to Y'$ be maps of sets. We have

$$(f \boxtimes_{X \times Y} g)_*(U \boxtimes_{X \times Y} V) = f_*(U) \boxtimes_{X' \times Y'} g_*(V)$$

for each $U \in \mathcal{P}(X)$ and each $V \in \mathcal{P}(Y)$.

17. *Relation to Direct Images With Compact Support.* We have

$$\begin{aligned} f_*(U) &= f_!(U^{\mathsf{c}})^{\mathsf{c}} \\ &\overset{\text{def}}{=} Y \setminus f_!(X \setminus U) \end{aligned}$$

for each $U \in \mathcal{P}(X)$.

Proof 6.1.5 ▶ Proof of Proposition 6.1.4

Item 1: Functoriality

Omitted.

Item 2: Triple Adjointness

This follows from Remark 6.1.3, Remark 6.2.2, Remark 6.3.3, and Kan Extensions, **??** of **??**.

Item 3: Interaction With Unions of Families of Subsets

We have

$$\begin{aligned}\bigcup_{V\in f_*(\mathcal{U})} V &= \bigcup_{V\in\{f_*(U)\in\mathcal{P}(X)\mid U\in\mathcal{U}\}} V \\ &= \bigcup_{U\in\mathcal{U}} f_*(U).\end{aligned}$$

This finishes the proof.

Item 4: Interaction With Intersections of Families of Subsets

We have

$$\begin{aligned}\bigcap_{V\in f_*(\mathcal{U})} V &= \bigcap_{V\in\{f_*(U)\in\mathcal{P}(X)\mid U\in\mathcal{U}\}} V \\ &= \bigcap_{U\in\mathcal{U}} f_*(U).\end{aligned}$$

This finishes the proof.

Item 5: Interaction With Binary Unions

See [Pro24n].

Item 6: Interaction With Binary Intersections

See [Pro24m].

Item 7: Interaction With Differences

See [**proof-wiki:image-of-set-difference-under-mapping**].

Item 8: Interaction With Complements

Applying Item 17 to $X \setminus U$, we have

$$\begin{aligned} f_*(U^{\mathsf{c}}) &= f_*(X \setminus U) \\ &= Y \setminus f_!(X \setminus (X \setminus U)) \\ &= Y \setminus f_!(U) \\ &= f_!(U)^{\mathsf{c}}. \end{aligned}$$

This finishes the proof.

Item 9: Interaction With Symmetric Differences

We have

$$\begin{aligned} f_*(U) \triangle f_*(V) &= (f_*(U) \cup f_*(V)) \setminus (f_*(U) \cap f_*(V)) \\ &\subset (f_*(U) \cup f_*(V)) \setminus (f_*(U \cap V)) \\ &= (f_*(U \cup V)) \setminus (f_*(U \cap V)) \\ &\subset f_*((U \cup V) \setminus (U \cap V)) \\ &= f_*(U \triangle V), \end{aligned}$$

where we have used:

1. Item 2 of Proposition 3.12.2 for the first equality.
2. Item 6 of this proposition together with Item 1 of Proposition 3.10.2 for the first inclusion.
3. Item 5 for the second equality.
4. Item 7 for the second inclusion.
5. Item 2 of Proposition 3.12.2 for the tchird equality.

Since $\mathcal{P}(Y)$ is posetal, naturality is automatic (**??**). This finishes the proof.

Item 10: Interaction With Internal Homs of Powersets

We have

$$f_*([U, V]_X) \overset{\text{def}}{=} f_*(U^{\mathsf{c}} \cup V)$$

$$
\begin{aligned}
&= f_*(U^{\mathsf{c}}) \cup f_*(V) \\
&= f_!(U)^{\mathsf{c}} \cup f_*(V) \\
&\overset{\text{def}}{=} [f_!(U), f_*(V)]_Y,
\end{aligned}
$$

where we have used:

1. Item 5 for the second equality.
2. Item 17 for the third equality.

Since $\mathcal{P}(Y)$ is posetal, naturality is automatic (**??**). This finishes the proof.

Item 11: Preservation of Colimits

This follows from Item 2 and **??**, **??** of **??**.[1]

Item 12: Oplax Preservation of Limits

The inclusion $f_*(X) \subset Y$ is automatic. See [Pro24m] for the other inclusions.

Item 13: Symmetric Strict Monoidality With Respect to Unions

This follows from Item 11.

Item 14: Symmetric Oplax Monoidality With Respect to Intersections

The inclusions in the statement follow from Item 12. Since $\mathcal{P}(Y)$ is posetal, the commutativity of the diagrams in the definition of a symmetric oplax monoidal functor is automatic (**??**).

Item 15: Interaction With Coproducts

Omitted.

Item 16: Interaction With Products

Omitted.

Item 17: Relation to Direct Images With Compact Support

Applying Item 16 of Proposition 6.3.6 to $X \setminus U$, we have

$$
\begin{aligned}
f_!(X \setminus U) &= B \setminus f_*(X \setminus (X \setminus U)) \\
&= B \setminus f_*(U).
\end{aligned}
$$

Taking complements, we then obtain

$$
f_*(U) = B \setminus (B \setminus f_*(U)),
$$

$$= B \setminus f_!(X \setminus U),$$

which finishes the proof.

[1] *Reference:* [Pro24n].

Proposition 6.1.6 ▶ Properties of Direct Images II

Let $f\colon X \to Y$ be a function.

1. *Functionality I.* The assignment $f \mapsto f_*$ defines a function

$$(-)_{*|X,Y}\colon \mathrm{Sets}(X,Y) \to \mathrm{Sets}(\mathcal{P}(X),\mathcal{P}(Y)).$$

2. *Functionality II.* The assignment $f \mapsto f_*$ defines a function

$$(-)_{*|X,Y}\colon \mathrm{Sets}(X,Y) \to \mathrm{Pos}((\mathcal{P}(X),\subset),(\mathcal{P}(Y),\subset)).$$

3. *Interaction With Identities.* For each $X \in \mathrm{Obj}(\mathrm{Sets})$, we have

$$(\mathrm{id}_X)_* = \mathrm{id}_{\mathcal{P}(X)}.$$

4. *Interaction With Composition.* For each pair of composable functions $f\colon X \to Y$ and $g\colon Y \to Z$, we have

$$(g \circ f)_* = g_* \circ f_*, \qquad \begin{tikzcd} \mathcal{P}(X) \arrow[r, "f_*"] \arrow[dr, "(g\circ f)_*"'] & \mathcal{P}(Y) \arrow[d, "g_*"] \\ & \mathcal{P}(Z). \end{tikzcd}$$

Proof 6.1.7 ▶ Proof of Proposition 6.1.6

Item 1: Functionality I

There is nothing to prove.

Item 2: Functionality II

This follows from Item 1 of Proposition 6.1.4.

Item 3: Interaction With Identities

This follows from Remark 6.1.3 and Kan Extensions, **??** of **??**.

Item 4: Interaction With Composition

This follows from Remark 6.1.3 and Kan Extensions, **??** of **??**. ∎

6.2 Inverse Images

Let $f\colon X \to Y$ be a function.

Definition 6.2.1 ▶ Inverse Images

The **inverse image function associated to** f is the function[1]

$$f^{-1}\colon \mathcal{P}(Y) \to \mathcal{P}(X)$$

defined by[2]

$$f^{-1}(V) \overset{\text{def}}{=} \{x \in X \mid \text{we have } f(x) \in V\}$$

for each $V \in \mathcal{P}(Y)$.

[1] *Further Notation:* Also written $f^*\colon \mathcal{P}(Y) \to \mathcal{P}(X)$.
[2] *Further Terminology:* The set $f^{-1}(V)$ is called the **inverse image of** V **by** f.

Remark 6.2.2 ▶ Unwinding Definition 6.2.1

Identifying $\mathcal{P}(Y)$ with $\mathsf{Sets}(Y, \{\mathsf{t}, \mathsf{f}\})$ via Item 2 of Proposition 5.1.4, we see that the inverse image function associated to f is equivalently the function

$$f^*\colon \mathcal{P}(Y) \to \mathcal{P}(X)$$

defined by

$$f^*(\chi_V) \overset{\text{def}}{=} \chi_V \circ f$$

for each $\chi_V \in \mathcal{P}(Y)$, where $\chi_V \circ f$ is the composition

$$X \xrightarrow{f} Y \xrightarrow{\chi_V} \{\text{true}, \text{false}\}$$

in Sets.

Proposition 6.2.3 ▶ Properties of Inverse Images I

Let $f\colon X \longrightarrow Y$ be a function.

1. *Functoriality.* The assignment $V \mapsto f^{-1}(V)$ defines a functor

$$f^{-1}\colon (\mathcal{P}(Y), \subset) \longrightarrow (\mathcal{P}(X), \subset).$$

In particular, for each $U, V \in \mathcal{P}(Y)$, the following condition is satisfied:

($\star$) If $U \subset V$, then $f^{-1}(U) \subset f^{-1}(V)$.

2. *Triple Adjointness.* We have a triple adjunction

$$\big(f_* \dashv f^{-1} \dashv f_!\big)\colon \quad \begin{tikzcd} \mathcal{P}(X) \arrow[r, bend left=60, "f_*"'] \arrow[r, bend right=60, "f_!"] \arrow[r, phantom, "\perp"] \arrow[r, phantom, "\perp"] & \mathcal{P}(Y) \arrow[l, "f^{-1}"'] \end{tikzcd},$$

witnessed by:

(a) Units and counits of the form

$$\begin{aligned} \mathrm{id}_{\mathcal{P}(X)} &\hookrightarrow f^{-1} \circ f_*, & \mathrm{id}_{\mathcal{P}(Y)} &\hookrightarrow f_! \circ f^{-1}, \\ f_* \circ f^{-1} &\hookrightarrow \mathrm{id}_{\mathcal{P}(Y)}, & f^{-1} \circ f_! &\hookrightarrow \mathrm{id}_{\mathcal{P}(X)}, \end{aligned}$$

having components of the form

$$\begin{aligned} U &\subset f^{-1}(f_*(U)), & V &\subset f_!\big(f^{-1}(V)\big), \\ f_*\big(f^{-1}(V)\big) &\subset V, & f^{-1}(f_!(U)) &\subset U \end{aligned}$$

indexed by $U \in \mathcal{P}(X)$ and $V \in \mathcal{P}(Y)$.

(b) Bijections of sets

$$\begin{aligned} \mathrm{Hom}_{\mathcal{P}(Y)}(f_*(U), V) &\cong \mathrm{Hom}_{\mathcal{P}(X)}\big(U, f^{-1}(V)\big), \\ \mathrm{Hom}_{\mathcal{P}(X)}\big(f^{-1}(U), V\big) &\cong \mathrm{Hom}_{\mathcal{P}(X)}(U, f_!(V)), \end{aligned}$$

natural in $U \in \mathcal{P}(X)$ and $V \in \mathcal{P}(Y)$ and (respectively) $V \in \mathcal{P}(X)$ and $U \in \mathcal{P}(Y)$. In particular:

i. The following conditions are equivalent:
   A. We have $f_*(U) \subset V$.
   B. We have $U \subset f^{-1}(V)$.

ii. The following conditions are equivalent:
   A. We have $f^{-1}(U) \subset V$.
   B. We have $U \subset f_!(V)$.

3. *Interaction With Unions of Families of Subsets.* The diagram

$$\begin{tikzcd}
\mathcal{P}(\mathcal{P}(Y)) \arrow[r, "(f^{-1})^{-1}"] \arrow[d, "\cup"'] & \mathcal{P}(\mathcal{P}(X)) \arrow[d, "\cup"] \\
\mathcal{P}(Y) \arrow[r, "f^{-1}"'] & \mathcal{P}(X)
\end{tikzcd}$$

commutes, i.e. we have

$$\bigcup_{V \in \mathcal{V}} f^{-1}(V) = \bigcup_{U \in f^{-1}(\mathcal{U})} U$$

for each $\mathcal{V} \in \mathcal{P}(Y)$, where $f^{-1}(\mathcal{V}) \overset{\text{def}}{=} \left(f^{-1}\right)^{-1}(\mathcal{V})$.

4. *Interaction With Intersections of Families of Subsets.* The diagram

$$\begin{tikzcd}
\mathcal{P}(\mathcal{P}(Y)) \arrow[r, "(f^{-1})^{-1}"] \arrow[d, "\cap"'] & \mathcal{P}(\mathcal{P}(X)) \arrow[d, "\cap"] \\
\mathcal{P}(Y) \arrow[r, "f^{-1}"'] & \mathcal{P}(X)
\end{tikzcd}$$

commutes, i.e. we have

$$\bigcap_{V \in \mathcal{V}} f^{-1}(V) = \bigcap_{U \in f^{-1}(\mathcal{U})} U$$

for each $\mathcal{V} \in \mathcal{P}(Y)$, where $f^{-1}(\mathcal{V}) \overset{\text{def}}{=} \left(f^{-1}\right)^{-1}(\mathcal{V})$.

5. *Interaction With Binary Unions.* The diagram

$$\begin{tikzcd}
\mathcal{P}(Y) \times \mathcal{P}(Y) \arrow[r, "f^{-1} \times f^{-1}"] \arrow[d, "\cup"'] & \mathcal{P}(X) \times \mathcal{P}(X) \arrow[d, "\cup"] \\
\mathcal{P}(Y) \arrow[r, "f^{-1}"'] & \mathcal{P}(X)
\end{tikzcd}$$

commutes, i.e. we have

$$f^{-1}(U \cup V) = f^{-1}(U) \cup f^{-1}(V)$$

for each $U, V \in \mathcal{P}(Y)$.

6. *Interaction With Binary Intersections.* The diagram

$$\begin{tikzcd}
\mathcal{P}(Y) \times \mathcal{P}(Y) \arrow[r, "f^{-1} \times f^{-1}"] \arrow[d, "\cap"'] & \mathcal{P}(X) \times \mathcal{P}(X) \arrow[d, "\cap"] \\
\mathcal{P}(Y) \arrow[r, "f^{-1}"'] & \mathcal{P}(X)
\end{tikzcd}$$

commutes, i.e. we have

$$f^{-1}(U \cap V) = f^{-1}(U) \cap f^{-1}(V)$$

for each $U, V \in \mathcal{P}(Y)$.

7. *Interaction With Differences.* The diagram

$$\begin{tikzcd}
\mathcal{P}(Y)^{\mathsf{op}} \times \mathcal{P}(Y) \arrow[r, "f^{\mathsf{op},-1} \times f^{-1}"] \arrow[d, "\setminus"'] & \mathcal{P}(X)^{\mathsf{op}} \times \mathcal{P}(X) \arrow[d, "\setminus"] \\
\mathcal{P}(Y) \arrow[r, "f^{-1}"'] & \mathcal{P}(X)
\end{tikzcd}$$

commutes, i.e. we have

$$f^{-1}(U \setminus V) = f^{-1}(U) \setminus f^{-1}(V)$$

for each $U, V \in \mathcal{P}(X)$.

8. *Interaction With Complements.* The diagram

$$\begin{tikzcd}
\mathcal{P}(Y)^{\mathsf{op}} \arrow[r, "f^{-1,\mathsf{op}}"] \arrow[d, "(-)^{\mathsf{c}}"'] & \mathcal{P}(X)^{\mathsf{op}} \arrow[d, "(-)^{\mathsf{c}}"] \\
\mathcal{P}(Y) \arrow[r, "f^{-1}"'] & \mathcal{P}(X)
\end{tikzcd}$$

commutes, i.e. we have

$$f^{-1}(U^{\mathsf{c}}) = f^{-1}(U)^{\mathsf{c}}$$

for each $U \in \mathcal{P}(X)$.

9. *Interaction With Symmetric Differences.* The diagram

$$\begin{tikzcd}
\mathcal{P}(Y)^{\mathsf{op}} \times \mathcal{P}(Y) \arrow[r, "f^{\mathsf{op},-1} \times f^{-1}"] \arrow[d, "\triangle"'] & \mathcal{P}(X)^{\mathsf{op}} \times \mathcal{P}(X) \arrow[d, "\triangle"] \\
\mathcal{P}(Y) \arrow[r, "f^{-1}"'] & \mathcal{P}(X)
\end{tikzcd}$$

i.e. we have

$$f^{-1}(U) \triangle f^{-1}(V) = f^{-1}(U \triangle V)$$

for each $U, V \in \mathcal{P}(Y)$.

10. *Interaction With Internal Homs of Powersets.* The diagram

$$\begin{tikzcd}
\mathcal{P}(Y)^{\mathsf{op}} \times \mathcal{P}(Y) \arrow[r, "f^{-1,\mathsf{op}} \times f^{-1}"] \arrow[d, "{[-_1,-_2]_Y}"'] & \mathcal{P}(X)^{\mathsf{op}} \times \mathcal{P}(X) \arrow[d, "{[-_1,-_2]_X}"] \\
\mathcal{P}(Y) \arrow[r, "f^{-1}"'] & \mathcal{P}(X)
\end{tikzcd}$$

commutes, i.e. we have an equality of sets

$$f^{-1}([U,V]_Y) = \left[f^{-1}(U), f^{-1}(V)\right]_X,$$

natural in $U, V \in \mathcal{P}(X)$.

11. *Preservation of Colimits.* We have an equality of sets

$$f^{-1}\left(\bigcup_{i\in I} U_i\right) = \bigcup_{i\in I} f^{-1}(U_i),$$

natural in $\{U_i\}_{i\in I} \in \mathcal{P}(Y)^{\times I}$. In particular, we have equalities

$$\begin{aligned} f^{-1}(U) \cup f^{-1}(V) &= f^{-1}(U \cup V), \\ f^{-1}(\emptyset) &= \emptyset, \end{aligned}$$

natural in $U, V \in \mathcal{P}(Y)$.

12. *Preservation of Limits.* We have an equality of sets

$$f^{-1}\left(\bigcap_{i\in I} U_i\right) = \bigcap_{i\in I} f^{-1}(U_i),$$

natural in $\{U_i\}_{i\in I} \in \mathcal{P}(Y)^{\times I}$. In particular, we have equalities

$$\begin{aligned} f^{-1}(U) \cap f^{-1}(V) &= f^{-1}(U \cap V), \\ f^{-1}(Y) &= X, \end{aligned}$$

natural in $U, V \in \mathcal{P}(Y)$.

13. *Symmetric Strict Monoidality With Respect to Unions.* The inverse image function of Item 1 has a symmetric strict monoidal structure

$$\left(f^{-1}, f^{-1,\otimes}, f_{\mathbb{1}}^{-1,\otimes}\right) \colon (\mathcal{P}(Y), \cup, \emptyset) \to (\mathcal{P}(X), \cup, \emptyset),$$

being equipped with equalities

$$\begin{aligned} f_{U,V}^{-1,\otimes} &\colon f^{-1}(U) \cup f^{-1}(V) \xrightarrow{=} f^{-1}(U \cup V), \\ f_{\mathbb{1}}^{-1,\otimes} &\colon \emptyset \xrightarrow{=} f^{-1}(\emptyset), \end{aligned}$$

natural in $U, V \in \mathcal{P}(Y)$.

14. *Symmetric Strict Monoidality With Respect to Intersections.* The inverse image function of Item 1 has a symmetric strict monoidal structure

$$\left(f^{-1}, f^{-1,\otimes}, f_{\mathbb{1}}^{-1,\otimes}\right) \colon (\mathcal{P}(Y), \cap, Y) \to (\mathcal{P}(X), \cap, X),$$

being equipped with equalities

$$f_{U,V}^{-1,\otimes} \colon f^{-1}(U) \cap f^{-1}(V) \xrightarrow{=} f^{-1}(U \cap V),$$
$$f_{\mathbb{1}}^{-1,\otimes} \colon X \xrightarrow{=} f^{-1}(Y),$$

natural in $U, V \in \mathcal{P}(Y)$.

15. *Interaction With Coproducts.* Let $f \colon X \to X'$ and $g \colon Y \to Y'$ be maps of sets. We have

$$(f \coprod g)^{-1}(U' \coprod V') = f^{-1}(U') \coprod g^{-1}(V')$$

for each $U' \in \mathcal{P}(X')$ and each $V' \in \mathcal{P}(Y')$.

16. *Interaction With Products.* Let $f \colon X \to X'$ and $g \colon Y \to Y'$ be maps of sets. We have

$$(f \boxtimes_{X' \times Y'} g)^{-1}(U' \boxtimes_{X' \times Y'} V') = f^{-1}(U') \boxtimes_{X \times Y} g^{-1}(V')$$

for each $U' \in \mathcal{P}(X')$ and each $V' \in \mathcal{P}(Y')$.

Proof 6.2.4 ► Proof of Proposition 6.2.3

Item 1: Functoriality

Omitted.

Item 2: Triple Adjointness

This follows from Remark 6.1.3, Remark 6.2.2, Remark 6.3.3, and Kan Extensions, **??** of **??**.

Item 3: Interaction With Unions of Families of Subsets

We have

$$\begin{aligned}\bigcup_{U\in f^{-1}(\mathcal{V})} U &= \bigcup_{U\in\{f^{-1}(V)\in\mathcal{P}(X)\mid V\in\mathcal{V}\}} U\\ &= \bigcup_{V\in\mathcal{V}} f^{-1}(V).\end{aligned}$$

This finishes the proof.

Item 4: Interaction With Intersections of Families of Subsets

We have

$$\begin{aligned}\bigcap_{U\in f^{-1}(\mathcal{V})} U &= \bigcap_{U\in\{f^{-1}(V)\in\mathcal{P}(X)\mid V\in\mathcal{V}\}} U\\ &= \bigcap_{V\in\mathcal{V}} f^{-1}(V).\end{aligned}$$

This finishes the proof.

Item 5: Interaction With Binary Unions

See [Pro24v].

Item 6: Interaction With Binary Intersections

See [Pro24u].

Item 7: Interaction With Differences

See [**proof-wiki:preimage-of-set-difference-under-mapping**].

Item 8: Interaction With Complements

See [**proof-wiki:complement-of-preimage-equals-preimage-of-complement**].

Item 9: Interaction With Symmetric Differences

We have

$$\begin{aligned}f^{-1}(U\triangle V) &= f^{-1}((U\cup V)\setminus(U\cap V))\\ &= f^{-1}(U\cup V)\setminus f^{-1}(U\cap V)\\ &= f^{-1}(U)\cup f^{-1}(V)\setminus f^{-1}(U\cap V)\\ &= f^{-1}(U)\cup f^{-1}(V)\setminus f^{-1}(U)\cap f^{-1}(V)\end{aligned}$$

$$= f^{-1}(U) \triangle f^{-1}(V),$$

where we have used:

1. Item 2 of Proposition 3.12.2 for the first equality.
2. Item 7 for the second equality.
3. Item 5 for the third equality.
4. Item 6 for the fourth equality.
5. Item 2 of Proposition 3.12.2 for the fifth equality.

This finishes the proof.

Item 10: Interaction With Internal Homs of Powersets

We have

$$\begin{aligned} f^{-1}([U,V]_Y) &\overset{\text{def}}{=} f^{-1}(U^{\mathsf{c}} \cup V) \\ &= f^{-1}(U^{\mathsf{c}}) \cup f^{-1}(V) \\ &= f^{-1}(U)^{\mathsf{c}} \cup f^{-1}(V) \\ &\overset{\text{def}}{=} \left[f^{-1}(U), f^{-1}(V)\right]_X, \end{aligned}$$

where we have used:

1. Item 8 for the second equality.
2. Item 5 for the third equality.

Since $\mathcal{P}(Y)$ is posetal, naturality is automatic (**??**). This finishes the proof.

Item 11: Preservation of Colimits

This follows from Item 2 and **??**, **??** of **??**.[1]

Item 12: Preservation of Limits

This follows from Item 2 and **??**, **??** of **??**.[2]

Item 13: Symmetric Strict Monoidality With Respect to Unions

This follows from Item 11.

Item 14: Symmetric Strict Monoidality With Respect to Intersections

This follows from Item 12.

Item 15: Interaction With Coproducts

Omitted.

Item 16: Interaction With Products

Omitted.

[1] *Reference:* [Pro24v].
[2] *Reference:* [Pro24u].

Proposition 6.2.5 ▶ Properties of Inverse Images II

Let $f\colon X \longrightarrow Y$ be a function.

1. *Functionality I.* The assignment $f \mapsto f^{-1}$ defines a function

$$(-)^{-1}_{X,Y}\colon \mathsf{Sets}(X,Y) \longrightarrow \mathsf{Sets}(\mathcal{P}(Y),\mathcal{P}(X)).$$

2. *Functionality II.* The assignment $f \mapsto f^{-1}$ defines a function

$$(-)^{-1}_{X,Y}\colon \mathsf{Sets}(X,Y) \longrightarrow \mathsf{Pos}((\mathcal{P}(Y),\subset),(\mathcal{P}(X),\subset)).$$

3. *Interaction With Identities.* For each $X \in \mathrm{Obj}(\mathsf{Sets})$, we have

$$\mathrm{id}_X^{-1} = \mathrm{id}_{\mathcal{P}(X)}\,.$$

4. *Interaction With Composition.* For each pair of composable functions $f\colon X \longrightarrow Y$ and $g\colon Y \longrightarrow Z$, we have

$$(g \circ f)^{-1} = f^{-1} \circ g^{-1}, \qquad \begin{tikzcd} \mathcal{P}(Z) \arrow[r, "g^{-1}"] \arrow[dr, "(g\circ f)^{-1}"'] & \mathcal{P}(Y) \arrow[d, "f^{-1}"] \\ & \mathcal{P}(X). \end{tikzcd}$$

Proof 6.2.6 ▶ Proof of Proposition 6.2.5

Item 1: Functionality I

There is nothing to prove.

Item 2: Functionality II

This follows from Item 1 of Proposition 6.2.3.

Item 3: Interaction With Identities

This follows from Remark 6.2.2 and Categories, **??** of **??**.

Item 4: Interaction With Composition

This follows from Remark 6.2.2 and Categories, **??** of **??**.

6.3 Direct Images With Compact Support

Let $f\colon X \to Y$ be a function.

Definition 6.3.1 ▶ Direct Images With Compact Support

The **direct image with compact support function associated to** f is the function

$$f_!\colon \mathcal{P}(X) \to \mathcal{P}(Y)$$

defined by[1,2]

$$\begin{aligned} f_!(U) &\overset{\text{def}}{=} \left\{ y \in Y \,\middle|\, \begin{array}{l} \text{for each } x \in X, \text{ if we have} \\ f(x) = y, \text{ then } x \in U \end{array} \right\} \\ &= \left\{ y \in Y \,\middle|\, \text{we have } f^{-1}(y) \subset U \right\} \end{aligned}$$

for each $U \in \mathcal{P}(X)$.

[1] *Further Terminology:* The set $f_!(U)$ is called the **direct image with compact support of** U **by** f.

[2] We also have

$$\begin{aligned} f_!(U) &= f_*(U^{\mathsf{c}})^{\mathsf{c}} \\ &\overset{\text{def}}{=} Y \setminus f_*(X \setminus U); \end{aligned}$$

see Item 16 of Proposition 6.3.6.

Notation 6.3.2 ► Further Notation for Direct Images With Compact Support

Sometimes one finds the notation

$$\forall_f \colon \mathcal{P}(X) \to \mathcal{P}(Y)$$

for f_*. This notation comes from the fact that the following statements are equivalent, where $y \in Y$ and $U \in \mathcal{P}(X)$:

- We have $y \in \forall_f(U)$.
- For each $x \in X$, if $y = f(x)$, then $x \in U$.

We will not make use of this notation in the present work.

Remark 6.3.3 ► Unwinding Definition 6.3.1

Identifying $\mathcal{P}(X)$ with $\mathsf{Sets}(X, \{\mathsf{t}, \mathsf{f}\})$ via Item 2 of Proposition 5.1.4, we see that the direct image with compact support function associated to f is equivalently the function

$$f_! \colon \mathcal{P}(X) \to \mathcal{P}(Y)$$

defined by

$$\begin{aligned} f_!(\chi_U) &\overset{\text{def}}{=} \mathsf{Ran}_f(\chi_U) \\ &= \lim\left(\left(\underline{(-_1)} \overset{\to}{\times} f\right) \overset{\text{pr}}{\twoheadrightarrow} X \xrightarrow{\chi_U} \{\text{true}, \text{false}\}\right) \\ &= \lim_{\substack{x \in X \\ f(x) = -_1}} (\chi_U(x)) \\ &= \bigwedge_{\substack{x \in X \\ f(x) = -_1}} (\chi_U(x)). \end{aligned}$$

where we have used Kan Extensions, **??** for the second equality. In other words, we have

$$[f_!(\chi_U)](y) = \bigwedge_{\substack{x \in X \\ f(x) = y}} (\chi_U(x))$$

$$= \begin{cases} \text{true} & \text{if, for each } x \in X \text{ such that} \\ & f(x) = y, \text{ we have } x \in U, \\ \text{false} & \text{otherwise} \end{cases}$$

$$= \begin{cases} \text{true} & \text{if } f^{-1}(y) \subset U \\ \text{false} & \text{otherwise} \end{cases}$$

for each $y \in Y$.

Definition 6.3.4 ► The Image and Complement Parts of $f_!$

Let U be a subset of X.[1,2]

1. The **image part of the direct image with compact support $f_!(U)$ of U** is the set $f_{!,\mathrm{im}}(U)$ defined by

$$\begin{aligned} f_{!,\mathrm{im}}(U) &\overset{\text{def}}{=} f_!(U) \cap \mathrm{Im}(f) \\ &= \left\{ y \in Y \,\middle|\, \begin{array}{l} \text{we have } f^{-1}(y) \subset U \\ \text{and } f^{-1}(y) \neq \emptyset. \end{array} \right\}. \end{aligned}$$

2. The **complement part of the direct image with compact support $f_!(U)$ of U** is the set $f_{!,\mathrm{cp}}(U)$ defined by

$$\begin{aligned} f_{!,\mathrm{cp}}(U) &\overset{\text{def}}{=} f_!(U) \cap (Y \setminus \mathrm{Im}(f)) \\ &= Y \setminus \mathrm{Im}(f) \\ &= \left\{ y \in Y \,\middle|\, \begin{array}{l} \text{we have } f^{-1}(y) \subset U \\ \text{and } f^{-1}(y) = \emptyset. \end{array} \right\} \\ &= \left\{ y \in Y \,\middle|\, f^{-1}(y) = \emptyset \right\}. \end{aligned}$$

[1] Note that we have

$$f_!(U) = f_{!,\mathrm{im}}(U) \cup f_{!,\mathrm{cp}}(U),$$

as

$$\begin{aligned} f_!(U) &= f_!(U) \cap Y \\ &= f_!(U) \cap (\mathrm{Im}(f) \cup (Y \setminus \mathrm{Im}(f))) \\ &= (f_!(U) \cap \mathrm{Im}(f)) \cup (f_!(U) \cap (Y \setminus \mathrm{Im}(f))) \end{aligned}$$

$$\stackrel{\text{def}}{=} f_{!,\mathrm{im}}(U) \cup f_{!,\mathrm{cp}}(U).$$

[2] In terms of the meet computation of $f_!(U)$ of Remark 6.3.3, namely

$$f_!(\chi_U) = \bigwedge_{\substack{x \in X \\ f(x) = -_1}} (\chi_U(x)),$$

we see that $f_{!,\mathrm{im}}$ corresponds to meets indexed over nonempty sets, while $f_{!,\mathrm{cp}}$ corresponds to meets indexed over the empty set.

Example 6.3.5 ► Examples of Direct Images With Compact Support

Here are some examples of direct images with compact support.

1. *The Multiplication by Two Map on the Natural Numbers.* Consider the function $f\colon \mathbb{N} \to \mathbb{N}$ given by

$$f(n) \stackrel{\text{def}}{=} 2n$$

for each $n \in \mathbb{N}$. Since f is injective, we have

$$\begin{aligned} f_{!,\mathrm{im}}(U) &= f_*(U) \\ f_{!,\mathrm{cp}}(U) &= \{\text{odd natural numbers}\} \end{aligned}$$

for any $U \subset \mathbb{N}$.

2. *Parabolas.* Consider the function $f\colon \mathbb{R} \to \mathbb{R}$ given by

$$f(x) \stackrel{\text{def}}{=} x^2$$

for each $x \in \mathbb{R}$. We have

$$f_{!,\mathrm{cp}}(U) = \mathbb{R}_{<0}$$

for any $U \subset \mathbb{R}$. Moreover, since $f^{-1}(x) = \left\{-\sqrt{x}, \sqrt{x}\right\}$, we have e.g.:

$$\begin{aligned} f_{!,\mathrm{im}}([0,1]) &= \{0\}, \\ f_{!,\mathrm{im}}([-1,1]) &= [0,1], \\ f_{!,\mathrm{im}}([1,2]) &= \emptyset, \\ f_{!,\mathrm{im}}([-2,-1] \cup [1,2]) &= [1,4]. \end{aligned}$$

3. *Circles.* Consider the function $f\colon \mathbb{R}^2 \longrightarrow \mathbb{R}$ given by

$$f(x,y) \overset{\text{def}}{=} x^2 + y^2$$

for each $(x,y) \in \mathbb{R}^2$. We have

$$f_{!,\mathsf{cp}}(U) = \mathbb{R}_{<0}$$

for any $U \subset \mathbb{R}^2$, and since

$$f^{-1}(r) = \begin{cases} \text{a circle of radius } r \text{ about the origin} & \text{if } r > 0, \\ \{(0,0)\} & \text{if } r = 0, \\ \emptyset & \text{if } r < 0, \end{cases}$$

we have e.g.:

$$f_{!,\mathsf{im}}([-1,1] \times [-1,1]) = [0,1],$$
$$f_{!,\mathsf{im}}(([-1,1] \times [-1,1]) \setminus [-1,1] \times \{0\}) = \emptyset.$$

Proposition 6.3.6 ► Properties of Direct Images With Compact Support I

Let $f\colon X \longrightarrow Y$ be a function.

1. *Functoriality.* The assignment $U \mapsto f_!(U)$ defines a functor

$$f_!\colon (\mathcal{P}(X), \subset) \longrightarrow (\mathcal{P}(Y), \subset).$$

In particular, for each $U, V \in \mathcal{P}(X)$, the following condition is satisfied:

($\star$) If $U \subset V$, then $f_!(U) \subset f_!(V)$.

2. *Triple Adjointness.* We have a triple adjunction

$$(f_* \dashv f^{-1} \dashv f_!)\colon \quad \begin{tikzcd} \mathcal{P}(X) \arrow[r, bend left=60, "f_*"'] \arrow[r, bend right=60, "f_!"] \arrow[r, phantom, "\perp"] & \mathcal{P}(Y) \arrow[l, "f^{-1}"] \end{tikzcd},$$

witnessed by:

(a) Units and counits of the form

$$\begin{aligned} \mathrm{id}_{\mathcal{P}(X)} &\hookrightarrow f^{-1} \circ f_*, & \mathrm{id}_{\mathcal{P}(Y)} &\hookrightarrow f_! \circ f^{-1}, \\ f_* \circ f^{-1} &\hookrightarrow \mathrm{id}_{\mathcal{P}(Y)}, & f^{-1} \circ f_! &\hookrightarrow \mathrm{id}_{\mathcal{P}(X)}, \end{aligned}$$

having components of the form

$$\begin{aligned} U &\subset f^{-1}(f_*(U)), & V &\subset f_!\big(f^{-1}(V)\big), \\ f_*\big(f^{-1}(V)\big) &\subset V, & f^{-1}(f_!(U)) &\subset U \end{aligned}$$

indexed by $U \in \mathcal{P}(X)$ and $V \in \mathcal{P}(Y)$.

(b) Bijections of sets

$$\begin{aligned} \mathrm{Hom}_{\mathcal{P}(Y)}(f_*(U), V) &\cong \mathrm{Hom}_{\mathcal{P}(X)}\big(U, f^{-1}(V)\big), \\ \mathrm{Hom}_{\mathcal{P}(X)}\big(f^{-1}(U), V\big) &\cong \mathrm{Hom}_{\mathcal{P}(X)}(U, f_!(V)), \end{aligned}$$

natural in $U \in \mathcal{P}(X)$ and $V \in \mathcal{P}(Y)$ and (respectively) $V \in \mathcal{P}(X)$ and $U \in \mathcal{P}(Y)$. In particular:

i. The following conditions are equivalent:
   A. We have $f_*(U) \subset V$.
   B. We have $U \subset f^{-1}(V)$.

ii. The following conditions are equivalent:
   A. We have $f^{-1}(U) \subset V$.
   B. We have $U \subset f_!(V)$.

3. *Interaction With Unions of Families of Subsets.* The diagram

$$\begin{tikzcd} \mathcal{P}(\mathcal{P}(X)) \arrow[r, "(f_!)_!"] \arrow[d, "\cup"'] & \mathcal{P}(\mathcal{P}(Y)) \arrow[d, "\cup"] \\ \mathcal{P}(X) \arrow[r, "f_!"'] & \mathcal{P}(Y) \end{tikzcd}$$

commutes, i.e. we have

$$\bigcup_{U \in \mathcal{U}} f_!(U) = \bigcup_{V \in f_!(\mathcal{U})} V$$

for each $\mathcal{U} \in \mathcal{P}(X)$, where $f_!(\mathcal{U}) \overset{\text{def}}{=} (f_!)_!(\mathcal{U})$.

4. *Interaction With Intersections of Families of Subsets.* The diagram

$$\begin{tikzcd}
\mathcal{P}(\mathcal{P}(X)) \arrow[r, "(f_!)_!"] \arrow[d, "\cap"'] & \mathcal{P}(\mathcal{P}(Y)) \arrow[d, "\cap"] \\
\mathcal{P}(X) \arrow[r, "f_!"'] & \mathcal{P}(Y)
\end{tikzcd}$$

commutes, i.e. we have

$$\bigcap_{U \in \mathcal{U}} f_!(U) = \bigcap_{V \in f_!(\mathcal{U})} V$$

for each $\mathcal{U} \in \mathcal{P}(X)$, where $f_!(\mathcal{U}) \overset{\text{def}}{=} (f_!)_!(\mathcal{U})$.

5. *Interaction With Binary Unions.* Let $f\colon X \to Y$ be a function. We have a natural transformation

$$\begin{tikzcd}
\mathcal{P}(X) \times \mathcal{P}(X) \arrow[r, "f_! \times f_!"] \arrow[d, "\cup"'] \arrow[dr, phantom, "\circlearrowright"] & \mathcal{P}(Y) \times \mathcal{P}(Y) \arrow[d, "\cup"] \\
\mathcal{P}(X) \arrow[r, "f_!"'] & \mathcal{P}(Y)
\end{tikzcd}$$

with components

$$f_!(U) \cup f_!(V) \subset f_!(U \cup V)$$

indexed by $U, V \in \mathcal{P}(X)$.

6. *Interaction With Binary Intersections.* The diagram

$$\begin{tikzcd}
\mathcal{P}(X) \times \mathcal{P}(X) \arrow[r, "f_! \times f_!"] \arrow[d, "\cap"'] & \mathcal{P}(Y) \times \mathcal{P}(Y) \arrow[d, "\cap"] \\
\mathcal{P}(X) \arrow[r, "f_!"'] & \mathcal{P}(Y)
\end{tikzcd}$$

commutes, i.e. we have

$$f_!(U) \cap f_!(V) = f_!(U \cap V)$$

for each $U, V \in \mathcal{P}(X)$.

7. *Interaction With Complements.* The diagram

$$\begin{tikzcd} \mathcal{P}(X)^{\mathsf{op}} \arrow[r, "f_*^{\mathsf{op}}"] \arrow[d, "(-)^{\mathsf{c}}"'] & \mathcal{P}(Y)^{\mathsf{op}} \arrow[d, "(-)^{\mathsf{c}}"] \\ \mathcal{P}(X) \arrow[r, "f_!"'] & \mathcal{P}(Y) \end{tikzcd}$$

commutes, i.e. we have

$$f_!(U^{\mathsf{c}}) = f_*(U)^{\mathsf{c}}$$

for each $U \in \mathcal{P}(X)$.

8. *Interaction With Symmetric Differences.* We have a natural transformation

$$\begin{tikzcd} \mathcal{P}(X)^{\mathsf{op}} \times \mathcal{P}(X) \arrow[r, "f_!^{\mathsf{op}} \times f_!"] \arrow[d, "\triangle"'] \arrow[dr, phantom, "\supset"] & \mathcal{P}(Y)^{\mathsf{op}} \times \mathcal{P}(Y) \arrow[d, "\triangle"] \\ \mathcal{P}(X) \arrow[r, "f_!"'] & \mathcal{P}(Y) \end{tikzcd}$$

with components

$$f_!(U \triangle V) \subset f_!(U) \triangle f_!(V)$$

indexed by $U, V \in \mathcal{P}(X)$.

9. *Interaction With Internal Homs of Powersets.* We have a natural transformation

$$\begin{tikzcd} \mathcal{P}(X)^{\mathsf{op}} \times \mathcal{P}(X) \arrow[r, "f_*^{\mathsf{op}} \times f_!"] \arrow[d, "{[-_1,-_2]_X}"'] \arrow[dr, phantom, "\supset"] & \mathcal{P}(Y)^{\mathsf{op}} \times \mathcal{P}(Y) \arrow[d, "{[-_1,-_2]_Y}"] \\ \mathcal{P}(X) \arrow[r, "f_!"'] & \mathcal{P}(Y) \end{tikzcd}$$

with components

$$[f_*(U), f_!(V)]_Y \subset f_!([U, V]_X)$$

indexed by $U, V \in \mathcal{P}(X)$.

10. *Lax Preservation of Colimits.* We have an inclusion of sets

$$\bigcup_{i\in I} f_!(U_i) \subset f_!\left(\bigcup_{i\in I} U_i\right),$$

natural in $\{U_i\}_{i\in I} \in \mathcal{P}(X)^{\times I}$. In particular, we have inclusions

$$f_!(U) \cup f_!(V) \hookrightarrow f_!(U \cup V),$$
$$\emptyset \hookrightarrow f_!(\emptyset),$$

natural in $U, V \in \mathcal{P}(X)$.

11. *Preservation of Limits.* We have an equality of sets

$$f_!\left(\bigcap_{i\in I} U_i\right) = \bigcap_{i\in I} f_!(U_i),$$

natural in $\{U_i\}_{i\in I} \in \mathcal{P}(X)^{\times I}$. In particular, we have equalities

$$f^{-1}(U \cap V) = f_!(U) \cap f^{-1}(V),$$
$$f_!(X) = Y,$$

natural in $U, V \in \mathcal{P}(X)$.

12. *Symmetric Lax Monoidality With Respect to Unions.* The direct image with compact support function of Item 1 has a symmetric lax monoidal structure

$$\left(f_!, f_!^{\otimes}, f_{!|\mathbb{1}}^{\otimes}\right) \colon (\mathcal{P}(X), \cup, \emptyset) \to (\mathcal{P}(Y), \cup, \emptyset),$$

being equipped with inclusions

$$f_{!|U,V}^{\otimes} \colon f_!(U) \cup f_!(V) \hookrightarrow f_!(U \cup V),$$
$$f_{!|\mathbb{1}}^{\otimes} \colon \emptyset \hookrightarrow f_!(\emptyset),$$

natural in $U, V \in \mathcal{P}(X)$.

13. *Symmetric Strict Monoidality With Respect to Intersections.* The direct image function of Item 1 has a symmetric strict monoidal structure

$$\left(f_!, f_!^{\otimes}, f_{!|\mathbb{1}}^{\otimes}\right) \colon (\mathcal{P}(X), \cap, X) \to (\mathcal{P}(Y), \cap, Y),$$

being equipped with equalities

$$\begin{aligned} f_{!|U,V}^{\otimes} &\colon f_!(U \cap V) \xrightarrow{=} f_!(U) \cap f_!(V), \\ f_{!|\mathbb{1}}^{\otimes} &\colon f_!(X) \xrightarrow{=} Y, \end{aligned}$$

natural in $U, V \in \mathcal{P}(X)$.

14. *Interaction With Coproducts.* Let $f \colon X \to X'$ and $g \colon Y \to Y'$ be maps of sets. We have

$$(f \coprod g)_!(U \coprod V) = f_!(U) \coprod g_!(V)$$

for each $U \in \mathcal{P}(X)$ and each $V \in \mathcal{P}(Y)$.

15. *Interaction With Products.* Let $f \colon X \to X'$ and $g \colon Y \to Y'$ be maps of sets. We have

$$(f \boxtimes_{X \times Y} g)_!(U \boxtimes_{X \times Y} V) = f_!(U) \boxtimes_{X' \times Y'} g_!(V)$$

for each $U \in \mathcal{P}(X)$ and each $V \in \mathcal{P}(Y)$.

16. *Relation to Direct Images.* We have

$$\begin{aligned} f_!(U) &= f_*(U^{\mathsf{c}})^{\mathsf{c}} \\ &= Y \setminus f_*(X \setminus U) \end{aligned}$$

for each $U \in \mathcal{P}(X)$.

17. *Interaction With Injections.* If f is injective, then we have

$$\begin{aligned} f_{!,\mathsf{im}}(U) &= f_*(U), \\ f_{!,\mathsf{cp}}(U) &= Y \setminus \mathrm{Im}(f), \\ f_!(U) &= f_{!,\mathsf{im}}(U) \cup f_{!,\mathsf{cp}}(U) \\ &= f_*(U) \cup (Y \setminus \mathrm{Im}(f)) \end{aligned}$$

for each $U \in \mathcal{P}(X)$.

18. *Interaction With Surjections.* If f is surjective, then we have

$$\begin{aligned} f_{!,\mathrm{im}}(U) &\subset f_*(U), \\ f_{!,\mathrm{cp}}(U) &= \emptyset, \\ f_!(U) &\subset f_*(U) \end{aligned}$$

for each $U \in \mathcal{P}(X)$.

Proof 6.3.7 ▶ Proof of Proposition 6.3.6

Item 1: Functoriality

Omitted.

Item 2: Triple Adjointness

This follows from Remark 6.1.3, Remark 6.2.2, Remark 6.3.3, and Kan Extensions, **??** of **??**.

Item 3: Interaction With Unions of Families of Subsets

We have

$$\begin{aligned} \bigcup_{V \in f_!(\mathcal{U})} V &= \bigcup_{V \in \{f_!(U) \in \mathcal{P}(X) \mid U \in \mathcal{U}\}} V \\ &= \bigcup_{U \in \mathcal{U}} f_!(U). \end{aligned}$$

This finishes the proof.

Item 4: Interaction With Intersections of Families of Subsets

We have

$$\begin{aligned} \bigcap_{V \in f_!(\mathcal{U})} V &= \bigcap_{V \in \{f_!(U) \in \mathcal{P}(X) \mid U \in \mathcal{U}\}} V \\ &= \bigcap_{U \in \mathcal{U}} f_!(U). \end{aligned}$$

This finishes the proof.

Item 5: Interaction With Binary Unions

We have

$$\begin{aligned} f_!(U) \cup f_!(V) &= f_*(U^c)^c \cup f_*(V^c)^c \\ &= (f_*(U^c) \cap f_*(V^c))^c \\ &\subset (f_*(U^c \cap V^c))^c \\ &= f_*\big((U \cup V)^c\big)^c \\ &= f_!(U \cup V), \end{aligned}$$

where:

1. We have used Item 16 for the first equality.
2. We have used Item 2 of Proposition 3.11.2 for the second equality.
3. We have used Item 6 of Proposition 6.1.4 for the third equality.
4. We have used Item 2 of Proposition 3.11.2 for the fourth equality.
5. We have used Item 16 for the last equality.

This finishes the proof.

Item 7: Interaction With Complements

Omitted.

Item 8: Interaction With Symmetric Differences

Omitted.

Item 9: Interaction With Internal Homs of Powersets

We have

$$\begin{aligned} \big[f_*(U), f^!(V)\big]_X &\overset{\text{def}}{=} f_*(U)^c \cup f_!(V) \\ &= f_!(U^c) \cup f_!(V) \\ &\subset f_!(U^c \cup V) \\ &\overset{\text{def}}{=} f_!([U, V]_X), \end{aligned}$$

where we have used:

1. Item 7 of Proposition 6.3.6 for the second equality.
2. Item 5 of Proposition 6.3.6 for the inclusion.

Since $\mathcal{P}(X)$ is posetal, naturality is automatic (**??**). This finishes the proof.

Item 6: Interaction With Binary Intersections

This follows from Item 11.

Item 10: Lax Preservation of Colimits

Omitted.

Item 11: Preservation of Limits

This follows from Item 2 and **??**, **??** of **??**.

Item 12: Symmetric Lax Monoidality With Respect to Unions

This follows from Item 10.

Item 13: Symmetric Strict Monoidality With Respect to Intersections

This follows from Item 11.

Item 14: Interaction With Coproducts

Omitted.

Item 15: Interaction With Products

Omitted.

Item 16: Relation to Direct Images

We claim that $f_!(U) = Y \setminus f_*(X \setminus U)$.

- *The First Implication.* We claim that

$$f_!(U) \subset Y \setminus f_*(X \setminus U).$$

  Let $y \in f_!(U)$. We need to show that $y \notin f_*(X \setminus U)$, i.e. that there is no $x \in X \setminus U$ such that $f(x) = y$.

  This is indeed the case, as otherwise we would have $x \in f^{-1}(y)$ and $x \notin U$, contradicting $f^{-1}(y) \subset U$ (which holds since $y \in f_!(U)$).

  Thus $y \in Y \setminus f_*(X \setminus U)$.

- *The Second Implication.* We claim that

$$Y \setminus f_*(X \setminus U) \subset f_!(U).$$

Let $y \in Y \setminus f_*(X \setminus U)$. We need to show that $y \in f_!(U)$, i.e. that $f^{-1}(y) \subset U$.

Since $y \notin f_*(X \setminus U)$, there exists no $x \in X \setminus U$ such that $y = f(x)$, and hence $f^{-1}(y) \subset U$.

Thus $y \in f_!(U)$.

This finishes the proof of Item 16.

Item 17: Interaction With Injections

Omitted.

Item 18: Interaction With Surjections

Omitted. ∎

Proposition 6.3.8 ▶ Properties of Direct Images With Compact Support II

Let $f\colon X \to B$ be a function.

1. *Functionality I.* The assignment $f \mapsto f_!$ defines a function

$$(-)_{!|X,Y}\colon \mathsf{Sets}(X, Y) \to \mathsf{Sets}(\mathcal{P}(X), \mathcal{P}(Y)).$$

2. *Functionality II.* The assignment $f \mapsto f_!$ defines a function

$$(-)_{!|X,Y}\colon \mathsf{Sets}(X, Y) \to \mathsf{Pos}((\mathcal{P}(X), \subset), (\mathcal{P}(Y), \subset)).$$

3. *Interaction With Identities.* For each $X \in \mathrm{Obj}(\mathsf{Sets})$, we have

$$(\mathrm{id}_X)_! = \mathrm{id}_{\mathcal{P}(X)}.$$

4. *Interaction With Composition.* For each pair of composable functions $f\colon X \to Y$ and $g\colon Y \to Z$, we have

$$(g \circ f)_! = g_! \circ f_!, \qquad \begin{tikzcd} \mathcal{P}(X) \arrow[r, "f_!"] \arrow[dr, "(g\circ f)_!"'] & \mathcal{P}(Y) \arrow[d, "g_!"] \\ & \mathcal{P}(Z). \end{tikzcd}$$

Item 1: Functionality I

There is nothing to prove.

Item 2: Functionality II

This follows from Item 1 of Proposition 6.3.6.

Item 3: Interaction With Identities

This follows from Remark 6.3.3 and Kan Extensions, **??** of **??**.

Item 4: Interaction With Composition

This follows from Remark 6.3.3 and Kan Extensions, **??** of **??**.

6.4 A Six-Functor Formalism for Sets

Remark 6.4.1 ► A Six-Functor Formalism for Sets

The assignment $X \mapsto \mathcal{P}(X)$ together with the functors $f_!$, f^{-1}, and f_* of Item 1 of Proposition 6.1.4, Item 1 of Proposition 6.2.3, and Item 1 of Proposition 6.3.6, and the functors

$$\begin{aligned} -_1 \cap -_2 &\colon \mathcal{P}(X) \times \mathcal{P}(X) \to \mathcal{P}(X), \\ [-_1, -_2]_X &\colon \mathcal{P}(X)^{\mathsf{op}} \times \mathcal{P}(X) \to \mathcal{P}(X) \end{aligned}$$

of Item 1 of Proposition 3.9.2 and Item 1 of Proposition 4.7.4 satisfy several properties reminiscent of a six functor formalism in the sense of **??**.
We collect these properties in Proposition 6.4.2 below.

Proposition 6.4.2 ► A Six Functor Formalism for Sets

Let X be a set.

1. *The Beck–Chevalley Condition.* Let

$$\begin{tikzcd} X \times_Z Y \arrow[r, "\mathrm{pr}_2"] \arrow[d, "\mathrm{pr}_1"'] \arrow[dr, phantom, "\lrcorner", very near start] & Y \arrow[d, "g"] \\ X \arrow[r, "f"'] & Z \end{tikzcd}$$

be a pullback diagram in Sets. We have

$$\begin{tikzcd} \mathcal{P}(X) \arrow[r, "\mathrm{pr}_1^{-1}"] \arrow[d, "f_*"'] & \mathcal{P}(X \times_Z Y) \arrow[d, "(\mathrm{pr}_2)_*"] \\ \mathcal{P}(Z) \arrow[r, "g^{-1}"'] & \mathcal{P}(Y), \end{tikzcd} \qquad \begin{aligned} g^{-1} \circ f_* &= (\mathrm{pr}_2)_* \circ \mathrm{pr}_1^{-1}, \\ f^{-1} \circ g_* &= (\mathrm{pr}_1)_* \circ \mathrm{pr}_2^{-1}, \end{aligned} \qquad \begin{tikzcd} \mathcal{P}(X) \arrow[r, "\mathrm{pr}_2^{-1}"] \arrow[d, "g_*"'] & \mathcal{P}(X \times_Z Y) \arrow[d, "(\mathrm{pr}_1)_*"] \\ \mathcal{P}(Z) \arrow[r, "f^{-1}"'] & \mathcal{P}(Y). \end{tikzcd}$$

2. *The Projection Formula I.* The diagram

$$\begin{tikzcd} & \mathcal{P}(X) \times \mathcal{P}(X) \arrow[dr, "\cap"] & \\ \mathcal{P}(X) \times \mathcal{P}(Y) \arrow[ur, "\mathrm{id}_{\mathcal{P}(X)} \times f^{-1}"] \arrow[dr, "f_* \times \mathrm{id}_{\mathcal{P}(Y)}"'] & & \mathcal{P}(X) \arrow[dl, "f_*"] \\ & \mathcal{P}(Y) \times \mathcal{P}(Y) \arrow[r, "\cap"'] & \mathcal{P}(Y), \end{tikzcd}$$

commutes, i.e. we have

$$f_*\big(U \cap f^{-1}(V)\big) = f_*(U) \cap V$$

for each $U \in \mathcal{P}(X)$ and each $V \in \mathcal{P}(Y)$.

3. *The Projection Formula II.* We have a natural transformation

$$\begin{tikzcd} & \mathcal{P}(X) \times \mathcal{P}(X) \arrow[dr, "\cap"] & \\ \mathcal{P}(X) \times \mathcal{P}(Y) \arrow[ur, "\mathrm{id}_{\mathcal{P}(X)} \times f^{-1}"] \arrow[dr, "f_! \times \mathrm{id}_{\mathcal{P}(Y)}"'] & \circlearrowright & \mathcal{P}(X) \arrow[dl, "f_!"] \\ & \mathcal{P}(Y) \times \mathcal{P}(Y) \arrow[r, "\cap"'] & \mathcal{P}(Y), \end{tikzcd}$$

with components

$$f_!(U) \cap V \subset f_!\big(U \cap f^{-1}(V)\big)$$

indexed by $U \in \mathcal{P}(X)$ and $V \in \mathcal{P}(Y)$.

4. *Strong Closed Monoidality.* The diagram

$$\begin{tikzcd}
\mathcal{P}(Y)^{\mathsf{op}} \times \mathcal{P}(Y) \arrow[r, "f^{-1,\mathsf{op}} \times f^{-1}"] \arrow[d, "{[-_1,-_2]_Y}"'] & \mathcal{P}(X)^{\mathsf{op}} \times \mathcal{P}(X) \arrow[d, "{[-_1,-_2]_X}"] \\
\mathcal{P}(Y) \arrow[r, "f^{-1}"'] & \mathcal{P}(X)
\end{tikzcd}$$

commutes, i.e. we have an equality of sets

$$f^{-1}([U,V]_Y) = \left[f^{-1}(U), f^{-1}(V)\right]_X,$$

natural in $U, V \in \mathcal{P}(X)$.

5. *The External Tensor Product.* We have an external tensor product

$$-_1 \boxtimes_{X\times Y} -_2 \colon \mathcal{P}(X) \times \mathcal{P}(Y) \to \mathcal{P}(X \times Y)$$

given by

$$\begin{aligned} U \boxtimes_{X\times Y} V &\overset{\text{def}}{=} \mathrm{pr}_1^{-1}(U) \cap \mathrm{pr}_2^{-1}(V) \\ &= \{(u,v) \in X \times Y \mid u \in U \text{ and } v \in V\}. \end{aligned}$$

This is the same map as the one in Item 5 of Proposition 4.1.4. Moreover, the following conditions are satisfied:

(a) *Interaction With Direct Images.* Let $f \colon X \to X'$ and $g \colon Y \to Y'$ be functions. The diagram

$$\begin{tikzcd}
\mathcal{P}(X) \times \mathcal{P}(Y) \arrow[r, "f_* \times g_*"] \arrow[d, "\boxtimes_{X\times Y}"'] & \mathcal{P}(X') \times \mathcal{P}(Y') \arrow[d, "\boxtimes_{X'\times Y'}"] \\
\mathcal{P}(X \times Y) \arrow[r, "f_* \times g_*"'] & \mathcal{P}(X' \times Y')
\end{tikzcd}$$

commutes, i.e. we have

$$[f_* \times g_*](U \boxtimes_{X\times Y} V) = f_*(U) \boxtimes_{X'\times Y'} g_*(V)$$

for each $(U,V) \in \mathcal{P}(X) \times \mathcal{P}(Y)$.

(b) *Interaction With Inverse Images.* Let $f\colon X \to X'$ and $g\colon Y \to Y'$ be functions. The diagram

$$\begin{tikzcd}
\mathcal{P}(X') \times \mathcal{P}(Y') \arrow[r, "f^{-1}\times g^{-1}"] \arrow[d, "\boxtimes_{X'\times Y'}"'] & \mathcal{P}(X) \times \mathcal{P}(Y) \arrow[d, "\boxtimes_{X\times Y}"] \\
\mathcal{P}(X' \times Y') \arrow[r, "f^{-1}\times g^{-1}"'] & \mathcal{P}(X \times Y)
\end{tikzcd}$$

commutes, i.e. we have

$$\left[f^{-1} \times g^{-1}\right](U \boxtimes_{X'\times Y'} V) = f^{-1}(U) \boxtimes_{X\times Y} g^{-1}(V)$$

for each $(U, V) \in \mathcal{P}(X') \times \mathcal{P}(Y')$.

(c) *Interaction With Direct Images With Compact Support.* Let $f\colon X \to X'$ and $g\colon Y \to Y'$ be functions. The diagram

$$\begin{tikzcd}
\mathcal{P}(X) \times \mathcal{P}(Y) \arrow[r, "f_!\times g_!"] \arrow[d, "\boxtimes_{X\times Y}"'] & \mathcal{P}(X') \times \mathcal{P}(Y') \arrow[d, "\boxtimes_{X'\times Y'}"] \\
\mathcal{P}(X \times Y) \arrow[r, "f_!\times g_!"'] & \mathcal{P}(X' \times Y')
\end{tikzcd}$$

commutes, i.e. we have

$$[f_! \times g_!](U \boxtimes_{X\times Y} V) = f_!(U) \boxtimes_{X'\times Y'} g_!(V)$$

for each $(U, V) \in \mathcal{P}(X) \times \mathcal{P}(Y)$.

(d) *Interaction With Diagonals.* The diagram

$$\begin{tikzcd}
\mathcal{P}(X) \times \mathcal{P}(X) \arrow[r, "\boxtimes_{X\times X}"] \arrow[dr, "\cap"'] & \mathcal{P}(X \times X) \arrow[d, "\Delta_X^{-1}"] \\
& \mathcal{P}(X),
\end{tikzcd}$$

i.e. we have

$$U \cap V = \Delta_X^{-1}(U \boxtimes_{X\times X} V)$$

for each $U, V \in \mathcal{P}(X)$.

6. *The Dualisation Functor.* We have a functor

$$D_X \colon \mathcal{P}(X)^{\mathsf{op}} \to \mathcal{P}(X)$$

given by

$$\begin{aligned} D_X(U) &\overset{\text{def}}{=} [U, \emptyset]_X \\ &\overset{\text{def}}{=} U^{\mathsf{c}} \end{aligned}$$

for each $U \in \mathcal{P}(X)$, as in Item 5 of Proposition 4.7.4, satisfying the following conditions:

(a) *Duality.* We have

$$D_X(D_X(U)) = U, \qquad \begin{tikzcd} \mathcal{P}(X) \arrow[r, "D_X"] \arrow[dr, "\mathrm{id}_{\mathcal{P}(X)}"'] & \mathcal{P}(X) \arrow[d, "D_X"] \\ & \mathcal{P}(X). \end{tikzcd}$$

(b) *Duality.* The diagram

$$\begin{tikzcd} & \mathcal{P}(X)^{\mathsf{op}} \times \mathcal{P}(X)^{\mathsf{op}} \arrow[r, "\cap^{\mathsf{op}}"] & \mathcal{P}(X)^{\mathsf{op}} \arrow[dr, "D_X"] & \\ \mathcal{P}(X)^{\mathsf{op}} \times \mathcal{P}(X) \arrow[ur, "\mathrm{id}_{\mathcal{P}(X)^{\mathsf{op}}} \times D_X"] \arrow[rrr, "{[-_1,-_2]_X}"'] & & & \mathcal{P}(X) \end{tikzcd}$$

commutes, i.e. we have

$$\underbrace{D_X(U \cap D_X(V))}_{\overset{\text{def}}{=}[U \cap [V,\emptyset]_X, \emptyset]_X} = [U, V]_X$$

for each $U, V \in \mathcal{P}(X)$.

(c) *Interaction With Direct Images.* The diagram

$$\begin{tikzcd} \mathcal{P}(X)^{\mathsf{op}} \arrow[r, "f_!^{\mathsf{op}}"] \arrow[d, "D_X"'] & \mathcal{P}(Y)^{\mathsf{op}} \arrow[d, "D_Y"] \\ \mathcal{P}(X) \arrow[r, "f_*"'] & \mathcal{P}(Y) \end{tikzcd}$$

commutes, i.e. we have

$$f_*(D_X(U)) = D_Y(f_!(U))$$

for each $U \in \mathcal{P}(X)$.

(d) *Interaction With Inverse Images.* The diagram

$$\begin{tikzcd}
\mathcal{P}(Y)^{\mathsf{op}} \arrow[r, "f^{-1,\mathsf{op}}"] \arrow[d, "D_Y"'] & \mathcal{P}(X)^{\mathsf{op}} \arrow[d, "D_X"] \\
\mathcal{P}(Y) \arrow[r, "f^{-1}"'] & \mathcal{P}(X)
\end{tikzcd}$$

commutes, i.e. we have

$$f^{-1}(D_Y(U)) = D_X\big(f^{-1}(U)\big)$$

for each $U \in \mathcal{P}(X)$.

(e) *Interaction With Direct Images With Compact Support.* The diagram

$$\begin{tikzcd}
\mathcal{P}(X)^{\mathsf{op}} \arrow[r, "f_*^{\mathsf{op}}"] \arrow[d, "D_X"'] & \mathcal{P}(Y)^{\mathsf{op}} \arrow[d, "D_Y"] \\
\mathcal{P}(X) \arrow[r, "f_!"'] & \mathcal{P}(Y)
\end{tikzcd}$$

commutes, i.e. we have

$$f_!(D_X(U)) = D_Y(f_*(U))$$

for each $U \in \mathcal{P}(X)$.

Proof 6.4.3 ▶ Proof of Proposition 6.4.2

Item 1: The Beck–Chevalley Condition

We have

$$\begin{aligned}
\left[g^{-1} \circ f_*\right](U) &\overset{\text{def}}{=} g^{-1}(f_*(U)) \\
&\overset{\text{def}}{=} \{y \in Y \mid g(y) \in f_*(U)\} \\
&= \left\{ y \in Y \;\middle|\; \begin{array}{l} \text{there exists some } x \in U \\ \text{such that } f(x) = g(y) \end{array} \right\} \\
&= \left\{ y \in Y \;\middle|\; \begin{array}{l} \text{there exists some} \\ (x,y) \in \{(x,y) \in X \times_Z Y \mid x \in U\} \end{array} \right\} \\
&= \left\{ y \in Y \;\middle|\; \begin{array}{l} \text{there exists some} \\ (x,y) \in \{(x,y) \in X \times_Z Y \mid x \in U\} \\ \text{such that } y = y \end{array} \right\} \\
&= \left\{ y \in Y \;\middle|\; \begin{array}{l} \text{there exists some} \\ (x,y) \in \{(x,y) \in X \times_Z Y \mid x \in U\} \\ \text{such that } \mathrm{pr}_2(x,y) = y \end{array} \right\} \\
&\overset{\text{def}}{=} (\mathrm{pr}_2)_*(\{(x,y) \in X \times_Z Y \mid x \in U\}) \\
&= (\mathrm{pr}_2)_*(\{(x,y) \in X \times_Z Y \mid \mathrm{pr}_1(x,y) \in U\}) \\
&\overset{\text{def}}{=} (\mathrm{pr}_2)_*\left(\mathrm{pr}_1^{-1}(U)\right) \\
&\overset{\text{def}}{=} \left[(\mathrm{pr}_2)_* \circ \mathrm{pr}_1^{-1}\right](U)
\end{aligned}$$

for each $U \in \mathcal{P}(X)$. Therefore, we have

$$g^{-1} \circ f_* = (\mathrm{pr}_2)_* \circ \mathrm{pr}_1^{-1}.$$

For the second equality, we have

$$\begin{aligned}
\left[f^{-1} \circ g_*\right](U) &\overset{\text{def}}{=} f^{-1}(g_*(U)) \\
&\overset{\text{def}}{=} \{x \in X \mid f(x) \in g_*(V)\} \\
&= \left\{ x \in X \;\middle|\; \begin{array}{l} \text{there exists some } y \in V \\ \text{such that } f(x) = g(y) \end{array} \right\}
\end{aligned}$$

$$
\begin{aligned}
&= \left\{ x \in X \;\middle|\; \begin{array}{l} \text{there exists some} \\ (x,y) \in \{(x,y) \in X \times_Z Y \mid y \in V\} \end{array} \right\} \\
&= \left\{ x \in X \;\middle|\; \begin{array}{l} \text{there exists some} \\ (x,y) \in \{(x,y) \in X \times_Z Y \mid y \in V\} \\ \text{such that } x = x \end{array} \right\} \\
&= \left\{ x \in X \;\middle|\; \begin{array}{l} \text{there exists some} \\ (x,y) \in \{(x,y) \in X \times_Z Y \mid y \in V\} \\ \text{such that } \mathrm{pr}_1(x,y) = x \end{array} \right\} \\
&\overset{\text{def}}{=} (\mathrm{pr}_1)_*(\{(x,y) \in X \times_Z Y \mid y \in V\}) \\
&= (\mathrm{pr}_1)_*(\{(x,y) \in X \times_Z Y \mid \mathrm{pr}_2(x,y) \in V\}) \\
&\overset{\text{def}}{=} (\mathrm{pr}_1)_*\left(\mathrm{pr}_2^{-1}(V)\right) \\
&\overset{\text{def}}{=} \left[(\mathrm{pr}_1)_* \circ \mathrm{pr}_2^{-1}\right](V)
\end{aligned}
$$

for each $V \in \mathcal{P}(Y)$. Therefore, we have

$$f^{-1} \circ g_* = (\mathrm{pr}_1)_* \circ \mathrm{pr}_2^{-1}.$$

This finishes the proof.

Item 2: The Projection Formula I

We claim that

$$f_*(U) \cap V \subset f_*\left(U \cap f^{-1}(V)\right).$$

Indeed, we have

$$
\begin{aligned}
f_*(U) \cap V &\subset f_*(U) \cap f_*\left(f^{-1}(V)\right) \\
&= f_*\left(U \cap f^{-1}(V)\right),
\end{aligned}
$$

where we have used:

1. Item 2 of Proposition 6.1.4 for the inclusion.
2. Item 6 of Proposition 6.1.4 for the equality.

Conversely, we claim that

$$f_*\big(U \cap f^{-1}(V)\big) \subset f_*(U) \cap V.$$

Indeed:

1. Let $y \in f_*\big(U \cap f^{-1}(V)\big)$.
2. Since $y \in f_*\big(U \cap f^{-1}(V)\big)$, there exists some $x \in U \cap f^{-1}(V)$ such that $f(x) = y$.
3. Since $x \in U \cap f^{-1}(V)$, we have $x \in U$, and thus $f(x) \in f_*(U)$.
4. Since $x \in U \cap f^{-1}(V)$, we have $x \in f^{-1}(V)$, and thus $f(x) \in V$.
5. Since $f(x) \in f_*(U)$ and $f(x) \in V$, we have $f(x) \in f_*(U) \cap V$.
6. But $y = f(x)$, so $y \in f_*(U) \cap V$.
7. Thus $f_*\big(U \cap f^{-1}(V)\big) \subset f_*(U) \cap V$.

This finishes the proof.

Item 3: The Projection Formula II

We have

$$\begin{aligned} f_!(U) \cap V &\subset f_!(U) \cap f_!\big(f^{-1}(V)\big) \\ &= f_!\big(U \cap f^{-1}(V)\big), \end{aligned}$$

where we have used:

1. Item 2 of Proposition 6.3.6 for the inclusion.
2. Item 6 of Proposition 6.3.6 for the equality.

Since $\mathcal{P}(Y)$ is posetal, naturality is automatic (**??**).

Item 4: Strong Closed Monoidality

This is a repetition of Item 19 of Proposition 4.7.4 and is proved there.

Item 5: The External Tensor Product

We have

$$\begin{aligned}
U \boxtimes_{X\times Y} V &\overset{\text{def}}{=} \mathrm{pr}_1^{-1}(U) \cap \mathrm{pr}_2^{-1}(V) \\
&\overset{\text{def}}{=} \{(x,y) \in X \times Y \mid \mathrm{pr}_1(x,y) \in U\} \\
&\quad \cup \{(x,y) \in X \times Y \mid \mathrm{pr}_2(x,y) \in V\} \\
&= \{(x,y) \in X \times Y \mid x \in U\} \\
&\quad \cup \{(x,y) \in X \times Y \mid y \in V\} \\
&= \{(x,y) \in X \times Y \mid x \in U \text{ and } y \in V\} \\
&\overset{\text{def}}{=} U \times V.
\end{aligned}$$

Next, we claim that Items 5a to 5d are indeed true:

1. *Proof of Item 5a:* This is a repetition of Item 16 of Proposition 6.1.4 and is proved there.

2. *Proof of Item 5b:* This is a repetition of Item 16 of Proposition 6.2.3 and is proved there.

3. *Proof of Item 5c:* This is a repetition of Item 15 of Proposition 6.3.6 and is proved there.

4. *Proof of Item 5d:* We have

$$\begin{aligned}
\Delta_X^{-1}(U \boxtimes_{X\times X} V) &\overset{\text{def}}{=} \{x \in X \mid (x,x) \in U \boxtimes_{X\times X} V\} \\
&= \{x \in X \mid (x,x) \in \{(u,v) \in X \times X \mid u \in U \text{ and } v \in V\}\} \\
&= U \cap V.
\end{aligned}$$

This finishes the proof.

Item 6: The Dualisation Functor

This is a repetition of Items 5 and 6 of Proposition 4.7.4 and is proved there.

Appendices

A Other Chapters

Sets

1. Sets
2. Constructions With Sets
3. Monoidal Structures on the Category of Sets
4. Pointed Sets
5. Tensor Products of Pointed Sets

Relations

6. Relations
7. Constructions With Relations
8. Equivalence Relations and Apartness Relations

Category Theory

9. Categories

Monoidal Categories

10. Constructions With Monoidal Categories

Bicategories

11. Types of Morphisms in Bicategories

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